

Learning and Decision-Making in Strategic Environments

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by

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To My Family

Declaration

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Abstract

Classical statistical learning typically assumes a stationary data-generating law and a fixed set of observed variables. Many practical applications, however, depart from this baseline. Strategic agents manipulate or selectively disclose information in response to deployed decision rules, while in other domains, features that are jointly predictive are never observed together during training. This dissertation investigates these two distinct structural departures.

Within the strategic framework, we examine three problems: strategic classification under non-uniform preferences, strategic feature revelation in regression, and strategic window dressing in Initial Public Offering (IPO) regulation. Formulating these settings as Stackelberg screening games, we derive structural properties of optimal decision rules and develop learning algorithms with provable generalization and regret guarantees. Notably, we show that accounting for strategic incentives not only provides robustness against manipulation, but can also elicit valuable private information that strictly improves performance over non-strategic baselines. In IPO regulation, we show that social welfare can attain a strict interior optimum, revealing an inherent regulatory trade-off.

For partitioned observations, which represent a structural rather than strategic departure, predictive features and labels are partitioned across training sets and never jointly observed. We characterize the resulting learning limits, develop an imputation-based empirical risk minimization framework, and identify conditions under which the optimal regularizer follows an interpretable three-candidate rule.

Together, these results show that data-driven decision making depends fundamentally on agent incentives and observation structure alongside the data-generating law, extending statistical learning theory to richer operational environments.

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List of Symbols

$\mathbb{R}, \mathbb{R}^d$	Real line, d -dimensional Euclidean space
$\mathbb{R}_+$	Set of non-negative real numbers $[0, \infty)$
$\mathbb{Q}$	Set of rational numbers
$\mathbb{N}$	Set of natural numbers $\{1, 2, 3, \dots\}$
$\mathbf{1}, I_d$	Vector of all ones, $d \times d$ identity matrix
$\ \cdot\ $	Standard Euclidean norm (ℓ_2 -norm)
$\langle \cdot, \cdot \rangle$	Standard Euclidean inner product
$\mathbb{1}_{\{\cdot\}}, \mathbb{1}_A$	Indicator function of an event or set A
$\mathbb{P}(\cdot)$	Probability measure of an event
$\mathbb{E}[\cdot]$	Mathematical expectation operator
$\text{Var}(\cdot)$	Variance of a random variable
$\text{Cov}(\cdot, \cdot)$	Covariance between random variables
$\mathcal{O}(\cdot), \tilde{\mathcal{O}}(\cdot)$	Big-O asymptotic notation, Big-O hiding logarithmic factors
$\mathcal{H}_k, \ \cdot\ _{\mathcal{H}_k}$	Reproducing Kernel Hilbert Space (RKHS) and RKHS norm for kernel k

Abbreviations

ERM	Empirical Risk Minimization
GMM	Gaussian Mixture Model
GP	Gaussian Process
GP-TS	Gaussian Process Thompson Sampling
<i>i.i.d.</i>	Independent and Identically Distributed
IPO	Initial Public Offering
OLS	Ordinary Least Squares
PAC	Probably Approximately Correct
PSIL	Partitioned Side Information Learning
RKHS	Reproducing Kernel Hilbert Space
SAR-ERM	Strategic Advantage Regression Empirical Risk Minimization
SERM	Strategy-aware Empirical Risk Minimization
VC	Vapnik–Chervonenkis

Chapter 1

Introduction

The standard framework of statistical learning deals with extracting underlying patterns from identically structured, stationary empirical observations [1, 2]. Under this baseline, past observations guide future actions with rigorous generalization guarantees. However, there exists an important class of real-world decision-making problems that depart from these core assumptions. This dissertation is motivated by the need to systematically study and extend statistical learning theory to two such distinct operational departures.

This dissertation investigates two key extensions of the classical baseline. The first extension considers strategic environments featuring an intelligent agent who actively influences the data-generating law, for instance, by manipulating feature values or selectively withholding specific feature components [3–5]. Consequently, statistical patterns learned from historical observations may no longer remain valid once the agent adapts to the deployed decision rule. Any effective data-driven decision-making procedure must therefore explicitly account for the incentive structure of the strategic agent in order to remain effective.

The second extension considers settings where observations depart structurally from a uniform feature baseline: predictive features and labels are partitioned across distinct training sets and never jointly observed during training [6, 7]. This represents a structural information constraint rather than an agent incentive problem. In the classical setting, collecting more samples simply increases the quantity of information. However, when observation structures vary, different samples contribute distinct qualities and quantities of information. Consequently, the total information available for learning depends fundamentally not only on the number of observations, but also on which variables are jointly observed.

The central objective unifying this dissertation is to characterize achievable performance and the fundamental limitations of learning under these two distinct departures. For strategic learning, we address core questions such as: *What is the optimal decision rule when every parameter of the environment is known (referred to as the decision problem in this dissertation)? How can this optimal structure be learned from finite data? How does the resulting performance compare with the non-strategic baseline?* For learning with partitioned information, we characterize the fundamental learning limits imposed by which variables are jointly observed, develop an imputation-based empirical risk minimization framework for such partitioned settings, and evaluate how the choice of imputation governs learning performance relative to the benchmark where complete joint observations are available.

To address these challenges, we adopt a unified two-stage methodology throughout the dissertation. We first analyze the underlying decision problem assuming complete knowledge of the environment, which unveils key structural properties of the optimal decision rule. These structural insights then directly guide the design of the learning algorithm. Rather than searching over a large, unstructured hypothesis space, the learning procedure searches over a structured space suggested by the analysis of the decision problem, yielding more sample-efficient and accurate learning. For strategic learning, the natural mathematical framework is game theory, specifically sender–receiver games where the decision maker acts as the leader committing to a rule, commonly known as screening games, evaluated under the Stackelberg equilibrium concept [3, 8]. Incorporating this game-theoretic structure explicitly accounts for the strategic behavior of agents during learning. The theoretical results in this dissertation either characterize structural properties of the optimal decision rule or reformulate the underlying decision problem into a structured optimization problem that is easier to analyze. For learning with partitioned observations, the central challenge arises from statistical dependencies that are never jointly observed during training. We address this through a family of imputation-based learning algorithms, where the choice of imputation directly determines the achievable learning performance.

The mathematical models studied in this dissertation are intentionally formulated with structural simplicity to maintain analytical tractability and physical interpretability. Nevertheless, the theoretical insights and results derived from these models serve as valuable guiding principles for more complex settings where exact analytical solutions may be unavailable. A recurring theme that emerges across strategic learning problems is that strategic behavior need not always be viewed purely as an obstacle to be eliminated. When modeled explicitly, strategic

actions can reveal additional information and, in certain settings, be leveraged to improve overall decision making.

Organization of the Dissertation

The chapters of this dissertation are organized as follows. Each chapter is self-contained and presents independent research contributions that can be read without prerequisite dependencies on the others.

Chapter 2 studies the strategic classification problem where agents have non-uniform preferences over labels. Existing works typically assume that all agents share a uniform preference for a single label. Relaxing this assumption fundamentally changes the structure of the strategic classification problem. The chapter investigates how strategic decision rules should differ from their non-strategic counterparts and whether deliberately making classification errors near the class boundary can reduce the overall strategic classification error. It then studies the associated learning problem and shows that explicitly incorporating the underlying game-theoretic structure can improve learnability in certain situations. The work presented in this chapter is based on our paper published in *Autonomous Agents and Multi-Agent Systems* (JAAMAS, 2025) [9], with preliminary portions appearing in NCC 2024 [10] and ICC 2022 [11].

Chapter 3 studies learning from partitioned datasets, where the target variable and the side information are never jointly observed during training but are both available at test time. Unlike the classical missing-feature setting, in which a limited number of joint observations still exist, the partitioned information structure prevents direct learning of the relationships between the target variable, the features, and the side information. The chapter characterizes the fundamental learning limits imposed by this information structure, examines a family of imputation-based learning algorithms, and investigates how the choice of imputation governs the achievable learning performance. It further shows that, under suitable conditions, the optimal bound-minimizing regularization strategy can be restricted to a small discrete set of candidates. The results in this chapter are based on our manuscript submitted to *Information and Inference: A Journal of the IMA* (Oxford University Press).

Chapter 4 studies a regression problem in which a strategic feature provider selectively reveals only a subset of its verifiable features to the learner. Unlike strategic manipulation,

where feature values are altered, the strategic action here lies in the choice of which features to disclose. The chapter establishes that, in certain cases, the information conveyed by the disclosure decision, together with the revealed features, can improve prediction performance relative to the non-strategic baseline. It then characterizes the optimal disclosure policy and studies the corresponding offline and online learning problems arising from this strategic interaction. The work in this chapter is based on our paper accepted at the *IEEE Conference on Decision and Control* (CDC, 2026) [12] and our manuscript submitted for publication [13].

Chapter 5 deals with the regulation of Initial Public Offerings (IPOs) [14], where the firm can strategically window dress its presentation to influence investor decisions, while the regulator determines the extent to which such manipulation should be permitted. The chapter establishes that, in certain cases, allowing a limited degree of strategic window dressing can improve market outcomes by correcting investors' bias against fundamentally good firms. It then studies the corresponding learning problem, where the regulator learns effective regulatory policies when the underlying market environment is unknown. The results in this chapter are based on our manuscript submitted for publication [15].

Publications

Journal Articles:

1. M. K. Singh and A. A. Kulkarni. “Strategic Classification for Non-uniform Preferences using Penalty Labels and Randomisation,” *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 2025.
2. M. K. Singh, S. S. Kowshik, and A. A. Kulkarni. “Learning with Side Information from Partitioned Data,” *Information and Inference: A Journal of the IMA* (Oxford University Press), (Submitted).
3. M. K. Singh, A. A. Kulkarni, and K. Mahajan. “Learning under Strategic Feature Revelation,” (Submitted for publication, 2026).
4. M. K. Singh and A. A. Kulkarni. “Strategic IPO Regulation under Complete and Incomplete Information,” (Submitted for publication, 2026).

Conference Articles:

1. M. K. Singh, A. A. Kulkarni, and K. Mahajan. “Learning under Strategic Feature Revelation,” *IEEE Conference on Decision and Control (CDC)*, 2026 (Accepted).
2. M. K. Singh and A. A. Kulkarni. “Optimal Stochastic Decision Rule for Strategic Classification,” *National Conference on Communications (NCC)*, 2024.
3. M. K. Singh and A. A. Kulkarni. “Strategic Multiclass Classification with Non-uniform Preferences and its Relation to Incentive Compatibility,” *Indian Control Conference (ICC)*, 2022.

Chapter 2

Strategic Classification for Non-uniform Preferences using Penalty Labels and Randomisation

Strategic classification is the problem of classifying feature vectors that can be manipulated at the testing phase of the classifier. The classifier aims for its decision rule to be robust against strategic manipulation and also be efficiently learnable. In this chapter, we present two main ideas for the classifier to achieve this goal. The first idea is to enrich the classifier’s decision-making capabilities in two ways: ignoring feature vectors and making randomized decisions. This approach helps the classifier classify more feature vectors correctly by limiting the sender’s options for manipulation. The second idea is to use the strategic structure of the problem during learning. This approach, in certain contexts, can simplify the learning process. Combining these ideas, we propose and compare two learning algorithms, Vanilla ERM and Strategy-aware ERM, and provide a heuristic for selecting the one that yields a classifier both robust to strategic manipulation and efficiently learnable.

2.1 Introduction

Consider the following classification situation. A university assigns students to various programs based on their scores in an aptitude test using a decision rule that maps scores to programs. For example, one such rule could be to assign the Physics program to students who have scored

well in the physics section of the exam, regardless of their total score. For transparency, the university must declare its decision rule before the students write the exam. Every student taking the exam has a *true aptitude* corresponding it with a *true program*. However, the students may have preferences for programs that may differ from their true programs. Knowing the decision rule, they can seek external help, such as coaching, to improve their scores, incurring some costs for such assistance. Now suppose the university offers two related programs (say theoretical and experimental physics programs), where the students with an aptitude in one program may prefer the other program and, therefore, prepare for the entrance exam accordingly. The university aims to create a question paper for the entrance exam in such a manner as to assign the maximum number of students to their true programs. The challenge is that the university cannot measure the true aptitude of a student and must rely on the exam scores, i.e., its *demonstrated aptitude*, to assign the program. Moreover, the university does not know the *true function* (which maps true aptitude to a true program) and must learn it from past data.

The university admission problem is an example of a *strategic* classification problem. The strategic classification problem extends the classification problem to the regime where, during the classifier's testing phase, a strategic data sender can manipulate feature vectors to obtain a preferred label while incurring some cost. A non-strategic or *naive classifier* places the decision boundary precisely at the class boundary. Since this classifier's decision rule is known to the strategic sender, it is prone to manipulation. A *strategic classifier* is robust to such manipulation and, in general, differs non-trivially from the naive classifier.

Existing works on the strategic classification problem typically assume that the sender has a uniform preference for the labels (e.g., [3]). The standard example of this setting is the email classification problem, where the strategic sender prefers the non-spam label over the spam label, irrespective of the underlying email content. The strategic classifier for this case is the one that shifts the decision boundary into the *non-spam* region of the feature space, thereby making the criterion for classification as the non-spam label more stringent.

This chapter considers a binary classification problem where the strategic sender has a possibly non-uniform preference for labels, as illustrated in the above example for admission to theoretical and experimental physics programs. This assumption poses a new challenge as preferred feature space for one class overlaps with the space not preferred by the other. In particular, the structure of the strategic classifier obtained by [3] no longer holds without a

universally preferred label. The problem of constructing a strategic classifier with non-uniform preferences results in two subproblems: a *decision problem* and a *learning problem*. In the decision problem, the university knows the *true function* (map between the true aptitude and the true program) and has to calculate the strategic classifier from it. In the learning problem, the university does not know the true function and has to learn the strategic classifier from a finite sample of the true function.

Our main findings are that classification performance improves significantly when the classifier is either allowed to make randomized decisions or allowed to *not* classify certain feature vectors. Moreover, in certain problem instances, particularly those involving complex hypothesis classes, learning can be made easier by incorporating the sender’s strategic behaviour in the classifier’s learning phase.

2.1.1 Contributions

We adopt a game theoretic viewpoint for our analysis. We model the binary classification problem with non-uniform preferences as a screening game[†] between a sender and a receiver (classifier), with the receiver as the leader. The sender wants to maximize the difference between its utility and its cost and the receiver wants to maximize the probability of correctly classifying the feature vectors. The strategic classifier corresponds to the receiver’s Stackelberg equilibrium strategy. For most of our results, we work with a scalar feature space. We show that under certain assumptions, the problem with the vector feature space with a linear class boundary can be reduced to the problem we consider.

We introduce two main ideas. The first is to expand the classifier’s capabilities in two ways. One way is by introducing a new penalty label Δ , by which means the receiver ignores a feature vector assigned to this label. From the sender’s perspective, this situation is equivalent to receiving a $-\infty$ utility for the label Δ . In the context of university admission problems, the student is the sender, and the university is the receiver. The exam paper comprises of a subset of questions from a question bank, with the requirement that the student has to attempt exactly one question. The university assigns a label to each question in the exam paper with the implication that a student who correctly answers that question gets assigned that label or

[†]In a screening game, the receiver commits to a strategy first; the sender then observes this commitment and chooses a strategy in response, after which the receiver plays the committed strategy.

program. Assignment of label Δ to a question amounts to *dropping* that question from the exam paper. This approach is in stark contrast to the uniform preference case, where *all* the questions in the question bank are included in the question paper, and only *cut-off marks* for the preferred program are increased. Ignoring feature vectors can also be considered a generalization of feature selection.

Another way to expand the classifier’s action set is by allowing randomized classification. In the university admission problem, this is analogous to saying that solving a particular question is correlated (but not with certainty) with admission to a specific program. These ideas for expanding the classifier’s action set are motivated from mechanism design, where penalties influence a strategic sender’s behavior, and game theory, where randomization allows for richer outcomes.

The second main idea involves incorporating the underlying game-theoretic structure of the problem by modifying the hypothesis class used for learning. Learning from the original hypothesis class will be inefficient if the university’s inductive bias does not consider students’ strategic nature. We show that incorporating the game’s strategic structure through hypothesis class modification can make an unlearnable hypothesis class learnable.

Using these two ideas, we address the decision and learning problems of strategic classification in the scalar feature space setting. For the decision problem, our first finding is that a deterministic strategic classifier using a penalty label makes fewer classification errors than any classifier that does not use a penalty label, at least strictly fewer errors in some instances of the problem. With probability $1 - \delta$ (where we will write this in sections), a stochastic strategic classifier makes fewer errors than any deterministic classifier, including those that use a penalty label, and at least strictly fewer errors than a classifier with a penalty label in some instances of the problem.

However, the stochastic classifier uses only randomization on the original label set; using the penalty label does not increase its accuracy. Nonetheless, the penalty label becomes relevant when it is constrained to use a deterministic strategic classifier, a common situation encountered in real-life applications. Hence, the penalty label can be thought of as partially compensating the classifier for the absence of randomization.

The classifier described above assigns the label Δ to feature vectors near the class boundary. Indeed, multiple such classifiers exist, all with the same classification error, but they differ in

the following three aspects. The first aspect is classifier complexity, measured by the number of decision boundaries. A decision boundary is a feature vector at which the classifier changes its assigned label. Thus, the number of decision boundaries corresponds to the number of times the classifier switches labels as the feature vector varies. We show that classifiers with fewer decision boundaries are easier to learn due to their lower Vapnik–Chervonenkis (VC) dimension. The second aspect is classifier fairness, which refers to fair distribution of errors across classes. If in addition to classification performance, fairness is also in consideration, then the receiver can pick among the many classification rules, the one which proportionately distributes the error across classes. The third aspect is the sender’s payoff under a strategic classifier. This aspect becomes relevant if a sender can opt-out and obtain a reservation payoff. Thus, a receiver may select a classifier that gives the sender a payoff higher than its reservation payoff. Although we work with scalar features, our problem suffices to demonstrate the structural aspects of strategic classification and the importance of a penalty label. Moreover, in many university admission problems like those we consider, the final admission decision has to be done based on a single scalar score, whereby the scalar problem is practically relevant.

We introduce two new algorithms called vanilla ERM and strategy-aware ERM to address the learning problem. In vanilla ERM, the classifier first uses ERM to learn the true function from the original hypothesis class. The decision problem is then applied to the learned hypothesis. In the strategy-aware ERM approach, the hypothesis class is modified to incorporate the underlying game-theoretic structure. This modification involves a two-step process. First, each hypothesis in the original class is replaced with its corresponding strategic classifier, which is obtained by applying the decision problem to each hypothesis. Second, the strategic classifier obtained in the previous step is further replaced with a new hypothesis, formed by composing the strategic classifier with the sender’s worst-case best response. The classifier then uses the Empirical Risk Minimization (ERM) algorithm to learn from this modified hypothesis class. What makes strategy-aware ERM novel is its ability to transform an otherwise unlearnable problem into a learnable one. We show that there exist hypothesis classes that are unlearnable under vanilla ERM but become learnable with strategy-aware ERM, illustrating how incorporating game-theoretic structure can fundamentally improve learnability.

While strategy-aware ERM can learn from a broader range of hypothesis classes, its advantage in the ease of learning over vanilla ERM only sometimes holds. This ease of learning is controlled by the VC dimension of the hypothesis class, which in turn determines the

sample complexity in PAC learning. The comparison, therefore, becomes context-dependent if a hypothesis class is learnable via vanilla ERM. We provide a heuristic argument that strategy-aware ERM generally makes learning easier if the original hypothesis class is too complex, as it can reduce the effective VC dimension. On the other hand, vanilla ERM proves more efficient if the original hypothesis class is simple. We show this with the help of an example. A formal result, even for scalar features, remains elusive.

2.1.2 Related Work

This chapter contributes to the literature on strategic classification and learning in the presence of strategic agents [8, 16–20], where interactions are commonly formulated as games between a strategic sender and a classifier. We model this interaction as a sender–receiver screening game and introduce a penalty label that expands the receiver’s strategy space, giving rise to equilibrium strategies robust against the sender’s manipulation. Preliminary versions of the results in this chapter appeared in the conference papers [10, 11]. This chapter is derived from our manuscript [9]. To the best of our knowledge, the setting of strategic classification under non-uniform preferences with penalty labels has not been previously studied, and was first explored in our work [9].

Works such as [3, 21–23] consider the impact of a strategic sender on the PAC learnability of the hypothesis class. In [3, 21, 22], the sender is assumed to have a uniform preference for positive labels, whereas in [23], the sender is assumed to have a non-uniform preference for labels for each feature vector. In [3, 21, 23], strategic manipulation incurs a cost, whereas in [22], strategic manipulation does not incur a cost. The works [3, 21–23] do not explicitly utilize the underlying sender-receiver game in the learning process. The learning process in these works can be equivalently described as using the ERM algorithm to optimize over a modified hypothesis class, obtained by replacing each hypothesis with the composition of the hypothesis and the sender’s best response to that hypothesis. In contrast, in this chapter, the sender has a non-uniform preference over labels for each feature vector, and the structure of this preference differs from that in [23]. Furthermore, in this chapter, there is a cost incurred for modifying a feature vector. We explicitly make use of the underlying sender-receiver game in the learning process by allowing the ERM algorithm to optimize over a modified hypothesis class, which is obtained by replacing each hypothesis with its corresponding Stackelberg equilibrium composed

with the sender's best response.

There are other interesting formulations of the problem that we do not consider in this chapter. For instance, some research explores scenarios where the sender lacks knowledge of the classifier. Notable works in this area include [16], [17], and [18]. Additionally, other studies introduce more structure to the sender by incorporating user types, as seen in [19] and [24].

Finally, the social welfare implications of strategic classification have been explored in works such as [25], which analyze the social costs imposed on the sender. In this chapter, we briefly address related issues such as fairness and participation constraints in Section 2.3.6.

2.2 Problem Formulation

In this section, we formulate the strategic classification problem. Recall that in strategic classification, the sender can alter the feature vector at a cost during the classifier's testing phase to obtain a preferred label.

We formalize the strategic classification problem within this framework. Let $\mathcal{X}$ be the set of feature vectors; for simplicity we take $\mathcal{X} := [0, 1]$. Consider an abstract probability space and let X be a $\mathcal{X}$ -valued random variable defined on that space. Let $\mathcal{L}(X)$ denote the probability law of X on $\mathcal{X}$. Let $\mathfrak{D}$ be the collection of laws of X such that the corresponding cumulative distribution function, denoted as $\psi(\cdot \cdot \cdot)$, is differentiable and has a differentiable inverse except at finitely many points. We use D to represent an element of $\mathfrak{D}$. Let $\mathcal{Y} := \{-, +\}$ be the set of labels. Let $h : \mathcal{X} \rightarrow \mathcal{Y}$ be a *true function* that maps each feature $x \in \mathcal{X}$ to its corresponding *true label* $y \in \mathcal{Y}$. We assume the true function to be given by

$$h(x) := \begin{cases} -, & x \in [0, \mathbf{h}] \\ +, & x \in (\mathbf{h}, 1] \end{cases}, \quad (2.1)$$

where $0 \leq \mathbf{h} \leq 1$. We refer to $\mathbf{h}$ as the class boundary. We call a label a *majority class* label if the corresponding feature vectors whose true label is the said label has a probability (under probability law $D \in \mathfrak{D}$) greater than or equal to half. The other label is called the *minority label*.

We model the strategic classification problem as a game between two players, a sender and a receiver (classifier), who engage in a sender-receiver game with the receiver acting as the leader. The receiver is interested in correctly classifying the feature vectors, whereas the sender

is interested in obtaining a preferred label for each feature vector by potentially misreporting the feature vector. The gameplay proceeds as follows. The sender observes a realized feature vector $x \in \mathcal{X}$ drawn according to a probability law $D \in \mathfrak{D}$ and modifies it to $s(x) \in \mathcal{X}$ according to a rule s . It then communicates $s(x)$ to the receiver, who then classifies it with a label $r(s(x))$ according to a rule r .

The sender's strategy set for modifying feature vectors is given by

$$\mathbf{S} := \{s \mid s : \mathcal{X} \rightarrow \mathcal{X}\} \quad (2.2)$$

and the receiver's strategy set for classification is given by

$$\mathbf{R} := \{r \mid r : \mathcal{X} \rightarrow \mathcal{Y} \cup \Delta\}. \quad (2.3)$$

Here, label Δ is an additional label, which we call the penalty label, whose interpretation will be provided later in this section. We refer to the set $\{x : r(x) = \Delta\}$ of feature vectors with label Δ as the Δ -region.

We analyze the above game as a screening game; the receiver first commits a strategy $r \in \mathbf{R}$, and then the sender, who knows the committed strategy, plays a strategy $s \in \mathbf{S}$, and finally the receiver plays his committed strategy r . Notice the asymmetry in the role of the sender and the receiver. The receiver can perform classification but cannot access the private information of the sender (the feature vector x). In contrast, the sender has access to private information but can not do classification and has to depend on the receiver to get his (preferred) label.

The sender's preference for the label for each feature vector is represented by a utility function $U : (\mathcal{Y} \cup \Delta) \times \mathcal{X} \rightarrow \mathbb{R} \cup \{-\infty\}$. We assume it to be given by

$$U(y, x) := \begin{cases} \mathbf{u}_+ & x \in [0, \mathbf{h}], y = + \\ \mathbf{u}_- & x \in (\mathbf{h}, 1], y = - \\ 0 & y = h(x) \\ -\infty & y = \Delta \end{cases}, \quad (2.4)$$

where $\mathbf{u}_-, \mathbf{u}_+ \in \mathbb{R}$. Note that our choice of the utility function is more general than the one considered in [3], as it allows the sender to have a non-uniform preference for labels at each feature vector, making the receiver's task more difficult. The cost incurred by the sender when modifying x to x' is given by $C(x, x') = |\phi(x') - \phi(x)|$, where the function $\phi : \mathcal{X} \rightarrow [0, 1]$,

called cost kernel, is a strictly monotonically increasing, invertible, continuous function such that $\phi(0) = 0$ and $\phi(1) = 1$.

In the strategic classification problem, for a strategy (r, s) and a feature vector $x \in \mathcal{X}$, the sender's payoff at x is given by

$$\mathbb{U}(x; r, s) := U(r(s(x)), x) - C(s(x), x). \quad (2.5)$$

For a probability law $D \in \mathfrak{D}$ on $\mathcal{X}$, the sender overall payoff, which it wants to maximize, is given by

$$\mathcal{U}(r, s) := \mathbb{E}_{x \sim D} [\mathbb{U}(x; r, s)]. \quad (2.6)$$

For a strategy (r, s) and a function $f : \mathcal{X} \rightarrow \mathcal{Y}$, the error set $E(r, s; f)$ be defined as

$$E(r, s; f) := \{x \in \mathcal{X} : r(s(x)) \neq f(x)\}. \quad (2.7)$$

For a probability law $D \in \mathfrak{D}$ with a differentiable distribution function $\psi(x)$, the receiver's error, which it wants to minimize, is given by

$$\mathcal{E}(r, s; h, D) = D(E(r, s; h)) = \int_0^1 \mathbb{1}_{E(r, s; h)} \psi'(x) dx. \quad (2.8)$$

Here $\mathbb{1}_A$ denotes an indicator function on the set A and $\psi'(x) = \frac{d\psi(x)}{dx}$

Remark. We assume the set $\{x \in \mathcal{X} | r(s(x)) \neq h(x)\}$ is measurable w.r.t. D .

2.2.1 Interpretation of the penalty label

We now provide an interpretation of the penalty label. If the receiver classifies a feature vector x as Δ , i.e., $r(x) = \Delta$, then the receiver ignores the feature vector x . We assume that ignoring x results in an error for the receiver. From (2.4), $U(\Delta) = -\infty$, which implies that the penalty label is the least preferred label for the sender.

The receiver can exploit this fact to improve its classification accuracy. This property, where the receiver ignores certain feature vectors, is one of the novel contributions of this chapter. It is not considered in [3], and we later show that it helps the receiver reduce overall misclassification.

2.2.2 Solution concept

We now describe the solution concept of our problem. For a given receiver strategy r , the best response strategies of the sender are defined as follows.

Definition 2.1. Let r be a receiver strategy. Let $B(r)$ be the set of sender strategies s which satisfy

$$s(x) \in \arg \max_{x' \in \mathcal{X}} U(r(x'), x) - C(x', x) \quad \forall x \in \mathcal{X}.$$

The set $B(r)$ is called the best response set.

There can be multiple best responses. We assume that the receiver anticipates that the sender will play a worst-case best response strategy, which is defined as follows.

Definition 2.2 (Worst case best response). A sender strategy s_r is called a worst-case best response to a receiver strategy r if

$$s_r \in \arg \min_{s' \in B(r)} D(\{x : r(s'(x)) = h(x)\}) \quad \text{and}$$

$$\{x : r(s'(x)) = h(x)\} \not\subset \{x : r(s_r(x)) = h(x)\} \quad \text{for all } s' \in B(r).$$

Remark. We always assume the receiver's strategy set $\mathbf{R}$ is such that $B(r)$ is non-empty and s_r is well defined.

Notation. We will generally drop various arguments in $\mathcal{E}(\cdot, \cdot, \cdot, \cdot)$ when they are apparent from the context. Specifically, when h and ψ are clear from the context, will sometimes be denoted as $\mathcal{E}(r) := \mathcal{E}(r, s_r; h, \psi)$.

The sender-receiver pair of strategies $(r, s_r), r \in \mathbf{R}$ which minimises the receiver's error is called a Stackelberg equilibrium strategy or strategic classifier and is defined as follows.

Definition 2.3 (D -Stackelberg equilibrium). Let D be a probability law on $\mathcal{X}$ and $h(x)$ be a true function. A receiver strategy $r_{h,D}^* \in \mathbf{R}$ is called a D -Stackelberg equilibrium strategy if

$$r_{h,D}^* := \arg \min_{r \in \mathbf{R}} \mathcal{E}(r, s_r; h, D)$$

and $\mathcal{E}(r, s_r; h, D)$ is called the Stackelberg error.

Notation. Sometimes, we will drop the subscript from $r_{h,D}$ if it is clear from the context.

Notation. For a Stackelberg equilibrium (r, s_r) , since the sender strategy is directly derived from the receiver strategy r , we sometimes refer to r alone as the Stackelberg equilibrium.

We now consider the two subproblems of strategic classification: the *decision problem* and the *learning problem*. The setup discussed so far in this section remains common to both of them. In the decision problem, both the sender and the receiver know the true function h , and the receiver has to calculate the Stackelberg equilibrium from it. In the learning problem, the receiver does not know the true function h . However, the receiver has access to a hypothesis class $\mathcal{F}$ and a sample $S = \{(x_i, h(x_i))\}_{i=1}^m$ from which it has to estimate the Stackelberg equilibrium. Here, x_i 's are the samples which are drawn in an iid manner from a probability law D on $\mathcal{X}$ and $h(x_i)$'s are the corresponding true labels.

2.3 Decision Problem

In this section, we will characterise the Stackelberg equilibrium for the decision problem.

2.3.1 Motivating example: discrete feature space

To motivate the solution of the decision problem, we digress and recall a related problem from [11]. The problem formulation is almost the same as described in Section 2.2, and we only highlight the differences. Let $\mathcal{X} = \{1, 2, \dots, n\}$ be a finite set, and $h : \mathcal{X} \rightarrow \mathcal{Y}$ be the true function. Let $U : \mathcal{Y} \times \mathcal{X} \rightarrow \mathbb{R}$ be the utility function such that $U(h(x), x) = 0$ and $C : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be the cost function such that $C(x, x) = 0$. We now characterise the Stackelberg equilibrium strategy in terms of the independence number of a sender graph. The *sender graph* is defined as follows.

Definition 2.4. Let $G_h = (V, E)$ be a graph whose node set is $V = \mathcal{X}^2$. The edge set $E \subset \mathcal{X}^2 \times \mathcal{X}^2$ contains an undirected edge between distinct nodes $(\mathbf{i}, \mathbf{k})$ and $(\mathbf{j}, \mathbf{l})$ if at least one of the following conditions holds:

- (a) Either $\mathbf{i} = \mathbf{j}$ and $\mathbf{k} \neq \mathbf{l}$; or $h(\mathbf{i}) \neq h(\mathbf{j})$ and $\mathbf{k} = \mathbf{l}$.
- (b) $h(\mathbf{i}) \neq h(\mathbf{j})$ and either

$$U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) \geq 0 \text{ or}$$

$$U(h(\mathbf{i}), \mathbf{j}) - C(\mathbf{k}, \mathbf{j}) + C(\mathbf{l}, \mathbf{j}) \geq 0.$$

Recall that a subset of nodes in a graph is called an independent set if no edge connects any two nodes in that subset. The next theorem characterises the Stackelberg equilibrium strategy

in terms of the maximum independent set of the sender graph.

Theorem 2.5. Let I be an independent set of the graph G_h and $r(\cdot; I)$ be a receiver strategy defined as

$$r(\mathbf{k}; I) := \begin{cases} h(\mathbf{i}) & \text{if } \exists \mathbf{i} \text{ such that } (\mathbf{i}, \mathbf{k}) \in I \\ \Delta & \text{otherwise.} \end{cases}$$

The following statements hold.

- (a) If $I^\star$ is a maximum independent set of G_h , then $r^\star(\cdot) := r(\cdot; I^\star)$ is a Stackelberg equilibrium.
- (b) Conversely, if $r^\star(\cdot)$ is a Stackelberg equilibrium, then $I^\star := \{(x, s_{r^\star}(x)) : r^\star(s_{r^\star}(x)) = h(x)\}$ is a maximum independent set in G_h .

Proof. The proof is given in Section 2.6 (Proof 2.6.1). □

To establish the equivalence between an independent set of graph G_h and the correct classification of feature vectors, we rely on the following structural lemma.

Lemma 2.6. Let I be an independent set of the graph G_h and $r(\cdot; I)$ be a receiver strategy defined as

$$r(\mathbf{k}; I) := \begin{cases} h(\mathbf{i}) & \text{if } \exists \mathbf{i} \text{ such that } (\mathbf{i}, \mathbf{k}) \in I \\ \Delta & \text{otherwise.} \end{cases}$$

By the definition of I , $r(\cdot; I)$ is well defined. The following statements hold:

- (a) If I is an independent set of G_h , then for the receiver strategy $r(\cdot) := r(\cdot; I)$, every feature vector $\mathbf{i}$ for which there exists $\mathbf{k}$ such that $(\mathbf{i}, \mathbf{k}) \in I$ is correctly classified, i.e., $r(s_r(\mathbf{i})) = h(\mathbf{i})$. In particular, $|\{x \in \mathcal{X} : r(s_r(x)) = h(x)\}| \geq |I|$.
- (b) If $r(x)$ is a receiver strategy, then $I := \{(x, s_r(x)) : r(s_r(x)) = h(x)\}$ is an independent set in G_h , and $|I| = |\{x \in \mathcal{X} : r(s_r(x)) = h(x)\}|$.

Proof. The proof is given in Section 2.6 (Proof 2.6.1). □

We have shown that for the decision problem for a discrete feature vector set, the Stackelberg equilibrium strategy r is better than the naive classifier h . One of the reasons for this is that r can strategically choose to ignore feature vectors that are manipulation-prone.

2.3.2 Deterministic strategies for the receiver

In this subsection, we address the decision problem formulated in Section 2.2. We analyze the problem for various cases of utility functions.

Case 1: $u_-, u_+ \leq 0$

We show that this case is uninteresting as the objectives of the sender and the receiver are aligned as seen in the next theorem.

Theorem 2.7. Let $u_+, u_- \leq 0$. Then, $r(x) \equiv h(x)$ is a Stackelberg equilibrium.

Proof. The proof is given in Section 2.6 (Proof 2.6.2). □

Case 2: $u_+ \geq 1 \geq u_-$

For this case we show that any receiver r whose range contains the $+$ label, can at most only correctly classify feature vectors whose true label is $+$.

Theorem 2.8. The receiver strategy $r(x) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label, is a Stackelberg equilibrium.

Proof. The proof is given in Section 2.6 (Proof 2.6.2). □

Case 3: $u_-, u_+ \geq 1$

For this case, we show that for any receiver strategy r whose range contains both $-$ and $+$ labels does not correctly classify any feature vector.

Theorem 2.9. The receiver strategy $r(x) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label is a Stackelberg equilibrium.

Proof. The proof is given in Section 2.6 (Proof 2.6.2). □

Case 4: $1 \geq \mathbf{u}_+ \geq 0$ and $-\mathbf{u}_+ < \mathbf{u}_- \leq \mathbf{u}_+$

This case includes scenarios where the sender has a strictly non-uniform preference ($\mathbf{u}_- \geq 0$) as well as scenarios where the sender has a uniform preference ($-\mathbf{u}_+ < \mathbf{u}_- < 0$). We show that in each of these cases, there exists a receiver Stackelberg equilibrium strategy that belongs to the family of strategies $\mathcal{R}$ defined as follows.

Let $\mathbf{h}^-, \mathbf{h}^+$ be such that $0 \leq \mathbf{h}^- \leq \mathbf{h} < \mathbf{h}^+ \leq 1$ and $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+$. Since ϕ is a continuously increasing function, such $\mathbf{h}^-, \mathbf{h}^+$ exist. Let $r(\cdot; \mathbf{h}^-, \mathbf{h}^+)$ denote a receiver strategy defined as follows

$$r(x; \mathbf{h}^-, \mathbf{h}^+) := \begin{cases} h(x) & \text{if } x \in [0, \mathbf{h}^-] \cup [\mathbf{h}^+, 1] \\ \Delta & \text{if } x \in (\mathbf{h}^-, \mathbf{h}^+) \end{cases}.$$

Let

$$\mathcal{R}_1 := \{r(\cdot; \mathbf{h}^-, \mathbf{h}^+) : 0 \leq \mathbf{h}^- \leq \mathbf{h} < \mathbf{h}^+ \leq 1, \phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+\}.$$

The strategy $r(\cdot, \mathbf{h}^-, \mathbf{h}^+) \in \mathcal{R}_1$ is identical to the true function h except in the interval $(\mathbf{h}^-, \mathbf{h}^+)$, where it maps the feature vectors to label Δ . Let

$$\mathcal{R}_2 := \{r(\cdot) : r(\cdot) \equiv -\} \cup \{r(\cdot) : r(\cdot) \equiv +\},$$

be the set of constant-label receiver strategies. Let $\mathcal{R} := \mathcal{R}_1 \cup \mathcal{R}_2$.

We define dominance between receiver strategies as follows: for a given probability law D on $\mathcal{X}$, a strategy r dominates another strategy $\tilde{r}$ if $\mathcal{E}(r) \leq \mathcal{E}(\tilde{r})$. The next theorem shows that the family of receiver strategies $\mathcal{R}$ dominates every other receiver strategy. Hence, a receiver can restrict its search for a Stackelberg equilibrium to the set $\mathcal{R}$ itself.

Theorem 2.10. Let $D \in \mathfrak{D}$ be a law on $\mathcal{X}$ and let $h(x)$ be a true function. Let ϕ be a cost kernel, with $0 < \mathbf{u}_+ \leq 1$ and $-\mathbf{u}_+ < \mathbf{u}_- \leq \mathbf{u}_+$. Then there exists a Stackelberg equilibrium strategy $r(\cdot) \in \mathcal{R}$.

Proof. The proof is given in Section 2.6 (Proof 2.6.2). □

Case 5: $1 \geq \mathbf{u}_+ \geq 0$ and $\mathbf{u}_- \leq -\mathbf{u}_+$

This case corresponds to the sender having a uniform preference towards $+$ label. The next theorem specifies a Stackelberg equilibrium.

Theorem 2.11. Let $D \in \mathfrak{D}$ be a probability law on $\mathcal{X}$ and let $h(x)$ be a true function. If $\phi(\mathbf{h}) + \mathbf{u}_+ \leq 1$, let $\mathbf{h}^+ \in \mathcal{X}$ be such that $\phi(\mathbf{h}^+) - \phi(\mathbf{h}) = \mathbf{u}_+$. Then the receiver strategy

$$r(x; \mathbf{h}, \mathbf{h}^+) := \begin{cases} - & \text{for } x \in [0, \mathbf{h}], \\ \Delta & \text{for } x \in (\mathbf{h}, \mathbf{h}^+), \\ + & \text{for } x \in [\mathbf{h}^+, 1], \end{cases}$$

is a Stackelberg equilibrium with $\mathcal{E}(r) = 0$.

Proof. The proof is given in Section 2.6 (Proof 2.6.1). □

Remark. For the case where $1 \geq \mathbf{u}_+ \geq 0$, $\mathbf{u}_- \leq -\mathbf{u}_+$, and $\phi(\mathbf{h}) + \mathbf{u}_+ > 1$, we do not yet have an analytical structure for the Stackelberg equilibrium that holds for all $D \in \mathfrak{D}$.

2.3.3 Multi-dimensional decision problem

So far, we have analyzed a decision problem with a feature vector space $\mathcal{X} = [0, 1]$, and derived structural results for the Stackelberg equilibrium in this context. In this section, we generalize our analysis to a higher-dimensional feature space. We show that, under certain assumptions on the feature space, this higher-dimensional setting can be reduced to the one described in Section 2.2.

We focus on decision problems where the class boundary is defined by a linear hyperplane. Such scenarios commonly arise when decisions are based on a weighted sum of features, for example, admission decisions in academic institutions often depend on weighted combinations of subject scores. Our objective is to compute the Stackelberg equilibrium for this multi-dimensional decision problem.

Let $w \in \mathbb{R}^n$ be a unit vector, and let $\mathcal{X} \subset \mathbb{R}^n$ be a bounded convex set denoting the set of feature vectors. Assume the distribution D on $\mathcal{X}$ is uniform. Let $\mathbf{u}_- = \mathbf{u}_+ = 2\mathbf{u} > 0$. Let the true function $h : \mathcal{X} \rightarrow \mathcal{Y}$ be given by

$$h(x) = \begin{cases} + & \text{if } w^\top x > h, \\ - & \text{if } w^\top x \leq h, \end{cases} \quad (2.9)$$

where $h \in \mathbb{R}$. Let $\mathcal{Z} := \{z \in \mathcal{X} : w^\top z = h\}$ denote the class boundary set. We assume that the feature space $\mathcal{X}$ is sufficiently wide along the direction of w , such that for any feature vector

$z \in \mathcal{Z}$ on the class boundary, the feature vectors $z + 2\mathbf{u}w$ and $z - 2\mathbf{u}w$ remain within $\mathcal{X}$. Since $\mathcal{X}$ is convex, the line segment $\{z + tw : t \in [-2\mathbf{u}, 2\mathbf{u}]\}$ lies entirely in $\mathcal{X}$ for every $z \in \mathcal{Z}$. Let the cost kernel be the Euclidean norm, i.e., $\phi(x, x') = \|x - x'\|$. All other assumptions from Section 2.2 remain unchanged.

Theorem 2.12. Let h given in (2.9) be the true function and let D be the uniform distribution on $\mathcal{X}$. Let $r : \mathcal{X} \rightarrow \mathcal{Y} \cup \Delta$ be the receiver's strategy. Then $\mathcal{E}(r) \geq 2\mathbf{u}\sigma(\mathcal{Z})$, where $\sigma(\mathcal{Z}) := \mathcal{H}^{n-1}(\mathcal{Z})/\text{Vol}(\mathcal{X})$ is the surface area of $\mathcal{Z}$ induced by D .

Proof. The proof is given in Section 2.6 (Proof 2.6.2). □

We now show that the strategy r described in the following theorem is a Stackelberg equilibrium strategy.

Theorem 2.13. Let h be the true function, and let D be the uniform distribution on $\mathcal{X}$. Let the receiver's strategy r be defined as

$$r(x) := \begin{cases} + & \text{if } w^\top x \geq \mathbf{h} + \mathbf{u} \\ - & \text{if } w^\top x \leq \mathbf{h} - \mathbf{u} \\ \Delta & \text{otherwise} \end{cases}$$

Then r is a Stackelberg equilibrium, and $\mathcal{E}(r) = 2\mathbf{u}\sigma(\mathcal{Z})$, where $\sigma(\mathcal{Z}) := \mathcal{H}^{n-1}(\mathcal{Z})/\text{Vol}(\mathcal{X})$ is the surface area of $\mathcal{Z}$.

Proof. The proof is given in Section 2.6 (Proof 2.6.3). □

This theorem shows that if the class boundary is a hyperplane, then the decision boundaries of the receiver's Stackelberg equilibrium are also hyperplanes parallel to the class boundary hyperplane. We now show that this problem can be reformulated into our setup (Section 2.3) such that the respective Stackelberg equilibria remain consistent with each other. Let $H_c := \{x \in \mathcal{X} : w^\top x = \mathbf{h}\}$ denote the class boundary, and let $H_+ := \{x \in \mathcal{X} : w^\top x = \mathbf{h} + \mathbf{u}\}$ and $H_- := \{x \in \mathcal{X} : w^\top x = \mathbf{h} - \mathbf{u}\}$ denote the decision boundaries of the corresponding Stackelberg equilibrium. Let

$$a_{\min} := \min_{x \in \mathcal{X}} w^\top x \quad \text{and} \quad a_{\max} := \max_{x \in \mathcal{X}} w^\top x.$$

It follows that $a_{\min} \leq \mathbf{h} \leq a_{\max}$. Define the mapping $\mathcal{M} : \mathcal{X} \rightarrow [0, 1]$ as

$$\mathcal{M}(x) = \frac{w^\top x - a_{\min}}{a_{\max} - a_{\min}}.$$

This mapping projects all feature vectors onto the scalar interval $[0, 1]$ based on their position along the direction of the weight vector w . We now calculate the mapping of the feature vectors lying on the class boundary and the decision boundaries of the Stackelberg equilibrium.

$$\begin{aligned}\mathcal{M}(H_c) &= \frac{\mathbf{h} - a_{\min}}{a_{\max} - a_{\min}}, \\ \mathcal{M}(H_+) &= \frac{\mathbf{h} + \mathbf{u} - a_{\min}}{a_{\max} - a_{\min}}, \\ \mathcal{M}(H_-) &= \frac{\mathbf{h} - \mathbf{u} - a_{\min}}{a_{\max} - a_{\min}}.\end{aligned}$$

Now consider the decision problem outlined in Section 2.3, with the class boundary at $\frac{\mathbf{h} - a_{\min}}{a_{\max} - a_{\min}}$ and $\mathbf{u}_- = \mathbf{u}_+ = \frac{2\mathbf{u}}{a_{\max} - a_{\min}}$. From Theorem 2.10 (for the symmetric case $\mathbf{u}_- = \mathbf{u}_+$), it follows that the decision boundaries of the Stackelberg equilibrium are at $\mathbf{h}^- = \frac{\mathbf{h} - \mathbf{u} - a_{\min}}{a_{\max} - a_{\min}}$ and $\mathbf{h}^+ = \frac{\mathbf{h} + \mathbf{u} - a_{\min}}{a_{\max} - a_{\min}}$. This precisely matches the image under $\mathcal{M}$ of the decision boundaries of the Stackelberg equilibrium derived in Theorem 2.13, as claimed.

2.3.4 Stochastic strategies for the receiver

In this section, we allow the receiver to use a randomized classifier and characterize the Stackelberg equilibrium of the game. This chapter improves upon our previous study in [10] by refining key assumptions and proofs. Let the cost kernel $\phi(x)$ be $\phi(x) = x$, and let D , the probability law of X , be uniform. Let $\mathcal{P}(\mathcal{Y} \cup \Delta)$ be the set of probability laws on $\mathcal{Y} \cup \Delta$. A receiver strategy r is a randomized classifier if it is of the form $r : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y} \cup \Delta)$. This means that for each x , $r(x)$ is a probability law on $\mathcal{Y} \cup \Delta$. In the strategic classification problem, the receiver observes the values of $\mathcal{X}$ as communicated by the sender. Let X' denote the random variable representing the receiver's observation of $\mathcal{X}$ that is communicated by the sender, and let D' denote the corresponding probability law. For a sender strategy $s \in \mathbf{S}$, the random variable X' is given by $X' = s(X)$, and the probability law D' is given by

$$D'_s(X' \in A) = D(X \in \{x : s(x) \in A\}) = \int_{x:s(x) \in A} dx, \quad \text{for } A \subseteq \mathcal{X}.$$

We define a joint random variable (X', Y) over $(\mathcal{X} \times \mathcal{Y})$ whose probability law is given by

$$\mathbb{P}_r(X' \in [x_i, x_j], Y = y) := \int_{x_i}^{x_j} \mathbb{P}_r(Y = y | X' = x) dD'$$

where

$$\mathbb{P}_r(Y = y | X' = x) := r(x)(\{y\}), y \in \mathcal{Y} \cup \Delta. \quad (2.10)$$

We further constrain r such that

$$\exists x \in \mathcal{X} \text{ for which } \mathbb{P}_r(\Delta | X' = x) = 0. \quad (2.11)$$

and the sets

$$\begin{aligned} \{x : \mathbb{P}_r(- | X' = x) + \mathbb{P}_r(+ | X' = x) = 1, x \leq \mathbf{h}\} \text{ and} \\ \{x : \mathbb{P}_r(- | X' = x) + \mathbb{P}_r(+ | X' = x) = 1, x \geq \mathbf{h}\} \text{ are closed.} \end{aligned} \quad (2.12)$$

The receiver's strategy set is defined as

$$\mathbf{R} := \{r \mid r : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y} \cup \Delta), r \text{ satisfies (2.11) and (2.12)}\}. \quad (2.13)$$

For a receiver strategy r , let

$$p(x) := \mathbb{P}_r(Y = - | X' = x) \text{ and } q(x) := \mathbb{P}_r(\{Y = + | X' = x\})$$

be the probabilities of assigning labels $-$ and $+$ to the feature vector x , respectively. Consequently, the strategy $r(x)$ be represented as the triplet $(p(x), q(x), 1 - p(x) - q(x))$. Let

$$\mathfrak{U}(x; r, s) := \sum_{y \in \mathcal{Y} \cup \Delta} \mathbb{P}_r(Y = y | X' = s(x)) U(y, x),$$

where U is defined in (2.4). Since r is a randomized classifier, the sender's payoff for the receiver's strategy r at the feature vector x is modified to

$$\mathbb{U}(x; r, s) := \mathfrak{U}(x; r, s) - |s(x), x|.$$

We further assume that $0 < \mathbf{u}_+ = \mathbf{u}_- = 2\mathbf{u} \ll \min\{\mathbf{h}, 1 - \mathbf{h}\}$. Let the error function $T(x; r, s, h) : \mathcal{X} \rightarrow [0, 1]$ be defined as:

$$T(x; r, s, h) := \mathbb{P}_r\{Y \neq h(x) \mid X' = s(x)\}. \quad (2.14)$$

The error function quantifies the error made at each feature vector x by the sender strategy s and the receiver strategy r under the true function h . Since r is a randomized classifier, the receiver's payoff is modified to

$$\mathcal{E}(r, s; h) := \int_{\mathcal{X}} T(x; r, s, h) dx. \quad (2.15)$$

Notation. We sometimes drop some of the arguments from $T(\cdot, \cdot, \cdot, \cdot)$ when it is apparent from the context.

For a receiver strategy r , the sender's worst-case lazy best response is defined as follows.

Definition 2.14 (Worst-case Lazy Best Response). Let r be a receiver strategy. Let the set $V(x; r)$ be given as

$$V(x; r) := \arg \max_{x' \in \mathcal{X}} \left(\sum_{y \in \mathcal{Y} \cup \Delta} \mathbb{P}_r(Y = y | X' = x') (U(y, x) - |x' - x|) \right).$$

Let $\bar{B}(r)$ be the set of sender strategies s defined as

$$\bar{B}(r) := \left\{ s : \mathcal{X} \rightarrow \mathcal{X} \mid s(x) \in \arg \min_{x' \in V(x; r)} |x' - x|, \forall x \in \mathcal{X} \right\}. \quad (2.16)$$

A sender strategy s_r is called a worst-case lazy best response to a receiver strategy r if

$$s_r \in \arg \max_{s' \in \bar{B}(r)} \mathcal{E}(r, s'; h),$$

and there does not exist $s' \in \bar{B}(r)$ such that

$$T(x; r, s') \geq T(x; r, s_r) \quad \forall x \in \mathcal{X},$$

with $T(x_i; r, s') > T(x_i; r, s_r)$ for some $x_i \in \mathcal{X}$.

Next, we define the Stackelberg equilibrium strategy.

Definition 2.15. Let h be the true function. A strategy $(r_h^*, s_{r_h^*})$ is a Stackelberg equilibrium if

$$r_h^* := \arg \min_{r \in \mathcal{R}} \mathcal{E}(r, s_r; h)$$

and $s_{r_h^*}$ is its worst-case lazy best response.

Notation. Sometimes, we refer r^* itself as Stackelberg equilibrium if s_{r^*} and h are apparent from the context.

There are several reasons for considering the setup where the sender adopts the worst-case lazy best response. First, the existence of a Stackelberg equilibrium under the worst-case best response is hard to guarantee. This stands in stark contrast to the worst-case lazy best response, where we explicitly construct a Stackelberg equilibrium. Second, the error in the Stackelberg equilibrium under the worst-case lazy best response provides a lower bound on the error of the Stackelberg equilibrium (if it exists) under the worst-case best response. This, combined with other results, allows us to calculate the error of the Stackelberg equilibrium under the worst-case best response without having to explicitly construct the equilibrium itself. The following proposition and corollary illustrate these points.

Proposition 2.16: *Let $r \in \mathbb{R}$ be a receiver strategy such that its worst-case best response and worst-case lazy best response exist. Let s denote the worst-case best response to r , and let s' denote the worst-case lazy best response to r . Then, the receiver's error satisfies*

$$\mathcal{E}(r, s; h) \geq \mathcal{E}(r, s'; h).$$

Proof. The proof is given in Section 2.6 (Proof 2.6.4). $\square$

Corollary 2.17. Suppose there exists a Stackelberg equilibrium (r^*, s_{r^*}) under the setup where the sender plays the worst-case best response. Then, $\mathcal{E}(r^*, s_{r^*}) = \frac{\mathbf{u}}{2}$.

Proof. The proof is given in Section 2.6 (Proof 2.6.4). $\square$

Notation. In this subsection, s_r denotes the worst-case lazy best response, not the worst-case best response unless mentioned otherwise.

The next theorem gives the structure of a Stackelberg equilibrium.

Theorem 2.18. Let the feature space be $\mathcal{X} = [0, 1]$, let D be the uniform distribution on $\mathcal{X}$, and let the cost kernel be $\phi(x) = x$. Let $h : \mathcal{X} \rightarrow \mathcal{Y}$ be the true function defined in (2.1) with class boundary $\mathbf{h} \in (0, 1)$. Assume the sender's utility parameters satisfy $0 < \mathbf{u}_+ = \mathbf{u}_- = 2\mathbf{u}$, where $\mathbf{u} < \min\{\mathbf{h}, 1 - \mathbf{h}\}$. Let $\mathbf{h}^- := \mathbf{h} - \mathbf{u}$ and $\mathbf{h}^+ := \mathbf{h} + \mathbf{u}$. Then the randomized receiver strategy $r(\cdot; \mathbf{h}^-, \mathbf{h}^+) : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y} \cup \Delta)$, defined for all $x \in \mathcal{X}$ by

$$r(x; \mathbf{h}^-, \mathbf{h}^+) := \begin{cases} (1, 0, 0) & \text{if } x \in [0, \mathbf{h}^-], \\ \left(1 - \frac{x - \mathbf{h}^-}{2\mathbf{u}}, \frac{x - \mathbf{h}^-}{2\mathbf{u}}, 0\right) & \text{if } x \in (\mathbf{h}^-, \mathbf{h}^+), \\ (0, 1, 0) & \text{if } x \in [\mathbf{h}^+, 1], \end{cases} \quad (2.17)$$

is a Stackelberg equilibrium under the sender's worst-case lazy best response, and the receiver's error is $\mathcal{E}(r) = \frac{\mathbf{u}}{2}$.

Proof. The proof is given in Section 2.6 (Proof 2.6.4). $\square$

2.3.5 Examples and comparison

A naive classifier is the one which uses the true function $h(x)$ for classification. In this subsection, we compare the error made by the naive strategy with the deterministic and the

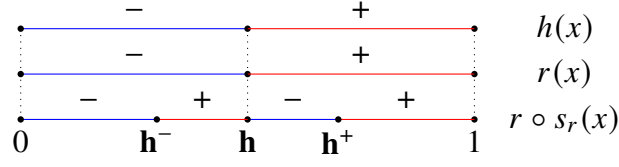


Figure 2.1: Naive classifier.

stochastic Stackelberg equilibrium. We consider a scenario where the true function is $h(x)$ given in (2.1), $\mathbf{u}_+ = \mathbf{u}_- := 2\mathbf{u}$ where $\mathbf{u}$ is sufficiently small and positive, the cost kernel is given by $\phi(x) = x$ and the probability law $D \in \mathfrak{D}$ is uniform.

In the first example, we calculate the error made by the naive classifier.

Example 2.19. The naive classifier $h(x)$ is given by

$$h(x) = \begin{cases} - & x \in [0, \mathbf{h}] \\ + & x \in (\mathbf{h}, 1]. \end{cases}$$

For a feature vector $x \in (\mathbf{h} - 2\mathbf{u}, \mathbf{h}]$, the sender can map it to a feature vector x' , where x' is such that $\mathbf{h} < x' < x + 2\mathbf{u}$, and obtain a positive payoff. This is better than not modifying x , as in that case, the sender receives zero payoff. Similarly, for a feature vector $x \in (\mathbf{h}, \mathbf{h} + 2\mathbf{u}]$, the sender can map it to a feature vector x' , where x' is such that $x - 2\mathbf{u} < x' \leq \mathbf{h}$, and obtain a positive payoff as opposed to a zero payoff when not modifying x . Hence $h(x)$ misclassifies the feature vectors belonging in the interval $(\mathbf{h} - 2\mathbf{u}, \mathbf{h} + 2\mathbf{u}]$ and $\mathcal{E}(h) = 4\mathbf{u}$.

In the next example, we calculate the error made by the deterministic Stackelberg equilibrium.

Example 2.20. Let r^* be a deterministic Stackelberg equilibrium obtained from Theorem 2.10. Then r^* is given by

$$r^*(x) = \begin{cases} h(x) & \text{if } x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1] \\ \Delta & \text{if } x \in [\mathbf{h}^-, \mathbf{h}^+], \end{cases}$$

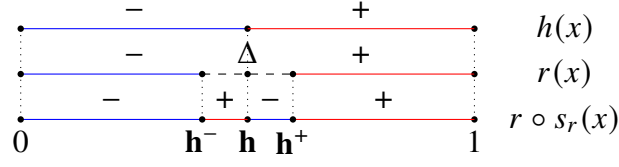


Figure 2.2: Deterministic Stackelberg equilibrium.

where $\mathbf{h}^- = \mathbf{h} - \mathbf{u}$ and $\mathbf{h}^+ = \mathbf{h} + \mathbf{u}$. The sender's worst-case best response is

$$s_{r^*}(x) = \begin{cases} x & \text{if } x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1] \\ \mathbf{h}^+ & \text{if } x \in [\mathbf{h}^-, \mathbf{h}) \\ \mathbf{h}^- & \text{if } x \in [\mathbf{h}, \mathbf{h}^+]. \end{cases}$$

Hence $E(r^*) = [\mathbf{h}^-, \mathbf{h}^+]$ and $\mathcal{E}(r^*) = 2\mathbf{u}$. This shows that ignoring feature vectors near the decision boundaries, which in any case will be misclassified under the naive decision rule, prevents other neighbouring feature vectors from getting misclassified.

In Section 2.3.2, we analyzed the setup where the sender adopts the worst-case best response. In contrast, Section 2.3.4 examined the setup where the sender employs the worst-case lazy best response. To ensure a fair comparison between the deterministic and stochastic Stackelberg equilibria, the sender's style of play should remain fixed. Therefore, in the following example, we calculate the classification error made by a receiver strategy, assuming the sender adopts the worst-case best response.

Example 2.21. Let $\epsilon > 0$. Let $r_\epsilon \equiv (1 - q_\epsilon(x), q_\epsilon(x), 0)$ be stochastic receiver strategy where q_ϵ is defined as

$$q_\epsilon(x) = \begin{cases} 0 & \text{if } x \in [0, \mathbf{h}_\epsilon^-) \\ \frac{x - \mathbf{h}_\epsilon^-}{2(\mathbf{u} + \epsilon)} & \text{if } x \in [\mathbf{h}_\epsilon^-, \mathbf{h}_\epsilon^+] \\ 1 & \text{if } x \in (\mathbf{h}_\epsilon^+, 1], \end{cases} \quad (2.18)$$

where $\mathbf{h}_\epsilon^- := \mathbf{h} - \mathbf{u} - \epsilon$ and $\mathbf{h}_\epsilon^+ := \mathbf{h} + \mathbf{u} + \epsilon$. It follows that the worst-case best response is $s_{r_\epsilon}(x) = x$ (the calculation is similar to what is done in Lemma 2.41 and is given in Section 2.6, Calculation 2.6.5) and $\mathcal{E}(r_\epsilon)$ is $\frac{\mathbf{u} + \epsilon}{2}$. This strategy shows that randomizing classifications near the decision boundaries can correctly classify feature vectors close to the decision boundaries, albeit with a low but positive probability. This happens because randomization ensures that the cost increase is offset by the increase in utility (which takes a fractional value due to randomization).

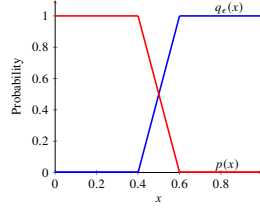


Figure 2.3: Stochastic decision ruler $q_e(x)$, $\mathbf{u} = 0.1$, $\mathbf{h} = 0.5$

It can be seen from these examples that both the deterministic and stochastic Stackelberg equilibria outperform the naive classifier. Furthermore, the stochastic classifier makes fewer errors than the deterministic classifier. The benefit of randomization in strategic interactions is well-established in game theory literature, and we now observe its advantages in the strategic classification as well.

One crucial point to note is that although the penalty label was useful in the deterministic classifier, it was not helpful in the stochastic classifier. Furthermore, the deterministic classifier is sub-optimal compared to the stochastic classifier. Thus, the randomization offers the receiver a richer source of leverage to make the sender behave appropriately than the penalty label. However, in many cases, such as university admissions, for transparency, the receivers are constrained to use only the deterministic classifier. Thus, in those cases, the penalty label can act as compensation for the lack of randomization.

The following example describes the deterministic Stackelberg equilibrium of the university admission problem introduced in the Section 2.1.

Example 2.22. (Stackelberg equilibrium of the university admission problem):

Recall the example from the introduction. Suppose a university offers two programs: Theoretical Physics and Experimental Physics. Admission to these programs is determined by each student's performance on an entrance exam. Each prospective student has a unique combination of theoretical and experimental skills, and their *true program* corresponds to the program that aligns with their stronger skill. However, students often prefer a program that does not align with their true program. The university's goal is to design the entrance exam to maximize the number of students admitted to their true program.

We model this as a strategic classification problem as follows: The university acts as the receiver, and the student as the sender. Let $\mathcal{X} := [0, 1]$ represent the set of feature vectors, where

$x \in \mathcal{X}$ denotes a student's true aptitude, with the interpretation that it has x theoretical skill and $(1 - x)$ experimental skill. We assume that true aptitudes are uniformly distributed across the student population; thus, D , the probability law on $\mathcal{X}$, is uniform. Let $\mathcal{Y} := \{-, +\}$ denote the label space, where $-$ indicates the Experimental Physics program and $+$ the Theoretical Physics program. The true function $h(x)$, which assigns each student to their true program, is given by the following equation:

$$h(x) = \begin{cases} + & \text{if } x > 0.5, \\ - & \text{if } x \leq 0.5. \end{cases} \quad (2.19)$$

This means $\mathbf{h} = 0.5$, and that for $x > 0.5$, a student's true program is Theoretical Physics, while for $x \leq 0.5$, it is Experimental Physics.

The university has access to a question bank containing questions of varying difficulties. We denote the difficulty level of a question by x , indicating that the question requires x theoretical knowledge and $(1 - x)$ experimental knowledge to solve. Note that there is a one-to-one correspondence between the feature vector set $\mathcal{X}$ and the question bank. Thus, a student may demonstrate an aptitude x' (not necessarily their true aptitude) by solving a question with difficulty level x' . The entrance exam comprises multiple questions selected from this bank, with each question associated with a program. A student can attempt only one question, and admission to the corresponding program is granted if they answer correctly.

We assume each student's preferred program differs from their true program. The student's utility function is given by

$$U(y, x) := \begin{cases} 0.04 & x \in [0, 0.5], y = + \\ 0.04 & x \in (0.5, 1], y = - \\ 0 & y = h(x) \\ -\infty & y = \Delta. \end{cases} \quad (2.20)$$

This means $\mathbf{u}_- = \mathbf{u}_+ = 0.04$, and a student gains a utility of 0.04 if assigned to their preferred program and 0 if assigned to their true program and $-\infty$ if not assigned any program. The cost kernel is given by $\phi(x) = x$, meaning that for a question with difficulty level x' , a student with true aptitude x incurs a cost of $|x - x'|$ while solving it.

The university's strategy set includes all possible question papers for the entrance exam. For a given question paper, each student selects the question that maximizes their utility at

the lowest cost, potentially demonstrating an aptitude x' different from their true aptitude to the university. The student's strategy set consists of mappings that specify which question to attempt for each question paper. We assume students answer their chosen question correctly. Consequently, the university, aware of students' strategic behaviour, must optimally design the question paper to maximize the number of students admitted to their true program.

According to Theorem 2.10, questions with difficulty levels in the interval $[\mathbf{h} - \mathbf{u}_-, \mathbf{h} + \mathbf{u}_+]$ should not be included in the exam. Substituting the value for $\mathbf{h}, \mathbf{u}_-, \mathbf{u}_+$ for this example, this interval turns out to be $[0.48, 0.52]$. Furthermore, the students who correctly answers questions with difficulty levels less than 0.48 should be admitted to the Experimental Physics program, and the students who correctly answers the questions with difficulty levels greater than 0.52 should be admitted to the Theoretical Physics program. The intuition for excluding questions in the interval $[0.48, 0.52]$ is as follows: if a question of difficulty level, say 0.51 (corresponding to Theoretical Physics) is included, a student with true aptitude 0.49 (whose true program is Experimental Physics) could solve it with only effort 0.02, leading to misclassification. However, this approach has a drawback: students whose true aptitudes lie within the interval $[0.48, 0.52]$ must attempt questions of difficulty level outside this range. Thus, by incurring a cost, they attempt questions linked to their preferred program and get misclassified by the university. Nonetheless, the overall benefits of this strategy outweigh its drawbacks, making it the optimal option for the university.

2.3.6 Other Stackelberg equilibria

In this section, we only consider deterministic Stackelberg equilibria. In many scenarios, the strategic classification problem may admit more than one Stackelberg equilibrium. While all such Stackelberg equilibria guarantee the same classification error for the receiver, they are not equivalent from the sender's perspective. Here, we discuss various considerations that a receiver may take into account besides minimizing the classification error.

Participation constraint for the sender

Consider the following instance of the problem. Let D be a uniform distribution on $\mathcal{X}$, with $\mathbf{h} = 0.5$ and $\mathbf{u}_- = \mathbf{u}_+ = 0.1$. It can be shown that both r_1 and r_2 , defined as

$$r_1(x) = \begin{cases} - & \text{if } x \in [0, 0.45] \\ \Delta & \text{if } x \in (0.45, 0.55) \\ + & \text{if } x \in [0.55, 1] \end{cases}, \quad r_2(x) = \begin{cases} - & \text{if } x = 0 \\ \Delta & \text{if } x \in (0, 1) \\ + & \text{if } x = 1 \end{cases}$$

are receiver Stackelberg equilibrium strategies. However, the situation differs from the sender's perspective. For $x \in \mathcal{X}$, the sender's payoff under r_1 and r_2 satisfies

$$\mathbb{U}(x; r_1, s_{r_1}) \geq \mathbb{U}(x; r_2, s_{r_2}) \text{ for all } x \in \mathcal{X}. \quad (2.21)$$

That is, for each $x \in \mathcal{X}$, the payoff for the sender under r_1 is greater than or equal to that under r_2 .

This scenario resembles a dilemma in the principal-agent problem, where the principal can impose an unfair contract on the agent, who ends up paying (instead of being paid) for their work. To remedy this in the principal-agent context, the agent is given the option to refuse the contract and receive a baseline payoff. This encourages the principal to offer a contract with a payoff greater than the baseline. This concept is known as the *participation constraint* for the agent.

Drawing a parallel to the principal-agent problem, the receiver can be viewed as the principal who offers a contract r , while the sender acts as the agent who, by making effort s on a feature vector x , gets a payoff of $\mathbb{U}(x; r, s)$. We allow the sender, after seeing the contract r , the option not to participate and instead receive a baseline payoff. There are many ways the sender can determine his decision to participate. Here, we assume that the sender does not participate if $\mathbb{U}(x; r, s_r)$ is lower than the baseline payoff for any $x \in \mathcal{X}$. We further assume that if the sender does not participate in the game, the receiver makes an error on the entire $\mathcal{X}$. This participation constraint of the sender would discourage the receiver from playing a degenerate strategy like r_2 , since the sender would reject such a game. Given this constraint, strategies like r_1 , with a Δ interval of length $\mathbf{u}$, provide justifiable non-degenerate Stackelberg equilibria, as we show in the following cases.

Case 1 (Negative participation constraint):

Suppose the sender incurs a fixed payoff $v \leq 0$ if it chooses not to participate in the game

after seeing the receiver's strategy. This means the sender can opt-out with a small penalty upon encountering an unfavourable receiver strategy. Consequently, many degenerate receiver strategies like r_2 will not satisfy the participation constraint and will no longer be a receiver Stackelberg equilibrium.

Now consider the strategy r_1 . For $x \in [0, 0.45) \cup (0.55, 1]$, the sender's payoff at x is $\mathbb{U}(x; r, s_r) = 0$. For $x \in [0.45, 0.5]$, the sender's payoff at x is $\mathbb{U}(x; r, s_r) = x - 0.45$ and for $x \in (0.55, 1]$, the sender's payoff at x is $\mathbb{U}(x; r, s_r) = 0.55 - x$. Thus, $\min_{x \in \mathcal{X}} \mathbb{U}(x; r, s_r) = 0$. This satisfies the sender's participation constraint, and hence the sender will participate in the game. Therefore, r_1 is a Stackelberg equilibrium even under this constraint.

Case 2 (Positive participation constraint):

Suppose the sender receives a fixed payoff $\nu > 0$ if it chooses not to participate in the game after seeing the receiver's strategy. We will show that every receiver strategy is a Stackelberg equilibrium strategy.

Let r be a receiver strategy. If $E(r, s_r, h) = \mathcal{X}$, then $\mathcal{E}(r) = 1$. If $E(r, s_r, h) \neq \mathcal{X}$, then there exists x such that $r(s_r(x)) = h(x)$. This implies $\mathbb{U}(x; r, s_r) = -|s_r(x) - x| \leq 0$. But this violates the sender's participation constraint, and thus the sender will ignore this strategy r . Therefore, $\mathcal{E}(r) = 1$. Hence, $\mathcal{E}(r) = 1$ for all $r \in \mathbb{R}$. Therefore, every receiver strategy $r \in \mathbb{R}$ is a receiver Stackelberg equilibrium strategy as claimed. More specifically, r_1 is a Stackelberg equilibrium even under this constraint.

Fairness and complexity of the classifier:

Let D be the uniform probability law on $[0, 1]$, $\mathbf{h} = 0.5$, and $\mathbf{u}_- = \mathbf{u}_+ = 0.1$. From Theorem 2.10, any strategy $r(x; \mathbf{h}^-, \mathbf{h}^+) \in \mathcal{R}$ such that $\mathbf{h}^- \leq 0.5 \leq \mathbf{h}^+$ and $|\mathbf{h}^- - \mathbf{h}^+| = 0.1$ is a Stackelberg equilibrium. These equilibria yield the same receiver error but differ in how errors are distributed across each label class.

Consider two such receiver strategies, r_1 and r_2 , with corresponding Δ -regions $(0.45, 0.55)$ and $(0.4, 0.5)$, respectively. The strategy r_1 appears fairer, as it balances the ratio of correctly classified labels, compared to r_2 , which misclassifies only the $-$ class. However, the receiver strategy r_2 is simpler, with 2 decision boundaries in $r_2 \circ s_{r_2}$ compared to 3 in $r_1 \circ s_{r_1}$. This simplicity is an advantage for learnability; the VC dimension of a hypothesis class increases with

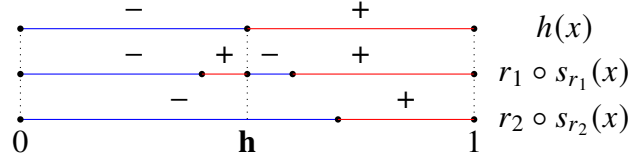


Figure 2.4: Fairness and complexity tradeoff.

the number of decision boundaries that functions in that class are allowed to have (see Proposition 2.28). This highlights a trade-off between fairness and efficient learnability. Depending on the application, one strategy may be preferable to the other.

2.4 Learning the Stackelberg equilibrium

In this section, we address the learning problem, where the receiver is unaware of the true function and is limited to employing only deterministic strategies. Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \rightarrow \mathcal{Y}\}$ denote a hypothesis class, and let

$$l(x, y; f) := \mathbb{1}_{f(x) \neq y}$$

be the *zero-one loss function*, where $x \in \mathcal{X}, y \in \mathcal{Y}$ and $f \in \mathcal{F}$. Let (X, Y) be a random variable over $(\mathcal{X}, \mathcal{Y})$ and D be its corresponding probability law. Let $L_D(f)$ denote the *risk function* corresponding to the zero-one loss function and be defined as

$$L_D(f) := \mathbb{E}_D[l(X, Y; f)]. \quad (2.22)$$

Suppose $Y = h(X)$, where h is the true function. Then we modify the definition of risk function given in (2.22). Let X random variable over $\mathcal{X}$ and D be it corresponding probability law. Then $L_D(f; h)$ denotes the *h-risk function* corresponding to the zero-one loss function and be defined as

$$L_D(f; h) := \mathbb{E}_D[l(X, h(X); f)]. \quad (2.23)$$

In the absence of strategic interactions between the sender and the receiver, a hypothesis class is PAC (Probably Approximately Correct) learnable if it satisfies the following definition (which is the same as Definition 3.1 in [2]).

Definition 2.23. A hypothesis class $\mathcal{F}$ is PAC learnable if there exists a function $m_{\mathcal{F}}^{\text{PAC}} : (0, 1)^2 \rightarrow \mathbb{N}$ and a learning algorithm $A : \cup_{n=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^n \rightarrow \mathcal{F}$ such that for every $(\epsilon, \delta) \in (0, 1)^2$, for

every true function $h \in \mathcal{F}$ and every probability law D over $\mathcal{X}$, the following holds: Let $S = \{(x_i, h(x_i))\}_{i=1}^m$ be a sample where x_i 's are drawn i.i.d. from D and $m \geq m_{\mathcal{F}}^{\text{PAC}}(\epsilon, \delta)$. Then

$$D^{\otimes m}(L_D(A(S); h) < \epsilon) > 1 - \delta.$$

There exists a more general notion of PAC learning called agnostic PAC learning which is defined in the following definition (which is the same as Definition 3.3 in [2]).

Definition 2.24. A hypothesis class $\mathcal{F}$ is agnostic PAC learnable if there exists a function $m_{\mathcal{F}}^{\text{agn}} : (0, 1)^2 \rightarrow \mathbb{N}$ and a learning algorithm $A : \cup_{n=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^n \rightarrow \mathcal{F}$ such that for every $(\epsilon, \delta) \in (0, 1)^2$ and every probability law D over $\mathcal{X} \times \mathcal{Y}$, the following holds: Let $S = \{(x_i, y_i)\}_{i=1}^m$ be a sample where (x_i, y_i) 's are drawn i.i.d. from D and $m \geq m_{\mathcal{F}}^{\text{agn}}(\epsilon, \delta)$. Then

$$D^{\otimes m} \left(L_D(A(S)) - \inf_{f \in \mathcal{F}} L_D(f) < \epsilon \right) > 1 - \delta.$$

An algorithm A is a *successful PAC/agnostic PAC learner* of $\mathcal{F}$ if $\mathcal{F}$ is PAC/agnostic PAC learnable and A is the corresponding learning algorithm defined in Definition 2.23/2.24. Consider the learning algorithm Empirical risk minimization (ERM)

$$\text{ERM} : \cup_{n=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^n \rightarrow \mathcal{F} \text{ defined as } \text{ERM}(\{(x_i, y_i)_{i=1}^m\}) := \arg \min_{f \in \mathcal{F}} \frac{1}{m} \sum_{i=1}^m l(x_i, y_i; f).$$

A hypothesis class $\mathcal{F}$ is said to have the *uniform convergence property* if there exists a function $m_{\mathcal{F}}^{\text{UC}} : (0, 1)^2 \rightarrow \mathbb{N}$ such that for every $(\epsilon, \delta) \in (0, 1)^2$ and every probability law D over $\mathcal{X} \times \mathcal{Y}$, if $S = \{(x_i, y_i)\}_{i=1}^m$ is a sample drawn i.i.d. from D with $m \geq m_{\mathcal{F}}^{\text{UC}}(\epsilon, \delta)$, then

$$D^{\otimes m} \left(\sup_{f \in \mathcal{F}} \left| \frac{1}{m} \sum_{i=1}^m l(x_i, y_i; f) - L_D(f) \right| < \epsilon \right) > 1 - \delta.$$

The following theorem shows that uniform convergence, PAC learnability, agnostic PAC learnability, and the success of ERM rules are all characterized by the finiteness of the VC dimension.

Theorem 2.25 (Fundamental theorem of statistical learning). Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \rightarrow \mathcal{Y}\}$ be a hypothesis class of functions from domain $\mathcal{X}$ to $\mathcal{Y} = \{0, 1\}$ and let the loss function be the 0 – 1 loss. Then, the following statements are equivalent:

- a) $\mathcal{F}$ has the uniform convergence property.
- b) Any ERM rule is a successful agnostic PAC learner for $\mathcal{F}$.
- c) $\mathcal{F}$ is agnostic PAC learnable.

- d) $\mathcal{F}$ is PAC learnable.
- e) Any ERM rule is a successful PAC learner for $\mathcal{F}$.
- f) $\mathcal{F}$ has finite VC dimension ($\text{VC dim}(\mathcal{F}) < \infty$).

Furthermore, if $\text{VC dim}(\mathcal{F}) = d < \infty$, there exist absolute constants $C_1, C_2 > 0$ such that for any $(\epsilon, \delta) \in (0, 1)^2$, the sample complexities satisfy:

1. $\mathcal{F}$ has the uniform convergence property with sample complexity

$$C_1 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon^2} \leq m_{\mathcal{F}}^{\text{UC}}(\epsilon, \delta) \leq C_2 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon^2}, \quad (2.24)$$

2. $\mathcal{F}$ is agnostic PAC learnable with sample complexity

$$C_1 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon^2} \leq m_{\mathcal{F}}^{\text{agn}}(\epsilon, \delta) \leq C_2 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon^2}, \quad (2.25)$$

3. $\mathcal{F}$ is PAC learnable with sample complexity

$$C_1 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon} \leq m_{\mathcal{F}}^{\text{PAC}}(\epsilon, \delta) \leq C_2 \frac{d \log\left(\frac{1}{\epsilon}\right) + \log\left(\frac{1}{\delta}\right)}{\epsilon}. \quad (2.26)$$

Proof. Refer to Theorems 6.7 and 6.8 in [2]. □

We now establish a similar learning criterion for a hypothesis class $\mathcal{F}$ in the context of strategic classification. Recall that for a true function $f \in \mathcal{F}$ and probability law $D \in \mathfrak{D}$, $r_{f,D}^*$ denotes the Stackelberg equilibrium as defined in Definition 2.3. Define $\tilde{\mathcal{F}} := \{r_{f,D}^* \circ s_{r_{f,D}^*} : f \in \mathcal{F}, D \in \mathfrak{D}\}$, referred to as the *Stackelberg hypothesis class*, which represents the space of receiver Stackelberg equilibria derived from $\mathcal{F}$ and $\mathfrak{D}$. A hypothesis class $\mathcal{F}$ is strategically learnable if it satisfies the following definition.

Definition 2.26 (Strategically learnable class). A hypothesis class $\mathcal{F}$ is strategically learnable if there exists a function $m_{\mathcal{F}} : (0, 1)^2 \rightarrow \mathbb{N}$ and a learning algorithm $A : \cup_{n=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^n \rightarrow \tilde{\mathcal{F}}$ such that for any true function $h \in \mathcal{F}$, any probability law $D \in \mathfrak{D}$, and any $(\epsilon, \delta) \in (0, 1)^2$, the following holds: Let $S = \{(x_i, h(x_i))_{i=1}^m\}$ be a sample where x_i are drawn i.i.d. from D and $m \geq m_{\mathcal{F}}(\epsilon, \delta)$, then

$$D^{\otimes m} \left(\mathcal{E}(A(S); h, D) - \mathcal{E}(r_{h,D}^*; h, D) < \epsilon \right) > 1 - \delta. \quad (2.27)$$

The function $m_{\mathcal{F}}(\epsilon, \delta)$ is called the sample complexity of the algorithm.

Our definition of strategic learnability is different from [23] because we let the algorithm search over the Stackelberg equilibrium hypothesis class (i.e., over $r_f, f \in \mathcal{F}$) rather than the original hypothesis class (i.e., over $f \in \mathcal{F}$).

We propose two algorithms for learning a strategic classifier: *vanilla ERM* and *strategy aware ERM* (defined in Section 2.4.2 and 2.4.3 respectively). We show that a hypothesis class is strategically learnable if and only if it is learnable under the strategy-aware ERM algorithm. However, there exist strategically learnable hypothesis classes which are not learnable under the vanilla ERM algorithm.

Assuming the hypothesis class is learnable by each algorithm, we compare their ease of learning (compared in Section 2.4.4). This is controlled by the VC dimension of the hypothesis class, which determines the sample complexity in PAC learning. We observe that when the hypothesis class is complex (i.e., contains functions with many decision boundaries), strategy-aware ERM makes learning easier, as it can reduce the VC dimension of the effective learning space. Conversely, for simpler hypothesis classes, vanilla ERM is often the more direct and efficient approach.

2.4.1 Some results on hypothesis class with multiple decision boundaries

A point is called a *decision boundary*, if it assigns different labels to its left and right neighbours. The hypothesis class we consider for our learning algorithms comprises of functions with multiple decision boundaries, requiring us to extend the results established in Theorem 2.10 (which only had one decision boundary). To achieve this, we examine a generalized version of a decision problem where the true function has multiple decision boundaries. Let $\mathcal{X} = [0, 1]$ represent the feature vector space, and let $\mathcal{Y} = \{-1, +1\}$ be the label space, and $D \in \mathfrak{D}$ a probability law on $\mathcal{X}$. Decompose $\mathcal{X}$ as $\mathcal{X} = O_1 \cup O_2 \cup \dots \cup O_{d-1}$, where each O_n is a disjoint interval defined as $O_n = [o_{(n)}, o^{(n)})$ for $n < d - 1$, $O_{d-1} = [o_{d-1}, 1]$, and $o^{(n)} = o_{(n+1)}$. Define the true function $h(x)$ as follows:

$$h(x) = \begin{cases} - & \text{if } x \in O_n, n \text{ is even,} \\ + & \text{if } x \in O_n, n \text{ is odd.} \end{cases} \quad (2.28)$$

Let the cost kernel be $\phi(x) = x$, and let Δ denote the penalty label (as considered in Section 2.2). Define the utility function $U(y, x)$ as:

$$U(y, x) := \begin{cases} \mathbf{u} & \text{if } x \in O_n, y = +, n \text{ is even,} \\ \mathbf{u} & \text{if } x \in O_n, y = -, n \text{ is odd,} \\ 0 & \text{if } y = h(x), \\ -\infty & \text{if } y = \Delta. \end{cases} \quad (2.29)$$

The structure of the Stackelberg equilibrium strategy generalizes Theorem 2.10 (when $\mathbf{u}_+ = \mathbf{u}_-$) and is stated in the following proposition.

Proposition 2.27. Let D be a probability law on $\mathcal{X}$, and let $h(x)$ be the true function. Suppose a Stackelberg equilibrium exists for the receiver. Then there exist intervals $\Theta_n \subseteq O_n$ for $n \in \{1, \dots, d-1\}$, such that the receiver strategy r defined by

$$r(x; \Theta_1, \dots, \Theta_{d-1}) = \begin{cases} h(x) & \text{if } x \in \bigcup_{n=1}^{d-1} \Theta_n, \\ \Delta & \text{if } x \notin \bigcup_{n=1}^{d-1} \Theta_n, \end{cases} \quad (2.30)$$

is a Stackelberg equilibrium, and $r \circ s_r$ is of the form

$$r(s_r(x)) = \begin{cases} h(x) & \text{if } x \in \bigcup_{n=1}^{d-1} \Theta_n, \\ -h(x) & \text{if } x \notin \bigcup_{n=1}^{d-1} \Theta_n. \end{cases}$$

Proof. The proof of the proposition is given in Section 2.6 (Proof 2.6.5). □

Our next theorem establishes the VC dimension of a hypothesis class containing hypotheses with multiple decision boundaries.

Proposition 2.28. Let the hypothesis class $\mathcal{F} \subset \{f|f : \mathcal{X} \rightarrow \mathcal{Y}\}$ be a collection of all functions with at most $d-1$ decision boundaries. Then $\text{VC dim}(\mathcal{F}) = d$.

Proof. The proof of the proposition is given in Section 2.6 (Proof 2.6.4). □

2.4.2 Vanilla ERM

One intuitive algorithm for the receiver to learn the strategic classifier is to estimate the true function from the sample data. The receiver then plays the screening game under this estimated

function, treating it as the true function, and outputs the corresponding Stackelberg equilibrium. The vanilla ERM algorithm expresses this idea. Let $D \in \mathfrak{D}$ be a probability law on $\mathcal{X}$, h be the true function and let $S = \{(x_i, h(x_i))\}_{i=1}^m$ be a sample where x_i are drawn i.i.d. from D . Let $D(S)$ be the empirical law of sample S defined as:

$$D(S)(B) := \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{x_i \in B} \quad (2.31)$$

where B is a Borel measurable set in $\mathcal{X}$.

Algorithm 2.1: Vanilla ERM algorithm

Data: Hypothesis class $\mathcal{F}$, Sample $S = \{(x_i, h(x_i))\}_{i=1}^m$,

Zero-one loss function $l_h(f, x) := \mathbb{1}_{\{f(x) \neq h(x)\}}$

Result: Hypothesis $\hat{r}_{erm}$

// Estimation

1 $A(S) := \arg \min_{f \in \mathcal{F}} \sum_{i=1}^m \mathbb{1}_{\{f(x_i) \neq h(x_i)\}}$

2 $D(S) :=$ Empirical law

// Decision

3 $\hat{r} := r_{A(S), D(S)}^*$, i.e., Stackelberg equilibrium under true function $\hat{h}$ and probability law $D(S)$

The Algorithm 1 is initialized by providing a hypothesis class $\mathcal{F}$ and a sample S . In the first step, the algorithm calculates $A(S)$, which estimates the true function h from the sample S via the ERM subroutine. In the second step, it calculates the empirical law $D(S)$ of sample S , which is an estimate of the input law D on $\mathcal{X}$ from which the x_i 's in S were chosen. In the third step, it calculates the Stackelberg equilibrium $r_{A(S), D(S)}^*$ (refer Definition 2.3) under the (estimated) true function $A(S)$ and the probability law $D(S)$ on $\mathcal{X}$, as given in Theorem 2.10.

We show that if the hypothesis class $\mathcal{F}$ has finite VC dimension, then it is strategically learnable under the Algorithm 1.

Theorem 2.29. Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \rightarrow \mathcal{Y}\}$ be a hypothesis class consisting of all functions with at most $d - 1$ decision boundaries, where $d < \infty$. Let $A : \cup_{n=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^n \rightarrow \mathcal{F}$ denote the Algorithm 1 (vanilla ERM algorithm) and let $B : \mathcal{F} \rightarrow \tilde{\mathcal{F}}$ map a hypothesis to its corresponding Stackelberg equilibrium. Then $\mathcal{F}$ is strategically learnable (cf. Definition 2.26) under $B \circ A$.

Proof. The proof is given in Section 2.6 (Proof 2.6.8). □

We prove the theorem with the help of the following two lemmas. The first lemma states that for a fixed true function $f \in \mathcal{F}$, the classification errors of the Stackelberg equilibrium strategies $r_{f,D}^*$ and $r_{f,D(S)}^*$ are similar for all $D \in \mathfrak{D}$.

Lemma 2.30. Let $(\epsilon, \delta) \in (0, 1)^2$. Let $f \in \mathcal{F}$ be a true function with at most K true-label intervals (i.e., at most $K - 1$ decision boundaries), and let $D \in \mathfrak{D}$ be a probability law on $\mathcal{X}$. Let $S = \{(x_i, h(x_i))\}_{i=1}^m$ be a sample of size m , where the x_i 's are drawn i.i.d. according to a probability law D , and let $D(S)$ be the empirical law of the sample S . Let $r_{D(S)}^*$ and r_D^* be the Stackelberg equilibrium under the true function f for probability law $D(S)$ and D respectively. Then, there exists an $\bar{m}(\epsilon, \delta)$ such that for all $m > \bar{m}(\epsilon, \delta)$ the following holds:

$$D^{\otimes m}(\mathcal{E}(r_{f,D(S)}^*; f, D) - \mathcal{E}(r_{f,D}^*; f, D) \leq 4K\epsilon) > 1 - 2e^{-2m\epsilon^2}.$$

Proof. The proof is given in Section 2.6 (Proof 2.6.6). □

The next lemma states that for a fixed probability law $D \in \mathfrak{D}$, the classification errors of the Stackelberg equilibrium strategies $r_{A(S),D}^*$ and $r_{h,D}^*$ are similar.

Remark. Notice that $r_{A(S),D}$ is the Stackelberg equilibrium when the estimated true function $A(S)$ is considered the true function. To avoid confusion with the actual true function h , which generated the samples, we will refer to h as the *ground true function* and $A(S)$ as the *estimated true function* whenever there is potential for ambiguity.

Lemma 2.31. Let $(\epsilon, \delta) \in (0, 1)^2$. Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \rightarrow \mathcal{Y}\}$ be a hypothesis class consisting of all functions with at most $d - 1$ decision boundaries. Let $A : \cup_{n=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^n \rightarrow \mathcal{F}$ denote the algorithm described in Algorithm 1 (vanilla ERM algorithm). For every ground true function $h \in \mathcal{F}$ and every probability law $D \in \mathfrak{D}$ on $\mathcal{X}$, the following holds. Let $S = \{(x_i, h(x_i))\}_{i=1}^m$ be a sample of size m , where x_i are drawn i.i.d. according to D . Let $r_{A(S),D}^*$ and $r_{h,D}^*$ be the Stackelberg equilibria under D and the estimated true function $A(S)$ and the ground true function $h(x)$, respectively. Then there exists $\underline{m}(\epsilon, \delta) := m_{\mathcal{F}}^{\text{PAC}}(\epsilon, \delta)$ such that for all $m \geq \underline{m}(\epsilon, \delta)$, the following holds.

(a) The following inequality holds uniformly over all receiver strategies $r \in \mathbb{R}$:

$$D^{\otimes m} \left(\sup_{r \in \mathbb{R}} |\mathcal{E}(r; h, D) - \mathcal{E}(r; A(S), D)| < \epsilon \right) > 1 - \delta.$$

(b) For receiver Stackelberg equilibria $r_{A(S),D}^*$ and $r_{h,D}^*$, we have

$$D^{\otimes m}(\mathcal{E}(r_{A(S),D}^*; h, D) - \mathcal{E}(r_{h,D}^*; h, D) < 2\epsilon) > 1 - \delta. \quad (2.32)$$

Furthermore, the sample complexity $\underline{m}(\epsilon, \delta)$ satisfies

$$C_1 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon} \leq \underline{m}(\epsilon, \delta) \leq C_2 \frac{d \log\left(\frac{1}{\epsilon}\right) + \log\left(\frac{1}{\delta}\right)}{\epsilon}. \quad (2.33)$$

Here $C_1, C_2 > 0$ are constants.

Proof. The proof is given in Section 2.6 (Proof 2.6.7). □

The previous theorem only provides sufficient conditions for strategic learnability. We will now present an algorithm that is both a necessary and sufficient condition for a hypothesis class to be strategically learnable.

2.4.3 Strategy aware ERM

The learning algorithm suggested in Algorithm 1 does not account for strategic behaviour during the testing phase. This is a potential source of inefficiency in determining the strategic learnability of a hypothesis class. An alternate algorithm which takes into account the strategic behaviour and which we show is both a necessary and sufficient condition for strategic learnability is described as follows. Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \rightarrow \mathcal{Y}\}$ be the hypothesis class consisting of all functions with at most $d - 1$ decision boundaries. Corresponding to this hypothesis class, consider the hypothesis class $\tilde{\mathcal{F}}$ defined as

$$\tilde{\mathcal{F}} := \{r \circ s_r \mid r := r_{f,D}^*, D \in \mathfrak{D}, f \in \mathcal{F}\}.$$

We learn the Stackelberg equilibrium directly from $\tilde{\mathcal{F}}$. The strategy-aware ERM algorithm expresses this idea.

Algorithm 2.2: Strategy aware ERM algorithm

Data: Hypothesis class $\mathcal{F}$, Sample $\{(x_i, y_i)\}_{i=1}^m$

Zero-one loss function $\ell_y(r \circ s_r, x) := \mathbb{1}_{\{r(s_r(x)) \neq y\}}$

Result: Hypothesis $\hat{f}$

// Decision

1 $\tilde{\mathcal{F}} := \{r \circ s_r \mid r := r_{f,D}^*, D \in \mathcal{D}, f \in \mathcal{F}\}$

// Estimation

2 $\hat{f}_{\text{ERM}} \circ s_{\hat{f}_{\text{ERM}}} := \arg \min_{r \circ s_r \in \tilde{\mathcal{F}}} \sum_{i=1}^m \mathbb{1}_{\{r(s_r(x_i)) \neq y_i\}}$

The algorithm 2 is initialised with a hypothesis class $\mathcal{F}$ and a sample S . In the first step, we modify the hypothesis class $\mathcal{F}$ to $\tilde{\mathcal{F}}$. In the second step, we run the ERM subroutine on the modified hypothesis class $\tilde{\mathcal{F}}$ and sample S to get a hypothesis $\hat{f}_{\text{ERM}} \circ s_{\hat{f}_{\text{ERM}}}$. We now show that $\text{VC dim}(\tilde{\mathcal{F}}) < \infty$ if and only if $\mathcal{F}$ is strategically learnable under Algorithm 2.

Theorem 2.32 (The fundamental theorem of strategic learning). Let $\mathcal{F} \subseteq \{f \mid f : \mathcal{X} \rightarrow \mathcal{Y}\}$ be a hypothesis class consisting of all functions with at most $d - 1$ decision boundaries. Let $A : \cup_{n=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^n \rightarrow \tilde{\mathcal{F}}$ denote the algorithm described in Algorithm 2. Then $\mathcal{F}$ is strategically learnable under A (refer Definition 2.26) iff $\text{VC dim}(\tilde{\mathcal{F}}) < \infty$.

Proof. The proof is given in Section 2.6 (Proof 2.6.9). □

From the findings of Theorem 2.29 and Theorem 2.32, we get the following corollary.

Corollary 2.33. Let $\mathcal{F}$ be a hypothesis class used in Algorithm 1 and $\tilde{\mathcal{F}}$ be the hypothesis class in Algorithm 2 obtained from $\mathcal{F}$. Then

$$\text{VC dim}(\mathcal{F}) < \infty \implies \text{VC dim}(\tilde{\mathcal{F}}) < \infty.$$

Proof. The proof is given in Section 2.6 (Proof 2.6.10). □

We now demonstrate that certain hypothesis classes are not strategically learnable via Algorithm 1 but become strategically learnable via Algorithm 2. Specifically, we show the existence of a hypothesis class $\mathcal{F}$ such that $\text{VC dim}(\mathcal{F}) = \infty$ but $\text{VC dim}(\tilde{\mathcal{F}}) < \infty$. This establishes that the converse of Corollary 2.33 does not hold.

Example 2.34. Consider the deterministic decision problem described in Section 2.2. Let h be the true function, D be the uniform probability law on $\mathcal{X}$ and $\mathbf{u}_- = \mathbf{u}_+ = 0.1$. Let $\mathbf{f} \in [0, 1)$, and let $F \subseteq ([\mathbf{f}, 1] \cap \mathbb{Q})$. Define the hypothesis $f_F : \mathcal{X} \rightarrow \mathcal{Y}$ as follows:

$$f_F(x) := \begin{cases} - & x \in [0, \mathbf{f}), \\ + & x \in F \setminus [0, \mathbf{f}), \\ - & x \in F^c \setminus [0, \mathbf{f}) \text{ where } F^c = X \setminus F. \end{cases}$$

Now, consider the following hypothesis class:

$$\mathcal{F} := \{f_F : \mathbf{f} \in (0, 1), F \subseteq ([\mathbf{f}, 1] \cap \mathbb{Q})\}$$

Then $\text{VC dim}(\mathcal{F}) = \infty$ because $\mathcal{F}$ can shatter arbitrarily large finite subsets of $[\mathbf{f}, 1] \cap \mathbb{Q}$. We now claim that $\text{VC dim}(\tilde{\mathcal{F}}) = 0$. Let $f_F \in \mathcal{F}$ be the true function. The receiver strategy $r(x) \equiv -$ is a Stackelberg equilibrium. This is because $\mathcal{E}(r) = 0$. Thus, we have $\tilde{\mathcal{F}} = \{r(\cdot) \equiv -\}$, implying that the VC dimension of $\tilde{\mathcal{F}}$ is 0. In conclusion, while $\mathcal{F}$ is not strategically learnable by Algorithm 1 due to its infinite VC dimension, it becomes strategically learnable by Algorithm 2, as the hypothesis space is effectively reduced to one with finite VC dimension.

2.4.4 Comparison between algorithms

We now consider an example of a learning problem and show how strategy-aware ERM compares with standard ERM in various situations. This example guides our intuition on when to choose which algorithm.

Example 2.35. Consider the decision problem where the true function has d decision boundaries. Let the probability measure $\mathfrak{D}$ be uniform on $[0, 1]$ and let $\mathbf{u}_+ = \mathbf{u}_- = \mathbf{u} := 0.2$. To keep things simple, we additionally assume that each sub-interval O_i has the same width $k\mathbf{u}$, where k is a fraction (this implies $n \approx \frac{1}{\mathbf{u}k}$ assuming an exact division). From Proposition 2.27, we know the structure of the receiver Stackelberg strategy r . Because of symmetry, we further assume that the Δ -region is equally shared between two adjacent intervals I_i and I_{i+1} , except when the entire I_i is a Δ -region. Then, the structure of $r \circ s_r$ for different numbers of decision boundaries is given as follows: *Case 1:* $k > 1$. In this case, the number of decision boundaries is $3n$.

Case 2: $k \leq 1$. In this case, the r assigns label Δ to the minority class. Hence, the number of decision boundaries is 1.

From the above example, we can see that as complexity increases, the number of decision boundaries increases to a point and then becomes 1. The threshold point, which is $\frac{1}{u}$, can be considered a boundary separating the simple hypothesis from the complex hypothesis. This means that the complexity of a hypothesis is with respect to a given game structure, which here is characterised by $\mathbf{u}$.

Now consider F_n , a set of all hypotheses with a number of decision boundaries up to n . Clearly, $\text{VC dim}(F_n) \geq \text{VC dim}(F_m)$ if $n \geq m$. Now, if $\mathcal{F} = F_n$, then from Example 2.35, we have $\tilde{\mathcal{F}} \subset F_n$ in Case 1 and $F_n \subset \tilde{\mathcal{F}}$ in Case 2. Therefore, $\text{VC dim}(\mathcal{F}) \geq \text{VC dim}(\tilde{\mathcal{F}})$ and $\text{VC dim}(\mathcal{F}) \leq \text{VC dim}(\tilde{\mathcal{F}})$ respectively. Heuristically, it means that if the $\text{VC dim}(\mathcal{F})$ is large with respect to $\mathbf{u}$, then choose Algorithm 2, or else choose Algorithm 1.

2.5 Conclusion, Limitations, and Future Work

We studied the strategic classification problem where a strategic sender of data has a non-uniform preferences over labels that a receiver can assign. We analyzed two versions of this problem: a decision problem and a learning problem. For the decision problem, we showed that expanding the classifier's action space by either introducing a penalty label that ignores certain feature vectors or introducing a randomized classification rule, significantly improves classification accuracy against a strategic sender. Our results show that an optimal deterministic classifier equipped with a penalty label makes fewer errors than one without a penalty label. Furthermore, the optimal stochastic classifier makes fewer errors than the optimal deterministic classifier with a penalty label. For the learning problem, we show that incorporating the sender's best response structure can enhance the learning process. We demonstrated this by introducing and comparing two algorithms, vanilla ERM and strategy-aware ERM, which integrate game theory and learning theory in distinct ways. Vanilla ERM is preferable for simpler hypothesis classes and strategy-aware ERM is preferable for complex hypothesis classes. This opens avenues for exploring more nuanced combinations of game theory and learning, potentially leading to more powerful strategic learning algorithms. While this chapter is limited to the assumption of a scalar feature space or a vector feature space with a linear class boundary, and of simple utility and cost functions, the structural insights it provides offer guidance for understanding strategic classification in more complex settings.

2.6 Proofs and Supporting Results

This section contains the proofs and supporting arguments for the results presented in this chapter.

2.6.1 Proofs for Discrete Feature Space

We begin by presenting the proof of Lemma 2.6.

Proof of Lemma 2.6. (a) Let I be an independent set of G_h . First, by condition (a) of Definition 2.4, for every feature vector $\mathbf{i} \in \mathcal{X}$, there is at most one action $\mathbf{k}$ such that $(\mathbf{i}, \mathbf{k}) \in I$ (since $\mathbf{i} = \mathbf{j}$ with $\mathbf{k} \neq \mathbf{l}$ creates an edge). Second, also by condition (a), if $(\mathbf{i}, \mathbf{k}) \in I$ and $(\mathbf{j}, \mathbf{k}) \in I$, then $h(\mathbf{i}) = h(\mathbf{j})$ (since $h(\mathbf{i}) \neq h(\mathbf{j})$ with $\mathbf{k} = \mathbf{l}$ creates an edge). Hence $r(\cdot; I)$ is well defined.

Now let $(\mathbf{i}, \mathbf{k}) \in I$. By definition of $r(\cdot; I)$, we have $r(\mathbf{k}) = h(\mathbf{i})$, which yields the sender at $\mathbf{i}$ a payoff of

$$U(r(\mathbf{k}), \mathbf{i}) - C(\mathbf{k}, \mathbf{i}) = U(h(\mathbf{i}), \mathbf{i}) - C(\mathbf{k}, \mathbf{i}) = -C(\mathbf{k}, \mathbf{i}) > -\infty.$$

Consequently, the sender's worst-case best response $s_r(\mathbf{i})$ must achieve a payoff at least $-C(\mathbf{k}, \mathbf{i})$, which implies $r(s_r(\mathbf{i})) \neq \Delta$ (since $U(\Delta, \mathbf{i}) = -\infty$).

From the definition of $r(\cdot; I)$, $r(s_r(\mathbf{i})) \neq \Delta$ implies that there exists some node $(\mathbf{j}, \mathbf{l}) \in I$ such that $s_r(\mathbf{i}) = \mathbf{l}$ and $r(\mathbf{l}) = h(\mathbf{j})$. Suppose for contradiction that $\mathbf{i}$ is misclassified, that is, $r(s_r(\mathbf{i})) \neq h(\mathbf{i})$. Since $r(s_r(\mathbf{i})) = h(\mathbf{j})$, this means $h(\mathbf{j}) \neq h(\mathbf{i})$. Furthermore, since $\mathbf{l} = s_r(\mathbf{i})$ is a best response for the sender at $\mathbf{i}$, reporting $\mathbf{l}$ must yield at least the payoff of reporting $\mathbf{k}$:

$$U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) \geq -C(\mathbf{k}, \mathbf{i}) \implies U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) \geq 0.$$

Since $h(\mathbf{i}) \neq h(\mathbf{j})$ and $U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) \geq 0$, condition (b) of Definition 2.4 holds. This implies that there is an edge between $(\mathbf{i}, \mathbf{k})$ and $(\mathbf{j}, \mathbf{l})$ in G_h , which contradicts the fact that I is an independent set. Therefore, we must have $r(s_r(\mathbf{i})) = h(\mathbf{i})$, showing that every feature vector $\mathbf{i}$ represented in I is correctly classified.

Since condition (a) ensures that each such feature vector $\mathbf{i}$ appears in at most one pair in I , there are exactly $|I|$ distinct feature vectors represented in I . Hence, $|\{x \in \mathcal{X} : r(s_r(x)) = h(x)\}| \geq |I|$.

(b) Let $r(x)$ be an arbitrary receiver strategy, and define $I := \{(x, s_r(x)) : r(s_r(x)) = h(x)\}$.

Since s_r is a function, each x is mapped to a unique best response $s_r(x)$. Thus, all pairs in I have distinct first coordinates, which implies $|I| = |\{x \in \mathcal{X} : r(s_r(x)) = h(x)\}|$.

We now show that I is an independent set in G_h . Let $(\mathbf{i}, \mathbf{k})$ and $(\mathbf{j}, \mathbf{l})$ be distinct elements of I , where $\mathbf{k} = s_r(\mathbf{i})$, $r(\mathbf{k}) = h(\mathbf{i})$ and $\mathbf{l} = s_r(\mathbf{j})$, $r(\mathbf{l}) = h(\mathbf{j})$. We verify that neither of the edge conditions in Definition 2.4 is satisfied:

- Condition (a): If $\mathbf{i} = \mathbf{j}$, then since s_r is a single-valued function, $\mathbf{k} = s_r(\mathbf{i}) = s_r(\mathbf{j}) = \mathbf{l}$, which makes the two nodes identical, a contradiction. Hence $\mathbf{i} = \mathbf{j}$ and $\mathbf{k} \neq \mathbf{l}$ cannot occur. If $\mathbf{k} = \mathbf{l}$, then $r(\mathbf{k}) = r(\mathbf{l})$, which implies $h(\mathbf{i}) = r(s_r(\mathbf{i})) = r(\mathbf{k}) = r(\mathbf{l}) = r(s_r(\mathbf{j})) = h(\mathbf{j})$. Hence $h(\mathbf{i}) \neq h(\mathbf{j})$ and $\mathbf{k} = \mathbf{l}$ cannot occur either. Thus condition (a) does not hold.
- Condition (b): Suppose $h(\mathbf{i}) \neq h(\mathbf{j})$. Since $\mathbf{k} = s_r(\mathbf{i})$ is a best response for sender $\mathbf{i}$ and $r(\mathbf{k}) = h(\mathbf{i})$, the payoff at $\mathbf{k}$ is $-C(\mathbf{k}, \mathbf{i})$. If sender $\mathbf{i}$ instead reported $\mathbf{l}$, the payoff would be $U(r(\mathbf{l}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) = U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i})$. If $U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) > 0$, sender $\mathbf{i}$ strictly prefers $\mathbf{l}$ over $\mathbf{k}$, contradicting the optimality of $\mathbf{k} = s_r(\mathbf{i})$. If $U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) = 0$, sender $\mathbf{i}$ is indifferent between $\mathbf{k}$ (which correctly classifies $\mathbf{i}$) and $\mathbf{l}$ (which misclassifies $\mathbf{i}$ since $h(\mathbf{j}) \neq h(\mathbf{i})$); under the worst-case best response tie-breaking (Definition 2.2), the sender would choose the misclassifying action $\mathbf{l}$, again contradicting $s_r(\mathbf{i}) = \mathbf{k}$. Hence $U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) < 0$. By symmetry, interchanging the roles of $\mathbf{i}$ and $\mathbf{j}$ gives $U(h(\mathbf{i}), \mathbf{j}) - C(\mathbf{k}, \mathbf{j}) + C(\mathbf{l}, \mathbf{j}) < 0$. Thus condition (b) does not hold.

Since neither condition holds, no edge connects $(\mathbf{i}, \mathbf{k})$ and $(\mathbf{j}, \mathbf{l})$. Thus, I is an independent set in G_h .

□

We now provide the proof of Theorem 2.5.

Proof of Theorem 2.5. (a) Let $I^\star$ be a maximum independent set of G_h , so that $|I^\star| = \alpha(G_h)$, where $\alpha(G_h)$ denotes the independence number of G_h . Consider the receiver strategy $r^\star(\cdot) := r(\cdot; I^\star)$. By Lemma 2.6(a), the number of feature vectors correctly classified by $r^\star$ satisfies

$$|\{x \in \mathcal{X} : r^\star(s_{r^\star}(x)) = h(x)\}| \geq |I^\star| = \alpha(G_h).$$

Conversely, for any arbitrary receiver strategy $r \in \mathbf{R}$, by Lemma 2.6(b), the set $I_r := \{(x, s_r(x)) : r(s_r(x)) = h(x)\}$ is an independent set in G_h . Hence,

$$|\{x \in \mathcal{X} : r(s_r(x)) = h(x)\}| = |I_r| \leq \alpha(G_h).$$

Therefore, no receiver strategy can correctly classify strictly more than $\alpha(G_h)$ feature vectors. Since $r^\star$ correctly classifies at least $\alpha(G_h)$ feature vectors, it correctly classifies exactly $\alpha(G_h)$ feature vectors, which is the maximum possible. Thus, $r^\star$ minimizes the classification error $\mathcal{E}(r, s_r; h, D)$, and is therefore a Stackelberg equilibrium.

- (b) Conversely, let $r^\star$ be a Stackelberg equilibrium. By Lemma 2.6(b), $I^\star := \{(x, s_{r^\star}(x)) : r^\star(s_{r^\star}(x)) = h(x)\}$ is an independent set in G_h , and $|I^\star| = |\{x \in \mathcal{X} : r^\star(s_{r^\star}(x)) = h(x)\}|$. Suppose for contradiction that $I^\star$ is not a maximum independent set. Then there exists an independent set J in G_h such that $|J| > |I^\star|$. By Lemma 2.6(a), the strategy $r(\cdot; J)$ satisfies

$$|\{x \in \mathcal{X} : r(s_{r(\cdot; J)}(x); J) = h(x)\}| \geq |J| > |I^\star| = |\{x \in \mathcal{X} : r^\star(s_{r^\star}(x)) = h(x)\}|.$$

This contradicts the optimality of $r^\star$ as a Stackelberg equilibrium. Therefore, $I^\star$ is a maximum independent set in G_h .

□

2.6.2 Proofs for Deterministic Strategies

We begin by presenting the proof of Theorem 2.7.

Proof of Theorem 2.7. First, we show that $s_h(x) = x$. Let $x \leq \mathbf{h}$, then $h(x) = -$. Since $\mathbf{u}_+ \leq 0$, it follows that

$$U(h(x), x) = 0 \geq \max_{x' \in \mathcal{X}} U(h(x'), x) - |\phi(x') - \phi(x)|.$$

Therefore $s_h(x) = x$. Similar reasoning holds for $x > \mathbf{h}$. Now, the set of feature vectors whose label is correctly classified by h is given by

$$E(h, s_h; h)^c = \{x \in \mathcal{X} \mid h(s_h(x)) = h(x)\} = \mathcal{X}.$$

Since h correctly classifies all feature vectors, it is a Stackelberg equilibrium.

□

Next, we provide the proof of Theorem 2.8.

Proof of Theorem 2.8. Let r be the receiver Stackelberg equilibrium. Then without loss of generality, we can assume that there exists $t \in \mathcal{X}$ such that $r(t) \neq \Delta$. Furthermore, either there does not exist any $x \in \mathcal{X}$ such that $r(x) = +$, or there exists some $t \in \mathcal{X}$ such that $r(t) = +$. Suppose there does not exist any $x \in \mathcal{X}$ such that $r(x) = +$. Then $r(s_r(x)) \equiv -$ for all $x \in \mathcal{X}$. Thus, $E(r) = (\mathbf{h}, 1]$. However, the receiver strategy $r'(x) \equiv -$ for all $x \in \mathcal{X}$ yields $E(r') = (\mathbf{h}, 1]$. Therefore r' is also a Stackelberg equilibrium. Now, suppose there exists $t \in \mathcal{X}$ such that $r(t) = +$. Since $\mathbf{u}_+ \geq 1$, it follows that for all $x \in [0, \mathbf{h}]$, $r(s_r(x)) = +$. Therefore, $E(r) \supseteq [0, \mathbf{h}]$. However, the receiver strategy $r'(x) \equiv +$ for all $x \in \mathcal{X}$ yields $E(r') = [0, \mathbf{h}]$. Therefore r' is also a Stackelberg equilibrium. Comparing these two cases, it follows that $r(x) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label, is a Stackelberg equilibrium. $\square$

We now present the proof of Theorem 2.9.

Proof of Theorem 2.9. Let r be a receiver Stackelberg equilibrium such that there exist $a, b \in \mathcal{X}$ with $r(a) = +$ and $r(b) = -$. Since $\mathbf{u}_+ \geq 1$, for all $x \in [0, \mathbf{h})$, the worst-case best response is $s_r(x) = a$, so $r(s_r(x)) = +$. Similarly, for $x \in [\mathbf{h}, 1)$, $s_r(x) = b$, so $r(s_r(x)) = -$. Therefore, $E(r) = \mathcal{X}$ and $\mathcal{E}(r) = 1$. However, for a receiver strategy $r'(x) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label, we have $\mathcal{E}(r') \leq 0.5$. This contradicts the optimality of the Stackelberg equilibrium strategy r . Therefore, no such a, b exist. This implies that the range of an optimal receiver strategy r can contain at most one non-penalty label. Thus, $r(x) \equiv c$ for all $x \in \mathcal{X}$, where c is the majority label, is a receiver Stackelberg equilibrium strategy. $\square$

We analyze the worst-case best response and error of a strategy $r \in \mathcal{R}_2$.

Lemma 2.36. Let $c \in \mathcal{Y}$ be a label and $r \in \mathcal{R}_2$ be a constant-label receiver strategy $r(\cdot) \equiv c$. Then $s_r(x) = x$ and $E(r)^c = \{x : h(x) = c\}$ (here c denotes the set complement).

Proof. Since the receiver uses a constant label strategy, for a given $x \in \mathcal{X}$, the sender's payoff is $U(r(s_r(x)), x) - C(s_r(x), x) = U(c, x) - C(s_r(x), x)$. Since, the utility term remains fixed irrespective of the sender strategy, the sender will focus on minimizing the cost term. Consequently, $s_r(x) = x$ incurs the least cost. The receiver strategy r correctly classify feature vectors whose true label is c , hence $E(r)^c = \{x : h(x) = c\}$. $\square$

We begin by analyzing the worst-case best response and error of a strategy r in $\mathcal{R}_1$.

Lemma 2.37. Let $r(\cdot; \mathbf{h}^-, \mathbf{h}^+) \in \mathcal{R}_1$.

(a) If

$$\phi(\mathbf{h}) - \phi(\mathbf{h}^+) \leq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h}),$$

then $s_r(\cdot)$ is given by

$$s_r(x) := \begin{cases} x & \text{if } x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1], \\ \mathbf{h}^- & \text{if } x \in (\mathbf{h}, \mathbf{g}], \\ \mathbf{h}^+ & \text{if } x \in [\mathbf{h}^-, \mathbf{h}] \cup (\mathbf{g}, \mathbf{h}^+], \end{cases}$$

where $\mathbf{g} := \phi^{-1}\left(\frac{\phi(\mathbf{h}^-) + \phi(\mathbf{h}^+) + \mathbf{u}_-}{2}\right)$. Furthermore, $E(r) = [\mathbf{h}^-, \mathbf{g}]$.

(b) If

$$\phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h}),$$

then $s_r(\cdot)$ is given by

$$s_r(x) := \begin{cases} x & \text{if } x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1], \\ \mathbf{h}^+ & \text{if } x \in [\mathbf{h}^-, \mathbf{h}^+], \end{cases}$$

Furthermore $E(r) = [\mathbf{h}^-, \mathbf{h}]$ and $\mathbf{u}_+ \geq \phi(\mathbf{h}) - \phi(\mathbf{h}^-) > \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$.

Proof. We divide the task of finding s_r over $\mathcal{X}$ into various subintervals of $\mathcal{X}$.

Case (i): $x \in [0, \mathbf{h}^-)$.

The sender's payoff is given by $U(r(s_r(x)), x) - |\phi(s_r(x)) - \phi(x)|$. In the utility term, $r(s_r(x))$ can result in three labels: $-$, $+$ and Δ . The Δ label is unpreferable because it gives a $-\infty$ utility. Hence the sender will choose $s_r(x)$ such that $r(s_r(x)) = -$ or $+$. For $r(s_r(x)) = -$, from the definition of r , $s_r(x) \in [0, \mathbf{h}^-]$. Among these options, the feature vector which incurs the least cost to the sender is x . For $r(s_r(x)) = +$, from the definition of r , $s_r(x) \in [\mathbf{h}^+, 1]$. Among these options, the feature vector which incurs the least cost to the sender is $\mathbf{h}^+$. Accordingly, the sender's payoff at x is

$$\mathbb{U}(x; r, s_r) := \begin{cases} \mathbf{u}_+ + \phi(x) - \phi(s_r(x)) & \text{if } s_r(x) = \mathbf{h}^+ \\ 0 & \text{if } s_r(x) = x. \end{cases}$$

Since ϕ is an increasing function and $r \in \mathcal{R}_1$ satisfies $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+ > 0$, therefore for $x \in [0, \mathbf{h}^-)$, we have

$$\mathbf{u}_+ + \phi(x) - \phi(s_r(x)) < \mathbf{u}_+ + \phi(\mathbf{h}^-) - \phi(\mathbf{h}^+) = \mathbf{u}_+ - \mathbf{u}_+ = 0.$$

Thus the payoff for modifying x to $\mathbf{h}^+$ is negative and hence $s_r(x) = x$.

Case (ii): $x \in (\mathbf{h}, \mathbf{h}^+]$.

Arguing as in Case (i), $s_r(x)$ can be equal to $\mathbf{h}^-$ or $\mathbf{h}^+$. Hence the sender's payoff at x is

$$\mathbb{U}(x; r, s_r) := \begin{cases} \phi(x) - \phi(\mathbf{h}^+) & \text{if } s_r(x) = \mathbf{h}^+ \\ \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x) & \text{if } s_r(x) = \mathbf{h}^-. \end{cases}$$

If $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})$, then since ϕ is strictly increasing, for all $x \in (\mathbf{h}, \mathbf{h}^+]$ we have $\phi(x) - \phi(\mathbf{h}^+) > \phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h}) \geq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x)$. Hence $s_r(x) = \mathbf{h}^+$. If $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) \leq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})$, then there exists $\mathbf{g} \in (\mathbf{h}, \mathbf{h}^+]$ (as defined in the statement of this lemma) such that

$$\phi(\mathbf{g}) - \phi(\mathbf{h}^+) = \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{g}).$$

Hence, if $\phi(x) > \phi(\mathbf{g})$, then $s_r(x) = \mathbf{h}^+$. If $\phi(x) \leq \phi(\mathbf{g})$, then $s_r(x) = \mathbf{h}^-$; note that at $x = \mathbf{g}$, the payoffs for reporting $\mathbf{h}^-$ and $\mathbf{h}^+$ are equal, and under worst-case tie-breaking (Definition 2.2), the sender selects $\mathbf{h}^-$ because $r(\mathbf{h}^-) = - \neq + = h(\mathbf{g})$ causes misclassification, whereas $r(\mathbf{h}^+) = + = h(\mathbf{g})$ would result in correct classification. Since ϕ is strictly increasing, this implies that

$$s_r(x) = \begin{cases} \mathbf{h}^- & \text{if } x \in (\mathbf{h}, \mathbf{g}], \\ \mathbf{h}^+ & \text{if } x \in (\mathbf{g}, \mathbf{h}^+]. \end{cases}$$

Case (iii): $x \in [\mathbf{h}^-, \mathbf{h}]$.

Arguing as in Case (i), $s_r(x)$ can be equal to $\mathbf{h}^-$ or $\mathbf{h}^+$. Accordingly, the sender's payoff at x is

$$\mathbb{U}(x; r, s_r) := \begin{cases} \mathbf{u}_+ + \phi(x) - \phi(\mathbf{h}^+) & \text{if } s_r(x) = \mathbf{h}^+ \\ \phi(\mathbf{h}^-) - \phi(x) & \text{if } s_r(x) = \mathbf{h}^-. \end{cases}$$

Since

$$\mathbf{u}_+ + \phi(x) - \phi(\mathbf{h}^+) \geq 0 \text{ and } \phi(\mathbf{h}^-) - \phi(x) \leq 0,$$

therefore $s_r(x) = \mathbf{h}^+$ (with worst-case tie-breaking at $x = \mathbf{h}^-$ where $r(\mathbf{h}^+) = + \neq - = h(\mathbf{h}^-)$).

Case (iv): Let $x \in (\mathbf{h}^+, 1]$. Arguing as in Case (i), $s_r(x)$ can be equal to $\mathbf{h}^-$ or x . Accordingly, the sender's payoff at x is

$$\mathbb{U}(x; r, s_r) := \begin{cases} \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x) & \text{if } s_r(x) = \mathbf{h}^- \\ 0 & \text{if } s_r(x) = x. \end{cases}$$

Since ϕ is an increasing function and $r \in \mathcal{R}_1$ satisfies $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+ > 0$, therefore for $x \in (\mathbf{h}^+, 1]$, we have

$$\mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(x) < \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h}^+) = \mathbf{u}_- - \mathbf{u}_+ < 0.$$

Therefore $s_r(x) = x$.

Hence, combining all cases (i)–(iv), we get that if $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) \leq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})$, then

$$s_r(x) := \begin{cases} x & \text{if } x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1], \\ \mathbf{h}^- & \text{if } x \in (\mathbf{h}, \mathbf{g}], \\ \mathbf{h}^+ & \text{if } x \in [\mathbf{h}^-, \mathbf{h}] \cup (\mathbf{g}, \mathbf{h}^+], \end{cases}$$

and $E(r) = [\mathbf{h}^-, \mathbf{g}]$, where $\mathbf{g} := \phi^{-1}\left(\frac{\phi(\mathbf{h}^-) + \phi(\mathbf{h}^+) + \mathbf{u}_-}{2}\right)$, which proves part (a).

Similarly combining all the cases, we get if $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})$, then

$$s_r(x) := \begin{cases} x & \text{if } x \in [0, \mathbf{h}^-) \cup (\mathbf{h}^+, 1], \\ \mathbf{h}^+ & \text{if } x \in [\mathbf{h}^-, \mathbf{h}^+], \end{cases}$$

and $E(r) = [\mathbf{h}^-, \mathbf{h}]$ and $\mathbf{u}_+ \geq \phi(\mathbf{h}) - \phi(\mathbf{h}^-) > \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$, which proves part (b). $\square$

Lemma 2.38. Let r be a receiver strategy. Let $h(x)$ be a true function, and let $x_1, x_2 \in E(r)^c$ (c denote the set complement) be feature vectors such that $x_1 \leq \mathbf{h} < x_2$. Let s_r be the worst-case best response of r . Let $z_1, z_2 \in \mathcal{X}$ be such that $z_1 = s_r(x_1)$ and $z_2 = s_r(x_2)$. Now consider the following two sender strategies

$$\hat{s}(t) := \begin{cases} z_2 & \text{for } t = x_1 \\ s_r(t) & \text{otherwise} \end{cases}, \quad \bar{s}(t) := \begin{cases} z_1 & \text{for } t = x_2 \\ s_r(t) & \text{otherwise} \end{cases}. \quad (2.34)$$

Then

$$\mathbb{U}(x_1; r, s_r) > \mathbb{U}(x_1; r, \hat{s}) \quad \text{and} \quad \mathbb{U}(x_2; r, s_r) > \mathbb{U}(x_2; r, \bar{s}), \quad (2.35)$$

which implies

$$-|\phi(z_1) - \phi(x_1)| > \mathbf{u}_+ - |\phi(z_2) - \phi(x_1)| \quad \text{and} \quad -|\phi(z_2) - \phi(x_2)| > \mathbf{u}_- - |\phi(z_1) - \phi(x_2)|.$$

Consequently,

$$(|\phi(z_2) - \phi(x_1)| - |\phi(z_2) - \phi(x_2)|) + (|\phi(z_1) - \phi(x_2)| - |\phi(z_1) - \phi(x_1)|) > \mathbf{u}_+ + \mathbf{u}_-. \quad (2.36)$$

Proof. Since s_r is a worst-case best response to r , we have

$$\mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, \hat{s}) \text{ and } \mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, \bar{s}) \text{ for all } x \in \mathcal{X}.$$

Since $\hat{s}(t) = s_r(t)$ for $t \in \mathcal{X} \setminus \{x_1\}$ and $\bar{s}(t) = s_r(t)$ for $t \in \mathcal{X} \setminus \{x_2\}$, we have

$$\mathbb{U}(t; r, s_r) = \mathbb{U}(t; r, \hat{s}) \text{ for } t \in \mathcal{X} \setminus \{x_1\} \text{ and } \mathbb{U}(t; r, s_r) = \mathbb{U}(t; r, \bar{s}) \text{ for } t \in \mathcal{X} \setminus \{x_2\}.$$

If $\mathbb{U}(x_1; r, s_r) = \mathbb{U}(x_1; r, \hat{s})$, then $\hat{s} \in B(r)$ and $E(r, \hat{s}) = E(r, s_r) \cup \{x_1\}$. This contradicts the definition of s_r given in Definition 2.2. Hence, $\mathbb{U}(x_1; r, s_r) > \mathbb{U}(x_1; r, \hat{s})$. Similarly, if $\mathbb{U}(x_2; r, s_r) = \mathbb{U}(x_2; r, \bar{s})$, then $\bar{s} \in B(r)$ and $E(r, \bar{s}) = E(r, s_r) \cup \{x_2\}$. This contradicts the definition of s_r . Hence, $\mathbb{U}(x_2; r, s_r) > \mathbb{U}(x_2; r, \bar{s})$, establishing (2.35). Substituting the payoffs under $r(z_1) = h(x_1) = -$ and $r(z_2) = h(x_2) = +$ into (2.35) yields (2.38). Adding the two inequalities in (2.38) and rearranging terms yields (2.36). $\square$

We say a strategy r dominates another receiver strategy $\tilde{r}$ if $E(r, s_r, h, D) \subseteq E(\tilde{r}, s_{\tilde{r}}, h, D)$.

We now show that the family of receiver strategies $\mathcal{R}$ dominates every other receiver strategy.

Lemma 2.39. Let $D \in \mathfrak{D}$ be a probability law on $\mathcal{X}$ and let $h(x)$ be a true function. Let $\tilde{r}$ be any receiver strategy. Then there exists a strategy $r \in \mathcal{R}$ such that $\mathcal{E}(\tilde{r}) \geq \mathcal{E}(r)$.

Proof. Suppose $E(\tilde{r})^c \cap [0, \mathbf{h}] = \emptyset$ or $E(\tilde{r})^c \cap (\mathbf{h}, 1] = \emptyset$, then $\tilde{r}$ misclassifies an entire class and thus is dominated by a constant majority-label strategy $r \in \mathcal{R}_2$. Therefore we can assume $E(\tilde{r})^c \cap [0, \mathbf{h}] \neq \emptyset$ and $E(\tilde{r})^c \cap (\mathbf{h}, 1] \neq \emptyset$. Let $\mathbf{h}^- := \sup(E(\tilde{r})^c \cap [0, \mathbf{h}])$ and $\tilde{\mathbf{g}} := \inf(E(\tilde{r})^c \cap (\mathbf{h}, 1])$ (here c represents the set complement) be well-defined. It follows that $\mathbf{h}^-$ represents the supremum of all the feature vectors with true label $-$ that get correctly classified by $\tilde{r}$. Similarly, $\tilde{\mathbf{g}}$ represents the infimum of all the feature vectors with true label $+$ that get correctly classified by $\tilde{r}$. Hence $E(\tilde{r}) \supset (\mathbf{h}^-, \tilde{\mathbf{g}})$.

Case (i): $\mathbf{h}^- = 0$.

In this case, $E(\tilde{r}) \supset (0, \mathbf{h})$, i.e., it misclassifies all the feature vectors with the true label $-$. Hence, for a receiver strategy $r \in \mathcal{R}_2$ which assigns the majority label, $\mathcal{E}(r) \leq \mathcal{E}(\tilde{r})$.

Case (ii): $\tilde{\mathbf{g}} = 1$.

In this case, $E(\tilde{r}) \supset (\mathbf{h}, 1)$ i.e., it misclassifies all the feature vectors with the true label $+$. Hence, for a receiver strategy $r \in \mathcal{R}_2$ which assigns the majority label, $\mathcal{E}(r) \leq \mathcal{E}(\tilde{r})$.

Case (iii): $\mathbf{h}^- > 0$, $\tilde{\mathbf{g}} < 1$, and $\phi(\mathbf{h}^-) > 1 - \mathbf{u}_+$.

Since $\mathbf{h}^- \leq \mathbf{h}$, for any $z \in (\mathbf{h}, 1]$ we have $\phi(z) - \phi(\mathbf{h}^-) \leq 1 - \phi(\mathbf{h}^-) < \mathbf{u}_+$. We show that no feature in $(\mathbf{h}, 1]$ can be correctly classified under $\tilde{r}$. Let $x_1 \in E(\tilde{r})^c \cap [0, \mathbf{h}]$ be a feature sufficiently close to $\mathbf{h}^-$. First, no $z \in (\mathbf{h}, 1]$ can satisfy $\tilde{r}(z) = +$, otherwise x_1 would report z and obtain the payoff $\mathbf{u}_+ - |\phi(z) - \phi(x_1)| > 0 \geq \mathbb{U}(x_1; \tilde{r}, s_{\tilde{r}})$, a contradiction. Therefore $\tilde{r}(s_{\tilde{r}}((\mathbf{h}, 1])) \in \{-, \Delta\}$. Second, no feature $x_2 \in (\mathbf{h}, 1]$ can be correctly classified by reporting $y_2 \leq \mathbf{h}$ with $\tilde{r}(y_2) = +$. To prevent x_1 from reporting y_2 , we must have $\phi(y_2) \leq \phi(\mathbf{h}^-) - \mathbf{u}_+ < \phi(x_1)$, so $y_2 < x_1 < x_2$. Thus, we have $|\phi(y_2) - \phi(x_1)| - |\phi(y_2) - \phi(x_2)| = -(\phi(x_2) - \phi(x_1))$, which together with $|\phi(y_1) - \phi(x_2)| - |\phi(y_1) - \phi(x_1)| \leq \phi(x_2) - \phi(x_1)$ makes the left-hand side of (2.36) in Lemma 2.38 at most 0, contradicting (2.36) since $\mathbf{u}_+ + \mathbf{u}_- > 0$. Hence, all features in $(\mathbf{h}, 1]$ are misclassified and therefore $\tilde{r}$ is dominated by the constant strategy $r(x) \equiv - \in \mathcal{R}_2$.

Case (iv): $\mathbf{h}^- > 0$, $\tilde{\mathbf{g}} < 1$, $\phi(\mathbf{h}^-) \leq 1 - \mathbf{u}_+$, and $\phi(\tilde{\mathbf{g}}) - \phi(\mathbf{h}^-) \geq \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$.

Let $\mathbf{h}^+ \in \mathcal{X}$ be such that $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+$. Such a $\mathbf{h}^+$ exists because $\phi(\mathbf{h}^-) \leq 1 - \mathbf{u}_+$, and it satisfies $\mathbf{h} < \mathbf{h}^+ \leq 1$. Consider the receiver strategy $r(x; \mathbf{h}^-, \mathbf{h}^+) \in \mathcal{R}_1$. By Lemma 2.37, if $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) \leq \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})$, then $E(r) = [\mathbf{h}^-, \mathbf{g}]$, where $\phi(\mathbf{g}) - \phi(\mathbf{h}^-) = \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$. Since $\phi(\tilde{\mathbf{g}}) - \phi(\mathbf{h}^-) \geq \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$, it follows that $\mathbf{g} \leq \tilde{\mathbf{g}}$, and hence $E(r) = [\mathbf{h}^-, \mathbf{g}] \subseteq [\mathbf{h}^-, \tilde{\mathbf{g}}]$. Otherwise, if $\phi(\mathbf{h}) - \phi(\mathbf{h}^+) > \mathbf{u}_- + \phi(\mathbf{h}^-) - \phi(\mathbf{h})$, then $E(r) = [\mathbf{h}^-, \mathbf{h}] \subseteq [\mathbf{h}^-, \tilde{\mathbf{g}}]$, since $\mathbf{h} \leq \tilde{\mathbf{g}}$. Thus, in either case, $E(r) \subseteq [\mathbf{h}^-, \tilde{\mathbf{g}}]$. Moreover, since $E(\tilde{r}) \supseteq (\mathbf{h}^-, \tilde{\mathbf{g}})$, we obtain $\mathcal{E}(r) \leq \mathcal{E}(\tilde{r})$.

Case (v): $\mathbf{h}^- > 0$, $\tilde{\mathbf{g}} < 1$, $\phi(\mathbf{h}^-) \leq 1 - \mathbf{u}_+$, and $\phi(\tilde{\mathbf{g}}) - \phi(\mathbf{h}^-) < \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}$.

We will show that this case cannot occur. Let ϵ be such that $0 < 2\epsilon < \frac{\mathbf{u}_+ + \mathbf{u}_-}{2} - \phi(\tilde{\mathbf{g}}) + \phi(\mathbf{h}^-)$. Let $\mathbf{h}^-, \tilde{\mathbf{g}}' \in E(\tilde{r})^c$ be such that $\mathbf{h}^- \leq \mathbf{h} < \tilde{\mathbf{g}} \leq \tilde{\mathbf{g}}'$ and

$$\phi(\mathbf{h}^-) - \phi(\mathbf{h}^{'}) < \epsilon \quad \text{and} \quad \phi(\tilde{\mathbf{g}}') - \phi(\tilde{\mathbf{g}}) < \epsilon. \quad (2.37)$$

Such feature vectors $\mathbf{h}^{'}, \tilde{\mathbf{g}}'$ exist by the definition of $\mathbf{h}^-$ and $\tilde{\mathbf{g}}$ and the continuity of ϕ . Adding

the inequalities in (2.37), and using the definition of ϵ , we get

$$\phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-'}) < \frac{\mathbf{u}_+ + \mathbf{u}_-}{2}. \quad (2.38)$$

Let $y_1 := s_{\tilde{r}}(\mathbf{h}^{-'})$ and $y_2 := s_{\tilde{r}}(\tilde{\mathbf{g}}')$. Since $\mathbf{h}^{-'} \leq \mathbf{h} < \tilde{\mathbf{g}}'$ with $\mathbf{h}^{-'}, \tilde{\mathbf{g}}' \in E(\tilde{r})^c$, applying (2.36) from Lemma 2.38 with $x_1 = \mathbf{h}^{-'}$ and $x_2 = \tilde{\mathbf{g}}'$ yields

$$(|\phi(y_2) - \phi(\mathbf{h}^{-'})| - |\phi(y_2) - \phi(\tilde{\mathbf{g}}')|) + (|\phi(y_1) - \phi(\tilde{\mathbf{g}}')| - |\phi(y_1) - \phi(\mathbf{h}^{-'})|) > \mathbf{u}_+ + \mathbf{u}_-. \quad (2.39)$$

By the reverse triangle inequality on $\mathbb{R}$, $|A| - |B| \leq |A - B|$, applied to each term in parentheses, we have for any reports $y_1, y_2 \in \mathcal{X}$:

$$|\phi(y_2) - \phi(\mathbf{h}^{-'})| - |\phi(y_2) - \phi(\tilde{\mathbf{g}}')| \leq |(\phi(y_2) - \phi(\mathbf{h}^{-'})) - (\phi(y_2) - \phi(\tilde{\mathbf{g}}'))| = \phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-'})$$

and

$$|\phi(y_1) - \phi(\tilde{\mathbf{g}}')| - |\phi(y_1) - \phi(\mathbf{h}^{-'})| \leq |(\phi(y_1) - \phi(\tilde{\mathbf{g}}')) - (\phi(y_1) - \phi(\mathbf{h}^{-'}))| = \phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-'}).$$

Summing these two inequalities bounds the left-hand side of (2.39) from above:

$$2(\phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-'})) \geq \text{LHS of (2.39)} > \mathbf{u}_+ + \mathbf{u}_- \implies \phi(\tilde{\mathbf{g}}') - \phi(\mathbf{h}^{-'}) > \frac{\mathbf{u}_+ + \mathbf{u}_-}{2},$$

which directly contradicts (2.38). Hence, Case (v) cannot occur.

Based on all five cases (i)–(v), there always exists a receiver strategy $r \in \mathcal{R}$ which dominates $\tilde{r}$. Therefore, the family of receiver strategies $\mathcal{R}$ dominates all other receiver strategies. $\square$

We now complete the proof of Theorem 2.10.

Proof of Theorem 2.10. From Lemma 2.39, every receiver strategy $\tilde{r}$ is dominated by a strategy in $\mathcal{R} = \mathcal{R}_1 \cup \mathcal{R}_2$. Hence, an optimal receiver Stackelberg equilibrium corresponds to a strategy in $\mathcal{R}$ that minimizes the classification error $\mathcal{E}(r)$. The finite set $\mathcal{R}_2 = \{r \equiv -, r \equiv +\}$ contains a minimum (the majority class under D). For $\mathcal{R}_1$, each strategy $r(\cdot; \mathbf{h}^-, \mathbf{h}^+)$ satisfies $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+$ and is uniquely parameterized by $\mathbf{h}^-$ on the compact interval $I := [\phi^{-1}(\max\{0, \phi(\mathbf{h}) - \mathbf{u}_+\}), \phi^{-1}(\min\{\phi(\mathbf{h}), 1 - \mathbf{u}_+\})]$. By Lemma 2.37, the error set of $r(x; \mathbf{h}^-, \mathbf{h}^+)$ is the interval $[\mathbf{h}^-, \min\{\mathbf{g}, \mathbf{h}\}]$, where $\mathbf{g} = \phi^{-1}(\frac{\phi(\mathbf{h}^-) + \phi(\mathbf{h}^+) + \mathbf{u}_-}{2}) = \phi^{-1}(\phi(\mathbf{h}^-) + \frac{\mathbf{u}_+ + \mathbf{u}_-}{2})$ depends continuously on $\mathbf{h}^-$. Because the probability law D has a continuous density function, the error mapping $\mathbf{h}^- \mapsto \mathcal{E}(r(\cdot; \mathbf{h}^-, \mathbf{h}^+)) = D([\mathbf{h}^-, \min\{\mathbf{g}, \mathbf{h}\}])$ is continuous on the

compact domain I . By the Extreme Value Theorem (Weierstrass), this continuous objective attains its minimum on I . Comparing the minimum errors across $\mathcal{R}_1$ and $\mathcal{R}_2$, an error-minimizing strategy in $\mathcal{R}$ exists, completing the proof. $\square$

Finally, we present the proof of Theorem 2.11.

Proof 2.6.1. (Proof of Theorem 2.11)

By the continuity of ϕ and the hypothesis $\phi(\mathbf{h}) + \mathbf{u}_+ \leq 1$, there exists $\mathbf{h}^+ \in \mathcal{X}$ such that $\phi(\mathbf{h}^+) - \phi(\mathbf{h}) = \mathbf{u}_+$. We now verify that every feature vector is correctly classified under the sender's worst-case best response to strategy $r(x; \mathbf{h}, \mathbf{h}^+)$.

Case (i): Let $x \in [0, \mathbf{h}]$. To obtain the $+$ label, the sender must report $y \geq \mathbf{h}^+$, yielding a payoff of $\mathbf{u}_+ - |\phi(y) - \phi(x)| \leq 0$. In contrast, if the sender reports x truthfully, the resulting payoff is 0. Thus, the best response is $s_r(x) = x$, which implies $E(r)^c \subseteq [0, \mathbf{h}]$.

Case (ii): Let $x \in (\mathbf{h}, 1]$. To obtain the $+$ label, the sender reports $y = \max\{x, \mathbf{h}^+\}$, which yields a payoff of $0 - |\phi(x) - \phi(y)|$. Conversely, obtaining the $-$ label yields a payoff of $\mathbf{u}_- - \text{cost} \leq \mathbf{u}_- \leq -\mathbf{u}_+$. If $x \in (\mathbf{h}, \mathbf{h}^+)$, the sender's payoff under $y = \mathbf{h}^+$ is $0 - |\phi(\mathbf{h}^+) - \phi(x)| \geq \phi(\mathbf{h}) - \phi(\mathbf{h}^+) = -\mathbf{u}_+ \geq \mathbf{u}_-$. If $x \in [\mathbf{h}^+, 1]$, the sender's payoff is $\mathbf{u}_+$. Thus, $s_r(x) = \max\{\mathbf{h}^+, x\}$. Therefore, $E(r) = \emptyset$, establishing that $r(\cdot; \mathbf{h}, \mathbf{h}^+)$ is a Stackelberg equilibrium.

2.6.3 Proofs for the Multi-Dimensional Decision Problem

We begin by presenting the proof of Theorem 2.12.

Proof 2.6.2. (Proof of Theorem 2.12)

Let r be a receiver strategy and let s_r be its worst-case best response. Define

$$K_+ = \{x : w^\top x > \mathbf{h} \text{ and } r(s_r(x)) = h(x) = +\}, \quad K_- = \{x : w^\top x \leq \mathbf{h} \text{ and } r(s_r(x)) = h(x) = -\}.$$

For each $z \in \mathcal{Z}$, define

$$f(z) := \inf \left(\{t > 0 : z + tw \in K_+\} \cup \{2\mathbf{u}\} \right), \quad g(z) := \sup \left(\{t \leq 0 : z + tw \in K_-\} \cup \{-2\mathbf{u}\} \right).$$

We claim that

$$f(z) - g(z) \geq 2\mathbf{u} \quad \text{for all } z \in \mathcal{Z}.$$

Suppose there exists $z \in \mathcal{Z}$ such that

$$f(z) - g(z) < 2\mathbf{u}. \tag{2.40}$$

This then rules out the possibility that either $f(z) = 2\mathbf{u}$ or $g(z) = -2\mathbf{u}$. By definition of infimum and supremum, for sufficiently small $\epsilon > 0$, there exist $t_1 \in (0, f(z) + \epsilon)$ and $t_2 \in (g(z) - \epsilon, 0]$ such that $\bar{x} := z + t_1 w \in K_+$ and $\underline{x} := z + t_2 w \in K_-$. Then

$$\|\bar{x} - \underline{x}\| = (t_1 - t_2)\|w\| < 2\mathbf{u}.$$

Suppose $\|\bar{x} - s_r(\bar{x})\| \leq \|\underline{x} - s_r(\underline{x})\|$ (similar reasoning applies for the opposite inequality). Then the payoff for $\underline{x}$ under s_r is

$$-\|\underline{x} - s_r(\underline{x})\|.$$

Now consider a new sender strategy s defined by

$$s(x) := \begin{cases} s_r(x) & \text{if } x \neq \underline{x} \\ s_r(\bar{x}) & \text{if } x = \underline{x} \end{cases}.$$

The payoff for $\underline{x}$ under s is

$$2\mathbf{u} - \|\underline{x} - s_r(\bar{x})\| \geq 2\mathbf{u} - \|\underline{x} - \bar{x}\| - \|\bar{x} - s_r(\bar{x})\| > -\|\bar{x} - s_r(\bar{x})\| \geq -\|\underline{x} - s_r(\underline{x})\|,$$

which contradicts the optimality of s_r . Hence, the assumption in (2.40) is false, and therefore

$$f(z) - g(z) \geq 2\mathbf{u}$$

for all $z \in \mathcal{Z}$. By the definitions of f and g , for $z \in \mathcal{Z}$, all points $z + tw$, with $t \in (g(z), f(z))$, are misclassified by r . Thus,

$$E(r) \supseteq \{x : x = z + tw, z \in \mathcal{Z}, t \in (g(z), f(z))\}$$

and

$$\mathcal{E}(r) = \int_{\mathcal{X}} \mathbb{1}_{E(r)} dx = \int_E dx.$$

We now evaluate this integral by using a change of variables. Let $\Phi : \mathcal{Z} \times \mathbb{R} \rightarrow \mathbb{R}^n$ be the change of variable map given by $\Phi(z, t) = z + tw$. It follows that $\det |J_\Phi| = \|w\| = 1$. Thus,

$$\int_E dx \geq \int_{\mathcal{Z}} \int_{g(z)}^{f(z)} \det |J_\Phi| d\sigma(z) dt = \int_{\mathcal{Z}} \int_{g(z)}^{f(z)} d\sigma(z) dt = \int_{\mathcal{Z}} (f(z) - g(z)) d\sigma(z) \geq 2\mathbf{u} \sigma(\mathcal{Z}),$$

where $\sigma(\mathcal{Z})$ is the surface area of $\mathcal{Z}$ induced by D .

We now provide the proof of Theorem 2.13.

Proof 2.6.3. (Proof of Theorem 2.13)

For $z \in \mathcal{Z}$, let

$$K_+ := \{x : w^\top x \geq \mathbf{h} + \mathbf{u}\} \quad \text{and} \quad K_- := \{x : w^\top x \leq \mathbf{h} - \mathbf{u}\}.$$

Let $x \in K_+$. We now calculate the worst-case best response for x . Let $s_r(x) = t$. If $t \in K_-$, then the payoff for the sender is $2\mathbf{u} - \|x - t\| < 0$ because K_+ and K_- are separated by $2\mathbf{u}$ units. Since $r(x) = \Delta$ for $x \in \mathcal{X} \setminus (K_+ \cup K_-)$, it follows that $t \notin \mathcal{X} \setminus (K_+ \cup K_-)$. Thus, $t = x$, as the sender gets 0 payoff, which is the maximum it could get. Hence, $s_r(x) = x$ for $x \in K_+$. Using similar reasoning for $x \in K_-$, we get $s_r(x) = x$ for $x \in K_-$ as well. Therefore,

$$E(r) \subset \{x : x = z + tw, z \in \mathcal{Z}, t \in (-\mathbf{u}, \mathbf{u})\}. \text{ and } \mathcal{E}(r) = \int_{E(r)} dx.$$

Evaluating this integral by using a change of variables from x to (z, t) , we get

$$\int_{E(r)} dx \leq \int_{\mathcal{Z}} \int_{-\mathbf{u}}^{\mathbf{u}} dt d\sigma(z) = \int_{\mathcal{Z}} 2\mathbf{u} d\sigma(z) = 2\mathbf{u} \sigma(\mathcal{Z}),$$

where $\sigma(\mathcal{Z})$ is the surface area of $\mathcal{Z}$ induced by D . Theorem 2.12 establishes that the receiver's error is lower-bounded by $2\mathbf{u} \sigma(\mathcal{Z})$. We conclude that $\mathcal{E}(r) = 2\mathbf{u} \sigma(\mathcal{Z})$ and r is a Stackelberg equilibrium.

2.6.4 Proofs for Stochastic Strategies

We begin by presenting the proof of Proposition 2.16.

Proof of Proposition 2.16. By definition, both s and s' are best responses to r . Suppose

$$\mathcal{E}(r, s'; h) > \mathcal{E}(r, s; h).$$

This contradicts the assumption that s is the worst-case best response, as it implies there exists another strategy s' that results in a strictly higher error for the receiver. Therefore, it follows that

$$\mathcal{E}(r, s; h) \geq \mathcal{E}(r, s'; h),$$

as claimed. □

Next, we provide the proof of Corollary 2.17.

Proof of Corollary 2.17. Let s' be the worst-case lazy best response to $r^\star$. Let $(\bar{r}^\star, s_{\bar{r}^\star})$ be the Stackelberg equilibrium under the worst-case lazy best response. From Proposition 2.16, we know that

$$\mathcal{E}(r^\star, s_{r^\star}) \geq \mathcal{E}(r^\star, s').$$

Additionally, by the definition of $\bar{r}^\star$, we have

$$\mathcal{E}(r^\star, s') \geq \mathcal{E}(\bar{r}^\star, s_{\bar{r}^\star}).$$

From Theorem 2.18, we know that

$$\mathcal{E}(\bar{r}^\star, s_{\bar{r}^\star}) = \frac{\mathbf{u}}{2}.$$

Combining these results, we conclude that

$$\mathcal{E}(r^\star, s_{r^\star}) \geq \frac{\mathbf{u}}{2}.$$

Furthermore, from Example 2.21 (which appears in Section 2.3.5), we know that for every $\epsilon > 0$,

$$\mathcal{E}(r_\epsilon, s_{r_\epsilon}) = \frac{\mathbf{u} + \epsilon}{2}.$$

Thus, if a Stackelberg equilibrium $r^\star$ exists, then it follows that

$$\mathcal{E}(r^\star, s_{r^\star}) = \frac{\mathbf{u}}{2}.$$

□

From the definition of $\mathbf{R}$ in (2.13), every receiver strategy $r \in \mathbf{R}$ has the structure $r(x) \equiv (p(x), q(x), 1 - p(x) - q(x))$. The first lemma shows that without loss of generality, we can assume the structure of r at each x to be either $r(x) \equiv (1 - q(x), q(x), 0)$ or $r(x) \equiv (0, 0, 1)$.

Lemma 2.40. Let $r \equiv (p, q, 1 - p - q) \in \mathbf{R}$ be a receiver strategy. Let $\Theta := \{x' : x' \in \mathcal{X}, p(x') + q(x') < 1\}$. Now consider another receiver strategy $\tilde{r} \equiv (\tilde{p}, \tilde{q}, 1 - \tilde{p} - \tilde{q})$ defined as

$$\tilde{r}(x) = \begin{cases} r(x) & \text{if } x \in \mathcal{X} \setminus \Theta, \\ (0, 0, 1) & \text{if } x \in \Theta. \end{cases}$$

Then $\mathcal{E}(r) = \mathcal{E}(\tilde{r})$.

Proof. Let $x \in \mathcal{X}$. We first show that $s_r(x), s_{\tilde{r}}(x) \notin \Theta$. From the definition of p, q and Θ , we have

$$1 - p(x') - q(x') = \mathbb{P}_r(\Delta | X' = x') > 0, \text{ for all } x' \in \Theta.$$

If $s_r(x) \in \Theta$, then

$$\mathbb{P}_r(y = \Delta | X'_{s_r} = s_r(x)) > 0$$

and the sender's payoff is $\mathbb{U}(x; r, s_r) = -\infty$. Similarly, if $s_{\tilde{r}}(x) \in \Theta$, then

$$\mathbb{P}_{\tilde{r}}(y = \Delta | X'_{s_{\tilde{r}}} = s_{\tilde{r}}(x)) = 1$$

and the sender's payoff is $\mathbb{U}(x; \tilde{r}, s_{\tilde{r}}) = -\infty$. But from (2.11), the feature vector set $\mathcal{X} \setminus \Theta \neq \emptyset$.

Therefore, there exists $t \in \mathcal{X} \setminus \Theta$ such that

$$\mathbb{P}_r(Y = \Delta | X' = t) = \mathbb{P}_{\tilde{r}}(Y = \Delta | X' = t) = 0.$$

Hence, for the sender strategy $s(x) = t$ for all $x \in \mathcal{X}$, we have

$$\mathbb{P}_r(Y = \Delta | X' = t) = \mathbb{P}_{\tilde{r}}(Y = \Delta | X' = t) = 0$$

and the sender's payoff at x for receiver strategies r and $\tilde{r}$ will be $\mathbb{U}(x, r, s) > -\infty$ and $\mathbb{U}(x, \tilde{r}, s) > -\infty$ respectively. But this contradicts that s_r and $s_{\tilde{r}}$ are the worst-case lazy best responses, as $\mathbb{U}(x, r, s) > \mathbb{U}(x, r, s_r)$ and $\mathbb{U}(x, \tilde{r}, s) > \mathbb{U}(x, \tilde{r}, s_{\tilde{r}})$. Hence,

$$s_r(x), s_{\tilde{r}}(x) \in \mathcal{X} \setminus \Theta \text{ for all } x \in \mathcal{X}. \quad (2.41)$$

From the construction of $\tilde{r}$, we know that $r(x) = \tilde{r}(x)$ for $x \in \mathcal{X} \setminus \Theta$. Therefore, combining it with (2.41), we get

$$r(s_r(x)) = \tilde{r}(s_r(x)) \text{ and } r(s_{\tilde{r}}(x)) = \tilde{r}(s_{\tilde{r}}(x)) \text{ for all } x \in \mathcal{X}. \quad (2.42)$$

We next show that s_r and $s_{\tilde{r}}$ are best responses to both r and $\tilde{r}$. Since s_r and $s_{\tilde{r}}$ are worst-case lazy best responses of r and $\tilde{r}$, respectively, we have

$$\mathbb{U}(x; r, s_r) \geq \mathbb{U}(x; r, s_{\tilde{r}}) \text{ and } \mathbb{U}(x; \tilde{r}, s_{\tilde{r}}) \geq \mathbb{U}(x; \tilde{r}, s_r) \text{ for all } x \in \mathcal{X}. \quad (2.43)$$

Substituting (2.42) in (2.43) and expanding it, we get

$$\mathbb{U}(x; r, s_r) - |s_r(x) - x| \geq \mathbb{U}(x; r, s_{\tilde{r}}) - |s_{\tilde{r}}(x) - x| = \mathbb{U}(x; \tilde{r}, s_{\tilde{r}}) - |s_{\tilde{r}}(x) - x| \quad \text{and}$$

$$\mathbb{U}(x; \tilde{r}, s_{\tilde{r}}) - |s_{\tilde{r}}(x) - x| \geq \mathbb{U}(x; \tilde{r}, s_r) - |s_r(x) - x| = \mathbb{U}(x; r, s_r) - |s_r(x) - x|.$$

which implies

$$|s_{\tilde{r}}(x) - x| - |s_r(x) - x| \geq \mathfrak{U}(x, \tilde{r}, s_{\tilde{r}}) - \mathfrak{U}(x, r, s_r) \geq |s_{\tilde{r}}(x) - x| - |s_r(x) - x|$$

or

$$\mathfrak{U}(x; \tilde{r}, s_{\tilde{r}}) - \mathfrak{U}(x; r, s_r) = |s_{\tilde{r}}(x) - x| - |s_r(x) - x|. \quad (2.44)$$

Thus, substituting (2.44) back in (2.43), we get

$$\mathbb{U}(x; r, s_r) = \mathbb{U}(x; r, s_{\tilde{r}}) \text{ and } \mathbb{U}(x; \tilde{r}, s_{\tilde{r}}) = \mathbb{U}(x; \tilde{r}, s_r). \quad (2.45)$$

This shows that $s_{\tilde{r}}$ and s_r are best responses to both r and $\tilde{r}$. Furthermore, since s_r and $s_{\tilde{r}}$ are both worst-case lazy best responses to r and $\tilde{r}$, respectively, using the laziness constraint on (2.45), we have

$$|s_r(x) - x| \leq |s_{\tilde{r}}(x) - x| \text{ and } |s_{\tilde{r}}(x) - x| \leq |s_r(x) - x| \text{ which implies } |s_r(x) - x| = |s_{\tilde{r}}(x) - x|. \quad (2.46)$$

As an additional consequence, (2.44) reduces to

$$\mathfrak{U}(r(s_r(x)), x) = \mathfrak{U}(\tilde{r}(s_{\tilde{r}}(x)), x). \quad (2.47)$$

Next, we show that s_r and $s_{\tilde{r}}$ are worst-case lazy best responses to both r and $\tilde{r}$. Since s_r and $s_{\tilde{r}}$ are worst-case lazy best responses of r and $\tilde{r}$, respectively, we have (using the worst-case constraint)

$$\begin{aligned} \mathcal{E}(r, s_r) &= \int_{\mathcal{X}} T(x; r, s_r) dx \geq \int_{\mathcal{X}} T(x; r, s_{\tilde{r}}) dx = \mathcal{E}(r, s_{\tilde{r}}) \quad \text{and} \\ \mathcal{E}(\tilde{r}, s_{\tilde{r}}) &= \int_{\mathcal{X}} T(x; \tilde{r}, s_{\tilde{r}}) dx \geq \int_{\mathcal{X}} T(x; \tilde{r}, s_r) dx = \mathcal{E}(\tilde{r}, s_r). \end{aligned} \quad (2.48)$$

Recall from (2.14) that T is defined as $T(x; r, s) = \mathbb{P}_r(Y \neq h(x) \mid X' = s(x))$. Furthermore, if $s(x) \in \mathcal{X} \setminus \Theta$, then

$$\mathfrak{U}(x; r, s) = 2\mathbf{u}\mathbb{P}_r(Y \neq h(x) \mid X' = s(x)) = 2\mathbf{u}T(x; r, s).$$

Hence, using this in (2.48), we get

$$\begin{aligned} \int_{\mathcal{X}} \mathfrak{U}(r(s_r(x)), x) dx &\geq \int_{\mathcal{X}} \mathfrak{U}(r(s_{\tilde{r}}(x)), x) dx, \quad \text{and} \\ \int_{\mathcal{X}} \mathfrak{U}(\tilde{r}(s_{\tilde{r}}(x)), x) dx &\geq \int_{\mathcal{X}} \mathfrak{U}(\tilde{r}(s_r(x)), x) dx. \end{aligned}$$

But combining (2.42) with (2.47), we get

$$\mathfrak{U}(\tilde{r}(s_r(x)), x) = \mathfrak{U}(r(s_r(x)), x) = \mathfrak{U}(\tilde{r}(s_{\tilde{r}}(x)), x) = \mathfrak{U}(r(s_{\tilde{r}}(x)), x).$$

Thus,

$$\int_{\mathcal{X}} T(x; r, s_r) dx = \int_{\mathcal{X}} T(x; r, s_{\tilde{r}}) dx \quad \text{and} \quad \int_{\mathcal{X}} T(x; \tilde{r}, s_{\tilde{r}}) dx = \int_{\mathcal{X}} T(x; \tilde{r}, s_r) dx.$$

Hence, $s_{\tilde{r}}$ and s_r are also worst-case lazy best responses of r and $\tilde{r}$, respectively. Therefore, $\mathcal{E}(r) = \mathcal{E}(\tilde{r})$ as claimed. $\square$

The next lemma calculates the worst-case lazy best response and classification error of a specific receiver strategy r , which will later be shown to be a Stackelberg equilibrium.

Lemma 2.41. Let $r \equiv (1 - q, q, 0)$, where

$$q(x) = \begin{cases} 0 & \text{if } x \in [0, \mathbf{h}^-], \\ \frac{x - \mathbf{h}^-}{2\mathbf{u}} & \text{if } x \in (\mathbf{h}^-, \mathbf{h}^+), \\ 1 & \text{if } x \in [\mathbf{h}^+, 1] \end{cases} \quad (2.49)$$

and, $\mathbf{h}^- = \mathbf{h} - \mathbf{u}$ and $\mathbf{h}^+ = \mathbf{h} + \mathbf{u}$. Then the worst-case lazy best response of r is $s_r(x) = x$, and the error function $T(x; r, s_r)$ is

$$T(x; r, s_r) = \begin{cases} 0 & \text{if } x \in [0, \mathbf{h}^-] \cup [\mathbf{h}^+, 1], \\ q(x) & \text{if } x \in (\mathbf{h}^-, \mathbf{h}], \\ 1 - q(x) & \text{if } x \in (\mathbf{h}, \mathbf{h}^+). \end{cases}$$

Furthermore, $\mathcal{E}(r) = \frac{\mathbf{u}}{2}$.

Proof. We divide the task of finding s_r over $\mathcal{X}$ into various subintervals of $\mathcal{X}$ and show that $s_r(x) = x$ in all of them. Let $s_r(x) = x'$.

Case (i): $x \in [0, \mathbf{h}^-]$.

We now calculate $\mathbb{U}(x; r, s_r)$ and show that in the current case, $x = x'$.

$$\mathbb{U}(x; r, s_r) = \begin{cases} -|x - x'| = 0 & \text{if } x' = x, \\ -|x - x'| < 0 & \text{if } x' \neq x \text{ and } x' \in [0, \mathbf{h}^-], \\ 2\mathbf{u}q(x') - x' + x \leq 0 & \text{if } x' \in (\mathbf{h}^-, \mathbf{h}^+), \\ 2\mathbf{u} - x' + x \leq 0 & \text{if } x' \in [\mathbf{h}^+, 1]. \end{cases}$$

Here we have used (2.49) to expand $q(x')$ and to show that $2\mathbf{u}q(x') - x' + x \leq 0$, and have used $\mathbf{h}^+ - \mathbf{h}^- = 2\mathbf{u}$ to show that $2\mathbf{u} - x' + x \leq 0$. Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) \leq 0$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = 0$. Furthermore, $x' = x$ is also the laziest option for s_r . Hence $s_r(x) = x$.

Case (ii): $x \in (\mathbf{h}^-, \mathbf{h}]$.

We now calculate $\mathbb{U}(x; r, s_r)$ and show that in the current case, $x = x'$.

$$\mathbb{U}(x; r, s_r) = \begin{cases} 2\mathbf{u}q(x') - x + x' = x - \mathbf{h}^- & \text{if } x' = x, \\ -x + x' < 0. & \text{if } x' \in [0, \mathbf{h}^-], \\ 2\mathbf{u}q(x') - x + x' \leq x - \mathbf{h}^- & \text{if } x' \in (\mathbf{h}^-, x] \text{ and } x' \neq x, \\ 2\mathbf{u}q(x') - x' + x \leq x - \mathbf{h}^- & \text{if } x' \in (x, \mathbf{h}^+), \\ 2\mathbf{u} - x' + x \leq x - \mathbf{h}^- & \text{if } x' \in [\mathbf{h}^+, 1]. \end{cases}$$

Here we have used (2.49) to expand $q(x')$ wherever it appears and have used $2\mathbf{u} = \mathbf{h}^+ - \mathbf{h}^-$ to show that $2\mathbf{u} - x' + x \leq x - \mathbf{h}^-$. Note that $x - \mathbf{h}^- \geq 0$. Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) \leq x - \mathbf{h}^-$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) \leq x - \mathbf{h}^-$. Furthermore, $x' = x$ is also the laziest option for s_r . Hence $s_r(x) = x$.

Case (iii): $x \in (\mathbf{h}, \mathbf{h}^+)$.

We now calculate $\mathbb{U}(x; r, s_r)$ and show that in the current case, $x = x'$.

$$\mathbb{U}(x; r, s_r) = \begin{cases} 2\mathbf{u}(1 - q(x')) - x' + x = \mathbf{h}^+ - x & \text{if } x' = x, \\ 2\mathbf{u} - x + x' \leq \mathbf{h}^+ - x & \text{if } x' \in [0, \mathbf{h}^-], \\ 2\mathbf{u}(1 - q(x')) - x + x' \leq \mathbf{h}^+ - x & \text{if } x' \in (\mathbf{h}^-, x], \\ 2\mathbf{u}(1 - q(x')) - x' + x \leq \mathbf{h}^+ - x & \text{if } x' \in (x, \mathbf{h}^+) \text{ and } x' \neq x, \\ -x' + x \leq -\mathbf{h}^+ + x & \text{if } x' \in [\mathbf{h}^+, 1]. \end{cases}$$

Here we have used (2.49) to expand $q(x')$ wherever it appears and have used $2\mathbf{u} = \mathbf{h}^+ - \mathbf{h}^-$ to show that $2\mathbf{u} - x + x' \leq \mathbf{h}^+ - x$. Note that $\mathbf{h}^+ - x \geq 0$. Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) \leq \mathbf{h}^+ - x$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = \mathbf{h}^+ - x$. Furthermore, $x' = x$ is also the laziest option for s_r . Hence $s_r(x) = x$.

Case (iv): $x \in [\mathbf{h}^+, 1]$.

We now calculate $\mathbb{U}(x; r, s_r)$ and show that in the current case, $x = x'$.

$$\mathbb{U}(x; r, s_r) = \begin{cases} -|x' - x| = 0 & \text{if } x' = x, \\ 2\mathbf{u} - x + x' \leq \mathbf{h}^+ - x \leq 0 & \text{if } x' \in [0, \mathbf{h}^-], \\ 2\mathbf{u}(1 - q(x')) - x + x' \leq \mathbf{h}^+ - x < 0 & \text{if } x' \in (\mathbf{h}^-, \mathbf{h}^+), \\ -|x' - x| < 0 & \text{if } x' \in [\mathbf{h}^+, 1] \text{ and } x' \neq x. \end{cases}$$

Here we have used (2.49) to expand $q(x')$ to show that $2\mathbf{u}(1 - q(x')) - x + x' < \mathbf{h}^+ - x < 0$. Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) \leq 0$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = 0$. Furthermore, $x' = x$ is also the laziest option for s_r . Hence $s_r(x) = x$. Hence, by combining all cases (i)–(iv), we conclude that $s_r(x) = x$. Therefore,

$$T(x; r, s_r) = \mathbb{P}_r\{Y \neq h(x) \mid X' = s_r(x)\} = \begin{cases} 0 & \text{if } x \in [0, \mathbf{h}^-] \cup [\mathbf{h}^+, 1], \\ q(x) & \text{if } x \in (\mathbf{h}^-, \mathbf{h}], \\ 1 - q(x) & \text{if } x \in (\mathbf{h}, \mathbf{h}^+). \end{cases}$$

Since $\mathcal{E}(r) = \int_{\mathcal{X}} T(x) dx$, it follows that $\mathcal{E}(r) = \frac{\mathbf{u}}{2}$. □

We now present the proof of Theorem 2.18.

Proof of Theorem 2.18. From Lemma 2.41, we have $\mathcal{E}(r) = \frac{\mathbf{u}}{2}$. Furthermore, we show that $\mathcal{E}(r') \geq \frac{\mathbf{u}}{2}$ for all $r' \in \mathbf{R}$. Let x and y be two features such that $x \leq \mathbf{h} < y$, and let $a := s_{r'}(x)$ and $b := s_{r'}(y)$. From the property of worst case best response we have

$$\mathbb{U}(x; r', a) \geq \mathbb{U}(x; r', b) \text{ and } \mathbb{U}(y; r', b) \geq \mathbb{U}(y; r', a).$$

Thus we get

$$2\mathbf{u}(q(b) - q(a)) \leq |b - x| - |a - x| \text{ and } 2\mathbf{u}(q(b) - q(a)) \leq |a - y| - |b - y|.$$

Summing them up and applying reverse triangle inequality ($|A| - |B| \leq |A - B|$), we get

$$q(b) - q(a) \leq \frac{y - x}{2\mathbf{u}}.$$

Therefore, error at x and y can be lower bounded as

$$T(x; r', s_{r'}) + T(y; r', s_{r'}) \geq 1 - \frac{y - x}{2\mathbf{u}}.$$

Now consider features $x(t) := \mathbf{h} - t$ and $y(t) := \mathbf{h} + t$ for $t \in [0, \mathbf{u}]$. Since we have assumed $2\mathbf{u} \ll \min\{\mathbf{h}, 1 - \mathbf{h}\}$, $x(t), y(t) \in \mathcal{X}$. The error from the interval $[\mathbf{h} - \mathbf{u}, \mathbf{h} + \mathbf{u}]$ is lower bounded as

$$\int_0^{\mathbf{u}} T(\mathbf{h} - t; r', s_{r'}) dt + \int_0^{\mathbf{u}} T(\mathbf{h} + t; r', s_{r'}) dt \geq \int_0^{\mathbf{u}} (1 - \frac{t}{\mathbf{u}}) dt = \frac{\mathbf{u}}{2}.$$

Hence, $\mathcal{E}(r') \geq \frac{\mathbf{u}}{2}$ for all $r' \in \mathbb{R}$, which implies r is a Stochastic Stackelberg equilibrium. $\square$

2.6.5 Calculations for Examples and Comparison

We now provide the steps of the calculations used in Example 2.21.

Calculation of Example 2.21: We divide the task of finding s_r over $\mathcal{X}$ into various subintervals of $\mathcal{X}$.

Case (i): $x \in [0, \mathbf{h}_\epsilon^-]$.

Let $s_r(x) = x'$. We will show that in the current case, $x = x'$. If $x' = x$, then

$$\mathbb{U}(x; r, s_r) = -|x - x'| = 0.$$

If $x' \neq x$ and $x' \in [0, \mathbf{h}_\epsilon^-]$, then

$$\mathbb{U}(x; r, s_r) = -|x - x'| < 0.$$

If $x' \in [\mathbf{h}_\epsilon^-, \mathbf{h}_\epsilon^+]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}q_\epsilon(x') - x' + x = \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x' - \mathbf{h}_\epsilon^-) - x' + x < x' - \mathbf{h}_\epsilon^- - x' + x = -\mathbf{h}_\epsilon^- + x \leq 0,$$

where we have used (2.18) to expand $q_\epsilon(x')$. If $x' \in (\mathbf{h}_\epsilon^+, 1]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u} - x' + x < \mathbf{h}_\epsilon^+ - \mathbf{h}_\epsilon^- - x' + x < -\mathbf{h}_\epsilon^- + x < 0,$$

where we have used the fact that $\mathbf{h}_\epsilon^+ - \mathbf{h}_\epsilon^- > 2\mathbf{u}$. Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) < 0$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = 0$. Hence $s_r(x) = x$.

Case (ii): $x \in [\mathbf{h}_\epsilon^-, \mathbf{h}]$.

Let $s_r(x) = x'$. We will show that in the current case, $x = x'$. If $x' = x$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}q_\epsilon(x') - x + x' = \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x' - \mathbf{h}_\epsilon^-) - x + x' = \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we have used (2.18) to expand $q_\epsilon(x')$. If $x' \in [0, \mathbf{h}_\epsilon^-]$, then

$$\mathbb{U}(x; r, s_r) = -x + x' < 0.$$

If $x' \in [\mathbf{h}_\epsilon^-, x]$ and $x' \neq x$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}q_\epsilon(x') - x + x' < \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we have used (2.18) to expand $q_\epsilon(x')$. If $x' \in (x, \mathbf{h}_\epsilon^+]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}q_\epsilon(x') - x' + x < \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we have used (2.18) to expand $q_\epsilon(x')$. If $x' \in (\mathbf{h}_\epsilon^+, 1]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u} - x' + x < x - \mathbf{h}_\epsilon^- - 2\epsilon < \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we used the fact that $\mathbf{h}_\epsilon^+ - \mathbf{h}_\epsilon^- = 2\mathbf{u} + 2\epsilon$. Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) < \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-)$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-)$. Hence $s_r(x) = x$.

Case (iii): $x \in [\mathbf{h}, \mathbf{h}_\epsilon^+]$.

Let $s_r(x) = x'$. We will show that in the current case, $x = x'$. If $x' = x$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}(1 - q_\epsilon(x')) - x' + x = \frac{-\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-) + 2\mathbf{u},$$

where we have used (2.18) to expand $q_\epsilon(x')$. If $x' \in [0, \mathbf{h}_\epsilon^-]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u} - x + x' \leq 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-).$$

If $x' \in (\mathbf{h}_\epsilon^-, x]$ and $x' \neq x$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}(1 - q_\epsilon(x')) - x + x' < 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-),$$

where we have used (2.18) to expand $q_\epsilon(x')$. If $x' \in (x, \mathbf{h}_\epsilon^+]$, then

$$\mathbb{U}(x; r, s_r) = 2\mathbf{u}(1 - q_\epsilon(x')) - x' + x < \frac{-\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-) + 2\mathbf{u},$$

where we have used (2.18) to expand $q_\epsilon(x')$. If $x' \in (\mathbf{h}_\epsilon^+, 1]$, then

$$\mathbb{U}(x; r, s_r) = -x' + x < 0.$$

Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) < 2\mathbf{u} - \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-)$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = 2\mathbf{u} - \frac{\mathbf{u}}{\mathbf{u} + \epsilon}(x - \mathbf{h}_\epsilon^-)$. Hence $s_r(x) = x$.

Case (iv): $x \in (\mathbf{h}_\epsilon^+, 1]$.

Let $s_r(x) = x'$. We will show that in the current case, $x = x'$. If $x' = x$, then

$$\mathbb{U}(x; r, s_r) = -|x' - x| = 0.$$

If $x' \in [0, \mathbf{h}_\epsilon^-]$, then

$$\mathbb{U}(x; r, s_r) < 2\mathbf{u} - \mathbf{h}_\epsilon^+ + \mathbf{h}_\epsilon^- = -2\epsilon < 0$$

If $x' \in [\mathbf{h}_\epsilon^-, \mathbf{h}_\epsilon^+]$, then

$$\begin{aligned} \mathbb{U}(x; r, s_r) &= 2\mathbf{u}(1 - q_\epsilon(x')) - x + x' = 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u} + \epsilon}(x' - \mathbf{h}_\epsilon^-) - x + x' \\ &< 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u} + \epsilon}(x' - \mathbf{h}_\epsilon^- + x - x') \leq 2\mathbf{u} + \frac{-\mathbf{u}}{\mathbf{u} + \epsilon}(\mathbf{h}_\epsilon^+ - \mathbf{h}_\epsilon^-) = 0. \end{aligned}$$

where we have used (2.18) to expand $q_\epsilon(x')$. If $x' \in (\mathbf{h}_\epsilon^+, 1]$ and $x' \neq x$, then

$$\mathbb{U}(x; r, s_r) = -|x' - x| < 0.$$

Thus for $x' \neq x$, we have $\mathbb{U}(x; r, s_r) < 0$ whereas for $x' = x$, $\mathbb{U}(x; r, s_r) = 0$. Hence $s_r(x) = x$.

Hence, by combining all the cases, we conclude that $s_r(x) = x$. Therefore,

$$T(x; r, s_r) = \mathbb{P}_r\{Y \neq h(x) \mid X' = s_r(x)\} = \begin{cases} 0 & \text{if } x \in [0, \mathbf{h}_\epsilon^-] \cup (\mathbf{h}_\epsilon^+, 1], \\ q_\epsilon(x) & \text{if } x \in [\mathbf{h}_\epsilon^-, \mathbf{h}_\epsilon^+], \\ 1 - q_\epsilon(x) & \text{if } x \in (\mathbf{h}_\epsilon^-, \mathbf{h}_\epsilon^+]. \end{cases}$$

Since $\mathcal{E}(r) = \int_{\mathcal{X}} T(x) dx$, it follows that $\mathcal{E}(r) = \frac{\mathbf{u} + \epsilon}{2}$.

2.6.6 Proofs for Multiple Decision Boundaries and Vanilla ERM

We begin with the proof of Proposition 2.28.

Proof 2.6.4. (Proof of Proposition 2.28) To show that $\text{VC dim}(\mathcal{F}) \geq d$, choose any d distinct points $0 \leq x_1 < x_2 < \dots < x_d \leq 1$ in $\mathcal{X}$. Let $(y_1, \dots, y_d) \in \mathcal{Y}^d$ be an arbitrary labeling. Let $J := \{j \in \{1, \dots, d-1\} : y_j \neq y_{j+1}\}$. Note that $|J| \leq d-1$. If $J = \emptyset$, all labels are identical, so setting $f(x) \equiv y_1$ satisfies $f \in \mathcal{F}$ (with 0 decision boundaries) and $f(x_k) = y_k$ for all $k \in \{1, \dots, d\}$. Let $k = |J| \geq 1$, and denote the elements of J in increasing order by $j_1 < j_2 < \dots < j_k$. For each $i \in \{1, \dots, k\}$, define the decision boundary

$$b_i := \frac{x_{j_i} + x_{j_i+1}}{2}.$$

Setting $b_0 := 0$ and $b_{k+1} := 1$, we have $0 \leq x_1 \leq x_{j_1} < b_1 < x_{j_1+1} \leq \dots \leq x_{j_k} < b_k < x_{j_k+1} \leq$

$x_d \leq 1$. Now define the hypothesis $f : \mathcal{X} \rightarrow \mathcal{Y}$ as:

$$f(x) := \begin{cases} y_1 & \text{if } x \in [0, b_1), \\ y_{j_i+1} & \text{if } x \in [b_i, b_{i+1}) \text{ for } i \in \{1, \dots, k-1\}, \\ y_d & \text{if } x \in [b_k, 1]. \end{cases}$$

By construction, the label assigned by f changes only at $b_1, \dots, b_k$, so f has exactly $k = |J| \leq d-1$ decision boundaries, which implies $f \in \mathcal{F}$. Moreover, $f(x_m) = y_m$ for all $m \in \{1, \dots, d\}$. Since the labeling $(y_1, \dots, y_d)$ was arbitrary, $\mathcal{F}$ shatters $\{x_1, \dots, x_d\}$, and thus $\text{VC dim}(\mathcal{F}) \geq d$.

To show that $\text{VC dim}(\mathcal{F}) \leq d$, consider an arbitrary set of $d+1$ points $0 \leq x_1 < x_2 < \dots < x_{d+1} \leq 1$ in $\mathcal{X}$. Assign alternating labels $y_j = (-1)^j$ so that $y_j \neq y_{j+1}$ for all $j \in \{1, \dots, d\}$. Any function f' satisfying $f'(x_j) = y_j$ for all $j \in \{1, \dots, d+1\}$ must change signs between every adjacent pair x_j and x_{j+1} , requiring at least d decision boundaries. Because every function in $\mathcal{F}$ has at most $d-1$ decision boundaries, no function in $\mathcal{F}$ can realize this labeling. Therefore, no set of $d+1$ points can be shattered by $\mathcal{F}$, proving $\text{VC dim}(\mathcal{F}) \leq d$. Combining both bounds yields $\text{VC dim}(\mathcal{F}) = d$.

Next, we present the proof of Proposition 2.27.

Proof 2.6.5. (Proof of Proposition 2.27) Let $r^\star$ be a Stackelberg equilibrium, and define:

$$\begin{aligned} \theta^{(n)} &:= \sup(E(r^\star)^c \cap O_n), \\ \theta_{(n)} &:= \inf(E(r^\star)^c \cap O_n), \\ \Theta_n &:= [\theta_{(n)}, \theta^{(n)}], \end{aligned}$$

where $\Theta_n := \emptyset$ if $E(r^\star)^c \cap O_n = \emptyset$. This implies $E(r^\star)^c \subseteq \bigcup_{n=1}^{d-1} \Theta_n$. We demonstrate that with this choice of Θ_n , the receiver strategy $r(x; \Theta_1, \dots, \Theta_{d-1})$ given in (2.30) is a Stackelberg equilibrium. Consider the following cases for Θ_n 's.

Case (i): All Θ_n 's are empty.

This case will not happen because if all Θ_n 's are empty, then $r^\star$ misclassifies all feature vectors. However, a majority-label (described in Section 2.2) receiver strategy will correctly classify at least half of them, contradicting that $r^\star$ is a Stackelberg equilibrium. Thus, at least one Θ_n is non-empty.

Case (ii): For all odd (or even) n , Θ_n is empty.

Suppose $\Theta_n = \emptyset$ for all odd n . For even n , denote the corresponding Θ_n as Θ_{2k} , where $k \in \{1, \dots, \lfloor \frac{d-1}{2} \rfloor\}$. Since $r(x) = \Delta$ for all $x \in \mathcal{X} \setminus \bigcup_k \Theta_{2k}$, it follows that for $x \in \mathcal{X}$, $s_r(x) \in \bigcup_k \Theta_{2k}$. Thus, $r(s_r(x)) \equiv h(s_r(x))$ from (2.30). From (2.28), we have $h(t) = -$ for $t \in \bigcup_k \Theta_{2k}$. Therefore, $r(s_r(x)) \equiv -$ and $r(s_r(x)) = h(x)$ for $x \in \bigcup_k \Theta_{2k}$. Hence

$$E(r)^c \supseteq \bigcup_k \Theta_{2k} \supseteq \bigcup_k \Theta_{2k}.$$

Since r^* is a Stackelberg equilibrium, it follows that $E(r^*)^c \subseteq \bigcup_k \Theta_{2k}$ by construction. Consequently, $E(r)^c = E(r^*)^c = \bigcup_k \Theta_{2k}$, and r is a Stackelberg equilibrium. A similar argument applies to show that r is a Stackelberg equilibrium when n for which $\Theta_n = \emptyset$ are all even.

Case (iii): There exist an odd k and even l such that $\Theta_k, \Theta_l \neq \emptyset$.

We will first show that $\theta_{(\max\{l,k\})} - \theta_{(\min\{l,k\})} \geq \mathbf{u}$. Suppose $k < l$ (similar reasoning holds when $k > l$). Suppose $\theta_{(l)} - \theta^{(k)} < \mathbf{u}$. Let $\epsilon > 0$ be such that it satisfies $0 < 2\epsilon < \mathbf{u} - (\theta_{(l)} - \theta^{(k)})$. Let $\theta^{(k')}, \theta_{(l')} \in \mathcal{X}$ be feature vectors such that $\theta^{(k')}, \theta_{(l')} \in E(r^*)^c$, $0 \leq \theta_{(l')} - \theta_{(l)} < \epsilon$, and $0 \leq \theta^{(k)} - \theta^{(k')} < \epsilon$. Such feature vectors exist by the definition of $\theta^{(k)}$ and $\theta_{(l)}$. It follows that

$$\theta_{(l')} - \theta^{(k')} < 2\epsilon + \theta_{(l)} - \theta^{(k)} < \mathbf{u}.$$

Note that $h(\theta_{(l')}) = -$ and $h(\theta^{(k')}) = +$. Suppose $\mathbb{U}(\theta^{(k')}; r^*, s_{r^*}) \leq \mathbb{U}(\theta_{(l')}; r^*, s_{r^*})$ (similar arguments applies when $\mathbb{U}(\theta^{(k')}; r^*, s_{r^*}) > \mathbb{U}(\theta_{(l')}; r^*, s_{r^*})$). Let s be a sender strategy defined as $s(x) := s_{r^*}(\theta_{(l')})$ for all $x \in \mathcal{X}$. Then

$$\begin{aligned} \mathbb{U}(\theta^{(k')}; r^*, s) &= \mathbf{u} - |\theta^{(k')} - s_{r^*}(\theta_{(l')})| \geq \mathbf{u} - |\theta_{(l')} - \theta^{(k')}| - |\theta_{(l')} - s_{r^*}(\theta_{(l')})| \\ &> -|\theta_{(l')} - s_{r^*}(\theta_{(l')})| = \mathbb{U}(\theta_{(l')}; r^*, s_{r^*}). \end{aligned}$$

This implies $\mathbb{U}(\theta^{(k')}; r^*, s) > \mathbb{U}(\theta^{(k')}; r^*, s_{r^*})$, which contradicts the fact that s_{r^*} is the worst-case best response. Therefore, $\theta_{(l)} - \theta^{(k)} \geq \mathbf{u}$, proving the claim.

Now consider the receiver strategy $r(x; \{\Theta_n\})$. We calculate its worst-case best response s_r . Let $t \in \Theta_l$. Since $r(x) = \Delta$ for $x \notin \bigcup_n \Theta_n$, we have $s_r(x) \in \bigcup_n \Theta_n$. Suppose i is even (similar arguments hold for odd i) and $s_r(t) \geq t$ (a similar arguments hold for $s_r(t) < t$). The sender's payoff is:

$$\mathbb{U}(t; r, s_r) = \begin{cases} -s_r(t) + t & \text{if } s_r(t) \in \Theta_j, j \text{ even,} \\ \mathbf{u} - s_r(t) + t & \text{if } s_r(t) \in \Theta_j, j \text{ odd.} \end{cases}$$

If j is even, $\mathbb{U}(t; r, s_r) = -s_r(t) + t \leq 0$, with the maximum value 0 at $s_r(t) = t$. If j is odd, then using $\theta_{(l)} - \theta^{(k)} \geq \mathbf{u}$, we get $s_r(t) - t \geq \mathbf{u}$, and hence $\mathbb{U}(t; r, s_r) \leq 0$. Thus $s_r(t) = t$

gives the maximum payoff to the sender. Therefore, $r(s_r(t)) = h(t)$ for all $t \in \cup_n \Theta_n$. This gives $E(r)^c \supseteq \cup_n \Theta_n \supseteq E(r^*)^c$. Since r^* minimizes classification error, r is also a Stackelberg equilibrium, and $E(r)^c = \cup_n \Theta_n$ (up to a set of measure zero), yielding $r(s_r(x)) = -h(x)$ for $x \notin \cup_n \Theta_n$.

Next, we provide the proof of Lemma 2.30.

Proof 2.6.6. (Proof of Lemma 2.30) Let $r_D := r_{f,D}^*$ and $r_m := r_{f,D(S)}^*$. Let $\psi(t)$ be the cumulative distribution function of the probability law D and $\psi_m(t)$ be the empirical cumulative distribution function for a sample S , i.e.,

$$\psi_m(t) := \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{\{x_i \leq t\}}.$$

According to the Dvoretzky–Kiefer–Wolfowitz (DKW) inequality ([26]), for any $(\epsilon, \delta) \in (0, 1)^2$, there exists an $\bar{m}(\epsilon, \delta)$ such that for all $m > \bar{m}(\epsilon, \delta)$, we have:

$$D^{\otimes m} \left(\sup_{t \in \mathcal{X}} |\psi(t) - \psi_m(t)| \leq \epsilon \right) > 1 - 2e^{-2m\epsilon^2}. \quad (2.50)$$

Let $\mathcal{S} := \{S : \sup_{t \in \mathcal{X}} |\psi(t) - \psi_m(t)| \leq \epsilon\}$. From (2.50), $D^{\otimes m}(\mathcal{S}) > 1 - 2e^{-2m\epsilon^2}$. Fix a sample $S \in \mathcal{S}$. From Proposition 2.27, for any target function f with at most $K - 1$ internal boundaries (K true-label intervals), the error set of any candidate strategy r is a union of at most K open intervals of the form $E_n = (a_n, b_n)$. For an open interval (a_n, b_n) , its probability under $D(S)$ is $D_m((a_n, b_n)) = \psi_m(b_n) - \psi_m(a_n)$. Because the uniform DKW bound holds for all t and their limits simultaneously, the difference in probability for any single interval is bounded by 2ϵ . Summing over the K intervals that make up the error set, we obtain a uniform bound on the difference between the empirical error and the population error for any strategy r :

$$\sup_r |\mathcal{E}(r; f, D) - \mathcal{E}(r; f, D(S))| \leq 2K\epsilon. \quad (2.51)$$

Using the optimality of r_m under the empirical distribution $D(S)$ (so $\mathcal{E}(r_m; f, D(S)) \leq \mathcal{E}(r_D; f, D(S))$), we can bound the excess population error of r_m

$$\begin{aligned} \mathcal{E}(r_m; f, D) - \mathcal{E}(r_D; f, D) &= [\mathcal{E}(r_m; f, D) - \mathcal{E}(r_m; f, D(S))] \\ &\quad + [\mathcal{E}(r_m; f, D(S)) - \mathcal{E}(r_D; f, D(S))] \\ &\quad + [\mathcal{E}(r_D; f, D(S)) - \mathcal{E}(r_D; f, D)] \\ &\leq 2K\epsilon + 0 + 2K\epsilon = 4K\epsilon. \end{aligned}$$

This holds for all $S \in \mathcal{S}$, so we get

$$D^{\otimes m}(\mathcal{E}(r_m; f, D) - \mathcal{E}(r_D; f, D) \leq 4K\epsilon) > 1 - 2e^{-2m\epsilon^2}, \quad (2.52)$$

as required.

Next, we provide the proof of Lemma 2.31.

Proof 2.6.7. (Proof of Lemma 2.31) Let $r_1 := r_{h,D}^*$, $r_2 := r_{A(S),D}^*$, and $E(r, f) := E(r, s_r, f)$. Since $\text{VC dim}(\mathcal{F}) < \infty$, $\mathcal{F}$ is PAC learnable from Theorem 2.25. Furthermore, from Theorem 2.25 we have

$$D^{\otimes m}(L_D(A(S); h) < \epsilon) > 1 - \delta, \quad (2.53)$$

where $m \geq \underline{m}(\epsilon, \delta) := m_{\mathcal{F}}^{\text{PAC}}(\epsilon, \delta)$ and $\underline{m}(\epsilon, \delta)$ satisfies

$$C_1 \frac{d + \log\left(\frac{1}{\delta}\right)}{\epsilon} \leq \underline{m}(\epsilon, \delta) \leq C_2 \frac{d \log\left(\frac{1}{\epsilon}\right) + \log\left(\frac{1}{\delta}\right)}{\epsilon}.$$

Here $C_1, C_2 \geq 0$ are constants and d is the VC dimension of $\mathcal{F}$. Let

$$\mathcal{S} := \{S : L_D(A(S); h) < \epsilon\}. \quad (2.54)$$

From (2.53), we know that $D^{\otimes m}(\mathcal{S}) > 1 - \delta$. Fix a $S \in \mathcal{S}$. Define $W(S) := \{x \mid h(x) \neq A(S)(x)\}$. For a receiver strategy r and (an arbitrary) true function f (recall from Definition 2.1 that s_r will depend upon the true function f v.i.a. utility function), the classification error set $E(r; A(S))$ (defined in (2.7)) can be decomposed as

$$E(r; A(S)) = (E(r; A(S)) \cap W(S)) \cup (E(r; A(S)) \cap W(S)^c).$$

Since $S \in \mathcal{S}$, using (2.54), we get $L(A(S); h) < \epsilon$, and therefore it follows that $D(W(S)) < \epsilon$. Hence, we get

$$\mathcal{E}(r; A(S), D) < \epsilon + D(E(r; A(S)) \cap W(S)^c).$$

From the definition of $W(S)$, the sender's worst case best response $s_r(x)$ under h is identical to that under $A(S)$ for all $x \in W(S)^c$. Therefore,

$$\begin{aligned} E(r; A(S)) \cap W(S)^c &= \{x : r(s_r(x)) \neq A(S)(x)\} \cap \{x : h(x) = A(S)(x)\} \\ &= \{x : r(s_r(x)) \neq h(x)\} \cap \{x : h(x) = A(S)(x)\} \\ &= E(r; h) \cap W(S)^c, \end{aligned} \quad (2.55)$$

and thus

$$\mathcal{E}(r; A(S), D) \leq D(E(r; A(S)) \cap W(S)) + D(E(r; h) \cap W(S)^c) \leq D(W(S)) + \mathcal{E}(r; h, D). \quad (2.56)$$

Similarly, we can decompose the classification error set $E(r; h)$ as

$$E(r; h) := (E(r; h) \cap W(S)) \cup (E(r; h) \cap W(S)^c).$$

Repeating the argument used for deriving (2.56), we obtain

$$\mathcal{E}(r; h, D) \leq D(W(S)) + \mathcal{E}(r; A(S), D). \quad (2.57)$$

Combining (2.56) and (2.57), we get

$$|\mathcal{E}(r; h, D) - \mathcal{E}(r; A(S), D)| \leq D(W(S)).$$

Since $D(W(S)) < \epsilon$ for all $S \in \mathcal{S}$, we have $\sup_{r \in \mathbb{R}} |\mathcal{E}(r; h, D) - \mathcal{E}(r; A(S), D)| < \epsilon$ for all $S \in \mathcal{S}$. But we have $D^{\otimes m}(\mathcal{S}) > 1 - \delta$, therefore we get

$$D^{\otimes m} \left(\sup_{r \in \mathbb{R}} |\mathcal{E}(r; h, D) - \mathcal{E}(r; A(S), D)| < \epsilon \right) > 1 - \delta. \quad (2.58)$$

This proves part (a). Using part (a), we now establish

$$D^{\otimes m}(0 \leq \mathcal{E}(r_2; h, D) - \mathcal{E}(r_1; h, D) < 2\epsilon) > 1 - \delta, \quad (2.59)$$

where $r_1 = r_{h,D}^*$ and $r_2 = r_{A(S),D}^*$. This is the same as the claim (given in (2.32)) we want to prove. Since r_2 is the minimizer of the classification error under $A(S)$, we have $\mathcal{E}(r_2; A(S), D) - \mathcal{E}(r_1; A(S), D) \leq 0$. Also, by the definition of optimality, $0 \leq \mathcal{E}(r_2; h, D) - \mathcal{E}(r_1; h, D)$. For any $S \in \mathcal{S}$, applying the uniform bound from part (a), we get

$$\begin{aligned} \mathcal{E}(r_2; h, D) - \mathcal{E}(r_1; h, D) &\leq \mathcal{E}(r_2; A(S), D) + \epsilon - \mathcal{E}(r_1; h, D) \\ &\leq \mathcal{E}(r_1; A(S), D) + \epsilon - \mathcal{E}(r_1; h, D) \\ &\leq \mathcal{E}(r_1; h, D) + \epsilon + \epsilon - \mathcal{E}(r_1; h, D) = 2\epsilon. \end{aligned} \quad (2.60)$$

This holds true for all $S \in \mathcal{S}$. Since $D^{\otimes m}(\mathcal{S}) > 1 - \delta$, we get

$$D^{\otimes m}(0 \leq \mathcal{E}(r_2; h, D) - \mathcal{E}(r_1; h, D) < 2\epsilon) > 1 - \delta$$

as claimed in part (b).

Finally, we present the proof of Theorem 2.29.

Proof 2.6.8. (Proof of Theorem 2.29) Let $D \in \mathfrak{D}$ be the probability law on $\mathcal{X}$ and let $(a, \delta) \in (0, 1)^2$. We aim to show that the excess strategic risk is at most a with probability at least $1 - \delta$. Let $h \in \mathcal{F}$ be the true function and let $S = \{(x_i, y_i)\}_{i=1}^m$ be an m -sized sample where x_i are drawn i.i.d. from D . Let $\epsilon = a/(2 + 4K)$, where K is the maximum number of true-label intervals for functions in $\mathcal{F}$. We allocate confidence $\delta/2$ to the statistical learning event and $\delta/2$ to the uniform DKW event. Let $m \geq \max\{m_{\mathcal{F}}^{\text{PAC}}(\epsilon, \delta/2), \bar{m}(\epsilon, \delta/2)\}$, where $m_{\mathcal{F}}^{\text{PAC}}(\epsilon, \delta/2)$ is the PAC sample complexity from Theorem 2.25 corresponding to confidence $\delta/2$, and $\bar{m}(\epsilon, \delta/2) = \frac{\log(4/\delta)}{2\epsilon^2}$ ensures the DKW bound holds with probability $1 - \delta/2$. Let $D(S)$ be the corresponding empirical distribution of S . Let $r_1 := r_{h,D}^*$, $r_2 := r_{A(S),D}^*$, and $r_3 := r_{A(S),D(S)}^* = B(A(S))$.

Since $\text{VC dim}(\mathcal{F}) < \infty$, $\mathcal{F}$ is PAC learnable. We have $A(S)$ as the estimated true function such that

$$D^{\otimes m}(L_D(A(S); h) < \epsilon) > 1 - \delta/2. \quad (2.61)$$

Let $\mathcal{S}_1 := \{S : L_D(A(S); h) < \epsilon\}$ and $\mathcal{S}_2 := \{S : \sup_{x \in \mathcal{X}} |\psi(x) - \psi_m(x)| \leq \epsilon\}$. From (2.61), we have $D^{\otimes m}(\mathcal{S}_1) > 1 - \delta/2$. From the DKW inequality and our choice of m , $D^{\otimes m}(\mathcal{S}_2) > 1 - 2e^{-2m\epsilon^2} \geq 1 - \delta/2$. By the union bound, for the intersection $\mathcal{S} := \mathcal{S}_1 \cap \mathcal{S}_2$, we have $D^{\otimes m}(\mathcal{S}) > 1 - \delta$.

Fix $S \in \mathcal{S}$. From Lemma 2.31(a) applied with $f = A(S)$ as the (estimated) true function, the error under D satisfies for any strategy r :

$$|\mathcal{E}(r; h, D) - \mathcal{E}(r; A(S), D)| < \epsilon. \quad (2.62)$$

Applying this to r_3 , we get

$$\mathcal{E}(r_3; h, D) < \mathcal{E}(r_3; A(S), D) + \epsilon. \quad (2.63)$$

Since $S \in \mathcal{S}_2$, Lemma 2.30 applied to the target $A(S)$ yields

$$\mathcal{E}(r_3; A(S), D) - \mathcal{E}(r_2; A(S), D) \leq 4K\epsilon. \quad (2.64)$$

By definition, r_2 minimizes the classification error for target $A(S)$ under distribution D , so $\mathcal{E}(r_2; A(S), D) \leq \mathcal{E}(r_1; A(S), D)$. Using this and applying (2.62) to r_1 , we obtain

$$\mathcal{E}(r_2; A(S), D) \leq \mathcal{E}(r_1; A(S), D) < \mathcal{E}(r_1; h, D) + \epsilon. \quad (2.65)$$

Summing the inequalities (2.63), (2.64), and (2.65), we get

$$\mathcal{E}(r_3; h, D) < \mathcal{E}(r_1; h, D) + 2\epsilon + 4K\epsilon = \mathcal{E}(r_1; h, D) + a.$$

This holds for all $S \in \mathcal{S}$. Thus, we have

$$D^{\otimes m}(\mathcal{E}(r_3; h, D) - \mathcal{E}(r_1; h, D) < a) > 1 - \delta.$$

The choice of D , (a, δ) , and h were arbitrary, establishing that $\mathcal{F}$ is strategically learnable under $B \circ A$.

2.6.7 Proofs for Strategy-Aware ERM

We begin with the proof of Theorem 2.32.

Proof 2.6.9. (Proof of Theorem 2.32) Let $D \in \mathfrak{D}$ be the probability law on X and let $(\epsilon, \delta) \in (0, 1)^2$. Let $h \in \mathcal{F}$ be the true function and let $S = \{(x_i, y_i)\}_{i=1}^m$ be an m -sized sample where x_i 's are drawn i.i.d. from D . Let $\mathcal{E}(r, f) := \mathcal{E}(r, s_r, f, D)$. The strategy-aware ERM algorithm is similar to the vanilla ERM algorithm but applied to the modified hypothesis class $\tilde{\mathcal{F}}$ under the zero-one loss function

$$\ell_h(r \circ s_r, x) = \mathbb{1}_{\{r(s_r(x)) \neq h(x)\}}.$$

We first prove the if part. Let $A(S)$ be the hypothesis obtained from the strategy-aware ERM algorithm. Since $\text{VC dim}(\tilde{\mathcal{F}}) < \infty$, we know that $\tilde{\mathcal{F}}$ is agnostic PAC learnable from Theorem 2.25. Furthermore, from Theorem 2.25, for $m \geq m_{\tilde{\mathcal{F}}}^{\text{agn}}(\epsilon, \delta)$, we have

$$D^{\otimes m} \left(\mathcal{E}(A(S), h) - \min_{r \circ s_r \in \tilde{\mathcal{F}}} \mathcal{E}(r, h) < \epsilon \right) > 1 - \delta. \quad (2.66)$$

Note that by construction of $\tilde{\mathcal{F}}$, we have

$$\min_{f \in \mathcal{F}} \mathcal{E}(r_{f,D}^*, s_{r_{f,D}^*}, h, D) = \min_{r \circ s_r \in \tilde{\mathcal{F}}} \mathcal{E}(r, s_r, h, D).$$

Substituting $\min_{r \circ s_r \in \tilde{\mathcal{F}}} \mathcal{E}(r, h) = \mathcal{E}(r_{h,D}^*; h, D)$ into (2.66), we obtain

$$D^{\otimes m} \left(\mathcal{E}(A(S), h) - \mathcal{E}(r_{h,D}^*; h, D) < \epsilon \right) > 1 - \delta. \quad (2.67)$$

Since the choice of $D \in \mathfrak{D}$, $(\epsilon, \delta) \in (0, 1)^2$, and $h \in \mathcal{F}$ were arbitrary, (2.67) shows that $\mathcal{F}$ is strategically learnable under A with sample complexity $m_{\mathcal{F}}(\epsilon, \delta) = m_{\tilde{\mathcal{F}}}^{\text{agn}}(\epsilon, \delta)$.

Next, we prove the only if part. Suppose $\mathcal{F}$ is strategically learnable under Algorithm A (Strategy-aware ERM). Then by Definition 2.26,

$$D^{\otimes m} \left(\mathcal{E}(A(S); h, D) - \mathcal{E}(r_{h,D}^*; h, D) < \epsilon \right) > 1 - \delta \quad (2.68)$$

holds for $m \geq m_{\mathcal{F}}(\epsilon, \delta)$. Note that from the definition of $r_{h,D}^*$, we have $\mathcal{E}(r_{h,D}^*; h, D) \leq \mathcal{E}(r; h, D)$ for all $r \circ s_r \in \tilde{\mathcal{F}}$, which means $\mathcal{E}(r_{h,D}^*; h, D) = \inf_{r \circ s_r \in \tilde{\mathcal{F}}} \mathcal{E}(r; h, D)$. Furthermore, from the definitions of $\mathcal{E}(\cdot; h)$ and the risk $L_D(\cdot; h)$, we have $\mathcal{E}(r \circ s_r; h, D) = L_D(r \circ s_r; h)$. Therefore, (2.68) reduces to

$$D^{\otimes m} \left(L_D(A(S); h) - \inf_{r \circ s_r \in \tilde{\mathcal{F}}} L_D(r \circ s_r; h) < \epsilon \right) > 1 - \delta. \quad (2.69)$$

Note that Algorithm A outputs $\hat{r}_{\text{serm}} \circ s_{\hat{r}_{\text{serm}}}$, which minimizes the empirical risk over $\tilde{\mathcal{F}}$ and is exactly the Empirical Risk Minimization (ERM) rule applied to the hypothesis class $\tilde{\mathcal{F}}$. For the ERM rule to guarantee this bound uniformly over all true functions $h \in \mathcal{F}$ and probability laws D , the class $\tilde{\mathcal{F}}$ must satisfy the uniform convergence property. By the fundamental theorem of statistical learning (Theorem 2.25), the uniform convergence of empirical risk to true risk for an ERM rule over a hypothesis class holds if and only if the class has a finite VC dimension. Consequently, we must have $\text{VC dim}(\tilde{\mathcal{F}}) < \infty$.

Next, we present the proof of Corollary 2.33.

Proof 2.6.10. (Proof of Corollary 2.33) From Theorem 2.29, having a finite VC dimension for $\mathcal{F}$ ($\text{VC dim}(\mathcal{F}) < \infty$) is a sufficient condition for $\mathcal{F}$ to be strategically learnable (via the vanilla ERM algorithm). On the other hand, by Theorem 2.32, $\text{VC dim}(\tilde{\mathcal{F}}) < \infty$ is a necessary and sufficient condition for strategic learnability. Consequently, if $\text{VC dim}(\mathcal{F}) < \infty$, then $\mathcal{F}$ is strategically learnable, which by necessity implies that $\text{VC dim}(\tilde{\mathcal{F}}) < \infty$.

Chapter 3

Learning with Side Information from Partitioned Data

We address the problem of classification where, at test time, side information—a label from a different task—is available alongside the primary input features. Our objective is to design a classifier that can effectively leverage this additional side information. This side information cannot be treated simply as an additional feature during training because the training dataset is partitioned: we do not have access to simultaneous joint observations of labels and side information together with the features. In this chapter, we propose an algorithm that operates under this setting by imputing the missing side information. Its behavior is governed by a regularization hyperparameter that determines the degree of trust placed in the imputed values. We derive an upper bound on the excess risk as a function of this hyperparameter. We characterize the bound-minimizing regularizer via a Three-Candidate Rule: the optimal choice selects between two boundary endpoints (fully trusting or entirely ignoring the side information) and a feasible interior candidate, exhibiting conditional threshold-like behavior depending on the underlying signal-to-imputation ratio. We further establish a minimax lower bound showing that the excess risk of any algorithm remains bounded away from zero, revealing a fundamental limitation of the partitioned-data setting.

3.1 Introduction

Consider the following scenario in university admissions. A university (the primary university) seeks to admit candidates with the right aptitude for its program. However, a candidate's aptitude is not directly observable and must instead be inferred from their profile, which includes exam scores, experience, and other attributes. Typically, admission decisions are based on historical data, profiles of past candidates and their subsequent performance in the program.

Now, suppose a candidate, in addition to their own profile, also discloses the outcome of their application at another university (the secondary university). This additional information, referred to as *side information*, raises a central question: How should the primary university leverage the decision of the secondary university to improve the accuracy of its own admission process?

The value of such side information depends crucially on the relation between the two universities. If the secondary university shares similar academic standards, subject focus, and reputation, the side information can be beneficial in assessing the candidate's aptitude. Conversely, if the two institutions differ significantly in their requirements or selectivity, the side information may be misleading and could result in suboptimal decisions. In general, the degree of alignment between the objectives and standards of the two universities determines how much weight the primary university should assign to the side information.

We consider the setting where, historically, no candidate has admission outcomes from both universities at the same time. Thus the past data includes rows with student features and admission decisions from either the primary university or the secondary university. Indeed if decisions of both universities were available, the secondary university decision could be absorbed readily as an additional feature in the learning problem. The data with the primary university decisions is also assumed to be more voluminous compared to that with secondary university decisions. Both these assumptions are motivated from real life experiences, students often apply to universities near their homes whereby past student data is often partitioned as above when the primary and secondary universities are geographically far apart. A university is also likely to have access to very limited data of other universities.

In this chapter, we formally study this problem and quantify how the error in admission decisions relates to the amount of trust placed in side information. We further characterize

how much trust to place in side information, assuming the true relationship between candidate features, the primary university's decision, and the secondary university's decision is known a priori.

We formalize this problem within a classification framework. The primary and secondary university classifiers predict the admission outcome for a candidate, referred to as the label and side information, respectively. The primary university trains its classifier using both its own training data and the secondary university's decision rule. This allows the primary university to incorporate the secondary university's knowledge into its own inference process. At test time, when both the candidate's profile and the secondary university's admission decision are available, the primary university aims to construct a composite classifier that combines its own prediction with the side information, ensuring performance at least as good as the primary classifier alone.

Recall that the two training datasets are completely disjoint, referred to as a *partitioned dataset*. The challenge with a partitioned dataset is that very different underlying relationships between candidate features, the primary university's decision, and the secondary university's decision can produce the same observed data. More precisely, the *joint distribution* of features, label and side information is *unlearnable* from partitioned data. As a result, there is a fundamental limitation to the achievable accuracy of prediction.

3.1.1 Contributions

A central contribution of this chapter is the design and analysis of the Partitioned Side Information Learning (PSIL) algorithm for combining the primary and secondary classifiers. The proposed algorithm first trains the secondary classifier and then uses it to impute the missing side information in the primary classifier's training dataset. However, the estimated side information may not be fully accurate, as its accuracy depends on both the size of the secondary classifier's training data and the complexity of the relationship between features and side information. Furthermore, since no historical data point contains both the true label and true side information (due to the partitioned dataset), any observed correlation between the label and the imputed side information could be spurious. To address this, we introduce a regularization mechanism that controls the influence of the estimated side information during training. The inverse of the regularization strength can be interpreted as a degree of trust in the imputed values. A high

regularization value corresponds to little trust on the side information, while a low regularization value corresponds to greater trust. The optimal choice of this regularization parameter depends on knowledge of the true relationship between candidate aptitude, the label, and side information.

We establish a theoretical upper bound on the excess risk of the resulting composite classifier as a function of the regularization parameter. This upper bound exhibits the expected limiting behaviors: in the case of zero trust (infinite regularization), the bound reduces to that of the primary classifier alone; with complete trust (zero regularization), it becomes (roughly) the sum of the bounds for the primary and secondary classifiers. Importantly, our analysis shows that this upper bound on excess risk does not necessarily vanish, even with infinite training data. Indeed we confirm the non-vanishingness of excess risk by establishing a corresponding non-vanishing minimax lower bound on the excess risk. This fundamental limitation arises from the partitioned data setting, where features, labels, and side information are never jointly observed.

We do an ex-post analysis of the upper bound to find the optimal regularization assuming the data distribution is known. We find that the upper bound exhibits a conditional threshold-like behavior: depending on the underlying signal-imputation ratio, the bound-minimizing regularizer may take an extreme value—either fully trusting or completely ignoring the side information—or settle at an interior value. Though this conclusion pertains only to the upper bound and may not reflect the behavior of the actual excess risk, our simulations for the actual excess risk suggest that a similar behavior is observed there too.

3.1.2 Related Work

This chapter contributes to the literature on learning with auxiliary and incomplete information, bridging Learning Using Privileged Information (LUPI), missing data imputation, and transfer learning. This chapter is derived from our manuscript [27]. To the best of our knowledge, the setting of training on partitioned datasets where side information is available only at test time has not been previously studied, and was first investigated in our work [27].

Learning Using Privileged Information (LUPI), introduced by [7], is a learning paradigm that leverages additional information available during training, but not at test time, to enhance

learning performance. This is analogous to a teacher who, in addition to providing answers, also offers hints or explanations to students. Subsequent works have extended the LUPI framework to various domains, including [28–31]. In contrast, this chapter considers a setting where side information is not observed jointly with labels during training (i.e., the dataset is partitioned), and becomes available only at test time, making it distinct from the above approaches.

Our problem can be viewed as learning from an incomplete training dataset, in our case, a partitioned dataset. A rich body of literature addresses missing data imputation, including [32–36]. Unlike these works, which primarily focus on the soundness of imputation methods, our approach treats imputation as an intermediate step, with the main objective of analyzing the excess risk of the proposed algorithm. Additionally, we apply regularization to the imputed values based on our confidence in their accuracy, introducing an added layer of caution. This emphasis on excess risk analysis distinguishes our approach from existing imputation-focused literature.

This chapter shares certain similarities with transfer learning, as discussed in [37], where knowledge from an auxiliary task is used to improve performance on a main task. However, in standard transfer learning, auxiliary task labels are typically available during both training and testing, and are jointly observed with the main task labels during training. In contrast, our setting involves side information that is not jointly observed with labels during training, making it structurally different from conventional transfer learning approaches.

Several other works, such as [38–41], define and utilize side information in their own ways across different contexts. A common aspect of these approaches is that side information is jointly observed with labels in the training data and is typically not available at test time (with the exception of [38], where side information is also available at test time). In contrast, our setting differs in formulation: we consider a partitioned training dataset and assume that side information is available only at test time.

3.2 Problem Formulation

In this section, we mathematically formulate the problem of learning with side information with partitioned dataset.

3.2.1 Notation Scheme

Random variables are denoted by capital letters (e.g., X), with their specific instances represented by the corresponding lowercase letters (e.g., x). The space over which a random variable is defined is denoted using the corresponding calligraphic letter (e.g., $\mathcal{X}$). Risks are represented by capital letters (e.g., R, L), while empirical risks are indicated by placing a hat over the corresponding letter (e.g., $\hat{r}, \hat{L}$). Model parameters, such as weights, are denoted by lowercase letters (e.g., w, u). The optimal weights minimizing the population risk are denoted with a superscript star (e.g., $w^\star$), and those minimizing the empirical risk are denoted with a hat (e.g., $\hat{w}$). We use sans-serif letters to denote important constants (w.r.t. this chapter, e.g., W). For random variables X and Y , we write $\mathbb{E}_X[f(X, Y) \mid Y]$ simply as $\mathbb{E}_X[f(X, Y)]$ when the conditioning variable is clear from context.

3.2.2 Modeling

Let $\mathcal{X} \subset \mathbb{R}^d$ denote the feature space, $\mathcal{Y} = \{-1, 1\}$ the label space, and $\mathcal{Z} = \{-1, 1\}$ the side information space. Let (X, Y, Z) be a random variable representing the true data generating process over $\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$. We assume the following properties:

(C1) Boundedness: $\exists \kappa > 0 \quad \|X\| \leq \kappa$ almost surely,

(C2) Regularity: X has a Lebesgue density and a Lipschitz cumulative distribution function; (X, Y, Z) has densities $p_{y,z}(x)$ with respect to Lebesgue measure in x and counting measure in the binary coordinates y, z .

(C3) Positive definiteness: $\mathbb{E}[(1, X)(1, X)^T] \succ 0$,

(C4) Non-separability: For all x, z ,

$$\mathbb{P}(Y = 1 \mid X = x, Z = z) \notin \{0, 1\}, \quad \mathbb{P}(Y = 1 \mid X = x) \notin \{0, 1\}, \quad \mathbb{P}(Z = 1 \mid X = x) \notin \{0, 1\}. \quad (3.1)$$

Consider a dataset $\{(x_k, y_k, z_k)\}_{k=1}^{n+m}$ consisting of i.i.d. samples drawn from the joint distribution of (X, Y, Z) . We partition this dataset as follows.

$$D = \{(x_i, y_i)\}_{i=1}^n \quad (\text{missing } z_i), \quad S = \{(x_j, z_j)\}_{j=n+1}^{n+m} \quad (\text{missing } y_j), \quad n \geq m. \quad (3.2)$$

Here, D contains feature-label pairs with missing side information, and S contains feature-side information pairs with missing labels. At test time, a sample (x, y, z) is drawn from the joint

distribution of (X, Y, Z) . The objective is to learn a classifier using both S and D in order to accurately predict the target label y based on the features x and side information z .

We employ a logistic regression classifier with an intercept term and use a decision threshold of 0.5 for a deterministic binary prediction. Define the sigmoid function,

$$\sigma(t) := \frac{1}{1 + \exp(-t)}, \quad t \in \mathbb{R}.$$

and let the logistic loss for a feature vector $a \in \mathbb{R}^k$, label $b \in \{-1, 1\}$ and a weight vector $c \in \mathbb{R}^{k+1}$ be given by

$$\ell(a, b; c) := -\ln(\sigma(b(c^\top(1, a))))$$

To characterize the excess risk of the hypothesis learned by our algorithm, we now define several classifiers together with their associated population risks.

Let $u := (u_c, u_x) \in \mathbb{R}^{d+1}$ parameterize a classifier $\phi_u : \mathcal{X} \rightarrow \mathcal{Z}$ given by

$$\phi_u(x) := 2\mathbb{1}_{\{\sigma(u_c + u_x^\top x) > 0.5\}} - 1.$$

The logistic population risk of a classifier parameterized by u is defined as (with $a = X$, $b = Z$, $c = (u_c, u_x)$ in the logistic loss)

$$L(u) := \mathbb{E}_{(X, Y, Z)}[\ell(X, Z; u)] = \mathbb{E}_{(X, Z)}[-\ln(\sigma(Z(u_c + u_x^\top X)))].$$

The optimal weight $u^\star := (u_c^\star, u_x^\star)$ minimizes the population risk:

$$u^\star := \arg \min_u L(u). \quad (3.3)$$

From the non-separability assumption, we have $\|u^\star\| < \infty$ (a consequence of Lemma 3.9). The population risk under the zero-one loss is defined as

$$L_{(0,1)}(u) := \mathbb{E}_{(X, Y, Z)}[\mathbb{1}_{\{Z(u_c + u_x^\top X) \leq 0\}}] = \mathbb{E}_{(X, Z)}[\mathbb{1}_{\{Z(u_c + u_x^\top X) \leq 0\}}].$$

The logistic empirical risk over S of a classifier parameterised by u is defined as (with $a = x_j$, $b = z_j$, $c = (u_c, u_x)$ in the logistic loss)

$$\hat{L}(u) := \frac{1}{m} \sum_{(x_j, z_j) \in S} \ell(x_j, z_j; u) = \frac{1}{m} \sum_{(x_j, z_j) \in S} -\ln(\sigma(z_j(u_c + u_x^\top x_j))).$$

For a regularization parameter $\mu > 0$, the regularized empirical minimizer $\hat{u} := (\hat{u}_c, \hat{u}_x)$ is given by

$$\hat{u} := \arg \min_u \hat{L}(u) + \mu \|u\|^2. \quad (3.4)$$

The corresponding classifiers associated with $u^\star$ and $\hat{u}$ are denoted by

$$\phi^\star(x) := \phi_{u^\star}(x) \text{ and } \hat{\phi}(x) := \phi_{\hat{u}}(x). \quad (3.5)$$

Let $v := (v_c, v_x) \in \mathbb{R}^{d+1}$ parameterize a classifier $\psi_v : \mathcal{X} \rightarrow \mathcal{Y}$ given by

$$\psi_v(x) := 2\mathbb{1}_{\{\sigma(v_c + v_x^\top x) > 0.5\}} - 1.$$

The logistic population risk of a classifier parameterised by v is defined as (with $a = X$, $b = Y$, $c = (v_c, v_x)$ in the logistic loss)

$$T(v) := \mathbb{E}_{(X,Y,Z)}[\ell(X, Y; v)] = \mathbb{E}_{(X,Y)}[-\ln(\sigma(Y(v_c + v_x^\top X)))].$$

and the optimal weight $v^\star := (v_c^\star, v_x^\star)$ is given by

$$v^\star := \arg \min_v T(v). \quad (3.6)$$

Note that from the non-separability assumption, we have $\|v^\star\| < \infty$ (a consequence of Lemma 3.9). Finally let $w := (w_c, w_x, w_z) \in \mathbb{R}^{d+2}$ parameterize a classifier $\pi_w : \mathcal{X} \times \mathcal{Z} \rightarrow \mathcal{Y}$ given by

$$\pi_w(x, z) := 2\mathbb{1}_{\{\sigma(w_c + w_x^\top x + w_z^\top z) > 0.5\}} - 1.$$

The logistic population risk of a classifier parameterised by w is defined as (with $a = (X, Z)$, $b = Y$, $c = (w_c, w_x, w_z)$ in the logistic loss)

$$R(w) := \mathbb{E}_{(X,Y,Z)}[\ell(X, Z, Y; w)] = \mathbb{E}_{(X,Y,Z)}[-\ln(\sigma(Y(w_c + w_x^\top X + w_z^\top Z)))].$$

The optimal weight $w^\star := (w_c^\star, w_x^\star, w_z^\star)$ is defined as

$$w^\star := \arg \min_w R(w). \quad (3.7)$$

Note that from the non-separability assumption, $\|w^\star\| < \infty$ (a consequence of Lemma 3.9). For a regularization parameter $\lambda := (\lambda_c, \lambda_x, \lambda_z) \in [0, \infty)^3$, the regularized logistic population risk is

$$R_\lambda(w) := R(w) + \lambda_c |w_c|^2 + \lambda_x \|w_x\|^2 + \lambda_z |w_z|^2,$$

with minimizer

$$w_\lambda^\star := \arg \min_{w \in \mathbb{R}^{d+2}} R_\lambda(w). \quad (3.8)$$

Given a mapping $\phi : \mathcal{X} \rightarrow \mathcal{Z}$, the *logistic ϕ -population risk* over (X, Y) is

$$R^\phi(w) := \mathbb{E}_{(X,Y)}[\ell(X, \phi(X), Y; w)].$$

The corresponding regularized risk and its minimizer are

$$R_\lambda^\phi(w) := R^\phi(w) + \lambda_c |w_c|^2 + \lambda_x \|w_x\|^2 + \lambda_z |w_z|^2,$$

and

$$w_{\phi,\lambda}^\star := \arg \min_w R_\lambda^\phi(w) \quad (3.9)$$

respectively.

The logistic empirical risk over D with imputed side information $\hat{\phi}(x_i)$ described in (3.5) for a classifier parameterised by w is defined as (with $a = (x_i, \hat{\phi}(x_i))$, $b = y_i$, $c = (w_c, w_x, w_z)$ in the logistic loss)

$$\hat{r}^\phi(w) := \frac{1}{n} \sum_{(x_i, y_i) \in D} \ell(x_i, \hat{\phi}(x_i), y_i; w) = \frac{1}{n} \sum_{(x_i, y_i) \in D} -\ln(\sigma(y_i(w_c + w_x^\top x_i + w_z \hat{\phi}(x_i)))).$$

The corresponding regularized empirical minimizer is

$$\hat{w}_\lambda := \arg \min_w [\hat{r}^\phi(w) + \lambda_c |w_c|^2 + \lambda_x \|w_x\|^2 + \lambda_z |w_z|^2]. \quad (3.10)$$

Our goal is to analyze the excess risk of the learned classifier $\hat{w}_\lambda$ in terms of the various population-optimal weights described earlier.

3.3 Algorithm

We present our Partitioned Side Information Learning (PSIL) algorithm for leveraging side information in classification problems with partitioned datasets. The procedure consists of two main stages:

1. Learning the Feature to Side Information mapping: Select a regularization parameter μ and learn a classifier $\hat{u} \equiv \hat{\phi} : \mathcal{X} \rightarrow \mathcal{Z}$ from the sample S using logistic regression (thresholded at 0.5). This classifier is used to impute the missing Z -values in the primary dataset D .
2. Learning the Feature and imputed Side Information to Label mapping with regularization: Choose regularization parameters $(\lambda_c, \lambda_x, \lambda_z)$. For each $(x_i, y_i) \in D$ we impute $z_i = \hat{\phi}(x_i)$ and train a regularized logistic regression model (thresholded at 0.5) to learn a classifier $\hat{w}_\lambda : \mathcal{X} \times \mathcal{Z} \rightarrow \mathcal{Y}$.

Algorithm 3.1 Partitioned Side Information Learning (PSIL)**Require:** $D = \{(x_i, y_i)\}_{i=1}^n$, $S = \{(x_j, z_j)\}_{j=1}^m$ **Ensure:** Hypothesis $\hat{w}_\lambda$

- 1: Select the regularization parameter $\mu := \frac{c}{\sqrt{m}}$, where $c > 0$ is chosen to minimize side-information prediction error on a validation set constructed from S .
- 2: Learn $\hat{u}$ by minimizing the regularized empirical risk over S :

$$\hat{u} := \arg \min_{u \in \mathbb{R}^{d+1}} \hat{L}(u) + \mu \|u\|^2$$
- 3: For each $(x_i, y_i) \in D$, impute $z_i := \hat{\phi}(x_i)$.
- 4: Set $\lambda_c := \lambda_x := \frac{1}{\sqrt{n}}$ and choose $\lambda_z \geq n^{-1/2}$.
- 5: Learn $\hat{w}_\lambda$ by minimizing the regularized empirical risk over D with imputed z_i :

$$\hat{w}_\lambda := \arg \min_{w \in \mathbb{R}^{d+2}} \hat{r}^{\hat{\phi}}(w) + \lambda_c |w_c|^2 + \lambda_x \|w_x\|^2 + \lambda_z |w_z|^2$$
- 6: At test time, given (x, z) (with observed z), predict $\hat{y} = 1$ if $\hat{w}_{\lambda,c} + \hat{w}_{\lambda,x}^\top x + \hat{w}_{\lambda,z} z > 0$, else predict $\hat{y} = -1$.

Among all regularization parameters, only λ_z requires non-trivial tuning and thus becomes the main focus of both theoretical and empirical analysis in this chapter. In principle, the optimal value of λ_z depends on the properties of the unknown joint distribution of (X, Z, Y) . In standard scenarios, one would set aside a validation set to empirically estimate this hyperparameter. With the partitioned data, a held-out sample of (X, Y) can validate the imputed-input risk $R^{\hat{\phi}}$, conditional on the fitted imputer. It cannot in general validate the target risk R , which uses observed Z , without a jointly observed labeled evaluation sample or additional identifying assumptions. Our theoretical selection criterion concerns this target risk.

3.4 Analysis of the Algorithm

In this section, we analyze the excess risk of the Partitioned Side Information Learning (PSIL) algorithm (Algorithm 3.1). Our main theoretical result, Theorem 3.1, provides an explicit upper bound on the excess risk as a function of the sample sizes, distribution-dependent quantities, and the regularization parameter. This bound characterizes how the choice of regularizer mediates the trade-off between relying on imputed side information and using only the primary features. Corollary 3.2 provides guidance for selecting the optimal regularization parameter from three candidates. Moreover, Corollary 3.3 establishes a conditional threshold-based result showing

that, for sufficiently large side information sample sizes, the bound-minimizing regularizer takes an extreme value when the side information is either very noisy or highly informative.

Next, Theorem 3.4 establishes a minimax lower bound on the excess risk for any algorithm, quantifying the intrinsic difficulty of learning from partitioned datasets in which features, labels, and side information are never observed jointly.

Finally, we conduct simulations on a variety of datasets to investigate the behavior of the value of λ_z that minimizes the excess risk of our proposed algorithm.

3.4.1 Main Result: Upper Bound on Excess Risk

We now state our main theorem, which provides an explicit upper bound on the excess risk as a function of the sample sizes m and n , relevant distribution-dependent quantities, and the choice of regularizer λ_z .

To ensure coercivity of the upper bound, we require a non-degeneracy condition on the expected score of the imputer.

(ND) **Non-degeneracy:** We assume that the boundary has measure zero and the optimal features are not perfectly collinear

$$P(s^*(X) = 0) = 0, \quad E[A_* A_*^T] \succ 0, \quad \text{where} \quad s^*(x) = u_c^* + u_x^{*T} x, \quad A_* = (1, X, \phi^*(X)). \quad (3.11)$$

Under an absolutely continuous X distribution, it suffices that ϕ^* takes both labels with positive probability. A nonconstant affine classifier has a zero-probability boundary. An affine relation involving $1, X$ and ϕ^* cannot vanish on both positive-volume label regions unless its coefficients all vanish.

Theorem 3.1. Let u^* and v^* be as defined in (3.3) and (3.6), respectively. Let w^* and w_λ^* be as defined in (3.7) and (3.8). Let W be the deterministic population envelope defined in Proposition 3.16. Consider the partitioned training datasets D of size n and S of size m , as defined in (3.2). Under the continuous density assumption (ND) and non-separability condition (C4), let $n \geq n_0$ and $m \geq m_0$. Let $0 < \delta < 1$. Take the regularizer λ as

$$\lambda = \left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \lambda_z \right) \in \mathbb{R}^3,$$

where $\lambda_z \geq \frac{1}{\sqrt{n}}$. Then, with probability at least $1 - \delta$, the following inequality holds simultaneously for all such λ

$$\begin{aligned} R(\hat{w}_\lambda) - R(w^\star) &\leq \frac{C_1}{\sqrt{n}} + \min \left\{ \lambda_z |w_z^\star|^2, T(v^\star) - R(w^\star) \right\} \\ &\quad + \min \left\{ \frac{1}{2\lambda_z}, W \right\} \frac{4}{\ln(2)} \left(\frac{C_2}{\sqrt{m}} + L(u^\star) \right), \end{aligned} \quad (3.12)$$

where $C_1 = C_1(W, \delta)$ and $C_2 = O(\sqrt{\ln(1/\delta)})$ are purely deterministic constants. Since $m < n$, the term $1/\sqrt{m}$ on the right-hand side does not get absorbed into the $1/\sqrt{n}$ term.

Proof Outline. To analyze the excess risk, we decompose the left-hand side of (3.12) into four terms, each capturing a distinct source of error

$$R(\hat{w}_\lambda) - R(w^\star) = \underbrace{R(\hat{w}_\lambda) - R^{\hat{\phi}}(\hat{w}_\lambda)}_{(A)} + \underbrace{R^{\hat{\phi}}(\hat{w}_\lambda) - R^{\hat{\phi}}(w_\lambda^\star)}_{(B)} + \underbrace{R^{\hat{\phi}}(w_\lambda^\star) - R(w_\lambda^\star)}_{(C)} + \underbrace{R(w_\lambda^\star) - R(w^\star)}_{(D)}.$$

The first and third terms, (A) and (C), quantify the estimation error resulting from the use of the imputed side information $\hat{\phi}$. This error depends on the complexity of the relationship between (X, Z) quantified by $L(u^\star)$, the sample size m of the side information dataset S , and the regularization parameter λ_z , which determines the extent to which the model relies on the imputed side information. These terms contribute the following to the upper bound

$$\min \left(\frac{1}{2\lambda_z}, W \right) \times \frac{4}{\ln(2)} \left(\frac{C_2}{\sqrt{m}} + L(u^\star) \right).$$

The second and fourth terms, (B) and (D), are bounded together. Empirical optimality retains the population penalty, and a population comparator argument then gives the contribution

$$\min \{ \lambda_z |w_z^\star|^2, T(v^\star) - R(w^\star) \}$$

together with $C_1/\sqrt{n}$ to the upper bound in (3.12). At $\lambda_z = \infty$, the constraint $w_z = 0$ yields the feature-only regularized population fit; its unregularized benchmark is $v^\star$. Thus, this term quantifies the proportion of the worst-case gap $T(v^\star) - R(w^\star)$ (with the worst at $\lambda_z = \infty$) that is realized, depending on the choice of the regularizer λ_z . The complete proof is given in Section 3.6 (Proof 3.6.4). $\square$

Interpretation of the Excess Risk Bound

To provide intuition for the upper bound in Theorem 3.1, consider the university admission scenario. Here, the feature $x \in \mathbb{R}^d$ represents a candidate's academic scores, the side information z is the admission decision of a secondary university, and the label y is the decision of the primary university. The excess risk $R(\hat{w}_\lambda) - R(w^*)$ represents the performance gap between our learned classifier $\hat{w}_\lambda$ and the benchmark optimal classifier w^* , which has access to the true joint distribution of (X, Y, Z) . This risk is controlled by the regularization parameter λ_z and consists of three distinct components.

The first term, $O(1/\sqrt{n})$, is the standard estimation error associated with the sample size n of the primary university's dataset. This error is unavoidable and decays as more primary data becomes available.

The second term captures the potential gain from incorporating z . It is proportional to the gap $T(v^*) - R(w^*)$, which measures how much knowing the true side information improves prediction accuracy over using features x alone. This term is controlled linearly by λ_z . If z carries meaningful information (i.e., the gap is significant), the upper bound is lowered by reducing λ_z , which encourages the model to assign larger weight to the side information.

The third term quantifies the error introduced by imputing z from x using the secondary university sample. It depends on the sample size m of the secondary university's data and the optimal prediction error $L(u^*)$, which reflects the complexity of the relationship between features and side information. This term is inversely controlled by λ_z (scaling as $1/\lambda_z$). If the features cannot accurately predict z , the imputed values are noisy, and a large λ_z is required to ignore the unreliable imputed data.

The interplay between the second and third terms creates a tension in selecting λ_z . If z is highly informative for predicting y (large gap $T(v^*) - R(w^*)$), a smaller λ_z is desirable to leverage this information. However, if x poorly predicts z (large $L(u^*)$), a larger λ_z is necessary to mitigate the noise from imperfect imputation. The regularizer λ_z must balance these two competing terms. In this case, the optimal λ_z must find a balance to extract the signal in z while damping the noise from imperfect imputation. As we will show in subsequent sections, despite this tension, when the side information sample size is large, the optimal λ_z can exhibit a conditional threshold-based behavior, taking on extreme values (either very large or very small)

depending on which factor is more limiting: the noise introduced by imperfectly imputed z , or the dependence of y on z . The direction of this behavior is determined by the underlying population distributional properties.

Retrospective Regularizer Selection

Theorem 3.1 provides an explicit upper bound on the excess risk as a function of the regularization parameter λ_z . We are interested in determining the value of λ_z that minimizes this upper bound. Since λ_z is a hyperparameter, its optimal value depends on the underlying population distribution. To analyze its optimal choice, we define the parameters of the bound as follows: Let $a = |w_z^\star|^2$ represent the signal strength, $\Delta = T(v^\star) - R(w^\star)$ represent the worst-case feature-only risk gap, and $b_m = \frac{4}{\ln 2}(L(u^\star) + \frac{C_2}{\sqrt{m}})$ represent the bounded imputation error. The upper bound on the excess risk (excluding the $O(1/\sqrt{n})$ term) as a function of $\lambda_z \geq \lambda_0 = 1/\sqrt{n}$ is given by

$$F(\lambda_z) := \min\{a\lambda_z, \Delta\} + b_m \min\left\{W, \frac{1}{2\lambda_z}\right\}. \quad (3.13)$$

Let $\lambda_z^\star := \arg \min_{\lambda_z \geq \lambda_0} F(\lambda_z)$ be the (retrospective) optimal regularizer that minimizes the theoretical upper bound. Because $F(\lambda_z)$ is a combination of two minima, it is a piecewise function. Its global minimum over the valid range must occur at one of three candidate points: the lower endpoint λ_0 , the upper endpoint ∞ , or an interior point where the derivative of the active branches is zero. We establish this exactly in the following corollary, replacing the regime analysis of the earlier work.

Corollary 3.2 (The Three-Candidate Rule): *For any distribution, the optimal regularizer $\lambda_z^\star$ that minimizes the upper bound $F(\lambda_z)$ on the interval $[\lambda_0, \infty)$ is selected from at most three candidates:*

1. *The lower endpoint: $\lambda_0 = 1/\sqrt{n}$.*
2. *The upper endpoint: $\lambda_\infty = \infty$.*
3. *The interior point: $\lambda_I = \sqrt{\frac{b_m}{2a}}$, provided it satisfies $\lambda_0 \leq \lambda_I$, $a\lambda_I \leq \Delta$, and $\frac{1}{2\lambda_I} \leq W$.*

Proof. The proof is given in Section 3.6.5 (Proof 3.6.5). □

To understand when each candidate is selected, we examine the asymptotic behavior as $m \rightarrow \infty$. Notice that $b_m \rightarrow \frac{4}{\ln 2}L(u^\star)$. If the true imputation loss is non-zero (i.e., $L(u^\star) > 0$),

then the interior candidate approaches a strictly positive constant

$$\lim_{m \rightarrow \infty} \lambda_I = \sqrt{\frac{2L(u^\star)}{a \ln 2}} > 0. \quad (3.14)$$

The optimal choice is determined by the balance between the signal strength of the side information ($a = |w_z^\star|^2$) and the noise level of the imputation ($L(u^\star)$). We formalize this by defining the signal-imputation ratio

$$\rho := \frac{|w_z^\star|^2}{L(u^\star)}. \quad (3.15)$$

Depending on this ratio, we establish the Conditional Threshold-Based Corollary, showing exactly when the endpoints strictly dominate.

Corollary 3.3 (Conditional Threshold-Based Regularization): *For sufficiently large m, n , there exist critical thresholds ρ_{low} and ρ_{high} such that*

- *If $\rho < \rho_{\text{low}}$ (the side information is too noisy), the optimal regularizer is strictly the upper endpoint: $\lambda_z^\star = \infty$.*
- *If $\rho > \rho_{\text{high}}$ (the side information is clean and strong), the optimal regularizer is strictly the lower endpoint: $\lambda_z^\star = \lambda_0 = 1/\sqrt{n}$.*
- *If $\rho \in (\rho_{\text{low}}, \rho_{\text{high}})$ (moderate signal-imputation ratio), the theoretical bound is strictly minimized by the interior point $\lambda_z^\star = \lambda_I$.*

Proof. The proof is given in Section 3.6.5 (Proof 3.6.5). □

3.4.2 Lower Bound on Excess Risk

In this section, we establish a fundamental limitation of learning from partitioned datasets. Specifically, we show that for any learning algorithm A , there exists a probability distribution of (X, Y, Z) such that if (D, S) is a partitioned data set defined in (3.2), the excess risk $R(A(D, S)) - R(w^\star)$ is bounded away from zero with non-vanishing probability. This demonstrates that, even with infinite training data, the inability to observe features, labels, and side information jointly imposes an irreducible barrier to learning performance.

Theorem 3.4 (Lower Bound on Excess Risk). Let (D, S) be a partitioned dataset defined in (3.2), and let $(D, S) \mapsto A(D, S) \in \mathbb{R}^{d+2}$ be any learning algorithm (possibly randomized). Then there exists a distribution of (X, Y, Z) satisfying conditions (C1)–(C4) generating the

partitioned training datasets (D, S) such that for $w^\star$ defined in (3.7), we have

$$\mathbb{P}_{(D,S)}(R(A(D, S)) - R(w^\star) > 0.19) \geq \frac{1}{2}, \quad (3.16)$$

where the probability is taken over the random draw of the partitioned training data (D, S) (and any internal randomization of the algorithm A).

Proof Outline. We construct two joint distributions, P_1 and P_2 over (X, Y, Z) satisfying conditions (C1)–(C4), such that the observable marginal laws (X, Y) and (X, Z) are identical under both distributions, while their full joint distributions differ.

Under the common training-data law Q , the partitioned dataset (D, S) provides zero statistical information to distinguish P_1 from P_2 . For any algorithm A , the sum of excess risks satisfies $(R_1(A(D, S)) - R_1^\star) + (R_2(A(D, S)) - R_2^\star) \geq 2(\ln 2 - H(0.2)) > 2 \times 0.19$ pointwise for every output. Defining the bad events $E_k = \{(D, S) : R_k(A(D, S)) - R_k^\star > 0.19\}$, this pointwise separation guarantees $E_1 \cup E_2 = \Omega$, so that under the common data law Q , $\max\{Q(E_1), Q(E_2)\} \geq 1/2$. Hence, for every learner, there exists a fixed distribution in the class under which the excess risk exceeds 0.19 with probability at least 1/2.

We note that while P_1 and P_2 satisfy conditions (C1)–(C4), they have optimal imputer weight $u^\star = 0$ (uninformative features for Z), which falls outside the Non-Degeneracy condition (ND) assumed for the upper bound. This lower bound establishes that across the broader partitioned learning class, non-vanishing excess risk is an intrinsic limitation arising from the unobserved joint relationship. The complete proof is given in Section 3.6 (Proof 3.6.6). $\square$

3.4.3 Simulation Studies

Corollary 3.2 provides theoretical guidance on the choice of λ_z to minimize the upper bound on excess risk. However, it does not guarantee that the same value of λ_z minimizes the excess risk in practice. To investigate this, we conduct simulations to identify the values of λ_z that minimize the risk of our algorithm on the test set (which is same as minimizing the excess risk).

Let

$$\lambda_z^\star := \arg \min_{\lambda_z} R(\hat{w}_\lambda),$$

where $R(\cdot)$ denotes the risk evaluated on the test set, serving as a proxy for the population risk.

Our simulation study evaluates the behavior of λ_z^* value, across two distinct scenarios: a collection of toy datasets and a collection of real-world datasets.

Toy DataSet: We generate 100 different toy distributions over (X, Y, Z) , each constructed as follows:

1. The features X are sampled from a Gaussian mixture model, whose parameters and mixing probabilities are randomly chosen.
2. Then the weight vectors u and w are sampled from standard normal distribution. Given fixed weights u , the binary variables Z is sampled via logistic models:

$$Z \sim \sigma((1, X)^\top u).$$

Once we obtain Z , for a fixed weight w , the binary variable Y is sampled via a logistic model:

$$Y \sim \sigma((1, X, Z)^\top w).$$

This process is repeated 100 times, each with a new random choice of parameters, resulting in a diverse family of distributions. For each distribution, we generate $m = 1000$ samples for the (X, Z) dataset (used for learning $X \rightarrow Z$), $n = 4000$ samples for the (X, Y) dataset (used for learning $X \times Z \rightarrow Y$), and $t = 20000$ samples for the test set (X, Y, Z) to compute the risk. For every $\lambda_z \in \{10^{-1}, \dots, 10^6\}$, a logistic regression model is trained and evaluated. The λ_z^* value for each distribution is chosen as the value that yields the lowest risk of the algorithm on the test set.

The frequency of each candidate λ value being chosen as λ_z^* across the 100 distributions is shown in Figure 3.1. The simulation results, summarized in the frequency plot, show a strong preference for the extreme values of the regularization parameter λ_z . This pattern provides empirical evidence supporting the conditional threshold-like behaviour of λ_z established in Corollary 3.3.

Importantly, our goal in this simulation is to evaluate the structural properties of the risk surface, not to propose an operational model-selection protocol. Therefore, rather than selecting λ_z via cross-validation on the training set (which would introduce finite-sample model-selection noise), we perform a retrospective scan using the test set risk to approximate the true population risk. While evaluating on the test set is a form of data snooping in an applied setting, here it

serves strictly as an exploratory oracle to compute the best-case λ_z^* . Because the threshold-like preference relies on the population signal-to-noise ratio ρ , this oracle approach confirms that the algorithm empirically exhibits the conditional threshold-like behavior in practice, reflecting the theoretical bound.

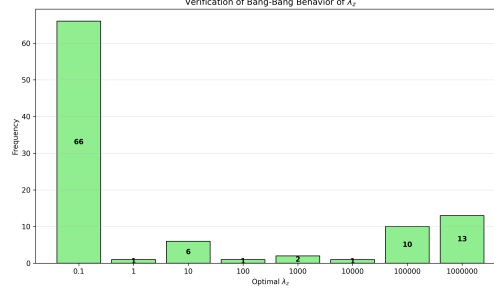


Figure 3.1: Frequency of λ_z^* across 100 distributions.

Real World Data Set: To validate the threshold-like behavior in practical settings, we examine the λ_z^* values across eight real-world datasets from the OpenML repository. For each dataset, we identify a primary label Y and a relevant binary side-information variable Z , while the remaining covariates are treated as features X . The datasets are summarized in Table 3.1.

Table 3.1: Real-world datasets used in the simulation study.

Dataset	Total Size	Y (Label)	Z (Side Info)	X (Features)
Adult Income	48,842	Income ($\leq \$50K$ / $> \$50K$)	Sex (Male / Female)	13 demographic features [42]
Bank Marketing	45,211	Class (1 / 2)	V7 (Yes / No)	15 client features (V1–V16) [43]
Mushroom	8,124	Edibility (Edible / Poisonous)	Bruises (Yes / No)	21 physical features [44]
Spambase	4,601	Email Type (Spam / Non-Spam)	Word “make” (Pres. / Abs.)	56 word/capital statistics [45]
Titanic Survivor	1,309	Survival (Survived / Died)	Sex (Male / Female)	3 features (class, age, fare) [46]
German Credit	1,000	Credit Risk (Good / Bad)	Foreign Worker (Yes / No)	19 credit/financial features [47]
Diabetes	768	Diabetes Status (+ / –)	Age (≤ 30 / > 30)	7 medical measurement features [48]
Breast Cancer	699	Diagnosis (Benign / Malignant)	Clump Thickness (≤ 5 / > 5)	8 cell measurement features [49]

We employ a 70/30 train-test split for each dataset. The training portion is then partitioned using two regimes: an 80/20 split (80% feature-label, 20% feature-side information) and a 50/50 split (equal feature-label and feature-side information). Within each regime, for every value of $\lambda_z \in \{10^{-1}, \dots, 10^6\}$, a model is trained and evaluated. The λ_z^* value for each distribution is chosen as the value that yields the lowest risk of the algorithm on the test set.

The results are reported as follows. For all eight datasets, we present a bar plot of the value

of λ_z^* under both the 80/20 and 50/50 regimes (Figure 3.2). The results broadly support the theoretical prediction: in both regimes, most datasets select λ_z^* near one of the two extreme values of the candidate grid rather than settling at an intermediate value, with only a few exceptions.

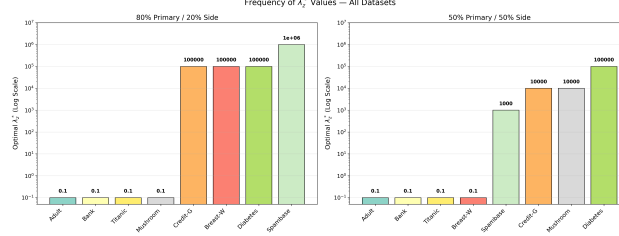


Figure 3.2: λ_z^* across datasets (80/20 and 50/50 regimes).

Furthermore, for the three largest datasets (Adult, Bank, and Mushroom), we present progressive risk plots (Figure 3.3) that illustrate how the test risk evolves as the training pool size increases, under both the 80/20 and 50/50 partitioning regimes. Each panel directly compares three approaches: *Full Side Info* (using both X and Z for prediction), *No Side Info* (using only X), and *Partitioned Side Info* (our Algorithm 3.1, evaluated at the smallest and largest candidate values of λ_z). The results in Figure 3.3 show that, for the partitioned side information algorithm, the minimum risk achieved over the smallest and largest candidate values of λ_z consistently lies between the Full Side Info and No Side Info plots.

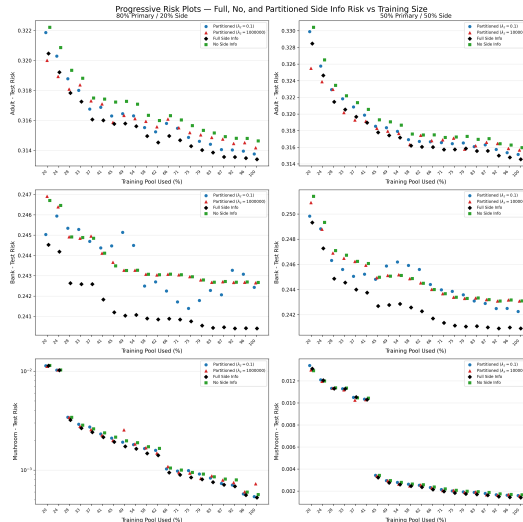


Figure 3.3: Risk plots for largest datasets (80/20 and 50/50 regimes).

In this section, we present simulation studies comparing the theoretical upper bound on excess risk (Theoretical RHS) with the actual observed test-set excess risk (Actual LHS), each

evaluated at the bound-minimizing regularization parameter λ_z^* . The simulation strictly implements the Partitioned Side Information Learning (PSIL) procedure defined in Algorithm 3.1, directly optimizing the regularized objectives for both the hard imputer $\hat{\phi}(x) \in \{-1, +1\}$ and the primary classifier $\hat{w}_\lambda$, and evaluating deployment risk using observed side information z at test time.

To investigate the theoretical guarantees across diverse distributional regimes, we first evaluate three synthetic toy distributions where the covariates $X \in \mathbb{R}^5$ are sampled from a three-component Gaussian mixture model truncated to a compact Euclidean ball to satisfy Assumption (C1). The distributions represent three distinct regimes: highly informative side information, a moderate signal-imputation balance, and noisy side information. For each distribution, we vary the primary sample size $n \in \{500, 1000, 2000, 5000\}$ with auxiliary sample size $m = n$. We calculate the theoretical upper bound evaluated at the optimal candidate λ_z^* using the envelope constants ($C_1 = C_2 = W = 1$) and compare it with the empirical excess risk achieved on an independent held-out test set of 20,000 samples across 20 independent repetitions, displaying 99% confidence bands. As shown in Figure 3.4, the actual excess risk closely tracks the shape and decay rate of the theoretical upper bound across all three regimes.

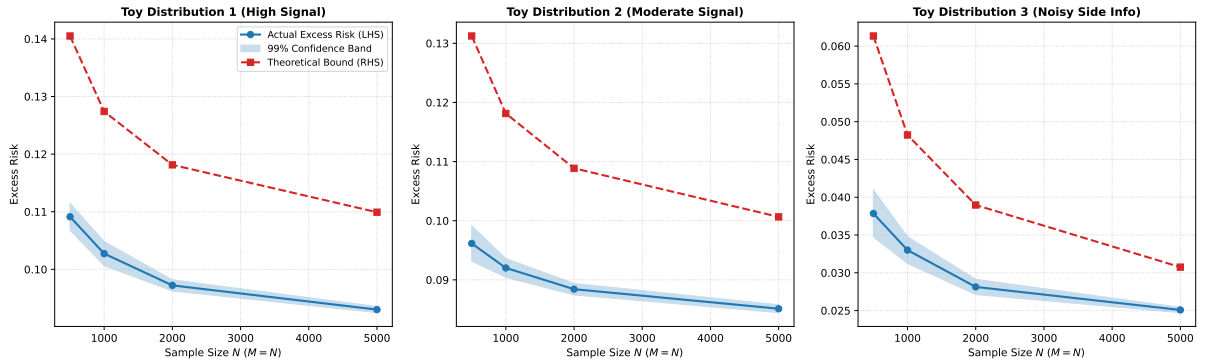


Figure 3.4: Theoretical Upper Bound (RHS) and Actual Excess Risk (LHS) with 99% confidence bands across sample size n ($m = n$) for three truncated Gaussian mixture toy distributions.

We conduct an evaluation on three real-world benchmark datasets from the OpenML repository: Adult Income, Bank Marketing, and Mushroom. To ensure strict separation between model selection and evaluation, each dataset is partitioned into three disjoint splits: a training pool (70%) from which (S, D) are sampled across varying sample sizes n ($m = n$), a separate oracle sample (15%) used exclusively to estimate population parameters and select the candidate regularizer λ_z^* , and a held-out test set (15%) used solely to evaluate test-set deployment risk. No

test observations influence model selection or regularizer tuning. As displayed in Figure 3.5, the actual excess risk decays steadily with increasing sample size, demonstrating consistent alignment with the trajectory of the theoretical upper bound across diverse tabular feature domains.

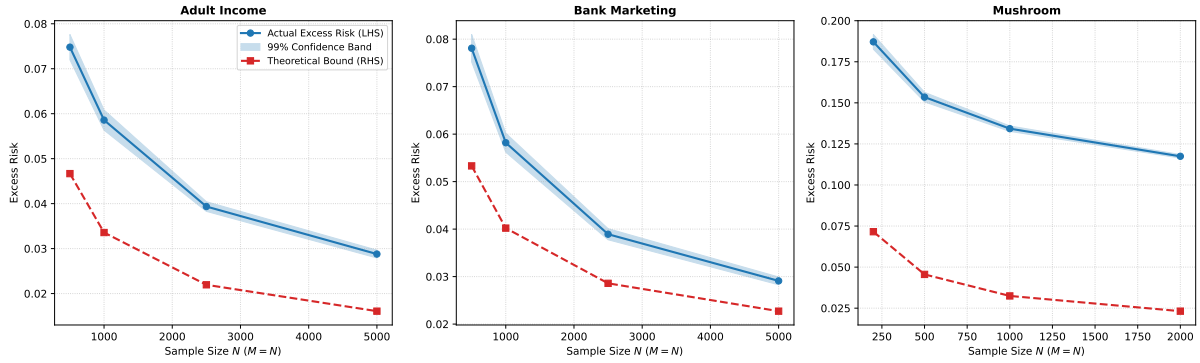


Figure 3.5: Theoretical Upper Bound (RHS) and Actual Excess Risk (LHS) with 99% confidence bands across sample size n ($m = n$) for Adult Income, Bank Marketing, and Mushroom under strict 3-way sample partitioning.

3.5 Conclusion, Limitations, and Future Work

In this chapter, we addressed the problem of utilizing test-time side information for classification when joint observations of side information and labels were absent from the training dataset. We proposed an algorithm that imputes the missing side information during training. The algorithm’s behavior is governed by a regularization hyperparameter, which quantifies the trust placed in the imputed data. We derived an upper bound on the excess risk as a function of this hyperparameter, noting that the optimal value depends on prior knowledge of the relationship between side information and labels. We established a Three-Candidate Rule for the bound-minimizing regularizer, showing that the optimal regularizer selects between two boundary endpoints and a feasible interior candidate, exhibiting conditional threshold-like behavior when the side information is either very noisy or highly informative. This choice cannot be inferred from the partitioned data alone and must instead be guided by theoretical considerations about the underlying ground-truth distribution. Furthermore, we established a fundamental limitation of this problem: the excess risk of any learning algorithm is lower-bounded by a non-vanishing term. This limitation arises directly from the lack of joint side information–label observations

during training.

Despite these contributions, this chapter has some limitations. First, our theoretical analysis and algorithmic development are currently restricted to the binary classification setting; extending these results to multi-label scenarios remains an open direction. Second, the theoretical analysis is restricted to the setting of logistic regression; extending these results to other models or loss functions remains another open direction. Third, the upper bound we derive may not be tight in all regimes, particularly due to the presence of the W term, which can lead to loose guarantees in practice. Addressing these limitations is an important avenue for future research. Additionally, another promising future direction is to explore the strategic aspect of this problem, where an agent decides whether to reveal side information to the classifier at test time.

3.6 Proofs and Supporting Results

This section contains all proofs deferred from the main text, together with the auxiliary lemmas they rely on. The complete proof of Theorem 3.1 is given in Section 3.6.4; the complete proof of Theorem 3.4 is given in Section 3.6.6. The remaining sections provide supporting lemmas used throughout.

3.6.1 Lemmas About Generalization Gap from Existing Literature

In this section, we review key results from [50] that will be used in the proof of the main theorem (Theorem 3.1). For convenience, we restate these theorems below. Before doing so, we summarize the assumptions and notation on which those results rely.

Let $\mathcal{H}$ be the hypothesis space, $\mathcal{H}^*$ (the dual space of $\mathcal{H}$) be the feature space, and $\mathcal{B}$ the label space. Let $l(b, r) \in \mathbb{R}$ denote the loss function, where $b \in \mathcal{B}$ and r is the score given by the inner product between the hypothesis and the feature vector. Consider the following assumptions.

- (AI) Let $\mathcal{H}$ be a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$, and let $\mathcal{B}$ be a Polish space. Let (A, B) be a random variable taking values in $\mathcal{H} \times \mathcal{B}$ with distribution P , and suppose there exists $\alpha > 0$ such that $\|A\| \leq \alpha$ almost surely.

- (AII) The loss function $\mathbf{l} : \mathcal{B} \times \mathbb{R} \rightarrow [0, \infty)$ is Lipschitz in its second argument, namely there exists $G > 0$ such that for all $b \in \mathcal{B}$ and $r, r' \in \mathbb{R}$, we have $|\mathbf{l}(b, r) - \mathbf{l}(b, r')| \leq G|r - r'|$.
- (AIII) Let $\mathfrak{L}(\mathbf{w}) := \int_{\mathcal{H} \times \mathcal{B}} \mathbf{l}(b, \langle \mathbf{w}, a \rangle) dP(a, b)$ denote the population risk. Suppose there exists $\mathbf{w}_* \in \mathcal{H}$ such that $\mathfrak{L}(\mathbf{w}_*) = \inf_{\mathbf{w} \in \mathcal{H}} \mathfrak{L}(\mathbf{w})$. For $\lambda > 0$, let $\widehat{\mathbf{w}}_\lambda \in \mathcal{H}$ be defined by $\widehat{\mathbf{w}}_\lambda := \arg \min_{\mathbf{w} \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \mathbf{l}(b_i, \langle \mathbf{w}, a_i \rangle) + \lambda \|\mathbf{w}\|^2$.

We first restate Theorem 1 from [50].

Lemma 3.5 (Theorem 1, [50]). Suppose the assumptions (AI), (AII), and (AIII) hold. Then for sample size n and for all $\lambda > 0$ and $0 < \delta < 1$, with probability at least $1 - \delta$, the following inequality holds:

$$\mathfrak{L}(\widehat{\mathbf{w}}_\lambda) - \mathfrak{L}(\mathbf{w}_*) \leq \frac{G^2 \alpha^2 \ln(1/\delta)}{\lambda n} + \lambda \|\mathbf{w}_*\|^2.$$

Hence letting $\lambda = \frac{G\alpha}{\|\mathbf{w}_*\|} \sqrt{\frac{\ln(1/\delta)}{n}}$ leads to a rate of $O\left(\|\mathbf{w}_*\| \sqrt{\frac{\ln(1/\delta)}{n}}\right)$.

We now use this theorem to establish an upper bound on the excess risk between $\hat{u}$ (which parameterizes $\hat{\phi}$) and $u^\star$ (which parameterizes $\phi^\star$).

Lemma 3.6. Let $\hat{u}$ and $u^\star$ be defined in (3.3) and (3.4), respectively. Let $S = \{(x_j, z_j)\}_{j=n+1}^{n+m}$ and $\delta \in (0, 1)$. Let $\mu = \frac{1}{\|u^\star\|} \sqrt{\frac{\ln(1/\delta)}{m}}$. Then, with probability at least $1 - \delta$, we have

$$L(\hat{u}) - L(u^\star) \leq \epsilon_u, \quad (3.17)$$

where $\epsilon_u = O\left(\|u^\star\| \sqrt{\frac{\ln(1/\delta)}{m}}\right)$.

Proof. Apply Theorem 1 in [50] with $G = 1$, $\mathbf{w}_* = u^\star$, $\widehat{\mathbf{w}}_\lambda = \hat{u}$, $n = m$, $\mathcal{H} = \mathbb{R}^{d+1}$, $\mathcal{B} = \mathcal{Z}$, and $\lambda = \mu$. $\square$

We next restate Lemma 2 from Appendix B of [50].

Lemma 3.7 (Lemma 2, [50]). Suppose the assumptions (AI), (AII), and (AIII) hold. Then for sample size n and every $\delta \in (0, 1)$, with probability at least $1 - \delta$, the following inequality holds simultaneously for all $\mathbf{w} \in \mathcal{H}$.

$$\mathfrak{L}(\mathbf{w}) - \hat{\mathfrak{L}}(\mathbf{w}) \leq \frac{4G(1 + \alpha\|\mathbf{w}\|)}{\sqrt{n}} \left(1 + \sqrt{\ln(2 + \log_2(1 + \alpha\|\mathbf{w}\|)) + \ln(1/\delta)}\right).$$

We use this lemma to quantify the gap between the empirical and population risk for any hypothesis w when side information is imputed by a classifier $\phi : \mathcal{X} \rightarrow \mathcal{Z}$.

Lemma 3.8. Let $\hat{\phi}$ be defined in (3.5) and let κ be as given in (C1) in (3.1). Let $D = \{(x_i, y_i)\}_{i=1}^n$ and $\delta \in (0, 1)$. Conditional on S , with probability at least $1 - \delta$ over the independent sample D , the following inequality holds

$$R^{\hat{\phi}}(w) - \hat{R}^{\hat{\phi}}(w) \leq \frac{4(1 + \sqrt{\kappa^2 + 2}\|w\|)}{\sqrt{n}} \left(1 + \sqrt{\ln(2 + \log_2(1 + \sqrt{\kappa^2 + 2}\|w\|)) + \ln(1/\delta)}\right).$$

Proof. Conditional on S , the map $\hat{\phi}$ is fixed and the observations in D remain i.i.d. Apply Lemma 2 in Appendix B of [50] with $\mathcal{H} = \mathbb{R}^{d+2}$, $A = (1, X, \hat{\phi}(X))$, $B = Y$, $G = 1$, and $\alpha = \sqrt{\kappa^2 + 2}$. $\square$

Lemma 3.9. Let $A = (1, X) \in \mathbb{R}^{k+1}$ be bounded, let $Y \in \{-1, 1\}$, and assume $\mathbb{E}[AA^\top] \succ 0$ and $0 < \mathbb{P}(Y = 1 \mid A) < 1$ almost surely. For

$$L(a) = \mathbb{E}[\ln(1 + \exp(-Ya^\top A))],$$

there exists $\chi > 0$ such that $L(a) \geq \chi\|a\|$ for every a . Consequently L attains its minimum at a finite weight $a^\star$, with $\|a^\star\| \leq \ln(2)/\chi < \infty$.

Proof. Write $[t]_+ = \max\{t, 0\}$. For a unit vector u , let $h(u) = \mathbb{E}[(-Yu^\top A)_+]$. The positive definiteness of $\mathbb{E}[AA^\top]$ implies that $\mathbb{P}(u^\top A \neq 0) > 0$. Furthermore, conditioned on this event, $-Yu^\top A > 0$ holds if and only if $Y = -\text{sign}(u^\top A)$. From the non-separability condition (C4) in (3.1), we have $\mathbb{P}(Y = -\text{sign}(u^\top A) \mid A) > 0$; thus, $h(u) > 0$. Applying the Lipschitz and Cauchy–Schwarz inequalities, we obtain

$$|h(u) - h(v)| \leq \mathbb{E}\|A\| \|u - v\|.$$

Thus, h is continuous on the compact unit sphere, and we define $\chi := \min_{\|u\|=1} h(u) > 0$. Since $\ln(1 + e^{-t}) \geq [-t]_+$, for any vector a we have $L(a) \geq \mathbb{E}[(-Ya^\top A)_+] = \|a\| h\left(\frac{a}{\|a\|}\right) \geq \chi\|a\|$. Hence, L is continuous and coercive, and it attains its minimum value at a point denoted by $a^\star$. Since $L(0) = \ln 2$, we conclude that $\|a^\star\| \leq \frac{\ln 2}{\chi} < \infty$. $\square$

3.6.2 Bounding Optimal Weights

Algorithm 3.1 trains on the partitioned datasets D and S and returns a classifier with weights $\hat{w}_\lambda \in \mathbb{R}^{d+2}$. In the proof of Theorem 3.1, we will show that the excess risk associated with $\hat{w}_\lambda$ admits an upper bound involving terms of order $\frac{\|\hat{w}_\lambda\|}{\sqrt{n}}$. Since $\hat{w}_\lambda$ is a random variable, we do not

know a priori how terms like $\frac{\|\hat{w}_\lambda\|}{\sqrt{n}}$ scale with the sample size n . We therefore show that, with high probability, $\|\hat{w}_\lambda\|$ lies in a bounded subset of $\mathbb{R}^{d+2}$. A crucial property required for this analysis is the non-separability assumption given in (C4) in (3.1). The lemmas in this section provide such a bound on $\|\hat{w}_\lambda\|$ and establish $W < \infty$, where W is defined in Proposition 3.16.

We now give the definition of strong convexity.

Definition 3.10. A differentiable function $f : \mathfrak{Dom} \subseteq \mathbb{R}^k \rightarrow \mathbb{R}$ on a convex domain is strongly convex if there exists a strong convexity parameter $\alpha > 0$ such that

$$f(w) \geq f(w') + \nabla f(w')^\top (w - w') + \frac{\alpha}{2} \|w - w'\|^2$$

for all $w, w' \in \mathfrak{Dom}$.

An equivalent definition in terms of Hessian is given as follows.

Definition 3.11. A twice continuously differentiable function f on an open convex domain $\mathfrak{Dom} \subseteq \mathbb{R}^k$ is strongly convex with parameter $\alpha > 0$ if and only if

$$z^\top (\nabla^2 f(w)) z \geq \alpha \|z\|^2,$$

for every $w \in \mathfrak{Dom}$ and every direction $z \in \mathbb{R}^k$.

To show that $\|\hat{w}_\lambda\|$ is upper-bounded with high probability, we establish that $\hat{\phi}(x)$ is close to $\phi^*(x)$ on average over the distribution of X . For this, we prove an intermediate result that shows that the logistic population risk $L(u)$ is strongly convex on a bounded deterministic domain. We first use empirical coercivity to define a deterministic envelope U for the imputer weights $\hat{u}$.

Lemma 3.12. Let $\hat{u}$ be defined in (3.4). Under the continuous density assumption (ND) and non-separability condition (C4), there exists a deterministic population radius $U < \infty$ and a sample size m_0 such that for all $m \geq m_0$, with probability at least $1 - \delta/4$, $\|\hat{u}\| \leq U$.

Proof. By the non-separability assumption (C4) on (X, Z) , there is no perfect separating hyperplane. Defining the unit-sphere coercivity constant $\chi_u = \min_{\|v\|=1} \mathbb{E}[(-Zv^\top(1, X))_+] > 0$, the expected logistic risk grows at least linearly: $L(r \cdot v) \geq r \cdot \chi_u$. By uniform convergence on the unit sphere, for sufficiently large $m \geq m_0$, the empirical coercivity constant satisfies $\hat{\chi}_u \geq \chi_u/2 > 0$ with probability at least $1 - \delta/4$. Since $\hat{u}$ minimizes the regularized empirical risk on S , its objective value is bounded by the value at the origin $u = 0$, which is exactly $\ln 2$.

Using the empirical coercivity bound, we obtain:

$$\|\hat{u}\| \cdot (\chi_u/2) \leq \ln 2 \implies \|\hat{u}\| \leq \frac{2 \ln 2}{\chi_u} := U.$$

This establishes that $\hat{u}$ remains inside the deterministic ball of radius U . $\square$

Next, we show that $L(u)$ is strongly convex on the deterministic ball $\mathcal{B}_U := \{u : \|u\| \leq U\}$.

Lemma 3.13. Let U be the deterministic radius from Lemma 3.12. Then the logistic risk $L(u)$ is strongly convex on $\mathcal{B}_U$.

Proof. The Hessian of the logistic risk $L(u)$ is

$$\nabla^2 L(u) = \mathbb{E}_{(X,Z)} [\sigma(Zs)(1 - \sigma(Zs))(1, X)(1, X)^\top],$$

where $s = u_c + u_x^\top X$. Set $K_S = \sqrt{1 + \kappa^2}$. For every $u \in \mathcal{B}_U$, $|Zs| \leq K_S U$. The positive continuous function $\sigma(t)(1 - \sigma(t))$ has a strictly positive minimum q_U on $[-K_S U, K_S U]$. Consequently

$$\nabla^2 L(u) \succeq q_U \mathbb{E}[(1, X)(1, X)^\top] \succeq \alpha I, \quad \alpha = q_U \lambda_{\min}(\mathbb{E}[(1, X)(1, X)^\top]) > 0.$$

The ball $\mathcal{B}_U$ is convex. This uniform bound establishes strong convexity on $\mathcal{B}_U$. $\square$

Using this, we now show in the following lemma that $\hat{\phi}(x)$ is close to $\phi^\star(x)$ on average over the distribution of X . Specifically, we quantify the expected disagreement between the learned classifier $\hat{\phi} : \mathcal{X} \rightarrow \mathcal{Z}$ and the optimal classifier $\phi^\star : \mathcal{X} \rightarrow \mathcal{Z}$ under the joint distribution of (X, Z) .

Lemma 3.14. Let $\hat{\phi}$ and $\phi^\star$ be defined as in equation (3.5) and let S be defined in (3.2). Let $\delta \in (0, 1)$. Then, with probability at least $1 - \delta/4$, we have

$$\mathbb{E}_X \left[|\hat{\phi}(X) - \phi^\star(X)| \right] \leq 2 \mathbb{P}_X \left(|s^\star(X)| \leq K_S \sqrt{2\epsilon_u/\alpha} \right),$$

where $s^\star(x) := u_c^\star + u_x^\star^\top x$, α is the strong convexity parameter of $L(u)$ on $\mathcal{B}_U$, and ϵ_u is the bound from Lemma 3.6.

Proof. From (3.5), $u^\star$ and $\hat{u}$ are the weights corresponding to $\phi^\star(x)$ and $\hat{\phi}(x)$. We first upper bound $\|\hat{u} - u^\star\|$ on the event from Lemma 3.6 (which holds with probability at least $1 - \delta/4$).

From Lemma 3.12, both $\hat{u} \in \mathcal{B}_U$ and $u^\star \in \mathcal{B}_U$ (since $u^\star$ is the population minimizer). From Lemma 3.13, $L(u)$ is strongly convex on $\mathcal{B}_U$ with parameter α . Since $\nabla L(u^\star) = 0$, from Definition 3.10 we have

$$L(\hat{u}) \geq L(u^\star) + \frac{\alpha}{2} \|\hat{u} - u^\star\|^2. \quad (3.18)$$

From Lemma 3.6, we have $L(\hat{u}) - L(u^\star) \leq \epsilon_u$. Substituting this gives $\|\hat{u} - u^\star\| \leq \sqrt{2\epsilon_u/\alpha}$.

Since $\hat{\phi}$ and $\phi^\star$ are step functions (hard classifiers), a disagreement $\hat{\phi}(X) \neq \phi^\star(X)$ only occurs if the difference in weights changes the sign of the score. A sign flip requires that the true margin $|s^\star(X)|$ is smaller than the perturbation caused by the estimated weights:

$$|s^\star(X)| = |u^{\star\top}(1, X)| \leq |(\hat{u} - u^\star)^\top(1, X)| \leq K_S \|\hat{u} - u^\star\| \leq K_S \sqrt{2\epsilon_u/\alpha},$$

where $K_S = \sqrt{1 + \kappa^2}$ is the feature bound.

This means disagreements are confined to the margin slab $\mathcal{E} := \{x : |s^\star(x)| \leq K_S \sqrt{2\epsilon_u/\alpha}\}$. Since the difference between the hard classifiers is exactly 2 when they disagree and 0 otherwise, we conclude:

$$\mathbb{E}_X \left[|\hat{\phi}(X) - \phi^\star(X)| \right] = 2\mathbb{P}_X(\hat{\phi}(X) \neq \phi^\star(X)) \leq 2\mathbb{P}_X(\mathcal{E}).$$

Under the non-degeneracy condition (ND), $\mathbb{P}_X(|s^\star(X)| = 0) = 0$, so this margin probability tends to zero as $m \rightarrow \infty$ (which drives $\epsilon_u \rightarrow 0$). $\square$

Finally, we use the non-separability condition (C4) to establish a uniform, deterministic population envelope W for the learned weights. This replaces any sample-dependent supremum with a unit-sphere coercivity argument.

Lemma 3.15. Let $\hat{w}_\lambda$ be defined in (3.10). Under the non-degeneracy condition (ND) and non-separability condition (C4), there exist deterministic population constants W and sample sizes n_0, m_0 such that for all $n \geq n_0$ and $m \geq m_0$, with probability at least $1 - \delta/4$,

$$\sup_{(\lambda_c, \lambda_x, \lambda_z)} \|\hat{w}_\lambda\| \leq 4 \ln 2 / \chi_* =: W.$$

Proof. By the non-degeneracy condition (ND), for the optimal imputed features $A_* = (1, X, \phi^\star(X))$, there is no separating hyperplane. On the compact unit sphere of directions $\{v : \|v\| = 1\}$, we define the population coercivity constant:

$$\chi_* = \min_{\|v\|=1} \mathbb{E} \left[\left(-Yv^\top A_* \right)_+ \right] > 0.$$

The logistic loss satisfies $\ln(1 + e^{-t}) \geq (-t)_+$. To transfer this to the empirical risk with the estimated imputer $\hat{\phi}$, we condition on S . Let $d_m = \mathbb{P}_X(\hat{\phi}(X) \neq \phi^*(X))$. By Lemma 3.14, on our high-probability event we have $d_m \leq g_m := \mathbb{P}_X(|s^*(X)| \leq K_S \sqrt{2\epsilon_u/\alpha})$. Under the regularity condition (C2) and non-degeneracy condition (ND), X has a continuous distribution with $\mathbb{P}_X(s^*(X) = 0) = 0$. Since $\epsilon_u \rightarrow 0$ as $m \rightarrow \infty$, the shrinking margin region collapses to a set of measure zero, ensuring $g_m \rightarrow 0$. For any $\|v\| \leq 1$, the population negative-margin expectations differ by at most $2d_m \leq 2g_m$.

Independently, over the primary sample D , with conditional probability at least $1 - \delta/4$, standard uniform bounds yield:

$$\sup_{\|v\| \leq 1} \left| (\hat{P}_n - P)(-Yv^\top A_{\hat{\phi}})_+ \right| \leq \zeta_n := \frac{4K}{\sqrt{n}} + K\sqrt{\frac{\ln(8/\delta)}{2n}},$$

where $K = \sqrt{1 + \kappa^2 + 1}$ bounds $\|A_{\hat{\phi}}\|$. Thus, the empirical negative margins are bounded below by $\chi_* - 2g_m - \zeta_n$. Since $g_m \rightarrow 0$ and $\zeta_n \rightarrow 0$, there exist sample sizes n_0, m_0 such that for all $n \geq n_0$ and $m \geq m_0$, we have $2g_m + \zeta_n \leq \chi_*/2$. Hence, the empirical objective satisfies $\hat{R}^{\hat{\phi}}(w) \geq (\chi_*/2)\|w\|$.

Algorithm 3.1 minimizes the regularized empirical risk $\hat{R}^{\hat{\phi}}(w) + \text{Reg}(w)$. At the origin $w = 0$, the empirical logistic loss is exactly $\ln 2$, and the penalty is zero. Since $\hat{w}_\lambda$ is the minimizer,

$$(\chi_*/2)\|\hat{w}_\lambda\| \leq \hat{R}^{\hat{\phi}}(\hat{w}_\lambda) \leq \hat{R}^{\hat{\phi}}(\hat{w}_\lambda) + \text{Reg}(\hat{w}_\lambda) \leq \ln 2 \implies \|\hat{w}_\lambda\| \leq \frac{2 \ln 2}{\chi_*} < \frac{4 \ln 2}{\chi_*} = W.$$

This radius W uniformly bounds the learned weights $\hat{w}_\lambda$ for all candidate values of λ . $\square$

To fully bound the objective terms in Theorem 3.1, we also require deterministic bounds on the true population paths w_λ^* (using features and true side information) and v^* (using features only).

Proposition 3.16. Let $\chi_R > 0$ and $\chi_Y > 0$ be the population coercivity constants for $(1, X, Z)$ and $(1, X)$ respectively, which are strictly positive under (C4). We define the aggregate deterministic bound:

$$W := \max \left\{ \frac{4 \ln 2}{\chi_*}, \frac{\ln 2}{\chi_R}, \frac{\ln 2}{\chi_Y} \right\}.$$

Then on the joint success event, the learned weights satisfy $\|\hat{w}_\lambda\| \leq W$, and the population comparators satisfy $\sup_\lambda \|w_\lambda^*\| \leq W$ and $\|v^*\| \leq W$. $W < \infty$ is a purely deterministic population constant.

3.6.3 Other Useful Lemmas

This section contains the remaining lemmas required for proving the main theorem (Theorem 3.1). The following lemma calculates the gap in risk when Z is substituted with $\phi : \mathcal{X} \rightarrow \mathcal{Z}$.

Lemma 3.17. Let $w := (w_c, w_x, w_z) \in \mathbb{R}^{d+2}$ and let $\hat{\phi}$ be defined in (3.5). Then

$$|R^{\hat{\phi}}(w) - R(w)| \leq 2|w_z|L_{(0,1)}(\hat{u}).$$

Proof. For fixed (x, y) and w , set $\ell(x, t, y; w) = \ln(1 + \exp(-y(w_c + w_x^\top x + w_z t)))$. Its derivative in t satisfies

$$\left| \frac{\partial}{\partial t} \ell(x, t, y; w) \right| = |yw_z| [1 - \sigma(y(w_c + w_x^\top x + w_z t))] \leq |w_z|.$$

We further have $\mathbb{E}_{X,Z}|Z - \hat{\phi}(X)| = 2\mathbb{P}_{X,Z}(Z \neq \hat{\phi}(X)) \leq 2L_{(0,1)}(\hat{u})$. Therefore

$$|R^{\hat{\phi}}(w) - R(w)| \leq \mathbb{E}_{X,Y,Z} |\ell(X, \hat{\phi}(X), Y; w) - \ell(X, Z, Y; w)| \leq 2|w_z|L_{(0,1)}(\hat{u}).$$

□

The next lemma establishes the effect of the regularizer on w_z .

Lemma 3.18. Let $\lambda_c, \lambda_x, \lambda_z > 0$ and let $\hat{w}_\lambda$ and w_λ^* be defined in (3.10) and (3.8). For either $q = \hat{w}_\lambda$ or $q = w_\lambda^*$,

$$|q_z| \leq \min \left\{ \sqrt{\frac{\ln 2}{\lambda_z}}, \frac{1}{2\lambda_z} \right\} \leq \frac{\ln 2}{\lambda_z}.$$

Proof. Suppose $q = \hat{w}_\lambda$. Using the property that $\hat{w}_\lambda$ is an optimal regularized empirical risk minimizing weights, we have

$$\lambda_z |\hat{w}_{\lambda,z}|^2 \leq \hat{R}^{\hat{\phi}}(\hat{w}_\lambda) + \lambda_c |\hat{w}_{\lambda,c}|^2 + \lambda_x \|\hat{w}_{\lambda,x}\|^2 + \lambda_z |\hat{w}_{\lambda,z}|^2 \leq \hat{R}^{\hat{\phi}}(0) + 0 = \ln 2.$$

Thus,

$$|\hat{w}_{\lambda,z}| \leq \sqrt{\frac{\ln 2}{\lambda_z}}.$$

Next, we apply the first-order optimality condition to get

$$\frac{\partial}{\partial w_z} \left(\hat{R}^{\hat{\phi}}(\hat{w}_\lambda) + \lambda_c |\hat{w}_{\lambda,c}|^2 + \lambda_x \|\hat{w}_{\lambda,x}\|^2 + \lambda_z |\hat{w}_{\lambda,z}|^2 \right) = 0 \implies 2\lambda_z |\hat{w}_{\lambda,z}| = \left| \frac{\partial \hat{R}^{\hat{\phi}}(\hat{w}_\lambda)}{\partial \hat{w}_{\lambda,z}} \right|.$$

Simplifying it and using triangle inequality we get

$$2\lambda_z |\hat{w}_{\lambda,z}| = \left| \frac{\partial \hat{R}^{\hat{\phi}}(\hat{w}_\lambda)}{\partial \hat{w}_{\lambda,z}} \right| \leq \frac{1}{n} \sum_{i=1}^n |y_i \hat{\phi}(x_i) (1 - \sigma(y_i (\hat{w}_{\lambda,c} + \hat{w}_{\lambda,x}^\top x_i + \hat{w}_{\lambda,z} \hat{\phi}(x_i))))|.$$

Since $y_i \in \{-1, 1\}$ and $\hat{\phi}(x_i) \in \{-1, 1\}$, we have $|y_i \hat{\phi}(x_i)| = 1$. Moreover, because the sigmoid function satisfies $0 < \sigma(\cdot) < 1$, we have $0 < 1 - \sigma(\cdot) < 1$. Therefore,

$$2\lambda_z |\hat{w}_{\lambda,z}| \leq \frac{1}{n} \sum_{i=1}^n 1 = 1 \implies |\hat{w}_{\lambda,z}| \leq \frac{1}{2\lambda_z}.$$

An identical argument works for $q = w_\lambda^\star$. Combining both bounds gives

$$|q_z| \leq \min \left\{ \sqrt{\frac{\ln 2}{\lambda_z}}, \frac{1}{2\lambda_z} \right\} \leq \frac{\ln(2)}{\lambda_z}.$$

□

The following lemma is an application of McDiarmid's inequality to $\hat{R}^{\hat{\phi}}(w)$ and $R^{\hat{\phi}}(w)$.

Lemma 3.19. Let $\hat{\phi}$ be defined in (3.5). Conditional on S , fix $w = (w_c, w_x, w_z) \in \mathbb{R}^{d+2}$. For $0 < \delta < 1$, with conditional probability at least $1 - \delta$ over D ,

$$|\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)| \leq (\ln 2 + \sqrt{\kappa^2 + 2}\|w\|) \sqrt{\frac{2 \ln(2/\delta)}{n}}. \quad (3.19)$$

Proof. Conditional on S , the imputer $\hat{\phi}$ is fixed and independent of D . For any observation $(x_i, y_i) \in D$, let $A_i = (1, x_i, \hat{\phi}(x_i)) \in \mathbb{R}^{d+2}$. By Assumption (C1) in (3.1), $\|x_i\| \leq \kappa$ almost surely, and since $\hat{\phi}(x_i) \in \{-1, 1\}$, we have

$$\|A_i\| = \sqrt{1 + \|x_i\|^2 + |\hat{\phi}(x_i)|^2} \leq \sqrt{\kappa^2 + 2}.$$

By the Cauchy–Schwarz inequality,

$$|w_c + w_x^\top x_i + w_z \hat{\phi}(x_i)| = |w^\top A_i| \leq \|w\| \|A_i\| \leq \sqrt{\kappa^2 + 2} \|w\|.$$

Since the logistic loss $\ell(x_i, \hat{\phi}(x_i), y_i; w) = \ln(1 + \exp(-y_i w^\top A_i))$ is 1-Lipschitz with $\ell(x_i, \hat{\phi}(x_i), y_i; 0) = \ln 2$, and $y_i \in \{-1, 1\}$, each loss is non-negative and bounded almost surely by

$$0 \leq \ell(x_i, \hat{\phi}(x_i), y_i; w) \leq \ln 2 + |w^\top A_i| \leq \ln 2 + \sqrt{\kappa^2 + 2} \|w\| =: M_w.$$

Replacing any single observation in D changes the empirical risk $\hat{R}^{\hat{\phi}}(w) = \frac{1}{n} \sum_{i=1}^n \ell(x_i, \hat{\phi}(x_i), y_i; w)$ by at most M_w/n . Since $\mathbb{E}_D[\hat{R}^{\hat{\phi}}(w) \mid S] = R^{\hat{\phi}}(w)$, applying McDiarmid's bounded differences inequality yields, for any $\epsilon > 0$,

$$\mathbb{P}(|\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)| \geq \epsilon \mid S) \leq 2 \exp\left(-\frac{2\epsilon^2}{n(M_w/n)^2}\right) = 2 \exp\left(-\frac{2n\epsilon^2}{M_w^2}\right).$$

Setting the right-hand side to δ and solving for ϵ gives

$$\epsilon = M_w \sqrt{\frac{\ln(2/\delta)}{2n}}.$$

Thus, with conditional probability at least $1 - \delta$ over D ,

$$|\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)| \leq M_w \sqrt{\frac{\ln(2/\delta)}{2n}} \leq M_w \sqrt{\frac{2 \ln(2/\delta)}{n}} = (\ln 2 + \sqrt{\kappa^2 + 2} \|w\|) \sqrt{\frac{2 \ln(2/\delta)}{n}},$$

which proves the claimed inequality. $\square$

The previous lemma gave a pointwise bound, we now specify a uniform bound.

Lemma 3.20. Let $B > 0$, $K = \sqrt{\kappa^2 + 2}$, and define for any $\eta \in (0, 1)$:

$$\Gamma(B, K, \eta) := 4KB + (\ln 2 + KB) \sqrt{\frac{\ln(2/\eta)}{2}}.$$

Conditional on S , with probability at least $1 - \delta$ over D ,

$$\sup_{\|w\| \leq B} |\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)| \leq \frac{\Gamma(B, K, \delta)}{\sqrt{n}}.$$

Remark. In our application, the radius B will be set to the deterministic population envelope W (defined in Proposition 3.16), ensuring that this uniform concentration bound is applied to a fixed, non-random hypothesis class.

Proof. Conditional on S , $\hat{\phi}$ is fixed. Without loss of generality, we can center the loss by subtracting the constant $\ell(x, \hat{\phi}(x), y; 0) = \ln 2$ without affecting the difference $\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)$. Using standard symmetrization with independent Rademacher variables $\sigma_1, \dots, \sigma_n \in \{-1, 1\}$ as shown in [2, Thm 26.2], we get

$$\mathbb{E}_D \left[\sup_{\|w\| \leq B} |\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)| \middle| S \right] \leq 2 \mathbb{E}_{D, \sigma} \left[\sup_{\|w\| \leq B} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i (\ell(x_i, \hat{\phi}(x_i), y_i; w) - \ln 2) \right| \middle| S \right].$$

Since the scalar loss $t \mapsto \ln(1 + e^{-t}) - \ln 2$ is 1-Lipschitz and vanishes at zero, the contraction principle combined with the Cauchy–Schwarz inequality yields

$$\mathbb{E}_{\sigma} \left[\sup_{\|w\| \leq B} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i (\ell(x_i, \hat{\phi}(x_i), y_i; w) - \ln 2) \right| \right] \leq \frac{2B}{n} \mathbb{E}_{\sigma} \left\| \sum_{i=1}^n \sigma_i A_i \right\| \leq \frac{2KB}{\sqrt{n}},$$

where $A_i = (1, x_i, \hat{\phi}(x_i))$ satisfies $\|A_i\| \leq \sqrt{\kappa^2 + 2} = K$ under Assumption (C1), and the final inequality follows from Jensen's inequality with $\mathbb{E}_{\sigma} \left\| \sum_{i=1}^n \sigma_i A_i \right\|^2 = \sum_{i=1}^n \|A_i\|^2 \leq nK^2$. Hence,

the expected uniform deviation is bounded by $\frac{4KB}{\sqrt{n}}$. To establish concentration around the expectation, define $\Phi(D) := \sup_{\|w\| \leq B} |\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)|$. From Lemma 3.19, each loss on the ball $\|w\| \leq B$ lies in $[0, \ln 2 + KB]$. Thus, replacing any single observation in D changes $\Phi(D)$ by at most $c_k = \frac{\ln 2 + KB}{n}$. Applying McDiarmid's bounded differences inequality, the deviation from its expectation satisfies, with conditional probability at least $1 - \delta$,

$$\Phi(D) - \mathbb{E}_D[\Phi(D) \mid S] \leq \frac{\ln 2 + KB}{\sqrt{n}} \sqrt{\frac{\ln(2/\delta)}{2}}.$$

Combining the expectation bound with this concentration bound yields

$$\sup_{\|w\| \leq B} |\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)| \leq \frac{4KB}{\sqrt{n}} + \frac{\ln 2 + KB}{\sqrt{n}} \sqrt{\frac{\ln(2/\delta)}{2}} = \frac{\Gamma(B, K, \delta)}{\sqrt{n}},$$

which completes the proof. $\square$

The next lemma examines how the optimal weights and risk behave as the regularization parameter varies.

Lemma 3.21. Let $\lambda = (\lambda_c, \lambda_x, \lambda_z)$ and $\theta = (\lambda_c, \lambda_x, \theta_z)$ with $\lambda_z \leq \theta_z$. Then

$$|w_{\theta,z}^*| \leq |w_{\lambda,z}^*|. \quad (3.20)$$

If $\lambda_c = \lambda_x = \lambda_0 = n^{-1/2}$, then

$$R(w_\lambda^*) \leq R(w_\theta^*) + \lambda_0(|w_{\theta,c}^*|^2 + \|w_{\theta,x}^*\|^2). \quad (3.21)$$

The same statements hold for $R^{\phi^*}(\cdot)$ and its corresponding optimal weights.

Proof. Write $P(q) = \lambda_c |q_c|^2 + \lambda_x \|q_x\|^2$. By the optimality of w_λ^* for the regularized objective with parameter λ ,

$$R(w_\lambda^*) + P(w_\lambda^*) + \lambda_z |w_{\lambda,z}^*|^2 \leq R(w_\theta^*) + P(w_\theta^*) + \lambda_z |w_{\theta,z}^*|^2.$$

Similarly, by the optimality of w_θ^* for the objective with parameter θ ,

$$R(w_\theta^*) + P(w_\theta^*) + \theta_z |w_{\theta,z}^*|^2 \leq R(w_\lambda^*) + P(w_\lambda^*) + \theta_z |w_{\lambda,z}^*|^2.$$

Adding these two inequalities and canceling common terms yields

$$(\theta_z - \lambda_z)(|w_{\theta,z}^*|^2 - |w_{\lambda,z}^*|^2) \leq 0. \quad (3.22)$$

Since $\theta_z \geq \lambda_z > 0$, this implies $|w_{\theta,z}^*| \leq |w_{\lambda,z}^*|$, which proves (3.20).

Using $|w_{\theta,z}^*|^2 \leq |w_{\lambda,z}^*|^2$ in the first optimality inequality gives

$$R(w_\lambda^*) + P(w_\lambda^*) \leq R(w_\theta^*) + P(w_\theta^*) + \lambda_z(|w_{\theta,z}^*|^2 - |w_{\lambda,z}^*|^2) \leq R(w_\theta^*) + P(w_\theta^*).$$

Since $P(w_\lambda^*) \geq 0$, we have $R(w_\lambda^*) \leq R(w_\theta^*) + P(w_\theta^*)$. Setting $\lambda_c = \lambda_x = \lambda_0 = n^{-1/2}$, we obtain $P(w_\theta^*) = \lambda_0(|w_{\theta,c}^*|^2 + \|w_{\theta,x}^*\|^2)$, which proves (3.21). The same argument holds verbatim for R^{ϕ^*} by replacing R with R^{ϕ^*} and using the corresponding regularized minimizers. $\square$

We next present a corollary which relates the risk of various optimal weights.

Corollary 3.22. Let $\lambda = (n^{-1/2}, n^{-1/2}, \lambda_z)$ with $\lambda_z > 0$ and set $J_\lambda(w) = n^{-1/2}(|w_c|^2 + \|w_x\|^2) + \lambda_z|w_z|^2$. Then

$$R(w_\lambda^*) + J_\lambda(w_\lambda^*) \leq \min \left\{ R(w^*) + J_\lambda(w^*), T(v^*) + \frac{\|v^*\|^2}{\sqrt{n}} \right\}. \quad (3.23)$$

In particular, $R(w_\lambda^*) \leq T(v^*) + \|v^*\|^2/\sqrt{n}$.

Proof. By the optimality of w_λ^* , we have

$$R(w_\lambda^*) + J_\lambda(w_\lambda^*) \leq R(w^*) + J_\lambda(w^*).$$

Similarly, evaluating at the feature-only candidate $w = (v_c^*, v_x^*, 0) \in \mathbb{R}^{d+2}$ yields $R(v_c^*, v_x^*, 0) = T(v^*)$ and $J_\lambda(v_c^*, v_x^*, 0) = \frac{\|v^*\|^2}{\sqrt{n}}$, which gives

$$R(w_\lambda^*) + J_\lambda(w_\lambda^*) \leq T(v^*) + \frac{\|v^*\|^2}{\sqrt{n}}.$$

Taking the minimum of these two upper bounds establishes (3.23). Since $J_\lambda(w_\lambda^*) \geq 0$, we further get $R(w_\lambda^*) \leq T(v^*) + \|v^*\|^2/\sqrt{n}$. $\square$

3.6.4 Proof of the Main Theorem

We now present the proof of Theorem 3.1.

Proof of Theorem 3.1. Let $\delta \in (0, 1)$. We apply a four-event probability union bound, allocating a failure budget of $\delta/4$ to each of the following events: 1. Imputer empirical coercivity holds on S (Lemma 3.12), ensuring $\|\hat{u}\| \leq U$. 2. Imputer uniform concentration holds on S (Lemma

3.6), ensuring $L(\hat{u}) - L(u^\star) \leq \epsilon_u$. We denote $Q_m := L(u^\star) + \epsilon_u$. 3. Primary classifier empirical coercivity holds on D (Lemma 3.15), ensuring $\|\hat{w}_\lambda\| \leq W \leq \mathbf{W}$ for all candidate λ . 4. Primary uniform concentration holds on D over the deterministic ball of radius $\mathbf{W}$ (Lemma 3.20), bounding empirical deviations.

By the union bound, all four events hold simultaneously with probability at least $1 - \delta$. Conditioned on this joint success event, we decompose the LHS of (3.12) as follows: The LHS of (3.12) can be decomposed as:

$$R(\hat{w}_\lambda) - R(w^\star) = \underbrace{R(\hat{w}_\lambda) - R^\hat{\phi}(\hat{w}_\lambda)}_{(A)} + \underbrace{R^\hat{\phi}(\hat{w}_\lambda) - R^\hat{\phi}(w_\lambda^\star)}_{(B)} + \underbrace{R^\hat{\phi}(w_\lambda^\star) - R(w_\lambda^\star)}_{(C)} + \underbrace{R(w_\lambda^\star) - R(w^\star)}_{(D)}.$$

We will first address the sum of terms (A) and (C).

Claim. Conditioned on the joint success event,

$$R(\hat{w}_\lambda) - R^\hat{\phi}(\hat{w}_\lambda) + R^\hat{\phi}(w_\lambda^\star) - R(w_\lambda^\star) \leq \frac{4}{\ln(2)} Q_m \min \left\{ \frac{1}{2\lambda_z}, \mathbf{W} \right\}.$$

Proof of Claim. From Lemma 3.17, we have,

$$|R(\hat{w}_\lambda) - R^\hat{\phi}(\hat{w}_\lambda)| \leq 2|\hat{w}_{\lambda,z}|L_{(0,1)}(\hat{u}), \text{ and } |R(w_\lambda^\star) - R^\hat{\phi}(w_\lambda^\star)| \leq 2|w_{\lambda,z}^\star|L_{(0,1)}(\hat{u}).$$

The pointwise inequality $\ln(1 + e^{-t}) \geq \ln(2)\mathbb{1}_{\{t \leq 0\}}$ gives $L_{(0,1)}(u) \leq L(u)/\ln(2)$. Thus, we get

$$\begin{aligned} R(\hat{w}_\lambda) - R^\hat{\phi}(\hat{w}_\lambda) + R^\hat{\phi}(w_\lambda^\star) - R(w_\lambda^\star) &\leq |R(\hat{w}_\lambda) - R^\hat{\phi}(\hat{w}_\lambda)| + |R(w_\lambda^\star) - R^\hat{\phi}(w_\lambda^\star)| \\ &\leq 2L_{(0,1)}(\hat{u})(|\hat{w}_{\lambda,z}| + |w_{\lambda,z}^\star|) \leq \frac{2}{\ln(2)} L(\hat{u})(|\hat{w}_{\lambda,z}| + |w_{\lambda,z}^\star|) \end{aligned}$$

From the joint success event, we have $L(\hat{u}) \leq Q_m$, yielding

$$R(\hat{w}_\lambda) - R^\hat{\phi}(\hat{w}_\lambda) + R^\hat{\phi}(w_\lambda^\star) - R(w_\lambda^\star) \leq \frac{2}{\ln(2)} Q_m (|\hat{w}_{\lambda,z}| + |w_{\lambda,z}^\star|).$$

Lemma 3.18 gives $|\hat{w}_{\lambda,z}|, |w_{\lambda,z}^\star| \leq 1/(2\lambda_z)$. From the joint success event, both magnitudes are also bounded by the deterministic population envelope $\mathbf{W}$. Taking the minimum of the two upper bounds, their sum is at most twice the common cap. Hence

$$(A) + (C) \leq \frac{4Q_m}{\ln 2} \min \left\{ \frac{1}{2\lambda_z}, \mathbf{W} \right\}.$$

This proves the claim. $\square$

We next address the sum of terms (B) and (D).

Claim. Conditioned on the joint success event, setting the radius to the deterministic envelope W , we obtain for all $\lambda_z \geq n^{-1/2}$,

$$(B) + (D) \leq \min\{\lambda_z |w_z^*|^2, T(v^*) - R(w^*)\} + \frac{C_1(W, \delta)}{\sqrt{n}},$$

where

$$C_1(W, \delta) = 2\Gamma(W, K_Z, \delta/4) + \max\{|w_c^*|^2 + \|w_x^*\|^2, \|v^*\|^2\},$$

where K_Z is the feature bound for $(1, X, Z)$.

Proof of Claim. Let $\varepsilon_n = \Gamma(W, K_Z, \delta/4)/\sqrt{n}$ and $J_\lambda(w) = n^{-1/2}(|w_c|^2 + \|w_x\|^2) + \lambda_z |w_z|^2$. By the empirical optimality of $\hat{w}_\lambda$, we have $\hat{R}^\phi(\hat{w}_\lambda) + J_\lambda(\hat{w}_\lambda) \leq \hat{R}^\phi(w_\lambda^*) + J_\lambda(w_\lambda^*)$. On our joint success event, both $\|\hat{w}_\lambda\|$ and $\|w_\lambda^*\|$ are bounded by W . Thus, applying the uniform deviation bound from Lemma 3.20 on the ball of radius W , we obtain

$$R^\phi(\hat{w}_\lambda) + J_\lambda(\hat{w}_\lambda) \leq R^\phi(w_\lambda^*) + J_\lambda(w_\lambda^*) + 2\varepsilon_n, \quad (3.24)$$

which rearranges to $R^\phi(\hat{w}_\lambda) - R^\phi(w_\lambda^*) \leq J_\lambda(w_\lambda^*) - J_\lambda(\hat{w}_\lambda) + 2\varepsilon_n$. Recalling that $(B) = R^\phi(\hat{w}_\lambda) - R^\phi(w_\lambda^*)$ and $(D) = R(w_\lambda^*) - R(w^*)$, adding them yields

$$(B) + (D) \leq R(w_\lambda^*) + J_\lambda(w_\lambda^*) - R(w^*) - J_\lambda(\hat{w}_\lambda) + 2\varepsilon_n. \quad (3.25)$$

Next, from Corollary 3.22, subtracting $R(w^*)$ from both sides of (3.23) gives

$$R(w_\lambda^*) + J_\lambda(w_\lambda^*) - R(w^*) \leq \min \left\{ \lambda_z |w_z^*|^2 + \frac{|w_c^*|^2 + \|w_x^*\|^2}{\sqrt{n}}, T(v^*) - R(w^*) + \frac{\|v^*\|^2}{\sqrt{n}} \right\}. \quad (3.26)$$

Substituting (3.26) into (3.25) and noting that $-J_\lambda(\hat{w}_\lambda) \leq 0$, we have

$$\begin{aligned} (B) + (D) &\leq \min \left\{ \lambda_z |w_z^*|^2 + \frac{|w_c^*|^2 + \|w_x^*\|^2}{\sqrt{n}}, T(v^*) - R(w^*) + \frac{\|v^*\|^2}{\sqrt{n}} \right\} + 2\varepsilon_n \\ &\leq \min \{ \lambda_z |w_z^*|^2, T(v^*) - R(w^*) \} + \frac{C_1(W, \delta)}{\sqrt{n}}, \end{aligned}$$

where $C_1(W, \delta) = 2\Gamma(W, K_Z, \delta/4) + \max\{|w_c^*|^2 + \|w_x^*\|^2, \|v^*\|^2\}$. This proves the claim. $\square$

Finally, combining the bound on $(A) + (C)$ from the first claim with the bound on $(B) + (D)$ from the second claim, we obtain the excess-risk inequality:

$$\begin{aligned} R(\hat{w}_\lambda) - R(w^*) &= (A) + (B) + (C) + (D) \\ &\leq \frac{C_1(W, \delta)}{\sqrt{n}} + \min \{ \lambda_z |w_z^*|^2, T(v^*) - R(w^*) \} \\ &\quad + \frac{4Q_m}{\ln(2)} \min \left\{ \frac{1}{2\lambda_z}, W \right\}, \end{aligned}$$

which completes the proof of Theorem 3.1. $\square$

3.6.5 Proofs of Corollaries for the Upper Bound

This subsection contains the proofs of Corollaries 3.2 and 3.3, which characterize the optimal regularization parameter λ_z^* and resulting excess risk bounds.

Proof of Corollary 3.2. Recall the upper bound on the excess risk from Theorem 3.1 (excluding the independent $O(1/\sqrt{n})$ term). Setting $a = |w_z^*|^2$, $\Delta = T(v^*) - R(w^*)$, and $b_m = \frac{4}{\ln 2}(L(u^*) + \frac{C_2}{\sqrt{m}})$, we aim to minimize:

$$F(\lambda_z) = \min\{a\lambda_z, \Delta\} + b_m \min\left\{W, \frac{1}{2\lambda_z}\right\}$$

over the extended domain $[\lambda_0, \infty]$, where $\lambda_0 = 1/\sqrt{n}$, and $F(\infty) = \Delta$. If $a = 0$, the oracle is feature-only and $\Delta = 0$, so $F(\lambda_z) = b_m \min\{W, 1/(2\lambda_z)\}$, which is minimized at λ_0 or ∞ . We thus assume $a > 0$ and $\Delta > 0$.

Let $t = 1/(2W)$ and $s = \Delta/a$. The function $F(\lambda_z)$ has two breakpoints. Below both breakpoints ($\lambda_z \leq \min\{s, t\}$), F increases since $F'(\lambda_z) = a > 0$. Above both breakpoints ($\lambda_z \geq \max\{s, t\}$), F decreases since $F'(\lambda_z) = -b_m/(2\lambda_z^2) < 0$. If $s < t$, the intervening plateau is $\Delta + b_m W > \Delta$. If $t < s$, the intervening expression is $a\lambda_z + b_m/(2\lambda_z)$, which has a stationary point at $\lambda_I = \sqrt{b_m/(2a)}$. The derivative jumps downward at both kinks, so neither kink introduces an additional strict local minimum. Therefore, a minimizer can be selected from at most three extended candidate points: the lower endpoint λ_0 , the upper endpoint ∞ , or the interior point λ_I . $\square$

Proof of Corollary 3.3. Assume $a > 0$, $\Delta > 0$, and $L(u^*) > 0$. We hold the population constants and confidence fixed, and let $m, n \rightarrow \infty$. Define the constants:

$$d = \Delta/a, \quad c_0 = 4/\ln 2, \quad A = 2c_0/d^2, \quad B = c_0 W^2/2, \quad T = c_0 W/d.$$

As $m, n \rightarrow \infty$, we evaluate F at the three candidates. The limiting values are $V_\infty = \Delta$, $V_0 = c_0 L(u^*) W$, and $V_I = L(u^*) \sqrt{2c_0 \rho}$, where $\rho = a/L(u^*)$ is the signal-imputation ratio. The interior candidate beats ∞ if and only if $\rho > A$, and it beats the lower endpoint if and only if $\rho < B$. The lower endpoint beats ∞ if and only if $\rho > T$. Therefore, we set the critical thresholds as:

$$\rho_{\text{low}} = \min\{A, T\}, \quad \rho_{\text{high}} = \max\{B, T\}.$$

- If $\rho < \rho_{\text{low}}$, then $\rho < A$ (interior loses to ∞) and $\rho < T$ (λ_0 loses to ∞), so the upper endpoint strictly dominates: $\lambda_z^* = \infty$.
- If $\rho > \rho_{\text{high}}$, then $\rho > B$ (interior loses to λ_0) and $\rho > T$ (∞ loses to λ_0), so the lower endpoint strictly dominates: $\lambda_z^* = \lambda_0$.
- If $Wd > 2$, then $A < T < B$. For $\rho \in (A, B) = (\rho_{\text{low}}, \rho_{\text{high}})$, the interior strictly beats both endpoints, giving $\lambda_z^* = \lambda_I$. If $Wd \leq 2$, then $B \leq T \leq A$, so $\rho_{\text{low}} = \rho_{\text{high}} = T$, and the interior is never strictly preferred.

For strict inequalities away from the thresholds, these comparisons persist for sufficiently large finite m, n . $\square$

3.6.6 Proof of the Lower Bound

We now prove Theorem 3.4.

Proof. We prove the lower bound by exhibiting two joint distributions that are indistinguishable from the partitioned training data, yet have distinct optimal predictors.

Let $X \sim \text{Unif}[0, 1]$ and fix $p \in (0, \frac{1}{2})$; we set $p = 0.2$ for a concrete numerical bound below. Define the two conditional joint laws of (Y, Z) given $X = x$ as follows.

- (I) Distribution P_1 : Conditional on $X = x$, (Y, Z) satisfy $\mathbb{P}(Y = Z) = 1 - p$. The joint distribution is:

	$Z = -1$	$Z = +1$
$Y = -1$	$(1 - p)/2$	$p/2$
$Y = +1$	$p/2$	$(1 - p)/2$

Consider score functions of the form $f(z) = \alpha z$. The population logistic risk is

$$R_1(\alpha) = \mathbb{E}[\ln(1 + \exp^{-Y\alpha Z})] = (1 - p) \ln(1 + \exp^{-\alpha}) + p \ln(1 + \exp^{\alpha}).$$

Setting $\frac{d}{d\alpha} R_1(\alpha) = 0$,

$$-\frac{1 - p}{1 + \exp^{\alpha}} + \frac{p \exp^{\alpha}}{1 + \exp^{\alpha}} = 0 \implies \exp^{\alpha^*} = \frac{1 - p}{p}.$$

Substituting back (using $1 + \exp^{\alpha^*} = 1/p$ and $1 + \exp^{-\alpha^*} = 1/(1 - p)$) gives

$$R_1^* = -(1 - p) \ln(1 - p) - p \ln p =: H(p).$$

(II) Distribution P_2 : Conditional on $X = x$, (Y, Z) satisfy $\mathbb{P}(Y = -Z) = 1 - p$. The joint distribution is:

	$Z = -1$	$Z = +1$
$Y = -1$	$p/2$	$(1 - p)/2$
$Y = +1$	$(1 - p)/2$	$p/2$

Consider the score function of the form $f(z) = \alpha z$ again. The population logistic risk is

$$R_2(\alpha) = p \ln(1 + \exp^{-\alpha}) + (1 - p) \ln(1 + \exp^{\alpha}).$$

This is minimized at $\exp^{\alpha^*} = \frac{p}{1-p}$. Substituting back gives

$$R_2^* = -p \ln p - (1 - p) \ln(1 - p) = H(p).$$

Indistinguishability: Under both P_1 and P_2 the marginal laws of Y and Z are each uniform on $\{-1, +1\}$:

$$\mathbb{P}(Y = y | X) = \frac{1}{2}, \quad \mathbb{P}(Z = z | X) = \frac{1}{2}.$$

Since the training data is partitioned into $D = \{(x_i, y_i)\}$ and $S = \{(x_j, z_j)\}$, let Q denote the distribution of this partitioned sample. The distribution Q is identically the product of these uniform marginals under both P_1 and P_2 . Hence no algorithm can reliably determine which distribution generated the data.

Lower bound: Let (D, S) be drawn from the common training law Q , together with any independent randomization seed Ξ of the algorithm. For any realization of the output $\hat{w} = A(D, S) \in \mathbb{R}^{d+2}$, consider the sum of excess risks under P_1 and P_2 :

$$(R_1(\hat{w}) - R_1^*) + (R_2(\hat{w}) - R_2^*) = R_1(\hat{w}) + R_2(\hat{w}) - 2H(p).$$

For the equal mixture of P_1 and P_2 , we have $\mathbb{P}(Y = 1 | X, Z) = \frac{1}{2}$ pointwise for every (X, Z) . Consequently, the mixture risk $\frac{1}{2}(R_1(\hat{w}) + R_2(\hat{w}))$ is minimized by the trivial score $\hat{w} = 0$, whose logistic loss is identically $-\ln(1/2) = \ln 2$. Therefore,

$$(R_1(\hat{w}) - R_1^*) + (R_2(\hat{w}) - R_2^*) \geq 2(\ln 2 - H(p)).$$

For $p = 0.2$, we have $\ln 2 - H(0.2) \approx 0.69315 - 0.50040 = 0.19275 > 0.19$. Hence, for every output $\hat{w}$,

$$(R_1(\hat{w}) - R_1^*) + (R_2(\hat{w}) - R_2^*) > 2 \times 0.19.$$

Now define the bad events $E_k := \{(D, S, \Xi) : R_k(A(D, S)) - R_k^* > 0.19\}$ for $k \in \{1, 2\}$. Since the sum of excess risks strictly exceeds 2×0.19 pointwise for every sample realization, every realization (D, S, Ξ) must belong to at least one of E_1 or E_2 , which implies $E_1 \cup E_2 = \Omega$. Because the distribution of (D, S) under both P_1 and P_2 is identically the common law Q , we have

$$Q(E_1) + Q(E_2) \geq Q(E_1 \cup E_2) = 1 \implies \max\{Q(E_1), Q(E_2)\} \geq \frac{1}{2}.$$

Therefore, there exists a fixed distribution $P_{k^*} \in \{P_1, P_2\}$ in the class such that under P_{k^*} , the algorithm incurs excess risk exceeding 0.19 with probability at least 1/2 over the draw of (D, S) and algorithm seed Ξ .

Finally, we note that under both P_1 and P_2 , the feature X is independent of Z and Y , so $u^* = 0$, which places this construction outside the Non-Degeneracy condition (ND) assumed in Theorem 3.1. This result establishes that across the broader class satisfying (C1)–(C4), non-vanishing excess risk is an intrinsic property of learning from partitioned observations. $\square$

Chapter 4

Learning under Strategic Feature Revelation

In this chapter, we study regression under strategic feature disclosure, where a sender strategically decides whether to voluntarily reveal or withhold features from a receiver. We model this interaction as a screening game and examine three problems that arise under different information structures available to the receiver. First, we study the decision problem, where the environment is known to the receiver. We define the strategic advantage $\Delta(\mathcal{E})$ as the receiver's performance gain from voluntary feature revelation over the mandatory baseline and establish a necessary and a sufficient conditions on the environment for this advantage to be positive. Second, we study the offline learning problem where the receiver has access to finite training data that includes the sender's preferences. We propose an ERM-based algorithm that estimates the strategic advantage and achieves a uniform generalization bound of $\mathcal{O}\left(\frac{d^3\sqrt{\log n}}{\sqrt{n}} + d^2\sqrt{\frac{\log(1/\delta)}{n}}\right)$, where d is the dimension of the feature space and n is the size of the training data. Third, we study the online learning problem under bandit feedback. We show that the expected risk functions arising in our setting lie in a Sobolev space $H^{\nu+d/2}$ with $\nu = 5/2$ whenever the joint density $p_{U,X,Y}$ is smooth and has compact support. Thus, the regularity assumptions underlying the standard Matérn-kernelized Gaussian-process Thompson sampling (GP-TS) algorithm are satisfied in our setting, allowing us to apply GP-TS and obtain its regret guarantee of $\tilde{\mathcal{O}}(T^{\frac{5+3d}{10+2d}})$.

4.1 Introduction

A standard assumption in regression is that feature omission is simply due to noise. However, in many real-world regression tasks, the sender has a preference over the regression outcomes. Because features are the sender's private information that the receiver (regressor) cannot directly observe, this information asymmetry allows the sender to selectively reveal only those features that get them closer to their desired outcome. The receiver must therefore account for this strategic behavior. This task is inherently challenging because the receiver cannot initially determine whether a missing feature is absent due to random channel noise or deliberate omission. Furthermore, even if the receiver can deduce that an omission was intentional, identifying the direction of the underlying value remains difficult if the sender has non-uniform preferences over the outcome.

To illustrate this problem, consider a policy planner collecting public health data. A citizen with poor health indicators might deliberately leave a column blank to present a lower risk profile if they want to purchase health insurance in the future. Conversely, an individual with good health indicators might also withhold their data to present a high-risk profile if they want to maximize a payout claim. This difference in motivations for leaving the column blank represents the non-uniform preferences of the senders. For the policymaker, the regression task is difficult not only due to the ambiguity of the omission's cause but also due to the ambiguous direction of the underlying feature value, as both high-risk and low-risk individuals have distinct incentives to conceal their data.

Even though strategic withholding complicates the regression task, it is not always unfavourable to the receiver's performance. We show that there exist environments where voluntary feature revelation actually improves prediction accuracy over a mandatory feature-revelation baseline. Nonetheless, our strategic-risk upper bound has a larger complexity term than the mandatory feature-revelation baseline.

4.1.1 Contributions

In this chapter, we study the problem of strategic feature disclosure for regression. We work under the structural assumption of an all-or-nothing revelation model, where the sender either reveals the feature vector or withholds it from the receiver. We examine three problems based

on the information structure available to the receiver.

First in the *Decision Problem*, where the environment is known to the receiver, we define the strategic advantage $\Delta(\mathcal{E})$ as the receiver's performance gain from voluntary feature revelation over the mandatory baseline, and derive a necessary and a sufficient conditions on the environment for this advantage to be positive.

Second in the *Offline Learning Problem*, where the receiver has access to a finite, fully revealed training dataset alongside the sender's preferences, we propose an ERM-based algorithm that estimates the strategic advantage and achieves a uniform generalization bound of $O\left(\frac{d^3\sqrt{\log n}}{\sqrt{n}} + d^2\sqrt{\frac{\log(1/\delta)}{n}}\right)$, where d is the dimension of the feature space.

Third in the *Online Learning Problem*, where the receiver learns sequentially under bandit feedback, we show that the expected risk functions arising in our setting lie in a Sobolev space $H^{\nu+d/2}$ with $\nu = 5/2$ whenever the joint density $p_{U,X,Y}$ is smooth and has compact support. Thus, the regularity assumptions underlying the standard Matérn-kernelized Gaussian-process Thompson sampling (GP-TS) algorithm are satisfied in our setting, allowing us to apply GP-TS over the joint action space of regimes and regression parameters and obtain its regret guarantee of $\tilde{O}(T^{\frac{5+3d}{10+2d}})$.

4.1.2 Related Work

This chapter contributes to the literature on learning in the presence of strategic agents [3, 8, 51], where interactions are commonly formulated as sender–receiver games. Existing studies have investigated robustness to strategic manipulation [52, 53], distributional changes induced by causal interventions [54], incentive mechanisms [55], and strategic precision degradation [56]. Most of this literature assumes that the sender can freely manipulate the disclosed information and has uniform preference. In contrast, we study environments in which disclosed features are verifiable, and the sender's strategic choice is limited to withholding information, while also allowing non-uniform preferences. Among the closest related works, [57] considers feature withholding but assumes uniform preferences and restricts attention to classification. The works of [9, 58] allow non-uniform preferences but instead permit strategic modification of features. In the economics literature, voluntary disclosure has been extensively studied [4, 5]; however, these models typically treat the sender as the Stackelberg leader. A preliminary version of the

results in this chapter appeared in the conference paper [12], which developed the decision and offline-learning foundations. This chapter is derived from our manuscript [13], which provides complete proofs and additionally investigates the online continuous-bandit learning setting. The online learning component of this chapter is essentially an application of Gaussian-process Thompson sampling for continuous bandits [59–61]. While the algorithm itself is standard, its regret guarantees require the expected risk functions to lie in an appropriate RKHS. We establish that this premise holds in our setting for environments with smooth joint densities over compact support, thereby justifying the application of the existing theory. To the best of our knowledge, the setting of jointly considering strategic feature withholding, non-uniform preferences, and a receiver-led game has not been previously studied, and was first explored in our conference work [12].

4.2 Problem Formulation

We model regression under strategic feature disclosure as a screening game between a sender (the feature provider) and a receiver (the decision maker). Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the feature space and let the outcome space be $\mathcal{Y} := \mathbb{R}$. Let (U, X, Y) be a tuple of random variables taking values in $\mathcal{Y} \times \mathcal{X} \times \mathcal{Y}$ with joint distribution $P_{(U,X,Y)}$. A realization (u, x, y) represents a sender type with private features x , a preferred outcome u , and a ground-truth outcome y . The receiver predicts the outcome using a regression model $r : \mathcal{X} \cup \{\emptyset\} \rightarrow \mathcal{Y}$, and both players evaluate performance using the squared loss $\mathcal{L}(\alpha, \beta) := (\alpha - \beta)^2$.

Nature draws a sender type $(u, x, y) \sim P_{(U,X,Y)}$. After observing its type, the sender chooses an action $s \in \{\emptyset, x\}$, where $\emptyset$ and x represent withholding or revealing the feature vector, respectively. The chosen action passes through an erasure channel with parameter $q \in [0, 1]$, arriving at the receiver as a random message $M \in \mathcal{X} \cup \{\emptyset\}$ with conditional distribution $P(m \mid s)$ given by

$$P(m \mid s) = \begin{cases} 1 & \text{if } s = \emptyset \text{ and } m = \emptyset, \\ 1 - q & \text{if } s = x \text{ and } m = x, \\ q & \text{if } s = x \text{ and } m = \emptyset. \end{cases}$$

The receiver applies a regression rule r to produce prediction $r(M)$. The sender incurs loss $\mathcal{L}(r(M), u) = (u - r(M))^2$ and the receiver incurs loss $\mathcal{L}(r(M), y) = (y - r(M))^2$.

The sender and receiver both seek to minimize their expected losses. The sender's strategy space is the collection of all functions mapping sender types to feasible actions, given by

$$\mathcal{S} := \{s : \mathcal{Y} \times \mathcal{X} \rightarrow \mathcal{X} \cup \{\emptyset\} \mid s(u, x) \in \{\emptyset, x\} \text{ for all } (u, x) \in \mathcal{Y} \times \mathcal{X}\},$$

and the receiver's strategy space is[†]

$$\mathcal{R} := \{r_b : \mathcal{X} \cup \{\emptyset\} \rightarrow \mathcal{Y} \mid r_b(\emptyset) = 0, r_b(x) = b^\top x \text{ for some } b \in \mathbb{R}^d\}.$$

In the screening game, the receiver first commits to a regression model $r_b \in \mathcal{R}$. The sender, after observing its type (u, x, y) and the receiver strategy r_b , best responds with

$$s_b(u, x) \in \arg \min_{a \in \{\emptyset, x\}} \mathbb{E}_{M \sim P(\cdot|a)} [\mathcal{L}(r_b(M), u)] \quad \text{for all } (u, x) \in \mathcal{Y} \times \mathcal{X}.$$

The receiver anticipates this behavior and optimally chooses

$$b^\star \in \arg \min_{b \in \mathbb{R}^d} \mathbb{E}_{(U, X, Y)} [\mathbb{E}_{M \sim P(\cdot|s_b(U, X))} [\mathcal{L}(r_b(M), Y)]] .$$

The receiver-sender strategy pair $(r_{b^\star}, s_{b^\star})$ constitutes a Stackelberg equilibrium of the screening game.

We denote the environment by $\mathcal{E} := (U, X, Y, q, \mathcal{S}, \mathcal{R})$. For an environment $\mathcal{E}$ and a receiver predictor $r_b \in \mathcal{R}$, the vanilla risk is defined as

$$R_V(b) := q \mathbb{E}_Y [\mathcal{L}(r_b(\emptyset), Y)] + (1 - q) \mathbb{E}_{(X, Y)} [\mathcal{L}(r_b(X), Y)].$$

This is the receiver's risk under the mandatory baseline (where the sender always reveals features, and omissions occur solely via channel erasure). Let $R_V^\star := \min_{b \in \mathbb{R}^d} R_V(b)$. For receiver predictor r_b and sender best response s_b , the strategic risk is

$$R_S(b) := \mathbb{E}_{(U, X, Y)} [\mathbb{E}_{M \sim P(\cdot|s_b(U, X))} [\mathcal{L}(r_b(M), Y)]] ,$$

which is the receiver's risk under voluntary feature disclosure. Let $R_S^\star := \min_{b \in \mathbb{R}^d} R_S(b)$. We define the strategic advantage of environment $\mathcal{E}$ as

$$\Delta(\mathcal{E}) := R_V^\star - R_S^\star.$$

A positive strategic advantage, $\Delta(\mathcal{E}) > 0$, indicates that voluntary feature revelation strictly improves receiver performance relative to the mandatory baseline.

[†]We restrict the receiver's strategy space to models without an intercept and where the default prediction is zero.

Based on the receiver's knowledge of the environment $\mathcal{E}$, we analyze strategic regression under three distinct information structures. The first information structure is when the receiver has complete knowledge of the joint distribution $P_{(U,X,Y)}$, referred to as the *Decision Problem*. The objective is to establish a necessary and a sufficient condition on the environment $\mathcal{E}$ and the receiver strategy space $\mathcal{R}$ under which $\Delta(\mathcal{E}) > 0$.

The second information structure is when the receiver does not know $P_{(U,X,Y)}$, but observes a finite training dataset $D_n = \{(u_k, x_k, y_k)\}_{k=1}^n$ of i.i.d. samples, referred to as the *Offline Learning Problem*. The objective is to learn $\Delta(\mathcal{E})$ from the training data using an algorithm and derive uniform generalization bounds for it.

The third information structure is when the receiver does not know the environment and sequentially interacts with the unknown environment over T rounds, observing only the realized bandit loss feedback after each action, referred to as the *Online Learning Problem*. The objective is to learn the optimal disclosure regime and regression model from the interaction using an algorithm and derive regret bounds for it.

4.3 Decision Problem

In this section, we study the decision problem under the regression framework established in Section 4.2, where the environment $\mathcal{E}$ (and in particular the joint distribution $P_{(U,X,Y)}$) is fully known to the receiver. We establish necessary and sufficient conditions on $\mathcal{E}$ under which voluntary feature disclosure strictly improves receiver performance relative to the mandatory baseline, i.e., $\Delta(\mathcal{E}) > 0$. To simplify calculations, we assume $\mathbb{E}[U] = \mathbb{E}[X] = \mathbb{E}[Y] = 0$ and define

$$F_b(u, x) := (2u - b^\top x)b^\top x \quad \text{and} \quad G_b(x, y) := (2y - b^\top x)b^\top x.$$

We begin by deriving the vanilla risk.

Lemma 4.1: *Let $\mathcal{E}$ be an environment. For any receiver strategy $r_b \in \mathcal{R}$, the vanilla risk is given by:*

$$R_V(b) = \text{Var}(Y) - (1 - q)\text{Var}(b_V^\top X) + (1 - q)\text{Var}((b - b_V)^\top X), \quad (4.1)$$

where $b_V = \text{Var}(X)^{-1}\text{Cov}(X, Y)$ is the ordinary least-squares (OLS) slope vector. The optimal

vanilla risk is given by

$$R_V^*(q) = \text{Var}(Y) - (1 - q)\text{Var}(b_V^\top X).$$

Proof. The proof is given in Section 4.7 (Proof 4.7.2). $\square$

Next, we characterize the sender's strategic disclosure rule.

Lemma 4.2 (Sender Best Response): *For a committed receiver strategy $r_b \in \mathcal{R}$, the sender reveals if and only if*

$$F_b(u, x) = (2u - b^\top x)b^\top x > 0.$$

Equivalently, with ties resolved by withholding,

$$s_b(u, x) = \begin{cases} x & \text{if } F_b(u, x) > 0, \\ \emptyset & \text{if } F_b(u, x) \leq 0. \end{cases}$$

Proof. The proof is given in Section 4.7 (Proof 4.7.2). $\square$

We now evaluate the strategic risk for a given receiver strategy.

Lemma 4.3: *Let $\mathcal{E}$ be an environment. For any receiver predictive rule $r_b \in \mathcal{R}$, the strategic risk simplifies to*

$$R_S(b) = \text{Var}(Y) - (1 - q)\mathbb{E}[G_b(X, Y)\mathbb{I}\{F_b(U, X) > 0\}], \quad (4.2)$$

where $G_b(x, y) = (2y - b^\top x)b^\top x$.

Proof. The proof is given in Section 4.7 (Proof 4.7.2). $\square$

Finally, using these lemmas we establish a necessary and a sufficient conditions on the environment $\mathcal{E}$ for which $\Delta(\mathcal{E}) > 0$. We begin by stating a sufficient condition.

Theorem 4.4 (Sufficient Condition for $\Delta(\mathcal{E}) > 0$): *Let $\mathcal{E}$ be an environment. If there exists a receiver strategy $r_b \in \mathcal{R}$ parameterized by a slope vector $b \in \mathbb{R}^d$ such that*

$$\|U - Y\|_2 < \frac{\mathbb{E}[|G_b(X, Y)|] - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X)}{4\sqrt{\text{Var}(b^\top X)}},$$

then $\Delta(\mathcal{E}) > 0$. Here, $\|U - Y\|_2 = \sqrt{\mathbb{E}[(U - Y)^2]}$ denotes the statistical alignment between the sender's preference value U and the true ground-truth label Y .

Proof. The proof is given in Section 4.7 (Proof 4.7.2). □

We finish by establishing a necessary condition for $\Delta(\mathcal{E}) > 0$.

Theorem 4.5 (Necessary Condition for $\Delta(\mathcal{E}) > 0$): *Let $\mathcal{E}$ be an environment. If for all receiver strategy $r_b \in \mathcal{R}$ whose parameter vector $b \in \mathbb{R}^d$ satisfies*

$$\mathbb{E}[|G_b(X, Y)|] - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) \leq 0,$$

then $\Delta(\mathcal{E}) \leq 0$. Hence, a necessary condition for $\Delta(\mathcal{E}) > 0$ is that there exists a receiver strategy $r_b \in \mathcal{R}$ whose parameter vector $b \in \mathbb{R}^d$ satisfies

$$\mathbb{E}[|G_b(X, Y)|] - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) > 0.$$

Proof. The proof is given in Section 4.7 (Proof 4.7.2). □

Remark 4.6. The channel noise q only affects the magnitude of $\Delta(\mathcal{E})$ and cannot change its sign.

Simulation Results for the Decision Problem. To empirically validate the decision problem and visualize the strategic advantage $\Delta(\mathcal{E})$, we consider a family of zero-mean jointly Gaussian distributions for the continuous feature, outcome, and preference variables $(Y, X, U) \in \mathbb{R}^{d+2}$ (with $d = 1$) specified by covariance matrix $\Sigma \in \mathbb{R}^{3 \times 3}$. The variances are normalized to $\text{Var}(X) = \text{Var}(U) = \text{Var}(Y) = 1.0$, and the feature-preference covariance is fixed at $\text{Cov}(X, U) = 0.1$. To show the effect of channel noise (Remark 4.6), we compare two noise levels: low noise $q = 0.15$ and high noise $q = 0.85$.

We fix $\text{Cov}(X, U) = 0.1$ and sweep the covariance parameter space over a dense 40×40 uniform grid for $\text{Cov}(U, Y) \in [-1, 1]$ and $\text{Cov}(X, Y) \in [-1, 1]$. At each valid grid point, we compute the analytic formula for the vanilla risk R_V^* . The strategic risk R_S^* is evaluated numerically using a Monte Carlo sample size of 100,000 and minimized over the receiver slope parameter $b \in [-5, 5]$. We plot the resulting strategic advantage, where the color indicates the sign and the intensity indicates the magnitude. Furthermore, we mark the regions satisfying the theoretical sufficient and necessary conditions (evaluated over candidate slopes $b \in [-5, 5]$) using dashed and solid boundaries, respectively.

The empirical findings presented in Figure 4.1 corroborate our theoretical analysis. First, the sufficient condition boundary (dashed line) holds near the region where the preference-outcome alignment $\text{Cov}(U, Y)$ is large. This higher alignment lowers $\|U - Y\|_2$ (the LHS of Theorem 4.4), making the inequality easier to satisfy, while the RHS depends only on the feature-outcome statistics (X, Y) and is independent of U . Second, the heatmaps at both noise levels q display identical color regions and differ only in color intensity, which aligns with Remark 4.6 that q only attenuates the magnitude of the strategic advantage without altering its sign.

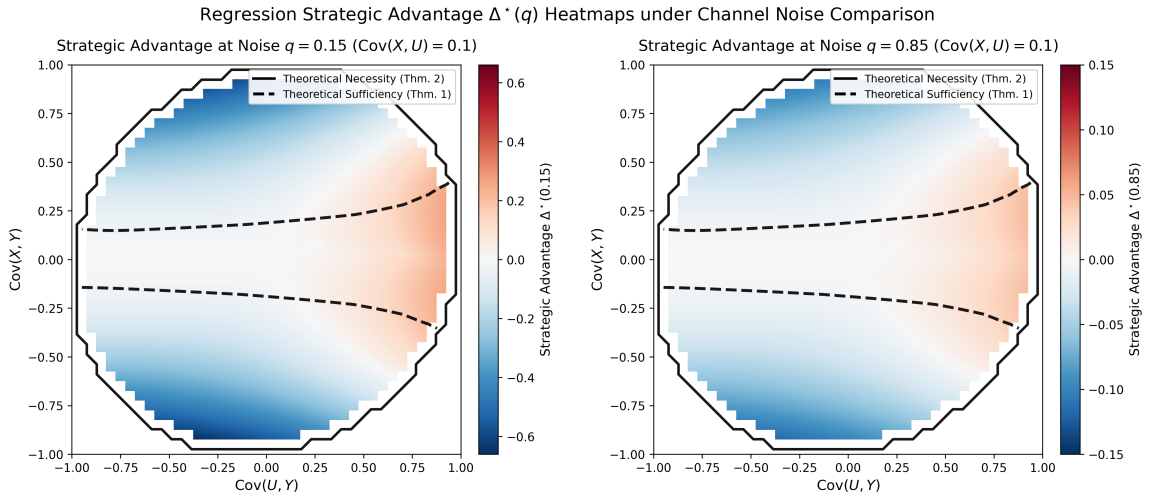


Figure 4.1: Strategic advantage $\Delta^*(q)$ under varying channel noise levels.

4.4 Offline Learning Problem

In the offline learning setting, the receiver does not know the joint distribution of (U, X, Y) , while the channel noise parameter q is assumed known. Instead, the receiver observes a fully revealed training dataset alongside the sender's preference values, given by

$$D_n = \{(u_k, x_k, y_k)\}_{k=1}^n,$$

where the triples (u_k, x_k, y_k) are drawn i.i.d. from the (unknown) distribution of (U, X, Y) .

We propose a learning algorithm for the receiver to estimate the strategic advantage $\Delta(\mathcal{E})$ from the training data and derive uniform generalization bounds for it. To simplify the calculations, assume $\|U\|, \|X\|, \|Y\| \leq 1$ almost surely, and restrict the receiver's strategy space, which

is also the learning algorithm's hypothesis class, to

$$\mathcal{R} := \{r : \mathcal{X} \cup \{\emptyset\} \rightarrow \mathcal{Y} \mid r(\emptyset) = 0, r(x) = b^\top x \text{ for some } b \in \mathbb{R}^d, \|b\| \leq 1\}.$$

[‡] For a given dataset D_n , define the empirical vanilla and strategic risks as

$$\widehat{R}_V(b) := (1 - q) \frac{1}{n} \sum_{k=1}^n (y_k - b^\top x_k)^2 + q \frac{1}{n} \sum_{k=1}^n y_k^2,$$

and

$$\widehat{R}_S(b) := \frac{1}{n} \sum_{k=1}^n \left[y_k^2 - (1 - q) G_b(x_k, y_k) \mathbb{I}\{F_b(u_k, x_k) > 0\} \right]$$

respectively. The corresponding empirical optimizers are denoted as[§]

$$\widehat{b}_V := \arg \min_{b \in \mathcal{R}} \widehat{R}_V(b), \quad \widehat{b}_S := \arg \min_{b \in \mathcal{R}} \widehat{R}_S(b).$$

Define the empirical strategic advantage as $\widehat{\Delta}(D_n) := \widehat{R}_V(\widehat{b}_V) - \widehat{R}_S(\widehat{b}_S)$, which estimates the population strategic advantage over the hypothesis class $\mathcal{R}$, $\Delta(\mathcal{E}) := \min_{b \in \mathcal{R}} R_V(b) - \min_{b \in \mathcal{R}} R_S(b)$. Algorithm 4.1 calculates $\widehat{\Delta}(D_n)$.

Algorithm 4.1 Strategic Advantage Regression Empirical Risk Minimization (SAR-ERM)

1: **Input:** Training set $D_n = \{(u_i, x_i, y_i)\}_{i=1}^n$, channel erasure probability q , parameterized hypothesis class $\mathcal{R} := \{b \in \mathbb{R}^d : \|b\|_2 \leq 1\}$.

2: **Compute and optimize using ERM:**

$$\begin{aligned} \widehat{R}_V(b) &\leftarrow (1 - q) \frac{1}{n} \sum_{i=1}^n (y_i - b^\top x_i)^2 + q \frac{1}{n} \sum_{i=1}^n y_i^2, \\ \widehat{R}_S(b) &\leftarrow \frac{1}{n} \sum_{i=1}^n y_i^2 - (1 - q) \frac{1}{n} \sum_{i=1}^n G_b(x_i, y_i) \mathbb{I}\{F_b(u_i, x_i) > 0\}, \\ \widehat{\Delta}(D_n) &\leftarrow \min_{b \in \mathcal{R}} \widehat{R}_V(b) - \min_{b \in \mathcal{R}} \widehat{R}_S(b) \end{aligned}$$

3: **Output:** Strategic advantage estimate $\widehat{\Delta}(D_n)$.

We now establish uniform generalization bounds for Algorithm 4.1. We begin by bounding the generalization gap between the empirical and population vanilla risks.

[‡]If the random variables and hypothesis class slopes were bounded by any constant $C > 0$ instead of 1, the generalization bounds change only by constant factors that are absorbed into the $\mathcal{O}(\cdot)$ notation.

[§]The population OLS slope $b_V = \text{Var}(X)^{-1} \text{Cov}(X, Y)$ admits an unconstrained closed-form formula. However, under the restricted domain constraint $\mathcal{R}$ and unknown population distribution, $\widehat{b}_V$ cannot be evaluated in closed form and is estimated numerically via Empirical Risk Minimization.

Lemma 4.7 (Uniform Generalization of Vanilla Risk): *Let $\delta \in (0, 1)$. With probability at least $1 - \delta$, we have*

$$\sup_{b \in \mathcal{R}} |R_V(b) - \widehat{R}_V(b)| \leq O \left(\frac{d^2}{\sqrt{n}} + d^2 \sqrt{\frac{\log(1/\delta)}{n}} \right). \quad (4.3)$$

Proof. The proof is given in Section 4.7 (Proof 4.7.3). $\square$

We next bound the generalization gap between the empirical and population strategic risks.

Lemma 4.8 (Uniform Generalization of Strategic Risk): *Let $\delta \in (0, 1)$. With probability at least $1 - \delta$ over the random draw of S*

$$\sup_{b \in \mathcal{R}} |R_S(b) - \widehat{R}_S(b)| \leq O \left(\frac{d^3 \sqrt{\log n}}{\sqrt{n}} + d^2 \sqrt{\frac{\log(1/\delta)}{n}} \right).$$

Proof. The proof is given in Section 4.7 (Proof 4.7.3). $\square$

Lemmas 4.7 and 4.8 show that our strategic-risk upper bound has a larger complexity term than the mandatory feature-revelation baseline (vanilla).

Finally, we combine these lemmas to establish a uniform generalization bound between the empirical strategic advantage $\widehat{\Delta}(D_n)$ and the population strategic advantage $\Delta(\mathcal{E})$.

Theorem 4.9 (Generalization Bound for Strategic Advantage): *Let $\delta \in (0, 1)$. With probability at least $1 - \delta$ over the random draw of D_n*

$$|\Delta(\mathcal{E}) - \widehat{\Delta}(D_n)| \leq O \left(\frac{d^3 \sqrt{\log n}}{\sqrt{n}} + d^2 \sqrt{\frac{\log(1/\delta)}{n}} \right).$$

Proof. The proof is given in Section 4.7 (Proof 4.7.3). $\square$

Simulation Results for Offline Learning. To evaluate our Algorithm 4.1 and empirically validate the uniform generalization bounds for strategic advantage from Theorem 4.9, we simulate the receiver offline learning problem across feature dimensions $d \in \{2, 5\}$. For each evaluation environment, the data generation process draws joint feature vectors, outcomes, and sender preferences $(X, Y, U) \in \mathbb{R}^{d+2}$ from a 2-component Gaussian Mixture Model (GMM) with mixture weights $\pi_1 = 0.6, \pi_2 = 0.4$. Component 1 represents a sub-population with mean vector $\mu_1 = (+0.8\mathbf{1}/\sqrt{d}, 1.0, 1.0)^\top$, while Component 2 represents a sub-population with mean vector $\mu_2 = (-0.8\mathbf{1}/\sqrt{d}, -0.8, -1.0)^\top$. Both components share a symmetric positive-definite

covariance matrix $\Sigma \in \mathbb{R}^{(d+2) \times (d+2)}$ specified by feature covariance $\Sigma_X = 0.64I_d$, outcome variance $\text{Var}(Y) = 0.6$, preference variance $\text{Var}(U) = 0.5$, and cross-covariances $\text{Cov}(X, Y) = \frac{0.5}{\sqrt{d}}\mathbf{1}$, $\text{Cov}(X, U) = \frac{0.3}{\sqrt{d}}\mathbf{1}$, and $\text{Cov}(Y, U) = 0.3$. The channel noise parameter is fixed at $q = 0.5$.

In each independent trial, a fresh empirical training set D_n of size n and a fresh held-out test dataset D_{test} of size $N_{\text{test}} = 10,000$ are independently sampled to evaluate population strategic risk. We evaluate the empirical vanilla and strategic risk learning across a logarithmic grid of 65 training sample sizes $n \in [50, 3000]$. For each training size n , we run $M = 15$ independent Monte Carlo trials. In each trial, an empirical training set D_n of size n is sampled, and the receiver hypotheses $\hat{b}_V$ and $\hat{b}_S$ are learned. For each trial, we evaluate the empirical strategic advantage $\hat{\Delta}(D_n) = \hat{R}_V(\hat{b}_V) - \hat{R}_S(\hat{b}_S)$ on training set D_n and the population strategic advantage $\Delta(\mathcal{E})$ on test set D_{test} , recording the absolute strategic advantage generalization gap $|\Delta(\mathcal{E}) - \hat{\Delta}(D_n)|$.

The empirical findings are presented in Figure 4.2 and corroborate our theoretical analysis. As the sample size n increases, the strategic advantage generalization gap decays at the rate $O(\sqrt{d/n})$ as claimed in Theorem 4.9. Furthermore, comparing $d = 2$ and $d = 5$ confirms that higher feature dimensionality causes the generalization gap to decay at a slower rate, as claimed in the d -dependency of the generalization gap in Theorem 4.9.

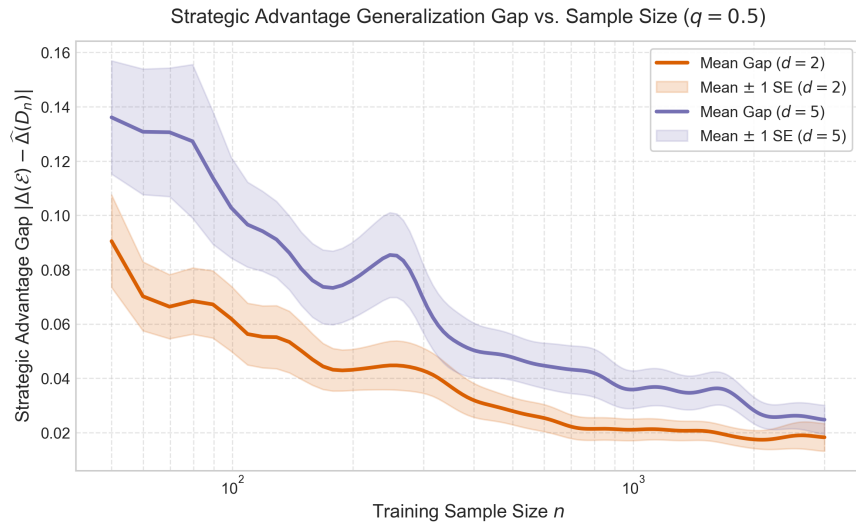


Figure 4.2: Generalization gap $|\Delta(\mathcal{E}) - \hat{\Delta}(D_n)|$ versus training sample size n under $q = 0.5$ (solid lines show mean gap, shaded bands indicate ± 1 standard error across $M = 15$ trials).

4.5 Online Learning Problem

In the online learning setting, the receiver does not know the joint distribution of (U, X, Y) , while the channel noise parameter $q \in [0, 1]$ is assumed known. Instead of observing a static offline training dataset, the receiver sequentially interacts with the unknown environment over rounds $t = 1, 2, \dots, T$. At each round t , a fresh sender sample (X_t, Y_t, U_t) is first drawn from the underlying environment distribution $P_{(U, X, Y)}$. The receiver then selects a disclosure regime $m_t \in \mathcal{M} := \{V, S\}$ and a regression parameter $b_t \in \mathcal{R} := \{b \in \mathbb{R}^d : \|b\| \leq 1\}$ (the compact strategy space defined in Section 4.4), forming the joint action $a_t = (m_t, b_t) \in \mathcal{A} := \mathcal{M} \times \mathcal{R}$. At the end of round t , the receiver observes scalar bandit loss feedback $Z_t = \ell(m_t, b_t, X_t, Y_t, U_t)$, which is the least-squares loss for the realized (U_t, X_t, Y_t) under model b_t and disclosure regime m_t . The receiver then updates both its chosen regime and parameter vector to minimize cumulative loss, aiming to match the performance of the optimal fixed regime and regression model in hindsight.

We pose this as a continuous bandit feedback problem and approach it through Gaussian Process Thompson Sampling (GP-TS) algorithm as proposed in Algorithm 2 in [60]. The receiver aims to optimize the unknown expected risk function $f(a)$ over the joint action space $\mathcal{A} = \mathcal{M} \times \mathcal{R}$, defined as

$$f(a) := \mathbb{E}_{(X, Y, U)}[\ell(m, b, X, Y, U)] = \begin{cases} R_V(b) & \text{if } m = \text{Vanilla}, \\ R_S(b) & \text{if } m = \text{Strategic}. \end{cases} \quad (4.4)$$

To address this, the receiver places a Gaussian Process prior over candidate loss functions, initialized with prior mean $\mu_0(a) = 0$ and prior covariance kernel $k_0(a, a') = k(a, a')$. For joint actions $a = (m, b)$ and $a' = (m', b')$, the kernel $k_0(a, a') = k(a, a')$ takes the Matérn-5/2 functional form:

$$k(a, a') := \mathbb{I}\{m = m'\} \left(1 + \frac{\sqrt{5}\|b - b'\|}{l} + \frac{5\|b - b'\|^2}{3l^2} \right) \exp\left(-\frac{\sqrt{5}\|b - b'\|}{l}\right), \quad (4.5)$$

where $l > 0$ is the kernel lengthscale parameter. For the direct-sum kernel $k((m, b), (m', b')) = \mathbb{I}\{m = m'\}k_0(b, b')$, the joint RKHS norm satisfies $B_k \leq \sqrt{2} \max\{B_{k,V}, B_{k,S}\}$. At each round t , the receiver maintains the updated posterior distribution $\text{GP}(\mu_{t-1}, v_t^2 k_{t-1})$, where v_t is an inflation factor given by $v_t = B_k + R\sqrt{2(\gamma_{t-1} + 1 + \log(2/\delta))}$ [60]. The receiver draws a virtual objective function $\tilde{f}_t \sim \text{GP}(\mu_{t-1}, v_t^2 k_{t-1})$ and selects action $a_t = (m_t, b_t) \in \arg \min_{a \in \mathcal{A}_t} \tilde{f}_t(a)$ over the decision set $\mathcal{A}_t = \{V, S\} \times \mathcal{R}_t$, where $\mathcal{R}_t$ is an ϵ_t -discretization net of $\mathcal{R}$ of mesh size $\epsilon_t = O(1/t^2)$ following [60].

Upon observing the scalar loss feedback Z_t , the receiver updates the GP posterior mean $\mu_t(a)$ and covariance kernel $k_t(a, a')$ according to the standard Gaussian process regression update rules [60]:

$$\mu_t(a) = \mathbf{k}_t(a)^\top (K_t + \lambda I_t)^{-1} \mathbf{Z}_t, \quad (4.6)$$

$$k_t(a, a') = k(a, a') - \mathbf{k}_t(a)^\top (K_t + \lambda I_t)^{-1} \mathbf{k}_t(a'), \quad (4.7)$$

where $\mathbf{Z}_t = [Z_1, \dots, Z_t]^\top \in \mathbb{R}^t$ is the vector of observed losses, $\mathbf{k}_t(a) = [k(a_1, a), \dots, k(a_t, a)]^\top \in \mathbb{R}^t$ is the kernel cross-covariance vector between action a and past actions $a_1, \dots, a_t$, and $K_t \in \mathbb{R}^{t \times t}$ is the kernel Gram matrix with entries $[K_t]_{i,j} = k(a_i, a_j)$ for $1 \leq i, j \leq t$, with $\lambda > 0$ serving as the regularization parameter. This procedure is detailed in Algorithm 4.2.

Algorithm 4.2 Gaussian Process Thompson Sampling (GP-TS)

- 1: **Input:** Matérn-5/2 covariance kernel $k_0(a, a') = k(a, a') = \mathbb{I}\{m = m'\} \left(1 + \frac{\sqrt{5}\|b-b'\|}{l} + \frac{5\|b-b'\|^2}{3l^2}\right) \exp\left(-\frac{\sqrt{5}\|b-b'\|}{l}\right)$, regularization $\lambda > 0$, confidence $\delta \in (0, 1)$.
 - 2: **Initialize:** Posterior mean $\mu_0(a) = 0$ and covariance $k_0(a, a') = k(a, a')$ for all $a \in \mathcal{A}$.
 - 3: **for** $t = 1, 2, \dots, T$ **do**
 - 4: Set inflation factor $v_t = B_k + R\sqrt{2(\gamma_{t-1} + 1 + \log(2/\delta))}$.
 - 5: Sample virtual objective function $\tilde{f}_t \sim \text{GP}(\mu_{t-1}, v_t^2 k_{t-1})$.
 - 6: Select action $a_t = (m_t, b_t) \in \arg \min_{a \in \mathcal{A}_t} \tilde{f}_t(a)$ over decision set $\mathcal{A}_t = \{V, S\} \times \mathcal{R}_t$.
 - 7: Observe loss feedback $Z_t = \ell(m_t, b_t, X_t, Y_t, U_t)$.
 - 8: Update posterior mean $\mu_t(a)$ and covariance $k_t(a, a')$ via (4.6) and (4.7).
 - 9: **end for**
-

We measure the performance of Algorithm 4.2 using cumulative regret relative to the minimal expected loss achievable in hindsight, namely $\min\{R_S^*, R_V^*\}$:

$$\text{Reg}_T := \sum_{t=1}^T (f(a_t) - f(a^*)), \quad f(a^*) = \min\{R_S^*, R_V^*\}, \quad (4.8)$$

where $a^* \in \arg \min_{a \in \mathcal{A}} f(a)$.

To establish a regret guarantee for GP-TS in our setting, we define the following class of regular environments.

Definition 4.10 (Sobolev Environments). Let $\mathcal{P}_{\text{Sob}}$ denote the set of environment joint distributions (U, X, Y) such that the expected risk functions $L(m, b)$ belong to the Sobolev space $H^{5/2+d/2}(\mathcal{R})$.

We next show that $\mathcal{P}_{\text{Sob}}$ contains all environments with a smooth joint density $p_{U,X,Y}(u, x, y)$ on compact support, and is therefore non-empty.

Lemma 4.11 (Smooth Environments): *Suppose the environment variables (U, X, Y) have a smooth, bounded joint density $p_{U,X,Y}(u, x, y)$ with compact support, and the receiver's action space $\mathcal{R} \subset \{b \in \mathbb{R}^d : \|b\|_2 \geq b_{\min}\}$ is a compact domain bounded away from the origin for some $b_{\min} > 0$. Then $L(m, b) \in H^{5/2+d/2}(\mathcal{R})$, and the environment is contained in $\mathcal{P}_{\text{Sob}}$.*

Proof. The proof is given in Section 4.7 (Proof 4.7.4). $\square$

We now state the theorem which gives the regret of Algorithm 4.2.

Theorem 4.12 (Regret Bound for Strategic GP-TS): *For any environment distribution in $\mathcal{P}_{\text{Sob}}$, let the receiver run GP-TS (Algorithm 4.2) with a Matérn- ν kernel (for smoothness parameter $\nu > 0$) over the joint action space $\mathcal{A} = \mathcal{M} \times \mathcal{R} \subset \{V, S\} \times \mathbb{R}^d$. Then, with probability at least $1 - \delta$, the cumulative regret satisfies*

$$\text{Reg}_T = \tilde{O}\left(T^{\frac{2\nu+3d}{4\nu+2d}}\right). \quad (4.9)$$

Specifically, substituting the practical Matérn-5/2 kernel ($\nu = 5/2$), the bound yields $\tilde{O}(T^{\frac{5+3d}{10+2d}})$, which is sublinear ($\text{Reg}_T/T \rightarrow 0$) for dimensions $d \in \{1, 2, 3, 4\}$.

Proof Outline. The proof of Theorem 4.12 applies the kernelized Thompson Sampling regret bound of [60], bounding the maximum information gain by $\gamma_T = O\left(T^{\frac{d}{5+d}}(\log T)^{\frac{5}{5+d}}\right)$ [61]. The complete proof is given in Section 4.7 (Proof 4.7.4). $\square$

Simulation Results for Online Learning. To evaluate Algorithm 4.2 and empirically validate the cumulative regret bounds from Theorem 4.12, we simulate the online learning problem across feature dimensions $d \in \{1, 2\}$. For each evaluation environment, the data generation process draws joint feature vectors, outcomes, and sender preferences $(X, Y, U) \in \mathbb{R}^{d+2}$ from a 2-component Gaussian Mixture Model (GMM) with mixture weights $\pi_1 = 0.6, \pi_2 = 0.4$. Component 1 represents a sub-population with mean vector $\mu_1 = (+0.81/\sqrt{d}, 1.0, 1.0)^\top$, while Component 2 represents a sub-population with mean vector $\mu_2 = (-0.81/\sqrt{d}, -0.8, -1.0)^\top$. Both components share a symmetric positive-definite covariance matrix $\Sigma \in \mathbb{R}^{(d+2) \times (d+2)}$ specified by feature covariance $\Sigma_X = 0.64I_d$, outcome variance $\text{Var}(Y) = 0.6$, preference

variance $\text{Var}(U) = 0.5$, and cross-covariances $\text{Cov}(X, Y) = \frac{0.5}{\sqrt{d}}\mathbf{1}$, $\text{Cov}(X, U) = \frac{0.3}{\sqrt{d}}\mathbf{1}$, and $\text{Cov}(Y, U) = 0.3$. The channel noise parameter is fixed at $q = 0.5$.

In each independent trial, the receiver sequentially interacts with the environment over $T = 300$ rounds using Algorithm 4.2 configured with a Matérn kernel (approximated practically using a Squared-Exponential kernel with lengthscale $l = 0.7$) and regularization $\lambda = 0.1$. At each round t , a sample function $\tilde{f}_t$ is drawn from the updated GP posterior to select action $a_t = (m_t, b_t) \in \mathcal{A}$. We average the cumulative regret Reg_T across $M = 10$ independent Monte Carlo simulation runs.

The empirical findings are presented in Figure 4.3. As predicted by Theorem 4.12, the empirical cumulative regret exhibits a sublinear growth rate matching the theoretical $\tilde{O}\left(T^{\frac{5+3d}{10+2d}}\right)$ scaling (which instantiates the general Matérn regret bound $\tilde{O}\left(T^{\frac{2\nu+3d}{4\nu+2d}}\right)$ for smoothness parameter $\nu = 5/2$), with shaded bands showing ± 1 standard error across $M = 10$ independent Monte Carlo trials. Furthermore, comparing $d = 1$ and $d = 2$ confirms that higher feature dimensionality leads to a larger regret growth rate due to the dimensional dependence, corroborating our theoretical analysis for environments with smooth joint densities $p_{U,X,Y}$ on compact support.

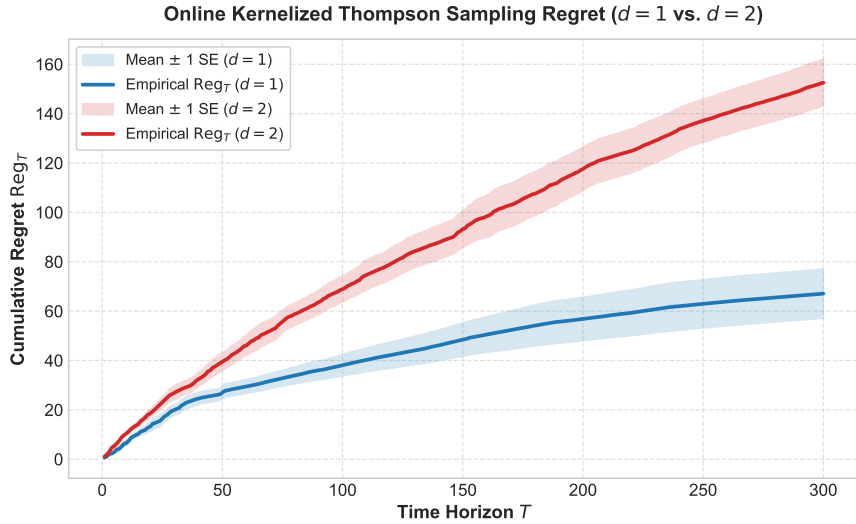


Figure 4.3: Empirical cumulative regret Reg_T versus time horizon T for $d = 1$ and $d = 2$ (solid lines depict mean cumulative regret, shaded bands indicate ± 1 standard error across $M = 10$ trials).

4.6 Conclusion, Limitations, and Future Work

In this chapter, we study the problem of strategic feature disclosure in the regression setting. We showed that under certain conditions on the environment, the receiver can achieve better performance when the sender strategically withholds features. We establish conditions on the environment for this to occur and demonstrate how to empirically estimate these conditions in both offline and online settings, providing uniform bounds on the estimation error and online regret bounds. Some limitations of this chapter include the restriction to linear hypothesis classes for both decision and learning problems, and the assumption of all-or-nothing feature revelation. A promising direction for future work is extending the analysis to non-linear hypothesis classes and to partial feature-revelation strategies.

4.7 Proofs and Supporting Results

This section provides standard learning-theoretic helper results and the proofs for the results presented in this chapter.

4.7.1 Supporting Results

In this section, we establish the supporting results required for the proofs of the theorems and lemmas presented in the chapter.

We now state the theorem characterizing generalization bounds via Rademacher complexity.

Theorem 4.13 (Rademacher Complexity Generalization Bound, [62]): *Let $\mathcal{F}$ be a family of real-valued functions mapping from a domain $\mathcal{Z}$ to a bounded interval $[a, b]$, and let $S = (z_1, \dots, z_n) \sim \mathcal{D}^n$ be an i.i.d. sample of size n . Then, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random draw of S , we have*

$$\sup_{f \in \mathcal{F}} \left| \mathbb{E}[f] - \widehat{\mathbb{E}}_S[f] \right| \leq 2\text{Rad}_n(\mathcal{F}) + (b - a) \sqrt{\frac{\log(2/\delta)}{2n}},$$

where $\widehat{\mathbb{E}}_S[f] := \frac{1}{n} \sum_{i=1}^n f(z_i)$ is the empirical average over S , $\widehat{\text{Rad}}_S(\mathcal{F}) := \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(z_i) \right]$ is the empirical Rademacher complexity on sample S , and $\text{Rad}_n(\mathcal{F}) := \mathbb{E}_S[\widehat{\text{Rad}}_S(\mathcal{F})]$ is the

expected Rademacher complexity over sample size n .

Proof. Let $g(S) := \sup_{f \in \mathcal{F}} (\mathbb{E}[f] - \widehat{\mathbb{E}}_S[f])$. Changing any single sample point $z_i \rightarrow z'_i$ alters $g(S)$ by at most $(b - a)/n$. Applying McDiarmid's concentration inequality (Lemma 26.4 of [62]) yields $g(S) \leq \mathbb{E}_S[g(S)] + (b - a)\sqrt{\frac{\log(2/\delta)}{2n}}$ with probability at least $1 - \delta/2$. By Lemma 26.2 of [62], the expected supremum satisfies $\mathbb{E}_S[g(S)] \leq 2\mathbb{E}_S[\widehat{\text{Rad}}_S(\mathcal{F})] = 2\text{Rad}_n(\mathcal{F})$. Applying the identical concentration bound to the negated class $-\mathcal{F} := \{-f : f \in \mathcal{F}\}$ (noting $\text{Rad}_n(-\mathcal{F}) = \text{Rad}_n(\mathcal{F})$) bounds the lower-tail supremum $\sup_{f \in \mathcal{F}} (\widehat{\mathbb{E}}_S[f] - \mathbb{E}[f]) = \sup_{h \in -\mathcal{F}} (\mathbb{E}[h] - \widehat{\mathbb{E}}_S[h])$ by the same threshold with confidence $1 - \delta/2$. By a union bound over both tail events holding with probability at least $1 - \delta$, the identity $\sup_{f \in \mathcal{F}} |\mathbb{E}[f] - \widehat{\mathbb{E}}_S[f]| = \max \left\{ \sup_{f \in \mathcal{F}} (\mathbb{E}[f] - \widehat{\mathbb{E}}_S[f]), \sup_{f \in \mathcal{F}} (\widehat{\mathbb{E}}_S[f] - \mathbb{E}[f]) \right\}$ yields the two-sided absolute supremum bound. $\square$

We now establish a useful property regarding the Rademacher complexity of a product function class, expressed in terms of the individual function classes.

Lemma 4.14 (Rademacher Complexity of Product Function Classes): *Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be two function classes mapping from domain $\mathcal{Z}$ to $\mathbb{R}$, bounded almost surely by $M_1 = \sup_{f_1 \in \mathcal{F}_1} \|f_1\|_\infty$ and $M_2 = \sup_{f_2 \in \mathcal{F}_2} \|f_2\|_\infty$ respectively. Define the product function class $\mathcal{F}_1 \cdot \mathcal{F}_2 := \{z \mapsto f_1(z)f_2(z) : f_1 \in \mathcal{F}_1, f_2 \in \mathcal{F}_2\}$. Then, for any sample $S = (z_1, \dots, z_n)$, the empirical Rademacher complexity satisfies*

$$\widehat{\text{Rad}}_S(\mathcal{F}_1 \cdot \mathcal{F}_2) \leq M_2 \widehat{\text{Rad}}_S(\mathcal{F}_1) + M_1 \widehat{\text{Rad}}_S(\mathcal{F}_2),$$

and taking expectation over $S \sim \mathcal{D}^n$ yields $\text{Rad}_n(\mathcal{F}_1 \cdot \mathcal{F}_2) \leq M_2 \text{Rad}_n(\mathcal{F}_1) + M_1 \text{Rad}_n(\mathcal{F}_2)$.

Proof. Fix a sample $S = (z_1, \dots, z_n) \in \mathcal{Z}^n$ and pick an arbitrary reference function $h_1 \in \mathcal{F}_1$. For any $f_1 \in \mathcal{F}_1$ and $f_2 \in \mathcal{F}_2$, writing $f_1(z_i)f_2(z_i) = (f_1(z_i) - h_1(z_i))f_2(z_i) + h_1(z_i)f_2(z_i)$ and applying sub-additivity of the supremum yields

$$\begin{aligned} \widehat{\text{Rad}}_S(\mathcal{F}_1 \cdot \mathcal{F}_2) &\leq \mathbb{E}_\sigma \left[\sup_{f_1 \in \mathcal{F}_1, f_2 \in \mathcal{F}_2} \frac{1}{n} \sum_{i=1}^n \sigma_i (f_1(z_i) - h_1(z_i)) f_2(z_i) \right] \\ &\quad + \mathbb{E}_\sigma \left[\sup_{f_2 \in \mathcal{F}_2} \frac{1}{n} \sum_{i=1}^n \sigma_i h_1(z_i) f_2(z_i) \right]. \end{aligned}$$

For the first term, since $|f_2(z_i)| \leq M_2$, the scalar mapping $t \mapsto t \cdot f_2(z_i)$ is M_2 -Lipschitz.

Applying Talagrand's contraction lemma (Lemma 26.9 of [62]) bounds this term by

$$M_2 \mathbb{E}_\sigma \left[\sup_{f_1 \in \mathcal{F}_1} \frac{1}{n} \sum_{i=1}^n \sigma_i (f_1(z_i) - h_1(z_i)) \right] = M_2 \widehat{\text{Rad}}_S(\mathcal{F}_1),$$

where the linear term $\frac{1}{n} \sum_{i=1}^n \sigma_i h_1(z_i)$ vanishes under expectation since $\mathbb{E}_{\sigma_i}[\sigma_i] = 0$.

For the second term, since $|h_1(z_i)| \leq M_1$, the scalar mapping $t \mapsto h_1(z_i) \cdot t$ is M_1 -Lipschitz.

Applying Talagrand's contraction lemma bounds this term by

$$M_1 \mathbb{E}_\sigma \left[\sup_{f_2 \in \mathcal{F}_2} \frac{1}{n} \sum_{i=1}^n \sigma_i f_2(z_i) \right] = M_1 \widehat{\text{Rad}}_S(\mathcal{F}_2).$$

Summing both bounds yields $\widehat{\text{Rad}}_S(\mathcal{F}_1 \cdot \mathcal{F}_2) \leq M_2 \widehat{\text{Rad}}_S(\mathcal{F}_1) + M_1 \widehat{\text{Rad}}_S(\mathcal{F}_2)$. Taking expectation over sample $S \sim \mathcal{D}^n$ establishes the population bound $\text{Rad}_n(\mathcal{F}_1 \cdot \mathcal{F}_2) \leq M_2 \text{Rad}_n(\mathcal{F}_1) + M_1 \text{Rad}_n(\mathcal{F}_2)$. $\square$

We now restate the regret bound theorem for Kernelized Thompson Sampling on Reproducing Kernel Hilbert Spaces (RKHS) under sub-Gaussian noise feedback from [60].

Theorem 4.15 (Kernelized Thompson Sampling Regret Bound, [60]): *Let $\mathcal{D} \subset [0, r]^d$ be a compact and convex domain, and let the objective function $f : \mathcal{D} \rightarrow \mathbb{R}$ belong to a Reproducing Kernel Hilbert Space $\mathcal{H}_k(\mathcal{D})$ associated with kernel k , satisfying $\|f\|_{\mathcal{H}_k} \leq B_k < \infty$. Suppose the observation noise $\eta_t = Z_t - f(a_t)$ is conditionally zero-mean and R -sub-Gaussian. Let $L = \sup_{x \in \mathcal{D}} \max_{j \in [d]} \left(\left| \frac{\partial^2 k(p, q)}{\partial p_j \partial q_j} \right|_{p=q=x} \right)^{1/2} < \infty$. Running GP-TS with round- t decision sets $\mathcal{D}_t \subset \mathcal{D}$ satisfying $|f(x) - f([x]_t)| \leq 1/t^2$ for all $x \in \mathcal{D}$ (via a discretization of mesh size $\epsilon_t \leq \frac{1}{B_k L t^2}$) guarantees that, with probability at least $1 - \delta$, the cumulative regret satisfies*

$$\text{Reg}_T = O \left(\sqrt{(\gamma_T + \log(2/\delta)) d \log(B_k d T)} \left(\sqrt{T \gamma_T} + B_k \sqrt{T \log(2/\delta)} \right) \right) = \tilde{O} \left(d^{1/2} \gamma_T \sqrt{T} \right),$$

where γ_T is the maximum information gain after T rounds for kernel k on domain $\mathcal{D}$.

Proof. This is Theorem 4 of [60]. We refer the reader to [60] for the full martingale concentration derivation. $\square$

4.7.2 Proofs for the Decision Problem

In this subsection, we present the proofs of the theoretical results for the decision problem.

We now present the proof of Lemma 4.1.

Proof. (Proof of Lemma 4.1) Under mandatory disclosure, the sender always reveals features ($s(U, X) = x$). The features X are successfully received with probability $1 - q$ and are erased with probability q . If erased, the receiver predicts the default value 0 and suffers quadratic loss Y^2 . If received, the receiver predicts $b^\top X$ and suffers $(Y - b^\top X)^2$. The vanilla risk is

$$R_V(b) = q\mathbb{E}[Y^2] + (1 - q)\mathbb{E}[(Y - b^\top X)^2].$$

Since $\mathbb{E}[Y] = 0$, we have $\mathbb{E}[Y^2] = \text{Var}(Y)$. To decompose the second term, we add and subtract the OLS predictor $b_V^\top X$:

$$\mathbb{E}[(Y - b^\top X)^2] = \mathbb{E}\left[\left((Y - b_V^\top X) + (b_V - b)^\top X\right)^2\right].$$

Expanding the square yields

$$\mathbb{E}[(Y - b^\top X)^2] = \mathbb{E}[(Y - b_V^\top X)^2] + \mathbb{E}[(b_V - b)^\top X]^2 + 2\mathbb{E}[(Y - b_V^\top X)(b_V - b)^\top X].$$

By the orthogonality principle of linear least squares, the error $Y - b_V^\top X$ is orthogonal to the feature space $\mathcal{X}$, meaning $\mathbb{E}[(Y - b_V^\top X)X] = 0$. Consequently, the cross-product term is zero:

$$2(b_V - b)^\top \mathbb{E}[(Y - b_V^\top X)X] = 0.$$

Furthermore, expanding the OLS residual risk yields

$$\mathbb{E}[(Y - b_V^\top X)^2] = \mathbb{E}[Y^2] - 2\mathbb{E}[Yb_V^\top X] + \mathbb{E}[(b_V^\top X)^2].$$

Applying orthogonality to $b_V^\top X$ gives $\mathbb{E}[(Y - b_V^\top X)b_V^\top X] = b_V^\top \mathbb{E}[(Y - b_V^\top X)X] = 0$, which implies $\mathbb{E}[Yb_V^\top X] = \mathbb{E}[(b_V^\top X)^2] = \text{Var}(b_V^\top X)$. Since $\mathbb{E}[Y] = 0$ and $\mathbb{E}[b_V^\top X] = 0$, we have

$$\mathbb{E}[(Y - b_V^\top X)^2] = \text{Var}(Y) - 2\text{Var}(b_V^\top X) + \text{Var}(b_V^\top X) = \text{Var}(Y) - \text{Var}(b_V^\top X),$$

and

$$\mathbb{E}[(b_V - b)^\top X]^2 = \text{Var}((b_V - b)^\top X) = \text{Var}((b - b_V)^\top X).$$

Substituting these back into the risk formula gives

$$\begin{aligned} R_V(b) &= q\text{Var}(Y) + (1 - q) (\text{Var}(Y) - \text{Var}(b_V^\top X) + \text{Var}((b - b_V)^\top X)) \\ &= \text{Var}(Y) - (1 - q)\text{Var}(b_V^\top X) + (1 - q)\text{Var}((b - b_V)^\top X), \end{aligned}$$

which matches (4.1).

The vanilla risk is minimized when $\text{Var}((b - b_V)^\top X) = 0$. Assuming $\text{Var}(X) \succ 0$, this occurs uniquely at $b = b_V$, yielding the optimal vanilla risk

$$R_V^*(q) = \text{Var}(Y) - (1 - q)\text{Var}(b_V^\top X).$$

□

We now present the proof of Lemma 4.2.

Proof. (Proof of Lemma 4.2) Consider a sender with preference realization u and features x under a committed receiver predictor parameter b . If the sender chooses to withhold ($s(u, x) = \emptyset$), the receiver predicts 0, and the sender suffers loss:

$$\ell_S(0, u) = u^2.$$

If the sender chooses to disclose ($s(u, x) = x$), the feature is successfully transmitted with probability $1 - q$ (resulting in prediction $b^\top x$ and loss $(b^\top x - u)^2$) and is erased with probability q (resulting in prediction 0 and loss u^2). The sender's expected loss from disclosing is

$$\mathbb{E}_\epsilon[\ell_S(\hat{Y}, u)] = (1 - q)(b^\top x - u)^2 + qu^2.$$

The sender prefers disclosure if and only if:

$$u^2 > (1 - q)(b^\top x - u)^2 + qu^2.$$

Since $q \in (0, 1)$, we divide by the positive constant $1 - q > 0$ to get

$$u^2 > (u - b^\top x)^2.$$

Expanding the right-hand side gives

$$u^2 > u^2 - 2ub^\top x + (b^\top x)^2 \iff (2u - b^\top x)b^\top x > 0.$$

Recalling the definition of $F_b(u, x)$, this simplifies to $F_b(u, x) > 0$. We resolve ties by withholding, which completes the proof. □

We now present the proof of Lemma 4.3.

Proof. (Proof of Lemma 4.3) If $s_b(U, X) = x$, the sender attempts revelation. With probability q , the channel erases the features, leading to prediction 0 and loss Y^2 . With probability $1 - q$, features are successfully received, resulting in prediction $b^\top X$ and loss $(Y - b^\top X)^2$. If $s_b(U, X) = \emptyset$, the feature is withheld, leading to prediction 0 and loss Y^2 . The strategic risk is

$$R_S(b) = \mathbb{E} \left[\mathbb{I}\{s_b(U, X) = \emptyset\} Y^2 + \mathbb{I}\{s_b(U, X) = x\} \left(qY^2 + (1 - q)(Y - b^\top X)^2 \right) \right].$$

Grouping the terms containing Y^2 yields

$$\begin{aligned} R_S(b) &= \mathbb{E} \left[Y^2 - (1 - q) \mathbb{I}\{s_b(U, X) = x\} Y^2 + (1 - q) \mathbb{I}\{s_b(U, X) = x\} (Y - b^\top X)^2 \right] \\ &= \mathbb{E} [Y^2] - (1 - q) \mathbb{E} \left[\mathbb{I}\{s_b(U, X) = x\} \left(Y^2 - (Y - b^\top X)^2 \right) \right]. \end{aligned}$$

Since $\mathbb{E}[Y] = 0$, we have $\mathbb{E}[Y^2] = \text{Var}(Y)$. Expanding the term $Y^2 - (Y - b^\top X)^2$ gives:

$$Y^2 - (Y^2 - 2Yb^\top X + (b^\top X)^2) = (2Y - b^\top X)b^\top X = G_b(X, Y).$$

Thus, the expression reduces to:

$$R_S(b) = \text{Var}(Y) - (1 - q) \mathbb{E} [G_b(X, Y) \mathbb{I}\{F_b(U, X) > 0\}],$$

which is the stated equation (4.2). □

We now present the proof of Theorem 4.4.

Proof. (Proof of Theorem 4.4) Let $\Delta(b) = R_V^* - R_S(b)$. By definition, if there exists $b \in \mathbb{R}^d$ such that $\Delta(b) > 0$, then $\Delta(\mathcal{E}) > 0$. Using the risk equations from Lemmas 4.1 and 4.3, the risk difference is

$$R_S(b) - R_V^* = (1 - q) \left[\text{Var}(b_V^\top X) - \mathbb{E} [G_b(X, Y) \mathbb{I}\{F_b(U, X) > 0\}] \right].$$

We rewrite the expectation as

$$\mathbb{E} [G_b(X, Y) \mathbb{I}\{F_b(U, X) > 0\}] = \mathbb{E} [G_b(X, Y)] - \mathbb{E} [G_b(X, Y) \mathbb{I}\{F_b(U, X) \leq 0\}].$$

Note that $G_b(X, Y) = (2Y - b^\top X)b^\top X = Y^2 - (Y - b^\top X)^2$. Taking expectations and using the vanilla risk calculations from Lemma 4.1, we have

$$\begin{aligned} \mathbb{E} [G_b(X, Y)] &= \mathbb{E} [Y^2] - \mathbb{E} [(Y - b^\top X)^2] \\ &= \text{Var}(Y) - (\text{Var}(Y) - \text{Var}(b_V^\top X) + \text{Var}((b - b_V)^\top X)) \end{aligned}$$

$$= \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X).$$

Thus, the risk difference is

$$R_S(b) - R_V^* = (1 - q) \left[\text{Var}((b - b_V)^\top X) + \mathbb{E} [G_b(X, Y) \mathbb{I}\{F_b(U, X) \leq 0\}] \right].$$

For this difference to be strictly negative, we require the bracketed term to be negative. We decompose the expectation term

$$\mathbb{E} [G_b \mathbb{I}\{F_b \leq 0\}] = \mathbb{E} [G_b \mathbb{I}\{G_b \leq 0\}] + \mathbb{E} [G_b (\mathbb{I}\{F_b \leq 0\} - \mathbb{I}\{G_b \leq 0\})].$$

For the first term, we use the standard algebraic identity $z \mathbb{I}\{z \leq 0\} = \frac{z - |z|}{2}$ for any real scalar z . Applying this to $G_b(X, Y)$ and substituting $\mathbb{E}[G_b(X, Y)] = \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X)$ yields

$$\begin{aligned} \mathbb{E} [G_b \mathbb{I}\{G_b \leq 0\}] &= \frac{\mathbb{E}[G_b] - \mathbb{E}[|G_b|]}{2} \\ &= \frac{\text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) - \mathbb{E}[|G_b(X, Y)|]}{2}. \end{aligned} \quad (4.10)$$

For the second term, we establish the pointwise upper bound

$$G_b (\mathbb{I}\{F_b \leq 0\} - \mathbb{I}\{G_b \leq 0\}) \leq 2|Y - U| |b^\top X|.$$

Hence:

$$G_b (\mathbb{I}\{F_b \leq 0\} - \mathbb{I}\{G_b \leq 0\}) \leq |G_b - F_b| = 2|Y - U| |b^\top X|.$$

Taking expectation and applying the Cauchy-Schwarz inequality gives:

$$\mathbb{E} [G_b (\mathbb{I}\{F_b \leq 0\} - \mathbb{I}\{G_b \leq 0\})] \leq 2\mathbb{E} [|Y - U| |b^\top X|] \leq 2\|U - Y\|_2 \sqrt{\text{Var}(b^\top X)}.$$

Combining these components, we get

$$\mathbb{E} [G_b \mathbb{I}\{F_b \leq 0\}] \leq \frac{\text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) - \mathbb{E}[|G_b|]}{2} + 2\|U - Y\|_2 \sqrt{\text{Var}(b^\top X)}.$$

Substitute this inequality back into the risk difference:

$$R_S(b) - R_V^* \leq (1 - q) \left[\frac{\text{Var}(b_V^\top X) + \text{Var}((b - b_V)^\top X) - \mathbb{E}[|G_b|]}{2} + 2\|U - Y\|_2 \sqrt{\text{Var}(b^\top X)} \right].$$

To guarantee that $R_S(b) - R_V^* < 0$, it is sufficient for the term in the brackets to be strictly negative:

$$2\|U - Y\|_2 \sqrt{\text{Var}(b^\top X)} < \frac{\mathbb{E}[|G_b|] - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X)}{2}.$$

Rearranging this inequality yields the sufficient condition. $\square$

We now present the proof of Theorem 4.5.

Proof. (Proof of Theorem 4.5) We analyze the risk difference:

$$R_S(b) - R_V^* = (1 - q) \left[\text{Var}((b - b_V)^\top X) + \mathbb{E}[G_b(X, Y) \mathbb{I}\{F_b(U, X) \leq 0\}] \right].$$

Pointwise, for any realizations, we have

$$G_b \mathbb{I}\{F_b \leq 0\} \geq G_b \mathbb{I}\{G_b \leq 0\}.$$

Taking expectations on both sides yields

$$\mathbb{E}[G_b \mathbb{I}\{F_b \leq 0\}] \geq \mathbb{E}[G_b \mathbb{I}\{G_b \leq 0\}] = \frac{\text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) - \mathbb{E}[|G_b|]}{2}.$$

Substitute this lower bound into the risk difference:

$$\begin{aligned} R_S(b) - R_V^* &\geq (1 - q) \left[\text{Var}((b - b_V)^\top X) + \frac{\text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) - \mathbb{E}[|G_b|]}{2} \right] \\ &= \frac{1 - q}{2} \left[\text{Var}(b_V^\top X) + \text{Var}((b - b_V)^\top X) - \mathbb{E}[|G_b|] \right]. \end{aligned}$$

Now, suppose that for all $b \in \mathbb{R}^d$, the necessary condition fails, meaning:

$$\mathbb{E}[|G_b(X, Y)|] - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X) \leq 0 \quad \forall b \in \mathbb{R}^d.$$

This implies that the term inside the brackets is always non-negative:

$$\text{Var}(b_V^\top X) + \text{Var}((b - b_V)^\top X) - \mathbb{E}[|G_b(X, Y)|] \geq 0 \quad \forall b \in \mathbb{R}^d.$$

Since $1 - q > 0$, this guarantees $R_S(b) - R_V^* \geq 0$ for all b , which means $\Delta(\mathcal{E}) \leq 0$. $\square$

4.7.3 Proofs for Offline Learning

In this subsection, we provide the proofs of the generalization bounds for offline learning.

We now present the proof of Lemma 4.7.

Proof. (Proof of Lemma 4.7) Let $L(b) = \mathbb{E}[(Y - b^\top X)^2]$ denote the expected quadratic loss under receiver linear strategy $b \in \mathcal{R}$, and let $\widehat{L}(b) = \frac{1}{n} \sum_{k=1}^n (y_k - b^\top x_k)^2$ denote the empirical loss on sample S . The difference between true and empirical vanilla risks is

$$R_V(b) - \widehat{R}_V(b) = q \left(\text{Var}(Y) - \frac{1}{n} \sum_{k=1}^n y_k^2 \right) + (1 - q)(L(b) - \widehat{L}(b)).$$

Applying the triangle inequality yields

$$\sup_{b \in \mathcal{R}} |R_V(b) - \widehat{R}_V(b)| \leq \left| \text{Var}(Y) - \frac{1}{n} \sum_{k=1}^n y_k^2 \right| + \sup_{b \in \mathcal{R}} |L(b) - \widehat{L}(b)|.$$

We allocate failure probability $\delta/2$ to each of these two terms using a union bound.

For the first term, since $Y \in [-1, 1]$ almost surely, $Y^2 \in [0, 1]$. Applying Hoeffding's concentration inequality yields $\left| \text{Var}(Y) - \frac{1}{n} \sum_{k=1}^n y_k^2 \right| \leq \sqrt{\frac{\log(4/\delta)}{2n}} = O\left(\sqrt{\frac{\log(1/\delta)}{n}}\right)$ with probability at least $1 - \delta/2$.

To bound the second term $\sup_{b \in \mathcal{R}} |L(b) - \widehat{L}(b)|$, we apply the Rademacher complexity generalization bound (Theorem 4.13) to the loss-composed strategy space $\ell \circ \mathcal{R} := \{(x, y) \mapsto (y - b^\top x)^2 : b \in \mathcal{R}\}$. First, we show that the scalar loss function is Lipschitz over the domain of predictions. For any strategy $b \in \mathcal{R}$ (where $\|b\|_\infty \leq 1$) and feature vector x (where $\|x\|_\infty \leq 1$), the absolute prediction is bounded by $|b^\top x| = |\sum_{i=1}^d b_i x_i| \leq \sum_{i=1}^d |b_i| |x_i| \leq d \cdot 1 = d$, so the scalar linear prediction $a = b^\top x$ lies in $[-d, d]$. Define the scalar loss function $\phi(a) = (y - a)^2$ for $a \in [-d, d]$ and $y \in [-1, 1]$. Differentiating $\phi(a)$ with respect to a gives $|\phi'(a)| = 2|y - a| \leq 2(1 + d)$. Hence, ϕ is L -Lipschitz over $[-d, d]$ with Lipschitz constant $L = 2(1 + d)$.

Next, by Talagrand's contraction lemma (Lemma 26.9 of [62]), the empirical Rademacher complexity of the loss-composed class $\ell \circ \mathcal{R}$ is bounded by

$$\widehat{\text{Rad}}_S(\ell \circ \mathcal{R}) \leq L \cdot \widehat{\text{Rad}}_S(\mathcal{R}) = 2(1 + d) \widehat{\text{Rad}}_S(\mathcal{R}),$$

where $\widehat{\text{Rad}}_S(\mathcal{R})$ is the empirical Rademacher complexity of the receiver strategy space $\mathcal{R} = \{x \mapsto b^\top x : \|b\|_\infty \leq 1\}$. By Lemma 26.10 of [62] (Rademacher complexity bound for linear classes), for any sample $S = (x_1, \dots, x_n)$ with $\|x_k\|_2 \leq \sqrt{d}$ and $\|b\|_2 \leq \sqrt{d}$:

$$\widehat{\text{Rad}}_S(\mathcal{R}) = \mathbb{E}_\sigma \left[\sup_{b \in \mathcal{R}} \frac{1}{n} \sum_{k=1}^n \sigma_k b^\top x_k \right] \leq \frac{\|b\|_2 \max_k \|x_k\|_2}{\sqrt{n}} \leq \frac{\sqrt{d} \cdot \sqrt{d}}{\sqrt{n}} = \frac{d}{\sqrt{n}}.$$

Thus, $\widehat{\text{Rad}}_S(\ell \circ \mathcal{R}) \leq 2(1 + d) \frac{d}{\sqrt{n}} = \frac{2d(1+d)}{\sqrt{n}} = O\left(\frac{d^2}{\sqrt{n}}\right)$, and taking expectation over $S \sim \mathcal{D}^n$ gives $\text{Rad}_n(\ell \circ \mathcal{R}) = O\left(\frac{d^2}{\sqrt{n}}\right)$.

Finally, applying Theorem 4.13 to class $\ell \circ \mathcal{R}$ (bounded in $[0, (1 + d)^2]$), with probability at least $1 - \delta/2$:

$$\sup_{b \in \mathcal{R}} |L(b) - \widehat{L}(b)| \leq 2\text{Rad}_n(\ell \circ \mathcal{R}) + (1 + d)^2 \sqrt{\frac{\log(4/\delta)}{2n}} = O\left(\frac{d^2}{\sqrt{n}} + d^2 \sqrt{\frac{\log(1/\delta)}{n}}\right).$$

Summing the bounds for both terms via a union bound completes the proof. $\square$

We now present the proof of Lemma 4.8.

Proof. (Proof of Lemma 4.8) *Notation abuse:* Except for R_S and $\widehat{R}_S$, where the subscript s denotes strategic, the subscript s elsewhere denotes the underlying sample set S .

The strategic risk and empirical strategic risk are

$$R_S(b) = \text{Var}(Y) - (1 - q)\mathbb{E}[h_b(U, X, Y)], \quad \widehat{R}_S(b) = \frac{1}{n} \sum_{k=1}^n y_k^2 - (1 - q) \frac{1}{n} \sum_{k=1}^n h_b(u_k, x_k, y_k),$$

where $h_b(u, x, y) = G_b(x, y)\mathbb{I}\{F_b(u, x) > 0\}$. Applying the triangle inequality:

$$\sup_{b \in \mathcal{R}} |R_S(b) - \widehat{R}_S(b)| \leq \left| \text{Var}(Y) - \frac{1}{n} \sum_{k=1}^n y_k^2 \right| + (1 - q) \sup_{b \in \mathcal{R}} |\mathbb{E}[h_b] - \widehat{\mathbb{E}}_S[h_b]|.$$

The first term is bounded by $O\left(\sqrt{\log(1/\delta)/n}\right)$ with probability at least $1 - \delta/2$ using Hoeffding's inequality.

To bound the second term $\sup_{b \in \mathcal{R}} |\mathbb{E}[h_b] - \widehat{\mathbb{E}}_S[h_b]|$, consider the function class $\mathcal{H} = \{h_b(u, x, y) = G_b(x, y)\mathbb{I}\{F_b(u, x) > 0\} : b \in \mathcal{R}\}$. Note that each $h_b = g_b \cdot k_b$ is a pointwise product of quadratic polynomial $g_b(x, y) = (2y - b^\top x)b^\top x$ from class $\mathcal{G}_1 = \{(x, y) \mapsto (2y - b^\top x)b^\top x : b \in \mathcal{R}\}$ and binary threshold indicator $k_b(u, x) = \mathbb{I}\{(2u - b^\top x)b^\top x > 0\}$ from class $\mathcal{K} = \{(u, x) \mapsto \mathbb{I}\{(2u - b^\top x)b^\top x > 0\} : b \in \mathcal{R}\}$. Since $\|b\|_\infty \leq 1$, $\|x\|_\infty \leq 1$, and $|y| \leq 1$, by Hölder's inequality $|b^\top x| \leq d$, which implies $\|g_b\|_\infty \leq (2 + d)d = d(d + 2)$ and $\|k_b\|_\infty \leq 1$. By Lemma 4.14, the empirical Rademacher complexity of $\mathcal{H} = \mathcal{G}_1 \cdot \mathcal{K}$ satisfies

$$\begin{aligned} \widehat{\text{Rad}}_S(\mathcal{H}) &\leq \sup_{b \in \mathcal{R}} \|g_b\|_\infty \cdot \widehat{\text{Rad}}_S(\mathcal{K}) + \sup_{b \in \mathcal{R}} \|k_b\|_\infty \cdot \widehat{\text{Rad}}_S(\mathcal{G}_1) \\ &\leq d(d + 2) \widehat{\text{Rad}}_S(\mathcal{K}) + \widehat{\text{Rad}}_S(\mathcal{G}_1). \end{aligned}$$

For the quadratic term $\mathcal{G}_1$, write $g_b(x_k, y_k) = \phi_k(b^\top x_k)$ where $\phi_k(t) = 2y_k t - t^2$ is a 1-variable L -Lipschitz scalar function with $L = 2(1 + d)$ over $t \in [-d, d]$. Applying Talagrand's contraction lemma (Lemma 26.9 of [62]) to $\widehat{\text{Rad}}_S(\mathcal{G}_1) = \mathbb{E}_\sigma \left[\sup_{b \in \mathcal{R}} \frac{1}{n} \sum_{k=1}^n \sigma_k \phi_k(b^\top x_k) \right]$ factors out the Lipschitz constant L

$$\widehat{\text{Rad}}_S(\mathcal{G}_1) \leq 2(1 + d) \widehat{\text{Rad}}_S(\mathcal{R}) \leq 2(1 + d) \frac{d}{\sqrt{n}} = O\left(\frac{d^2}{\sqrt{n}}\right),$$

where $\widehat{\text{Rad}}_S(\mathcal{R}) \leq \frac{d}{\sqrt{n}}$ is the linear predictor bound (Lemma 26.10 of [62]).

For the indicator class $\mathcal{K}$, each function $\mathbb{I}\{F_b(u, x) > 0\}$ is a thresholded quadratic polynomial $F_b(u, x) = 2u(b^\top x) - (b^\top x)^2$ in variables $(u, x) \in \mathbb{R}^{d+1}$. The vector space V of

degree-2 polynomials in $d + 1$ variables has dimension $\dim(V) = \frac{(d+3)(d+2)}{2} = O(d^2)$. By Theorem 3.5 of [63] (VC-dimension of thresholded vector spaces of functions), $\text{VC}(\mathcal{K}) \leq \dim(V) = \frac{(d+3)(d+2)}{2} = O(d^2)$. For sample $S = (z_1, \dots, z_n)$, the finite set of binary evaluation vectors $A_S = \{(k(z_1), \dots, k(z_n)) \in \{0, 1\}^n : k \in \mathcal{K}\}$ has size bounded by the growth function $\tau_{\mathcal{K}}(n) := \max_{S'} |A_{S'}|$. By the Sauer-Shelah-Perles Lemma (Lemma 6.10 of [62]), $|A_S| \leq \tau_{\mathcal{K}}(n) \leq (en/\text{VC}(\mathcal{K}))^{\text{VC}(\mathcal{K})}$. Since each binary vector $a \in A_S$ has L_2 -norm $\|a\|_2 = \sqrt{\sum_{i=1}^n a_i^2} \leq \sqrt{n}$, applying Massart's Finite Class Lemma (Lemma 26.8 of [62]) to A_S yields

$$\begin{aligned} \widehat{\text{Rad}}_S(\mathcal{K}) &= \widehat{\text{Rad}}_S(A_S) = \mathbb{E}_\sigma \left[\frac{1}{n} \sup_{a \in A_S} \sum_{i=1}^n \sigma_i a_i \right] \leq \frac{\sqrt{n} \sqrt{2 \log |A_S|}}{n} \\ &\leq \sqrt{\frac{2 \text{VC}(\mathcal{K}) \log(en/\text{VC}(\mathcal{K}))}{n}} \leq \sqrt{\frac{2 \text{VC}(\mathcal{K}) \log(en)}{n}} = O\left(\frac{d \sqrt{\log n}}{\sqrt{n}}\right). \end{aligned}$$

Combining both bounds gives $\widehat{\text{Rad}}_S(\mathcal{H}) \leq d(d+2)O\left(\frac{d \sqrt{\log n}}{\sqrt{n}}\right) + O\left(\frac{d^2}{\sqrt{n}}\right) = O\left(\frac{d^3 \sqrt{\log n}}{\sqrt{n}}\right)$, and taking expectation yields $\text{Rad}_n(\mathcal{H}) = O\left(\frac{d^3 \sqrt{\log n}}{\sqrt{n}}\right)$.

Finally, applying Theorem 4.13 to class $\mathcal{H}$ (bounded by $\|h_b\|_\infty \leq d(d+2)$), with probability at least $1 - \delta/2$

$$\begin{aligned} \sup_{b \in \mathcal{R}} |\mathbb{E}[h_b] - \widehat{\mathbb{E}}_S[h_b]| &\leq 2\text{Rad}_n(\mathcal{H}) + d(d+2) \sqrt{\frac{\log(4/\delta)}{2n}} \\ &= O\left(\frac{d^3 \sqrt{\log n}}{\sqrt{n}} + d^2 \sqrt{\frac{\log(1/\delta)}{n}}\right). \end{aligned}$$

Summing these bounds via a union bound completes the proof. $\square$

We now present the proof of Theorem 4.9.

Proof. (Proof of Theorem 4.9) Let $\Delta(\mathcal{E}) = \min_{b \in \mathcal{R}} R_V(b) - \min_{b \in \mathcal{R}} R_S(b)$ and $\widehat{\Delta}(D_n) = \min_{b \in \mathcal{R}} \widehat{R}_V(b) - \min_{b \in \mathcal{R}} \widehat{R}_S(b)$. We bound the difference using

$$\begin{aligned} \left| \min_{b \in \mathcal{R}} f(b) - \min_{b \in \mathcal{R}} g(b) \right| &\leq \sup_{b \in \mathcal{R}} |f(b) - g(b)|. \\ |\Delta(\mathcal{E}) - \widehat{\Delta}(D_n)| &\leq \left| \min_{b \in \mathcal{R}} R_V(b) - \min_{b \in \mathcal{R}} \widehat{R}_V(b) \right| + \left| \min_{b \in \mathcal{R}} R_S(b) - \min_{b \in \mathcal{R}} \widehat{R}_S(b) \right| \\ &\leq \sup_{b \in \mathcal{R}} |R_V(b) - \widehat{R}_V(b)| + \sup_{b \in \mathcal{R}} |R_S(b) - \widehat{R}_S(b)|. \end{aligned}$$

Using a union bound, both generalization events hold simultaneously with probability at least $1 - \delta$. Summing the bounds yields the advantage generalization bound. $\square$

4.7.4 Proofs for Online Learning

In this subsection, we provide the proofs for the online learning problem.

We now present the proof of Lemma 4.11.

Proof. (Proof of Lemma 4.11) We establish Sobolev space membership by showing that both the expected vanilla risk $R_V(b)$ and strategic risk $R_S(b)$ are infinitely differentiable (C^∞) on the compact domain $\mathcal{R} \subset \mathbb{R}^d$, which would imply $L(m, b)$ is infinite differentiable on compact domain which further implies their Sobolev norms of any order are finite.

Vanilla Risk $R_V(b)$: Recall the formula for the vanilla risk

$$R_V(b) = \text{Var}(Y) - (1 - q)\text{Var}(b_V^\top X) + (1 - q)\text{Var}((b - b_V)^\top X).$$

Since $R_V(b)$ is a quadratic polynomial in b , all its partial derivatives of order $|\alpha| \geq 3$ vanish identically. Thus, $R_V(b) \in C^\infty(\mathcal{R})$.

Strategic Risk $R_S(b)$ Smoothness on $\mathcal{R}$: Recall the strategic risk formula:

$$R_S(b) = \text{Var}(Y) - (1 - q)\mathbb{E}[G_b(X, Y)\mathbb{I}\{F_b(U, X) > 0\}],$$

where $F_b(u, x) = (2u - b^\top x)b^\top x$ and $G_b(x, y) = 2yb^\top x - (b^\top x)^2$. We show that $I(b) := \mathbb{E}[G_b(X, Y)\mathbb{I}\{F_b(U, X) > 0\}] \in C^\infty(\mathcal{R})$.

For any $b \in \mathcal{R}$ with $\|b\|_2 \geq b_{\min} > 0$, perform the elementary change of variables $v = u - \frac{1}{2}b^\top x$. The revelation condition $(2u - b^\top x)b^\top x > 0$ simplifies to $v(b^\top x) > 0$. Let $\hat{b} = b/\|b\|_2$, and let $R_b \in \text{SO}(d)$ be a smooth local rotation mapping $\hat{b}$ to the first standard basis vector e_1 . In rotated coordinates $z = R_b x$ (so $x = R_b^\top z$), the first coordinate satisfies $z_1 = \hat{b}^\top x$, which implies $b^\top x = \|b\|_2 z_1$.

Because $\|b\|_2 > 0$, the revelation condition $v(b^\top x) > 0$ is equivalent to $v z_1 > 0$, which defines the fixed region

$$\Omega_0 := \{(v, z_1) \in \mathbb{R}^2 : v z_1 > 0\} \times \mathbb{R}^{d-1} \times \mathbb{R}.$$

Crucially, the domain of integration Ω_0 has zero dependence on b . Since the Jacobian of the transformation $(u, x, y) \mapsto (v, z, y)$ is identically 1, change of variables yields

$$I(b) = \int_{\Omega_0} G_b(R_b^\top z, y) p_{U, X, Y} \left(v + \frac{1}{2}\|b\|_2 z_1, R_b^\top z, y \right) dv dz dy.$$

Because $p_{U,X,Y}$ is C^∞ with compact support, G_b is a polynomial in (z, y, b) , and the mapping $b \mapsto (\|b\|_2, R_b)$ is smooth for all $\|b\|_2 \geq b_{\min} > 0$, the integrand is an infinitely differentiable function of b whose partial derivatives of all orders are uniformly bounded on the fixed integration domain Ω_0 . Applying the standard Leibniz integral rule (dominated convergence theorem) directly justifies differentiating under the integral sign to any order k . Therefore, $I(b) \in C^\infty(\mathcal{R})$, and consequently $R_S(b) \in C^\infty(\mathcal{R})$.

Sobolev Space Membership: Since $\mathcal{R} \subset \mathbb{R}^d$ is compact and both $R_V(b), R_S(b) \in C^\infty(\mathcal{R})$, every partial derivative $D^\alpha L(m, b)$ is continuous on the compact set $\mathcal{R}$ and therefore bounded: $\sup_{b \in \mathcal{R}} |D^\alpha L(m, b)| \leq M_\alpha < \infty$. Consequently, the Sobolev norm of order $m_0 = 5/2 + d/2$ is finite:

$$\begin{aligned} \|L(m, \cdot)\|_{H^{5/2+d/2}(\mathcal{R})}^2 &= \sum_{|\alpha| \leq \lceil 5/2+d/2 \rceil} \int_{\mathcal{R}} |D^\alpha L(m, b)|^2 db \\ &\leq \text{Vol}(\mathcal{R}) \sum_{|\alpha| \leq \lceil 5/2+d/2 \rceil} M_\alpha^2 < \infty. \end{aligned}$$

Thus, $L(m, b) \in H^{5/2+d/2}(\mathcal{R})$. □

We now present the proof of Theorem 4.12.

Proof. (Proof of Theorem 4.12) The proof proceeds by verifying the premises required to apply the kernelized Thompson Sampling regret bound of [60] (stated as Theorem 4.15) to our strategic setting over $\mathcal{A} = \{V, S\} \times \mathcal{R}$.

1. Noise Sub-Gaussianity: Because the domain of features, outcomes, and decision vectors is bounded ($\|X\|_2 \leq X_0, |Y| \leq Y_0, \|b\|_2 \leq 1$), the realized loss $Z_t = \ell(m_t, b_t, X_t, Y_t, U_t)$ is uniformly bounded in a finite interval $[0, M_0]$. For any joint action $a_t = (m_t, b_t)$, both Z_t and its conditional expectation $\mathbb{E}[Z_t | a_t]$ are bounded in $[0, M_0]$. By Hoeffding's Lemma, the zero-mean observation noise $\eta_t = Z_t - \mathbb{E}[Z_t | a_t]$ is bounded in $[-M_0, M_0]$ and thus conditionally R -sub-Gaussian with $R = M_0/2$.

2. RKHS Membership and Direct-Sum Norm Bound: By Lemma 4.11, the expected risk functions under each regime belong to the Sobolev space $H^{5/2+d/2}(\mathcal{R})$, which is norm-equivalent to the RKHS $\mathcal{H}_{k_0}(\mathcal{R})$ of the Matérn-5/2 kernel: $\|R_V\|_{\mathcal{H}_{k_0}} \leq B_{k,V} < \infty$ and $\|R_S\|_{\mathcal{H}_{k_0}} \leq B_{k,S} < \infty$. For the direct-sum kernel $k((m, b), (m', b')) = \mathbb{I}\{m = m'\}k_0(b, b')$, the joint RKHS decomposes orthogonally as $\mathcal{H}_k = \mathcal{H}_{k_0} \oplus \mathcal{H}_{k_0}$. Consequently, the joint objective $f(m, b) = \mathbb{I}\{m = V\}R_V(b) +$

$\mathbb{I}\{m = S\}R_S(b)$ belongs to $\mathcal{H}_k(\mathcal{A})$ with bounded joint RKHS norm

$$\|f\|_{\mathcal{H}_k} = \sqrt{\|R_V\|_{\mathcal{H}_{k_0}}^2 + \|R_S\|_{\mathcal{H}_{k_0}}^2} \leq \sqrt{B_{k,V}^2 + B_{k,S}^2} \leq \sqrt{2} \max\{B_{k,V}, B_{k,S}\} =: B_k < \infty.$$

3. Lipschitz Regularity and Action Discretization Net: For the Matérn-5/2 kernel, k_0 is twice continuously differentiable on the compact set $\mathcal{R}$, so $L := \sup_{b \in \mathcal{R}} \max_{j \in [d]} \left(\left. \frac{\partial^2 k_0(p, q)}{\partial p_j \partial q_j} \right|_{p=q=b} \right)^{1/2} < \infty$. By standard RKHS properties [60], any $f \in \mathcal{H}_k$ is Lipschitz with constant $L_f \leq B_k L$. Evaluating candidate actions over the round- t decision set $\mathcal{A}_t = \{V, S\} \times \mathcal{R}_t$, where $\mathcal{R}_t$ is an ϵ_t -net of $\mathcal{R}$ with mesh size $\epsilon_t \leq \frac{1}{B_k L t^2}$, guarantees that $|f(a) - f([a]_t)| \leq B_k L \epsilon_t \leq 1/t^2$ for all $a \in \mathcal{A}$, satisfying the exact decision-set premise of Theorem 4.15. The cumulative discretization error across all T rounds is bounded by $\sum_{t=1}^T 1/t^2 \leq \pi^2/6 = O(1)$.

4. Regret Derivation via Information Gain: Applying Theorem 4.15 with sub-Gaussian noise parameter R , RKHS norm bound B_k , and round- t decision net $\mathcal{A}_t$ guarantees that, with probability at least $1 - \delta$, the cumulative regret satisfies

$$\text{Reg}_T = O\left(d^{1/2} \gamma_T \sqrt{T \log(dT/\delta)}\right).$$

For the direct-sum Matérn-5/2 kernel across the two regimes, the maximum information gain satisfies $\gamma_T \leq 2\gamma_{T,\mathcal{R}} = O\left(T^{\frac{d}{5+d}} (\log T)^{\frac{5}{5+d}}\right)$ [61]. Substituting this rate yields

$$\text{Reg}_T = O\left(d^{1/2} T^{\frac{5+3d}{10+2d}} (\log T)^{\frac{5}{5+d}} \sqrt{\log(dT/\delta)}\right) = \tilde{O}\left(T^{\frac{5+3d}{10+2d}}\right).$$

For any dimension $d \in \{1, 2, 3, 4\}$, the exponent $\frac{5+3d}{10+2d} < 1$, guaranteeing sublinear cumulative regret $\text{Reg}_T = o(T)$. $\square$

Chapter 5

Strategic IPO Regulation under Complete and Incomplete Information

In this chapter, we study the strategic interaction between a financial regulator, responsible for balancing market growth and investor protection, and an issuing firm preparing for an initial public offering (IPO). The regulator prescribes an enforcement tolerance level ε , which bounds the budget by which the firm can window-dress its fundamental feature vector into a presented feature vector while jointly choosing the presentation and share price. The market consists of $k \geq 2$ heterogeneous investor subpopulations with distinct personal valuation functions over the presented features. While the firm seeks to maximize total capital raised, the regulator aims to maximize social welfare by facilitating capital raising by viable firms while curbing misallocations to non-viable firms. We model this interaction as a Stackelberg screening game with the regulator as the leader and the firm as the follower. We analyze this game under two distinct information structures: the *Decision Problem*, in which the feature-valuation distributions are known to the regulator, and the *Learning Problem*, in which these distributions are unknown. In the Decision Problem, we characterize the structure of the firm's optimal pricing and window-dressing decisions and show, through an example, that the optimal regulatory enforcement level can be attained at an interior point. In the Learning Problem, we show that a regulator using a Gaussian Process Thompson Sampling (GP-TS) algorithm with a Matérn-5/2 kernel learns the optimal enforcement policy with a sublinear cumulative regret bound of $\tilde{O}(T^{2/3})$.

5.1 Introduction

Initial Public Offerings (IPOs) serve as a crucial mechanism for private companies to raise public capital and foster financial market growth. However, significant information asymmetries exist between issuing firms and prospective retail investors. Issuing firms frequently possess incentives to window-dress or strategically modify their presented financial metrics and business features to project a favorable valuation. To prevent investor exploitation and curb capital misallocation to unviable ventures, financial regulatory bodies mandate reporting standards and audit tolerance. Yet, overly stringent regulatory oversight can impose prohibitive compliance burdens, deterring viable firms from entering the capital market. Finding an optimal balance between market growth and investor protection is therefore a fundamental challenge in regulatory economics.

We model this interaction between a welfare-maximizing regulator and a capital-maximizing issuing firm, where the regulator sets an enforcement tolerance limit ε that constrains the Euclidean distance by which the firm can window-dress its fundamental feature vector x into a presented feature vector z , while jointly choosing the presentation and share price. The market consists of heterogeneous investor subpopulations with distinct personal valuation profiles over presented features. The firm seeks to maximize the total capital raised, whereas the regulator seeks to maximize social welfare by facilitating capital raising by viable firms while curbing misallocations to non-viable firms. We formulate this interaction as a Stackelberg screening game, with the regulator as the leader and the issuing firm as the follower.

5.1.1 Contributions

We analyze this regulatory game under two distinct information paradigms: the *Decision Problem* (where feature-valuation distributions are known to the regulator) and the *Learning Problem* (where distributions are unobserved and must be learned through online market interactions).

For the Decision Problem, we prove that for any fixed enforcement level ε , the firm's optimal presentation and pricing strategy reduces to solving $2^k - 1$ constrained convex optimization subproblems corresponding to candidate buyer subsets. We further show that expected social welfare need not be monotone in enforcement tolerance. In particular, an overly strict regime ($\varepsilon = 0$) can prevent viable firms with low perceived valuations at their baseline features from

raising sufficient capital, whereas an overly lax regime ($\varepsilon = \infty$) can enable non-viable firms to mislead investors. An intermediate enforcement level can therefore allow viable firms with low perceived valuations to improve their market prospects while limiting the ability of non-viable firms to mislead investors.

For the Learning Problem, the regulator must learn the optimal enforcement tolerance over a finite time horizon T in a continuous-armed bandit setting. Under market distributions with smooth probability density function with compact support, we show that the expected social welfare function resides in the Reproducing Kernel Hilbert Space (RKHS) associated with a Matérn- $\frac{5}{2}$ kernel. Consequently, employing the established Gaussian Process Thompson Sampling (GP-TS) algorithm guarantees a sublinear cumulative regret of $\tilde{O}(T^{2/3})$, proving that optimal regulatory oversight can be learned efficiently online.

5.1.2 Related Work

This chapter contributes to the literature on strategic decision making and regulatory policy, bridging strategic feature modification, market segmentation, IPO pricing, and online bandit learning.

Strategic Feature Modification and Information Design. Strategic feature modification is studied in [3, 9, 12, 57, 64] in screening games where agents modify features or withhold data under decision rules, while information design [65, 66] explores strategic disclosure. While our prior work [12] studied strategic feature revelation where verifiable features are selectively withheld, here the issuing firm actively manipulates (window-dresses) continuous features within a regulatory Euclidean ball.

Market Segmentation and IPO Pricing. Classical market segmentation [67] studies price discrimination across buyer groups. In our model, although the firm posts a single uniform price, internal subpopulation valuations are used to strategically target investor groups via feature presentation. Traditional IPO literature [68, 69] tracks investor surplus and firm capital under market frictions and information asymmetries; we evaluate analogous welfare and capital objectives, but with an issuing firm that strategically window-dresses features under regulatory enforcement limits.

Online Continuum-Armed and GP Bandits. Online bandit literature [59–61, 70–73] es-

establishes regret bounds under regularity (RKHS) conditions. In our learning setup, we show that market distributions with smooth probability density functions over compact support satisfy these regularity conditions, guaranteeing the $\tilde{O}(T^{2/3})$ regret bound. This chapter is derived from our manuscript [15]. To the best of our knowledge, the problem of regulating strategic window dressing under heterogeneous market valuations across decision and online bandit settings has not been previously studied, and was first explored in our work [15].

5.2 Problem Formulation

We consider a financial market consisting of a regulator, an issuing firm preparing for an initial public offering (IPO), and a population of prospective retail investors. We model the interaction between the regulator and the issuing firm as a Stackelberg screening game, with the regulator acting as the leader and the issuing firm as the follower. Let $\mathcal{X} \subset \mathbb{R}^d$ be a compact, convex feature space representing fundamental business and financial attributes of the firm, and let X be a random variable defined on $\mathcal{X}$. The total investor population $\mathcal{N}$ is partitioned into $k \geq 2$ heterogeneous investor subpopulations $\{N_j\}_{j=1}^k$ of random sizes such that $\sum_{j=1}^k N_j = \mathcal{N}$. Each subpopulation $j \in \{1, \dots, k\}$ possesses a bounded, concave valuation function $V_j(z)$ over presented features $z \in \mathcal{X}$. The true stock valuation $Y(X)$ is a random variable given by $Y(X) = V_j(X)$ with probability $\gamma_j(X)$, where $\sum_{j=1}^k \gamma_j(X) = 1$ for all $X \in \mathcal{X}$.

Let $\mathcal{R} := [0, \infty)$ denote the domain of enforcement tolerance levels available to the regulator. Given an enforcement level $\varepsilon \in \mathcal{R}$, the firm's feasible presentation space is bounded within an ε -ball around its true feature vector, $\mathcal{F}_Z(x, \varepsilon) := \{z \in \mathcal{X} : \|x - z\|_2 \leq \varepsilon\}$, while its pricing space is $\mathcal{F}_P := \mathbb{R}_+$. A strategy profile $(\varepsilon, \mathcal{Z}, \mathcal{P})$ maps each feature-enforcement pair $(x, \varepsilon) \in \mathcal{X} \times \mathcal{R}$ to a presentation choice $\mathcal{Z}(x, \varepsilon) \in \mathcal{F}_Z(x, \varepsilon)$ and a posted price $\mathcal{P}(x, \varepsilon) \in \mathcal{F}_P$.

The issuing firm offers a total share supply of $S = \eta\mathcal{N}$, where $\eta \in (0, 1]$ represents the supply ratio. Given a presented feature $Z = \mathcal{Z}(X, \varepsilon)$ and posted price $P = \mathcal{P}(X, \varepsilon)$, investors in subpopulation j purchase a share if and only if their personal valuation satisfies $V_j(Z) \geq P$. The aggregate market demand is given by $D(Z, P, \{N_j\}) = \sum_{j=1}^k N_j \mathbf{1}\{V_j(Z) \geq P\}$. Because total supply is capped at $\eta\mathcal{N}$, in the event of demand oversubscription, shares are allocated uniformly among interested buyers. The actual traded volume is given by $Q(Z, P, \{N_j\}, \varepsilon) := \min\{\eta\mathcal{N}, D(Z, P, \{N_j\})\}$. Total capital raised by the firm and surplus realized by investors are

$\Pi_F = P \cdot Q$ and $\Pi_I = (Y(X) - P) \cdot Q$ respectively. The realized per-capita social welfare is defined as

$$W(X, Y, \{N_j\}, \varepsilon; Z, P) := \frac{\Pi_F + \Pi_I}{N} = \frac{Q(Z, P, \{N_j\}, \varepsilon) Y(X)}{N}. \quad (5.1)$$

The sequence of events in the market proceeds chronologically as follows

- (i) *Regulatory Commitment*: The regulator publicly commits to an enforcement tolerance limit $\varepsilon \in \mathcal{R}$.
- (ii) *Nature's Move*: Nature samples the firm's true feature $X \sim P_X$, subpopulation sizes $\{N_j\}_{j=1}^k$, and valuation realization $Y(X) = V_j(X)$ with probability $\gamma_j(X)$.
- (iii) *Firm Strategy*: The firm observes baseline feature X and regulatory limit ε , choosing an optimal presentation $Z = \mathcal{Z}(X, \varepsilon) \in \mathcal{F}_Z(X, \varepsilon)$ and price $P = \mathcal{P}(X, \varepsilon) \in \mathcal{F}_P$.
- (iv) *Market Clearing & Welfare*: Investors evaluate $V_j(Z)$ and buy if $V_j(Z) \geq P$. Traded volume $Q = \min\{\eta N, D\}$ yields firm revenue Π_F and investor surplus Π_I , generating realized social welfare W .

We examine this formulation under two distinct regulatory information structures: the *Decision Problem* and the *Learning Problem*.

5.2.1 Decision Problem

In the Decision Problem, the subpopulation valuation functions $\{V_j(x)\}_{j=1}^k$ and the joint probability distribution of $(X, Y, \{N_j\})$ are common knowledge. The regulator knows the joint distribution but not the realized instance $(X, Y, \{N_j\})$, whereas the firm observes its realized baseline feature $X = x$.

For any regulatory enforcement action $\varepsilon \in \mathcal{R}$, the firm selects a presentation and pricing strategy to maximize its expected capital raised at realization $X = x$. This *best response* strategy solves

$$\begin{aligned} & (\mathcal{Z}^*(x, \varepsilon), \mathcal{P}^*(x, \varepsilon)) \\ & \in \arg \max_{z \in \mathcal{F}_Z(x, \varepsilon), p \geq 0} \mathbb{E}_{\{N_j\}} [\Pi_F(z, p, \{N_j\}, \varepsilon)]. \end{aligned} \quad (5.2)$$

The regulator seeks to maximize overall expected social welfare. Anticipating the firm's optimal best response $(\mathcal{Z}^*, \mathcal{P}^*)$ for any enforcement tolerance ε , the regulator solves the Stack-

elberg optimization problem

$$\varepsilon^* \in \arg \max_{\varepsilon \in \mathcal{R}} \mathbb{E}_{X, \{N_j\}, Y} \left[W(X, Y, \{N_j\}, \varepsilon; \mathcal{Z}^*(X, \varepsilon), \mathcal{P}^*(X, \varepsilon)) \right]. \quad (5.3)$$

The resulting strategy tuple $(\varepsilon^*, \mathcal{Z}^*, \mathcal{P}^*)$ constitutes the *Stackelberg equilibrium* of the game.

Let

$$\mathcal{W}(\varepsilon) := \mathbb{E}_{X, \{N_j\}, Y} \left[W(X, Y, \{N_j\}, \varepsilon; \mathcal{Z}^*(X, \varepsilon), \mathcal{P}^*(X, \varepsilon)) \right] \quad (5.4)$$

denote the expected social welfare objective function in (5.3). To guarantee that $\mathcal{W}(\varepsilon)$ exhibits regular structural properties, we impose the following regularity condition on the market distribution.

Assumption 5.1 (Regularity of Market Distribution). The baseline feature vector X has a smooth, bounded probability density function $f_X(x)$ supported on the compact convex domain $\mathcal{X} \subset \mathbb{R}^d$, the discrete valuation types $Y(X) \in \{V_1(X), \dots, V_k(X)\}$ are governed by smooth conditional probabilities $\gamma_j(x)$, and the resulting expected social welfare function $\mathcal{W}(\varepsilon)$ belongs to the RKHS $\mathcal{H}_k([0, \varepsilon_{\max}])$.

Remark 5.2 (Continuous Density on the Subpopulation Simplex). For tractability, we model the realized subpopulation sizes $(N_1, \dots, N_k)$ by first drawing $(\pi_1, \dots, \pi_k)$ from a continuous distribution over the positive orthant $\mathbb{R}_+^k$ and then setting

$$N_j = \frac{\pi_j}{\sum_{i=1}^k \pi_i} \mathcal{N}, \quad j = 1, \dots, k,$$

so that $\sum_{j=1}^k N_j = \mathcal{N}$.

5.2.2 Learning Problem

In the Learning Problem, the underlying joint distribution of (X, Y) is unknown to the regulator. To preserve analytical tractability, we consider $k = 2$ subpopulations with deterministic, equal subpopulation sizes $N_j = \mathcal{N}/2$ almost surely. The concave valuation functions $\{V_j\}_{j=1}^k$ remain common knowledge. The firm knows the marginal distribution of X and observes its realized feature X_t at each round t , whereas the regulator must learn the optimal enforcement tolerance ε^* through repeated market interactions over a finite horizon T . Each round consists of the

regulator prescribing tolerance ε_t , the firm choosing best response $(\mathcal{Z}_t^*, \mathcal{P}_t^*)$, and the regulator observing the realized social welfare W_t .

A *learning algorithm* for the regulator is a sequence of decision rules $\mathcal{A} = \{\mathcal{A}_t\}_{t=1}^T$, where $\mathcal{A}_t : \mathcal{H}_{t-1} \rightarrow \mathcal{R}$ maps historical interactions $\mathcal{H}_{t-1} = \{(\varepsilon_\tau, W_\tau)\}_{\tau=1}^{t-1}$ to enforcement level ε_t . The performance of algorithm $\mathcal{A}$ over horizon T is evaluated by its cumulative pseudo-regret:

$$R_T := \sum_{t=1}^T (\mathcal{W}(\varepsilon^*) - \mathcal{W}(\varepsilon_t)), \quad (5.5)$$

and its realized regret:

$$\text{Reg}_{\text{real}}(T) := R_T - \sum_{t=1}^T \xi_t, \quad (5.6)$$

where $\xi_t = W_t - \mathcal{W}(\varepsilon_t)$ is the zero-mean observation noise. A learning algorithm $\mathcal{A}$ achieves *sublinear pseudo-regret* if $R_T = o(T)$.

5.3 Decision Problem

In this section, we derive structural results characterizing the Stackelberg equilibrium strategy profile $(\varepsilon^*, \mathcal{Z}^*, \mathcal{P}^*)$ under complete information. We establish that finding the firm's optimal response reduces to a finite set of tractable convex optimization subproblems, and show how the regulator optimizes social welfare over the enforcement domain.

5.3.1 Firm's Optimal Strategy

We first characterize the firm's optimal pricing strategy for any given presentation feature $z \in \mathcal{X}$.

Lemma 5.3: *For a fixed baseline feature $x \in \mathcal{X}$, regulatory enforcement tolerance $\varepsilon \in \mathcal{R}$, and chosen presentation feature $z \in \mathcal{F}_Z(x, \varepsilon)$, the firm's optimal non-negative pricing strategy satisfies $p^*(z) \in \{0\} \cup \{V_j(z) : V_j(z) \geq 0\}$. If all investor valuations are strictly negative ($V_j(z) < 0$ for all j), no trade occurs and the firm earns zero capital.*

Proof. The proof is given in Section 5.6 (Proof 5.6.1). □

Leveraging this structural property of the optimal price, we now characterize the firm's joint presentation and pricing strategy by targeting candidate buyer subpopulations.

Theorem 5.4: *Assume that the subpopulation valuation functions $V_j(z)$ are concave for all $j \in \{1, \dots, k\}$. For any baseline feature $x \in X$ and enforcement tolerance $\varepsilon \in \mathcal{R}$, the firm's optimal presentation strategy $\mathcal{Z}^*(x, \varepsilon)$ is determined by solving at most $2^k - 1$ constrained convex optimization subproblems (corresponding to non-empty candidate buyer subsets). For fixed small k , this yields an exact and tractable reduction to convex programming.*

Proof. The proof is given in Section 5.6 (Proof 5.6.1). □

5.3.2 Regulator's Optimization and Social Welfare Smoothness

To determine the optimal enforcement tolerance ε^* , the regulator must anticipate the firm's best-response policy $(\mathcal{Z}^*(X, \varepsilon), \mathcal{P}^*(X, \varepsilon))$ characterized by Theorem 5.4. We first establish structural properties of the resulting expected social welfare function $\mathcal{W}(\varepsilon)$.

First, we show that social welfare saturates beyond a finite enforcement threshold, effectively compactifying the relevant domain.

Proposition 5.5: *For the regulatory enforcement domain $\mathcal{R} = [0, \infty)$, expected social welfare $\mathcal{W}(\varepsilon)$ becomes constant for all ε beyond a finite saturation threshold $\varepsilon_{\max} < \infty$. Consequently, optimizing $\mathcal{W}(\varepsilon)$ over $\mathcal{R}$ is equivalent to optimizing over the compact domain $[0, \varepsilon_{\max}]$.*

Proof. The proof is given in Section 5.6 (Proof 5.6.1). □

Having established that the effective enforcement domain is compact, we characterize the regulator's optimization problem over expected social welfare $\mathcal{W}(\varepsilon)$.

Regulator's Optimization and Tractability Anticipating the firm's optimal best-response policy $(\mathcal{Z}^*(X, \varepsilon), \mathcal{P}^*(X, \varepsilon))$ from Theorem 5.4, the regulator determines the optimal enforcement tolerance by solving:

$$\varepsilon^* \in \arg \max_{\varepsilon \in [0, \varepsilon_{\max}]} \mathcal{W}(\varepsilon). \quad (5.7)$$

Because $\mathcal{W}(\varepsilon)$ is smooth and the domain $[0, \varepsilon_{\max}]$ is compact, finding the global optimum ε^* reduces to maximizing a continuous, smooth objective over a compact interval. For any specified approximation tolerance $\delta > 0$, compactness guarantees that the domain $[0, \varepsilon_{\max}]$ can be covered by a finite number of δ -balls (or evaluated on a finite grid of mesh size δ), ensuring that ε^* can be efficiently computed to within tolerance δ using only finitely many evaluations.

5.3.3 Illustrative Example: Finding optimal regulation for affine valuation functions

To characterize optimal regulatory policy, we consider an instance of the IPO problem with $k = 2$ investor subpopulations, denoted by α and β . The market consists of equal population shares ($\pi_\alpha = \pi_\beta = 1/2$), so that $N_\alpha = N_\beta = \mathcal{N}/2$. The feature space $\mathcal{X} \subseteq \mathbb{R}^d$ is convex and compact, with 0 in its interior. Baseline feature vectors $X \in \mathcal{X}$ are distributed according to a smooth probability density function $f_X(x)$. The personal valuation functions for subpopulations α and β are:

$$V_\alpha(z) = w_\alpha^\top z + b_\alpha \quad \text{and} \quad V_\beta(z) = w_\beta^\top z + b_\beta \quad (5.8)$$

respectively. Conditional on baseline realization $X = x$, the true asset value per share $Y \in \mathbb{R}$ realizes as $V_\alpha(x)$ with probability $\gamma(x) \in [0, 1]$ and as $V_\beta(x)$ with probability $1 - \gamma(x)$. Furthermore we assume $\eta = 1$, i.e., total number of shares available is $\mathcal{N}$. Let

$$\mu(x) := \mathbb{E}[Y \mid X = x] = \gamma(x)V_\alpha(x) + (1 - \gamma(x))V_\beta(x). \quad (5.9)$$

Given regulatory enforcement tolerance $\varepsilon \in [0, \varepsilon_{\max}]$ and baseline feature realization $x \in \mathcal{X}$, the issuing firm selects a presentation vector $z = x + \Delta \in \mathcal{F}_Z(x, \varepsilon) = \{z \in \mathbb{R}^d : \|\Delta\|_2 \leq \varepsilon\}$ and posted price $P \geq 0$. From Theorem 5.4, the firm identifies its optimal response by maximizing capital raised $\Pi_F = P \cdot Q$ across the $2^2 - 1 = 3$ possibilities:

- (i) Targeting Subpopulation α : To maximize the perceived valuation of subpopulation α , the firm chooses its manipulation vector with $\Delta_\alpha^* = \varepsilon \frac{w_\alpha}{\|w_\alpha\|_2}$. The optimal posted price is

$$P_\alpha^*(x, \varepsilon) = V_\alpha(x + \Delta_\alpha^*) = w_\alpha^\top x + b_\alpha + \varepsilon \|w_\alpha\|_2. \quad (5.10)$$

At this price, only subpopulation α purchases shares, yielding traded volume $Q_\alpha = N_\alpha = \mathcal{N}/2$. The firm's private capital raised is:

$$\Pi_{F,\alpha}(x, \varepsilon) = \frac{\mathcal{N}}{2} (w_\alpha^\top x + b_\alpha + \varepsilon \|w_\alpha\|_2). \quad (5.11)$$

Investors in subpopulation α earn a profit of $\Pi_I = Q_\alpha(Y(x) - P_\alpha^*)$. The realized per-capita social welfare is $\frac{\Pi_{F,\alpha} + \Pi_I}{\mathcal{N}} = \frac{Q_\alpha Y(x)}{\mathcal{N}} = \frac{1}{2}Y(x)$ and the expected social welfare is $\mathbb{E}[\frac{\mu(x)}{2}]$.

- (ii) Targeting Subpopulation β : Just as in for targetting subpopulation α , the firm sets its manipulation vector with $\Delta_\beta^* = \varepsilon \frac{w_\beta}{\|w_\beta\|_2}$, and posts price

$$P_\beta^*(x, \varepsilon) = w_\beta^\top x + b_\beta + \varepsilon \|w_\beta\|_2. \quad (5.12)$$

Traded volume is $Q_\beta = N_\beta = \mathcal{N}/2$, and the firm raises capital:

$$\Pi_{F,\beta}(x, \varepsilon) = \frac{\mathcal{N}}{2} \left(w_\beta^\top x + b_\beta + \varepsilon \|w_\beta\|_2 \right). \quad (5.13)$$

Investors in subpopulation β earn a profit of $\Pi_I = Q_\beta(Y(x) - P_\beta^*)$, yielding conditional expected per-capita social welfare $\mathbb{E}[\frac{\mu(x)}{2}]$.

- (iii) Targeting Both Subpopulations: Since the entire stock of IPO has to be sold ($Q_{\alpha\beta} = N_\alpha + N_\beta = \mathcal{N}$), both subpopulations must be presented with the same manipulated feature vector $z = x + \Delta \in \mathcal{F}_Z(x, \varepsilon)$. The market-clearing price that induces both groups to purchase is constrained by the lower of their two valuations under the common presentation. Thus, the optimal common price is obtained by solving the max–min program:

$$P_{\alpha\beta}^*(x, \varepsilon) = \max_{\|\Delta\|_2 \leq \varepsilon} \min \{V_\alpha(x + \Delta), V_\beta(x + \Delta)\}. \quad (5.14)$$

The firm's capital raised is

$$\Pi_{F,\alpha\beta}(x, \varepsilon) = \mathcal{N} P_{\alpha\beta}^*(x, \varepsilon), \quad (5.15)$$

and the investors' profit is $\Pi_I = \mathcal{N}(Y(x) - P_{\alpha\beta}^*)$. The conditional expected per-capita social welfare is $\frac{\Pi_F + \Pi_I}{\mathcal{N}} = \mu(x)$.

Remark 5.6 (Optimal Common Presentation for Affine Valuations). For affine valuations $V_j(z) = w_j^\top z + b_j$, the max–min problem (5.14) seeks a common manipulation Δ satisfying $\|\Delta\|_2 \leq \varepsilon$ that maximizes $\min\{w_\alpha^\top(x + \Delta) + b_\alpha, w_\beta^\top(x + \Delta) + b_\beta\}$. Posing this as a standard convex program yields the optimal manipulation direction parameterized by a Lagrange multiplier $\lambda \in [0, 1]$:

$$\Delta_{\alpha\beta}^* = \varepsilon \frac{\lambda w_\alpha + (1 - \lambda) w_\beta}{\|\lambda w_\alpha + (1 - \lambda) w_\beta\|_2}. \quad (5.16)$$

When the firm balances both groups ($\lambda = 1/2$), the common manipulation aligns with the angular bisector $\frac{w_\alpha + w_\beta}{\|w_\alpha + w_\beta\|_2}$. Under equal sensitivity norms $\|w_\alpha\|_2 = \|w_\beta\|_2 = w_0$, if $w_\alpha \perp w_\beta$ (orthogonal preferences), the common presentation yields a per-group manipulation gain of $\frac{\varepsilon w_0}{\sqrt{2}}$, reflecting the geometric cost of a shared presentation compared to the separate single-group gain εw_0 .

The firm's optimal targeting rule partitions the fundamental feature space $\mathbb{R}^d$ into three mutually disjoint decision regions:

$$\mathcal{R}_\alpha(\varepsilon) \triangleq \{x \in \mathbb{R}^d : \Pi_{F,\alpha}(x, \varepsilon) > \max(\Pi_{F,\beta}(x, \varepsilon), \Pi_{F,\alpha\beta}(x, \varepsilon))\}, \quad (5.17)$$

$$\mathcal{R}_\beta(\varepsilon) \triangleq \{x \in \mathbb{R}^d : \Pi_{F,\beta}(x, \varepsilon) > \max(\Pi_{F,\alpha}(x, \varepsilon), \Pi_{F,\alpha\beta}(x, \varepsilon))\}, \quad (5.18)$$

$$\mathcal{R}_{\alpha\beta}(\varepsilon) \triangleq \{x \in \mathbb{R}^d : \Pi_{F,\alpha\beta}(x, \varepsilon) \geq \max(\Pi_{F,\alpha}(x, \varepsilon), \Pi_{F,\beta}(x, \varepsilon))\}. \quad (5.19)$$

The expected social welfare is given by:

$$\begin{aligned} \mathcal{W}(\varepsilon) &= \int_{\mathcal{R}_{\alpha}(\varepsilon)} \frac{1}{2} \mu(x) f_X(x) dx + \int_{\mathcal{R}_{\beta}(\varepsilon)} \frac{1}{2} \mu(x) f_X(x) dx + \int_{\mathcal{R}_{\alpha\beta}(\varepsilon)} 1 \cdot \mu(x) f_X(x) dx \\ &= \frac{1}{2} \mathbb{E}[\mu(X)] + \frac{1}{2} \int_{\mathcal{R}_{\alpha\beta}(\varepsilon)} \mu(x) f_X(x) dx. \end{aligned} \quad (5.20)$$

The regulation level ε influences social welfare exclusively through the set $\mathcal{R}_{\alpha\beta}(\varepsilon) = \{x \in \mathbb{R}^d : \Pi_{F,\alpha\beta}(x, \varepsilon) \geq \max(\Pi_{F,\alpha}(x, \varepsilon), \Pi_{F,\beta}(x, \varepsilon))\}$. We now characterise this set into two sectors.

- (a) When $P_\alpha^*(x, \varepsilon) \geq P_\beta^*(x, \varepsilon)$, we have $P_{\alpha\beta}^* = P_\beta^*$. The firm prefers serving both groups over targeting group α alone if and only if $\Pi_{F,\alpha\beta} \geq \Pi_{F,\alpha} \iff \mathcal{N}P_\beta^* \geq \frac{\mathcal{N}}{2}P_\alpha^*$, which yields the linear half-space condition

$$\left(w_\beta^\top x + b_\beta + \varepsilon \|w_\beta\|_2\right) - \frac{1}{2} \left(w_\alpha^\top x + b_\alpha + \varepsilon \|w_\alpha\|_2\right) = a_\alpha^\top x - c_\alpha + \rho_\alpha \varepsilon \geq 0, \quad (5.21)$$

where

$$a_\alpha := w_\beta - \frac{1}{2}w_\alpha \in \mathbb{R}^d, \quad c_\alpha := \frac{1}{2}b_\alpha - b_\beta \in \mathbb{R}, \quad \rho_\alpha := \|w_\beta\|_2 - \frac{1}{2}\|w_\alpha\|_2 \in \mathbb{R}. \quad (5.22)$$

- (b) When $P_\beta^*(x, \varepsilon) > P_\alpha^*(x, \varepsilon)$, we have $P_{\alpha\beta}^* = P_\alpha^*$. The condition $\Pi_{F,\alpha\beta} \geq \Pi_{F,\beta} \iff \mathcal{N}P_\alpha^* \geq \frac{\mathcal{N}}{2}P_\beta^*$ similarly gives

$$\left(w_\alpha^\top x + b_\alpha + \varepsilon \|w_\alpha\|_2\right) - \frac{1}{2} \left(w_\beta^\top x + b_\beta + \varepsilon \|w_\beta\|_2\right) = a_\beta^\top x - c_\beta + \rho_\beta \varepsilon \geq 0, \quad (5.23)$$

where

$$a_\beta := w_\alpha - \frac{1}{2}w_\beta \in \mathbb{R}^d, \quad c_\beta := \frac{1}{2}b_\beta - b_\alpha \in \mathbb{R}, \quad \rho_\beta := \|w_\alpha\|_2 - \frac{1}{2}\|w_\beta\|_2 \in \mathbb{R}. \quad (5.24)$$

Thus $\mathcal{R}_{\alpha\beta}(\varepsilon)$ partitions into two disjoint boundary sectors $\mathcal{R}_{\alpha\beta}(\varepsilon) = \mathcal{R}_{\alpha\beta,\alpha}(\varepsilon) \cup \mathcal{R}_{\alpha\beta,\beta}(\varepsilon)$, where

$$\mathcal{R}_{\alpha\beta,\alpha}(\varepsilon) := \{x \in \mathbb{R}^d : a_\alpha^\top x \geq c_\alpha - \rho_\alpha \varepsilon\} \cap C_\alpha, \quad (5.25)$$

$$\mathcal{R}_{\alpha\beta,\beta}(\varepsilon) := \{x \in \mathbb{R}^d : a_\beta^\top x \geq c_\beta - \rho_\beta \varepsilon\} \cap C_\beta, \quad (5.26)$$

with price-ordering sectors $C_\alpha := \{x \in \mathbb{R}^d : P_\alpha^*(x, \varepsilon) \geq P_\beta^*(x, \varepsilon)\}$ and $C_\beta := \{x \in \mathbb{R}^d : P_\beta^*(x, \varepsilon) > P_\alpha^*(x, \varepsilon)\}$. Under equal biases $b_\alpha = b_\beta = b_0$ and equal sensitivity norms $\|w_\alpha\|_2 =$

$\|w_\beta\|_2 = w_0$, these sectors simplify to $C_\alpha = \{x : (w_\alpha - w_\beta)^\top x \geq 0\}$ and $C_\beta = \{x : (w_\alpha - w_\beta)^\top x < 0\}$, which are static and form a partition of the feature space X .

Let $S_j := a_j^\top X \in \mathbb{R}$ denote the scalar random variable with marginal probability density function $f_{S_j}(s)$. Defining the sector-restricted conditional expected valuation as $\bar{\mu}_j(s) := \mathbb{E}[\mu(X)\mathbf{1}_{C_j}(X) \mid S_j = s]$, the Law of Iterated Expectations yields

$$\int_{\mathcal{R}_{\alpha\beta,j}(\varepsilon)} \mu(x) f_X(x) dx = \mathbb{E}\left[\mu(X)\mathbf{1}_{C_j}(X)\mathbf{1}_{\{S_j \geq c_j - \rho_j \varepsilon\}}\right] = \int_{c_j - \rho_j \varepsilon}^{\infty} \bar{\mu}_j(s) f_{S_j}(s) ds. \quad (5.27)$$

Because $C_\alpha \cap C_\beta = \emptyset$ and $C_\alpha \cup C_\beta = X$, upon full saturation when $c_j - \rho_j \varepsilon \leq s_{\min,j}$ for both sectors, $\sum_{j \in \{\alpha, \beta\}} \int_{-\infty}^{\infty} \bar{\mu}_j(s) f_{S_j}(s) ds = \mathbb{E}[\mu(X)(\mathbf{1}_{C_\alpha}(X) + \mathbf{1}_{C_\beta}(X))] = \mathbb{E}[\mu(X)]$, cleanly verifying the full-sale limit without double counting. Substituting into (5.20), we obtain expected social welfare as

$$\mathcal{W}(\varepsilon) = \frac{1}{2} \mathbb{E}[\mu(X)] + \frac{1}{2} \sum_{j \in \{\alpha, \beta\}} \int_{c_j - \rho_j \varepsilon}^{\infty} \bar{\mu}_j(s) f_{S_j}(s) ds. \quad (5.28)$$

Since ε appears solely in the lower limit of integration, the Fundamental Theorem of Calculus yields

$$\mathcal{W}'(\varepsilon) = \frac{1}{2} \sum_{j \in \{\alpha, \beta\}} \rho_j \cdot \bar{\mu}_j(c_j - \rho_j \varepsilon) \cdot f_{S_j}(c_j - \rho_j \varepsilon), \quad (5.29)$$

$$\mathcal{W}''(\varepsilon) = -\frac{1}{2} \sum_{j \in \{\alpha, \beta\}} \rho_j^2 \left[\bar{\mu}'_j(c_j - \rho_j \varepsilon) f_{S_j}(c_j - \rho_j \varepsilon) + \bar{\mu}_j(c_j - \rho_j \varepsilon) f'_{S_j}(c_j - \rho_j \varepsilon) \right]. \quad (5.30)$$

Simplification under Equal Biases and Equal Weight Norms To characterize the qualitative behavior of optimal regulation, we assume equal baseline biases $b_\alpha = b_\beta = b_0 \in \mathbb{R}$ and equal sensitivity norms $\|w_\alpha\|_2 = \|w_\beta\|_2 = w_0 > 0$ with $w_\alpha \neq w_\beta$. This assumption implies $c_\alpha = c_\beta = c := -\frac{1}{2}b_0$ and $\rho_\alpha = \rho_\beta = \rho := \frac{1}{2}w_0 > 0$. Recall that the feature space X is convex and compact with $0 \in \text{int}(X)$ and $a_j \neq 0$, therefore each random variable $S_j = a_j^\top X$ has bounded support $[s_{\min,j}, s_{\max,j}]$, where $s_{\min,j} := \min_{x \in X} a_j^\top x < 0$ and $s_{\max,j} := \max_{x \in X} a_j^\top x > 0$. For each boundary sector, the lower integration limit $c - \rho \varepsilon$ sweeps across the support of S_j as ε increases. By Proposition 5.5, expected social welfare saturates once $c - \rho \varepsilon \leq s_{\min,j}$ for all $j \in \{\alpha, \beta\}$, thus $\varepsilon_{\max} = \max_{j \in \{\alpha, \beta\}} ((c - s_{\min,j})/\rho) < \infty$. Therefore, the expected welfare and its derivatives simplify to

$$\mathcal{W}(\varepsilon) = \frac{1}{2} \mathbb{E}[\mu(X)] + \frac{1}{2} \sum_{j \in \{\alpha, \beta\}} \int_{c - \rho \varepsilon}^{s_{\max,j}} \bar{\mu}_j(s) f_{S_j}(s) ds, \quad (5.31)$$

$$\mathcal{W}'(\varepsilon) = \frac{\rho}{2} \sum_{j \in \{\alpha, \beta\}} \bar{\mu}_j(c - \rho\varepsilon) f_{S_j}(c - \rho\varepsilon), \quad (5.32)$$

$$\mathcal{W}''(\varepsilon) = -\frac{\rho^2}{2} \sum_{j \in \{\alpha, \beta\}} \left[\bar{\mu}'_j(c - \rho\varepsilon) f_{S_j}(c - \rho\varepsilon) + \bar{\mu}_j(c - \rho\varepsilon) f'_{S_j}(c - \rho\varepsilon) \right]. \quad (5.33)$$

We now establish a sufficient condition for $\mathcal{W}(\varepsilon)$ to be optimal at an interior point.

Proposition 5.7 (Non-monotonicity of the welfare function): *Let baseline feature space $\mathcal{X} \subset \mathbb{R}^d$ be compact and convex with $0 \in \text{int}(\mathcal{X})$. Assume equal biases $b_\alpha = b_\beta = b_0$, equal sensitivity norms $\|w_\alpha\|_2 = \|w_\beta\|_2 = w_0 > 0$, and marginal densities satisfy $f_{S_j}(s) > 0$ on $(s_{\min,j}, s_{\max,j})$. If the conditional expected valuation functions $\bar{\mu}_j(s) := \mathbb{E}[\mu(X) \mid S_j = s]$ satisfy*

- (i) *Strict monotonicity: $\bar{\mu}'_j(s) > 0$ for all $s \in [s_{\min,j}, s_{\max,j}]$ and each $j \in \{\alpha, \beta\}$,*
- (ii) *Viability at baseline cutoff: $\bar{\mu}_j(c) > 0$ for each $j \in \{\alpha, \beta\}$, where $c = -\frac{1}{2}b_0$,*
- (iii) *Non-viability of lowest baseline features: $\bar{\mu}_j(s_{\min,j}) < 0$ for each $j \in \{\alpha, \beta\}$,*

then the optimal regulatory enforcement policy ε^ lies strictly in the interior of the enforcement domain, $\varepsilon^* \in (0, \varepsilon_{\max})$, satisfying $\mathcal{W}'(\varepsilon^*) = 0$.*

Proof. The proof is given in Section 5.6 (Proof 5.6.1). □

Verification of Non-Emptiness To confirm that the sufficient conditions of Proposition 5.7 are non-empty, consider a 2D market instance with equal baseline biases $b_\alpha = b_\beta = 1$ (so $c = -\frac{1}{2}$) and orthogonal investor sensitivity vectors $w_\alpha = (2, 0)^\top$ and $w_\beta = (0, 2)^\top$, so that $\|w_\alpha\|_2 = \|w_\beta\|_2 = 2$ and $\rho = 1$. The investor valuations are $V_\alpha(x) = 2x_1 + 1$ and $V_\beta(x) = 2x_2 + 1$. Assuming equal mixture probabilities $\gamma_\alpha = \gamma_\beta = 1/2$, the true fundamental expected valuation is the valid convex combination $\mu(x) = \frac{1}{2}V_\alpha(x) + \frac{1}{2}V_\beta(x) = x_1 + x_2 + 1$. The vectors $a_\alpha = (-1, 2)^\top$ and $a_\beta = (2, -1)^\top$ define the scores $S_\alpha = -X_1 + 2X_2$ and $S_\beta = 2X_1 - X_2$, which satisfy $S_\alpha + S_\beta = X_1 + X_2$. We define the feature space as the compact convex domain $\mathcal{X} = \{x \in \mathbb{R}^2 : -3 \leq -x_1 + 2x_2 \leq 3 \text{ and } -3 \leq 2x_1 - x_2 \leq 3\}$ with uniform distribution $X \sim U(\mathcal{X})$. Because this linear transformation maps $\mathcal{X}$ bijectively onto the product domain $[-3, 3]^2$, the variables S_α and S_β are independent uniform random variables on $[-3, 3]$ with support bounds $s_{\min} = -3 < 0 < 3 = s_{\max}$ and saturation threshold $\varepsilon_{\max} = c - s_{\min} = 2.5$. Within each symmetric sector, $\bar{\mu}_j(s) = \frac{1}{2}(s + \mathbb{E}[S_{j'}]) + \frac{1}{2} = \frac{1}{2}s + \frac{1}{2}$. This directly satisfies all three conditions: (i) $\bar{\mu}'_j(s) = 1/2 > 0$, (ii) $\bar{\mu}_j(c) = \bar{\mu}_j(-0.5) = 0.25 > 0$, and (iii) $\bar{\mu}_j(-3) = -1 < 0$. Solving $\mathcal{W}'(\varepsilon^*) = 0$ yields the unique interior optimal enforcement policy $\varepsilon^* = 2.0 \in (0, 2.5)$,

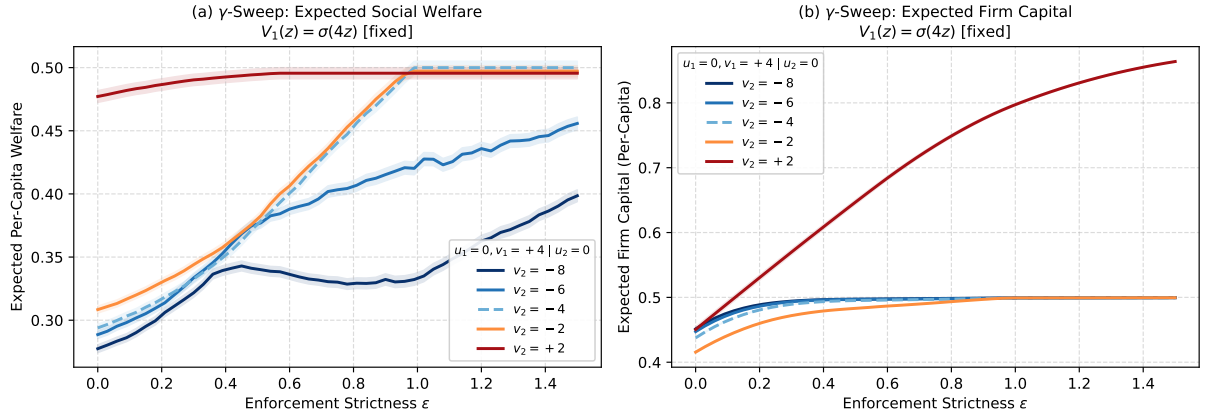


Figure 5.1: Decision problem simulation across the γ -sweep and δ -sweep.

with $\mathcal{W}'''(\epsilon^*) < 0$, confirming strict concavity and global interior maximality.

We extend beyond the linear setting by conducting Monte Carlo simulations (5,000 independent market realizations per enforcement level ϵ , ± 1 standard error bands) with logistic valuation functions:

$$V_j(z) = \sigma(u_j + v_j z), \quad \sigma(s) = \frac{1}{1 + e^{-s}}. \quad (5.34)$$

We fix $N_1 = N_2 = 50$, $\eta = 1$, $X \sim U[-1, 1]$, and Group 1 parameters at $u_1 = 0$, $v_1 = +4$ throughout. Group 2 parameters are varied across two parameter families:

- (i) γ -Sweep (Panels (a) and (b), Figure 5.1): $u_2 = 0$ fixed; $v_2 \in \{-8, -6, -4, -2, +2\}$.
- (ii) δ -Sweep (Panels (c) and (d), Figure 5.1): $v_2 = -4$ fixed; $u_2 \in \{-4, -2, 0, +2, +4\}$.

Figure 5.1 displays the simulation results, illustrating the non-monotonic trajectory of expected social welfare alongside expected firm capital in representative instances, as well as the asymptotic saturation of curves as ϵ increases across all configurations, empirically corroborating our theoretical findings.

5.4 Learning Problem

In this section, we examine the regulatory setting under incomplete information, where the underlying joint market distribution $(X, Y, \{N_j\})$ is unknown to the regulator. We formulate the regulator's search for optimal enforcement tolerance ϵ^* as an online continuum-armed bandit problem. We apply the well-established framework of Gaussian Process Thompson Sampling (GP-TS) [60] to analyze learning performance in our strategic IPO regulation setting.

At each round $t = 1, \dots, T$, the regulator selects an enforcement tolerance $\varepsilon_t \in [0, \varepsilon_{\max}]$ and subsequently observes the realized social welfare $W_t = \mathcal{W}(\varepsilon_t) + \xi_t$, where $\xi_t = W_t - \mathcal{W}(\varepsilon_t)$ represents zero-mean observation noise arising from market realization randomness.

Under Assumption 5.1, the expected social welfare function $\mathcal{W}(\varepsilon)$ is C^∞ over the compact domain $[0, \varepsilon_{\max}]$. Because $C^\infty([0, \varepsilon_{\max}])$ is densely embedded in the Sobolev space $H^3([0, \varepsilon_{\max}])$, which is norm-equivalent to the Reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}_k$ of the Matérn-5/2 kernel (for comprehensive background on Matérn kernels and RKHS properties, see, e.g., [60, 73]):

$$k_{5/2}(\varepsilon, \varepsilon') = \sigma_0^2 \left(1 + \frac{\sqrt{5} |\varepsilon - \varepsilon'|}{l} + \frac{5(\varepsilon - \varepsilon')^2}{3l^2} \right) \times \exp\left(-\frac{\sqrt{5} |\varepsilon - \varepsilon'|}{l}\right), \quad (5.35)$$

it follows directly that $\mathcal{W} \in \mathcal{H}_k$ with a bounded RKHS norm $\|\mathcal{W}\|_{\mathcal{H}_k} \leq B < \infty$. Here $\sigma_0^2 > 0$ is the signal variance and $l > 0$ is the characteristic length-scale parameter (for detailed information about Matérn kernels and RKHS refer [60]).

The regulator employs GP-TS (Algorithm 5.1) to adaptively select ε_t . By placing a GP prior $\mathcal{GP}(0, k_{5/2})$ over $\mathcal{W}$, the regulator maintains posterior mean $\mu_{t-1}(\cdot)$ and covariance $k_{t-1}(\cdot, \cdot)$ updated via standard Gaussian process regression equations over past history $\mathcal{H}_{t-1} = \{(\varepsilon_\tau, W_\tau)\}_{\tau=1}^{t-1}$:

$$\begin{aligned} \mu_{t-1}(\varepsilon) &= k_{t-1}(\varepsilon)^\top (K_{t-1} + \sigma^2 I)^{-1} \mathbf{W}_{t-1}, \\ k_{t-1}(\varepsilon, \varepsilon') &= k_{5/2}(\varepsilon, \varepsilon') - k_{t-1}(\varepsilon)^\top (K_{t-1} + \sigma^2 I)^{-1} k_{t-1}(\varepsilon'), \end{aligned} \quad (5.36)$$

where $k_{t-1}(\varepsilon) = [k_{5/2}(\varepsilon_1, \varepsilon), \dots, k_{5/2}(\varepsilon_{t-1}, \varepsilon)]^\top$, K_{t-1} is the $(t-1) \times (t-1)$ kernel matrix with entries $[K_{t-1}]_{ij} = k_{5/2}(\varepsilon_i, \varepsilon_j)$, and $\mathbf{W}_{t-1} = [W_1, \dots, W_{t-1}]^\top$.

Applying the established regret analysis of GP-TS for kernelized continuum-armed bandits [60] directly specializes the general sublinear regret guarantee to our strategic IPO regulation problem.

Theorem 5.8 (Regret Guarantee for IPO enforcement tolerance): *Under Assumption 5.1, if the regulator employs GP-TS (Algorithm 5.1) with the Matérn-5/2 kernel $k_{5/2}$, the cumulative regret $R_T = \sum_{t=1}^T (\mathcal{W}(\varepsilon^*) - \mathcal{W}(\varepsilon_t))$ is bounded with probability at least $1 - \delta$ by*

$$R_T = \tilde{O}\left(T^{2/3}\right). \quad (5.37)$$

Algorithm 5.1 Gaussian Process Thompson Sampling (GP-TS) for IPO Regulation

-
- 1: **Input:** Matérn-5/2 kernel $k_{5/2}(\cdot, \cdot)$, GP prior $\mathcal{GP}(0, k_{5/2})$, domain $[0, \varepsilon_{\max}]$, horizon T , noise variance σ^2 , confidence $\delta \in (0, 1)$.
 - 2: **for** $t = 1, 2, \dots, T$ **do**
 - 3: Set inflation factor $v_t = B + R\sqrt{2(\gamma_{t-1} + 1 + \log(2/\delta))}$ [60].
 - 4: Compute GP posterior mean $\mu_{t-1}(\varepsilon)$ and covariance $k_{t-1}(\varepsilon, \varepsilon')$ using (5.36) based on history $\mathcal{H}_{t-1}$.
 - 5: Sample function candidate $\widetilde{W}_t \sim \mathcal{GP}(\mu_{t-1}, v_t^2 k_{t-1})$.
 - 6: Select regulatory enforcement tolerance $\varepsilon_t \in \arg \max_{\varepsilon \in \mathcal{D}_t} \widetilde{W}_t(\varepsilon)$ over round- t net $\mathcal{D}_t \subset [0, \varepsilon_{\max}]$.
 - 7: Apply ε_t to the market and observe realized social welfare W_t .
 - 8: Update interaction history $\mathcal{H}_t = \mathcal{H}_{t-1} \cup \{(\varepsilon_t, W_t)\}$ and update posterior parameters.
 - 9: **end for**
-

Proof. The proof is given in Section 5.6 (Proof 5.6.2). □

To empirically validate the learning performance of the GP-TS algorithm in regulatory enforcement design, we simulate the continuous bandit problem across independent trials. We consider an underlying market environment with $k = 2$ investor subpopulations of equal size ($N_1 = N_2 = 50$, $N = 100$, $\eta = 1$), baseline feature $X \sim U[-1/3, 1/3]$, and linear valuation profiles $V_1(z) = 1 + 3z$ and $V_2(z) = 1 - 3z$. The firm's true valuation $Y(X)$ takes value in $\{V_1(X), V_2(X)\}$ according to probabilities $\gamma_1(X)$ and $\gamma_2(X)$, inducing observation noise in realized social welfare W_t ; we evaluate learning performance under two noise conditions: low noise ($\sigma = 0.05$) and high noise ($\sigma = 0.25$). At each round t , after selecting enforcement strictness $\varepsilon_t \in [0, 1.2]$, the regulator observes realized social welfare W_t .

The cumulative regret trajectories (R_T) displayed in Figure 5.2 exhibit sublinear growth bounded by the theoretical $\widetilde{O}(T^{2/3})$ rate established in Theorem 5.8 across both noise levels (shaded bands denote ± 1 standard error across 40 independent trials).

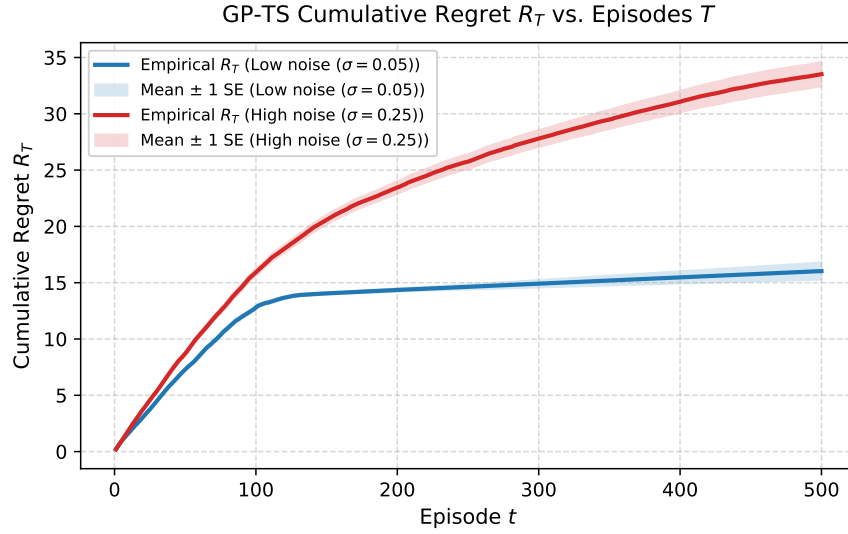


Figure 5.2: Cumulative regret R_T of GP-TS online regulatory learning under low ($\sigma = 0.05$) and high ($\sigma = 0.25$) noise (solid curves show mean cumulative regret, shaded bands indicate ± 1 standard error across 40 trials).

5.5 Conclusion, Limitations, and Future Work

In this chapter, we presented a game-theoretic framework for strategic IPO regulation, modeling the interaction between a welfare-maximizing financial regulator and a capital-maximizing issuing firm capable of feature window-dressing under a regulatory enforcement budget ε .

Under the *Decision Problem*, we established that the issuing firm's optimal presentation and pricing strategy reduces to solving $2^k - 1$ constrained convex optimization subproblems corresponding to candidate buyer subpopulations. We showed that expected social welfare need not be monotone in regulatory tolerance: overly rigid enforcement can prevent viable firms with low perceived valuations at their baseline features from raising sufficient capital, while excessively lax enforcement can enable non-viable firms to mislead investors. An intermediate enforcement level can therefore allow viable firms to improve their market prospects while limiting the ability of non-viable firms to mislead investors.

Under the *Learning Problem*, we modeled online regulatory search as a continuum-armed bandit problem. We established that when the market distribution of $(X, Y, \{N_j\})$ has compact support and admits a smooth probability density function with bounded density and derivatives, the expected welfare function belongs to the Reproducing Kernel Hilbert Space (RKHS) of a Matérn-5/2 kernel. Using the established Gaussian Process Thompson Sampling (GP-TS)

algorithm, we obtained a provably sublinear cumulative regret of $\tilde{O}(T^{2/3})$.

A key limitation of our formulation is the assumption of naive face-value investors who purchase based directly on their personal valuation of the presented features. Promising directions for future work include considering more sophisticated investors and richer dynamics in which the regulator and the issuing firm adaptively learn the market distribution and respond strategically to one another over time.

5.6 Proofs and Supporting Results

This section provides the proofs for the theoretical results presented in this chapter.

5.6.1 Proofs for the Decision Problem

In this subsection, we provide the proofs of the theoretical results for the decision problem.

We now provide the proof of Lemma 5.3.

Proof of Lemma 5.3: Optimal Pricing Strategy. For any fixed presentation feature z , the valuation of each investor subpopulation j is $V_j(z)$. The aggregate market demand $D(z, p, \{N_j\}) = \sum_{j=1}^k N_j \mathbf{1}\{V_j(z) \geq p\}$ is a piecewise-constant step function that is non-increasing in price $p \geq 0$, with jump discontinuities occurring exclusively at price points $p \in \{V_1(z), \dots, V_k(z)\}$. If all investor valuations are strictly negative ($\max_j V_j(z) < 0$), no buyer enters at any non-negative price $p \geq 0$, yielding zero demand and zero capital (the no-trade option). If at least one subpopulation has non-negative valuation, the expected traded volume $\mathbb{E}_{\{N_j\}}[Q(z, p, \{N_j\}, \varepsilon)]$ is constant on each interval $(v_m, v_{m+1}]$, where v_m and v_{m+1} are adjacent values in the sorted set $\{0\} \cup \{V_j(z) : V_j(z) \geq 0\}$. The firm's expected capital raised $p \cdot \mathbb{E}_{\{N_j\}}[Q]$ is strictly increasing on (v_m, v_{m+1}) , so the supremum on any such interval is attained at its right endpoint. Hence, the optimal non-negative price satisfies $p^*(z) \in \{0\} \cup \{V_j(z) : V_j(z) \geq 0\}$. $\square$

We now provide the proof of Theorem 5.4.

Proof of Theorem 5.4: Firm's Optimal Presentation Strategy. For any target subset $S \subseteq \{1, \dots, k\}$, setting price $p_S(z) = \min_{j \in S} V_j(z)$ ensures participation of at least all groups in S , establishing

a lower bound $\Pi_F^{(S)}(z) \leq \Pi_F(z, \min_{j \in S} V_j)$ given by

$$\Pi_F^{(S)}(z) = \left(\min_{j \in S} V_j(z) \right) \cdot \mathbb{E}_{\{N_j\}} \left[\min \left\{ \eta \mathcal{N}, \sum_{j \in S} N_j \right\} \right]. \quad (5.38)$$

At the optimal strategy (z^*, p^*) , Lemma 5.3 guarantees $p^* = V_{m^*}(z^*)$. For the true buyer set $S^* = \{j : V_j(z^*) \geq p^*\}$, market demand equals $\sum_{j \in S^*} N_j$ exactly, making the lower bound tight, i.e., $\Pi_F^{(S^*)}(z^*) = \Pi_F(z^*, p^*)$. Because $\Pi_F^{(S)}(z) \leq \max_{z,p} \Pi_F(z, p)$ for all S and equality is achieved at (S^*, z^*) , maximizing these lower bounds over all S and z recovers the exact global maximum. Since each $\Pi_F^{(S)}(z)$ is concave in z (as the pointwise minimum of concave functions), evaluating $\max_z \Pi_F^{(S)}(z)$ over $z \in \mathcal{F}_Z(x, \varepsilon)$ defines a constrained convex optimization subproblem. Therefore, solving these $2^k - 1$ convex subproblems and selecting the maximum yields the global optimum. $\square$

We now provide the proof of Proposition 5.5.

Proof of Proposition 5.5: Welfare Saturation Threshold. Consider the unconstrained presentation problem where the firm faces no regulatory budget constraints ($\varepsilon = \infty$), yielding optimal presentation $z_\infty^*(x) = \arg \max_{z \in \mathcal{X}} \max_S \Pi_F^{(S)}(z, \infty)$. The minimal regulatory budget required to accommodate this unconstrained move is $\varepsilon_{\max}(x) = \|x - z_\infty^*(x)\|_2$. Because the baseline feature space $\mathcal{X}$ is a compact subset of $\mathbb{R}^d$, the continuous distance mapping $\varepsilon_{\max}(x)$ achieves a finite supremum $\varepsilon_{\max} = \sup_{x \in \mathcal{X}} \varepsilon_{\max}(x) < \infty$. For any tolerance $\varepsilon \geq \varepsilon_{\max}$, the unconstrained optimal presentation $z_\infty^*(x)$ lies within the feasible space $\mathcal{F}_Z(x, \varepsilon)$ for every realization $x \in \mathcal{X}$. Consequently, the firm's optimal presentation and pricing strategies $(\mathcal{Z}^*(x, \varepsilon), \mathcal{P}^*(x, \varepsilon))$ and expected social welfare $\mathcal{W}(\varepsilon)$ remain constant for all $\varepsilon \geq \varepsilon_{\max}$, restricting the optimization domain to $[0, \varepsilon_{\max}]$ without loss of generality. $\square$

We now provide the proof of Proposition 5.7.

Proof of Proposition 5.7: Non-Monotonicity and Interior Optimality. Evaluating $\mathcal{W}'(\varepsilon)$ at $\varepsilon = 0$ yields $\mathcal{W}'(0) = \frac{\rho}{2} [\bar{\mu}_\alpha(c) f_{S_\alpha}(c) + \bar{\mu}_\beta(c) f_{S_\beta}(c)]$. Since $\rho > 0$, $f_{S_j}(c) > 0$, and $\bar{\mu}_j(c) > 0$ by condition (ii), it follows that $\mathcal{W}'(0) > 0$, showing that social welfare strictly increases locally at $\varepsilon = 0$. Next, since $\varepsilon_{\max} = \max_{j \in \{\alpha, \beta\}} ((c - s_{\min, j})/\rho)$, as $\varepsilon \rightarrow \varepsilon_{\max}^-$, we have $c - \rho\varepsilon \rightarrow s_{\min, j}^+$. Because $\bar{\mu}_j(s_{\min, j}) < 0$ by condition (iii), $f_{S_j}(s) > 0$ on $(s_{\min, j}, s_{\max, j})$, and $\rho > 0$, taking the left-hand limit yields $\mathcal{W}'(\varepsilon_{\max}^-) < 0$, showing that social welfare strictly decreases locally at

$\varepsilon = \varepsilon_{\max}$. Because $\mathcal{W}'(\varepsilon)$ is continuous on the compact interval $[0, \varepsilon_{\max}]$ with $\mathcal{W}'(0) > 0$ and $\mathcal{W}'(\varepsilon_{\max}^-) < 0$, the Intermediate Value Theorem guarantees the existence of an interior stationary point $\varepsilon^* \in (0, \varepsilon_{\max})$ satisfying $\mathcal{W}'(\varepsilon^*) = 0$. Since the boundary points $\varepsilon \in \{0, \varepsilon_{\max}\}$ yield strictly lower welfare than nearby interior points, the global maximum on $[0, \varepsilon_{\max}]$ must be attained at an interior policy $\varepsilon^* \in (0, \varepsilon_{\max})$. Furthermore, condition (i) ensures strict monotonicity $\bar{\mu}'_j(s) > 0$, guaranteeing the second-order sufficient condition for a maximum, $\mathcal{W}''(\varepsilon^*) = -\frac{\rho^2}{2} \sum_j \bar{\mu}'_j(c - \rho\varepsilon^*) f_{S_j}(c - \rho\varepsilon^*) < 0$. $\square$

5.6.2 Proofs for the Learning Problem

In this subsection, we provide the proof of the regret bound for the regulatory learning problem.

We now provide the proof of Theorem 5.8.

Proof of Theorem 5.8: Regret Bound for Regulatory GP-TS. Under Assumption 5.1, the expected social welfare function $\mathcal{W}(\varepsilon)$ is smooth on $[0, \varepsilon_{\max}]$ and belongs to the RKHS $\mathcal{H}_k$ of $k_{5/2}$ with bounded norm $\|\mathcal{W}\|_{\mathcal{H}_k} \leq B < \infty$. Furthermore, because traded share volume is bounded by total supply $\eta\mathcal{N}$ and valuations are bounded in magnitude by $Y_{\max}$, realized social welfare satisfies the symmetric envelope $|W_t| \leq \eta Y_{\max}$. Consequently, the observation noise $\xi_t = W_t - \mathcal{W}(\varepsilon_t)$ is conditionally R -sub-Gaussian with $R = \eta Y_{\max}$ by Hoeffding's Lemma.

Applying the GP-TS regret bound of [60] on the 1D compact domain $[0, \varepsilon_{\max}]$, cumulative regret satisfies $R_T = O(B\sqrt{T\gamma_T} + \gamma_T\sqrt{T})$, where γ_T is the maximum information gain for kernel $k_{5/2}$. For the Matérn-5/2 kernel on a 1D compact domain ($d = 1, \nu = 5/2$), the maximum information gain is bounded by $\gamma_T = \tilde{O}(T^{\frac{d}{2\nu+d}}) = \tilde{O}(T^{1/6})$ [61]. Substituting $\gamma_T = \tilde{O}(T^{1/6})$ yields the sublinear cumulative regret bound $R_T = \tilde{O}(T^{2/3})$. $\square$

Chapter 6

Conclusion and Future Directions

This dissertation extends the classical framework of statistical learning to settings where the data-generating process is influenced by strategic agents or where which variables are jointly observed departs from the traditional fixed-feature baseline. Together, these settings show that the performance of a learning system depends not only on the underlying environment, but also on the strategic incentives present in that environment and the structural composition of the observations through which it is perceived. Modern data-driven decision-making systems must therefore account for both strategic incentives and observational structure as integral components of the learning problem.

Across the strategic learning problems considered in this dissertation, a recurring theme is that strategic behavior need not always be viewed as an obstacle to be eliminated. When explicitly modeled, strategic behavior can reveal useful private information and induce structural properties that simplify the underlying decision problem. For example, in strategic feature revelation (Chapter 4), the sender’s voluntary disclosure decision conveys information that, under non-erasure conditions, strictly improves predictive accuracy relative to the non-strategic baseline. Similarly, in IPO regulation (Chapter 5), permitting a bounded degree of presentation flexibility can align investor participation and strictly improve traded social welfare relative to complete prohibition or unregulated manipulation.

In the learning from partitioned observations problem (Chapter 3), which represents a structural rather than strategic departure, predictive performance depends fundamentally on which variables are observed simultaneously. When statistical dependencies between features and targets are never jointly observed during training, fundamental limitations arise on achievable

risk. Within these limitations, the choice of imputation strategy governs learning performance. We show that, under specified environmental conditions, the regularizer that minimizes the theoretical bound on excess risk follows a three-candidate rule, settling at an interior value or taking one of two endpoints depending on the signal-to-imputation ratio.

The models studied throughout this dissertation are intentionally kept analytically tractable to expose transparent economic and statistical mechanisms. Although these stylized assumptions necessarily scope the immediate domain of application, they enable clean structural characterizations that provide useful principles for more complex systems.

Several directions remain open for future research. Beyond extending the proposed models to richer valuation and cost structures, an important direction is to explore non-stationary environments where agent preferences or observation partitions evolve dynamically over time. Such investigations naturally lie at the intersection of statistical learning and algorithmic game theory, with the goal of developing algorithms that are statistically efficient, robust to manipulation, and aware of observational constraints.

In conclusion, this dissertation demonstrates that statistical learning can be rigorously extended beyond its classical assumptions by explicitly modeling strategic incentives and partitioned observation structures. Doing so not only clarifies the fundamental limitations imposed by these departures, but also reveals new opportunities for designing decision systems that effectively exploit the underlying game-theoretic and structural properties of the problem.

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